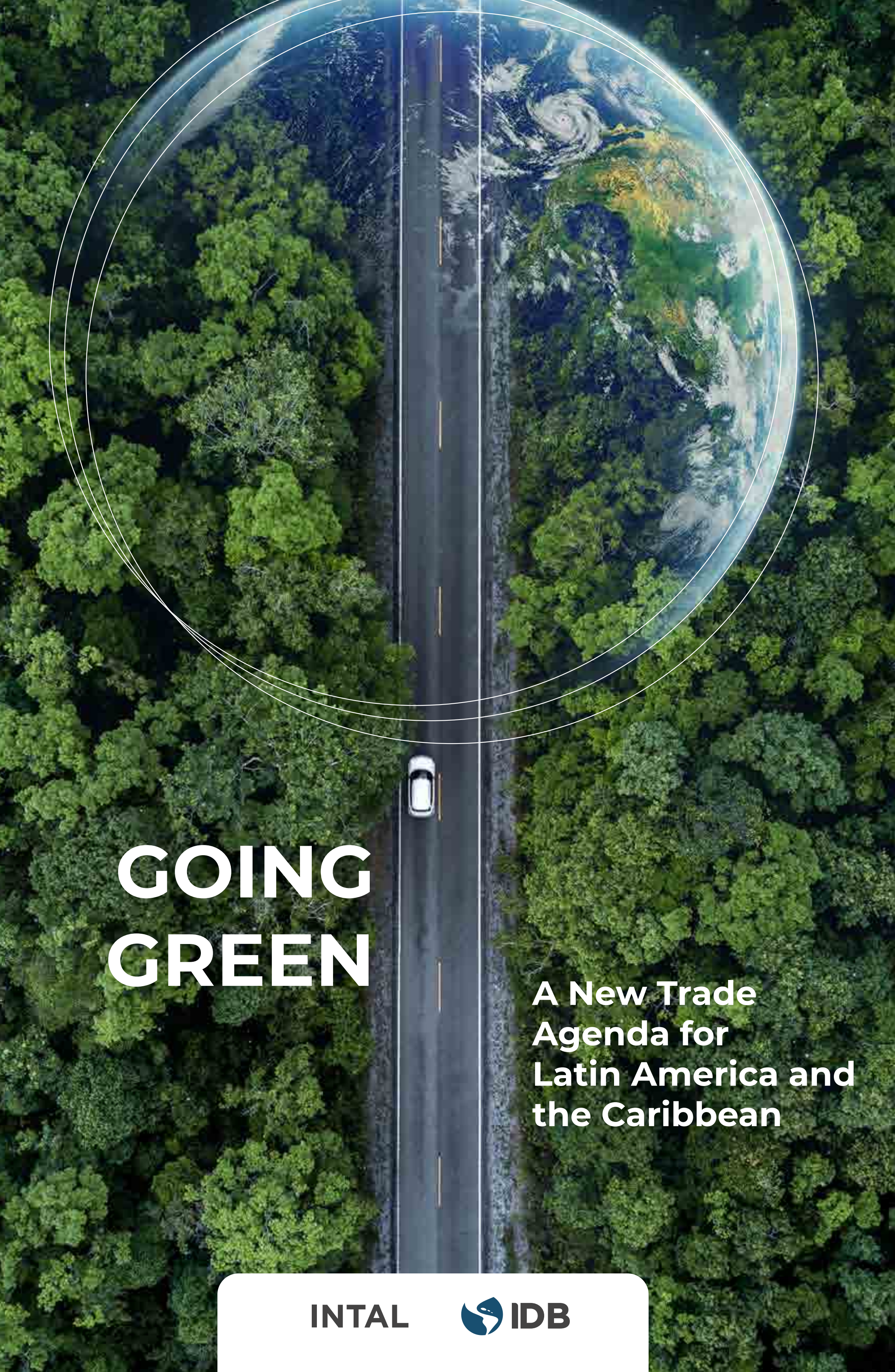

GOING GREEN

A New Trade Agenda for Latin America and the Caribbean

INTAL IDB

GOING GREEN

A New Trade Agenda for Latin America and the Caribbean

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INTRODUCTION

International trade is an essential part of the journey toward green, resilient, inclusive development in the coming years. Trade can contribute to reducing the impacts of climate change and related adaptation and mitigation efforts. It can also help countries achieve the Sustainable Development Goals (part of the 2030 Agenda for Sustainable Development) and meet the national commitments taken on under the Paris Agreement (Brenton and Chemutai, 2021).

Latin America and the Caribbean (LAC) is facing significant challenges in relation to environmental issues and trade. External trade flows account for 19.8% to 26.5% of the region's total emissions, depending on whether household emissions and land use are included in the calculation (Dolabella and Mesquita Moreira, 2022). Climate change is already affecting the region's export supply due to droughts and extreme climate events, which impact crops, yields, and agricultural and livestock production. Cavallo, Becerra, and Acevedo (2021) estimate that these impacts represent a loss of 3% of LAC's GDP. Planning and implementing initiatives to respond early to climate contingencies—such as the effects of El Niño (flooding) or La Niña (droughts)—could help reduce volatility and the negative impact of adverse climate factors on food security (FAO, 2023; Davis et al., 2021).

To further decouple emissions from production and international trade and reduce the carbon intensity of LAC's economies and trade, appropriate incentives need to be created regarding what goods and services are produced, distributed, and consumed and how this is done. In short, environmental issues need to be actively addressed to reconcile the need for sustainable, inclusive growth and global integration with the imperative to adapt to and mitigate negative impacts on production and the climate.

During the transition to greener growth, trade could help create new comparative advantages in LAC countries through the use of cleaner technologies, energy sources, and processes (Ramos and Chisari, 2021); the development and exchange of environmental goods and services; and by making greater use of the potential of the circular economy, among other initiatives. There are numerous opportunities for LAC in this area. One such example is making the most of the region's energy matrix, which is one of the cleanest in the world, as almost 30% of its sources are renewable, far higher than the world average (14%). Another latent opportunity is the fact that LAC is the world's largest producer of environmental services (Dolabella and Mesquita Moreira, 2022).

Similarly, LAC countries have significant mining capacity and reserves of minerals that are critical to the energy transition. Indeed, LAC is one of the world's leading exporters of silver (accounting for 60% of global exports), copper (59%), and lithium (37%), among other resources. It also has large deposits of minerals that are essential for developing green technologies, including wind turbines, photovoltaic panels, batteries and motors for electric vehicles, and electrolyzers for producing green hydrogen (INTAL IDB, 2023).[1]

Trade policy has been called on to play a major role in the climate and environmental crisis, which has become one of the most pressing public policy challenges of our time (Branstetter and Pizer, 2013). In response, many countries have adopted environmental measures that directly affect international trade, including border taxes related to carbon content (Bellora and Fontagné, 2023), technical regulations, labeling requirements, provisions on containers and packaging, performance standards, goods transportation requirements, and private standards (Dolabella and Mesquita Moreira, 2022). Such measures are permitted, provided that they do not create unnecessary barriers to trade.

More recently, measures have also been introduced to limit trade in products from deforested areas. For example, the new European Union (EU) environmental regulations prohibiting products from deforested areas from accessing the EU market would impact LAC exports more than the implementation of a border carbon adjustment mechanism. The share of LAC exports to the EU covered by the deforestation regulation is equivalent to 2% of the region's total sales to the world; the figure affected by border carbon taxes is significantly lower (0.11%) (INTAL IDB, 2023).

Conversely, policies could seek to reduce trade barriers, as can be seen in discussions at the World Trade Organization on eliminating tariffs on environmental goods, eliminating fossil fuel subsidies, and promoting the transfer of new, less polluting production technologies.

Against this backdrop, in December 2022, INTAL IDB and the Interdisciplinary Institute of Political Economy at the University of Buenos Aires (IIEP, UBA/

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[1] ·Nevertheless, further development of the value chains associated with these products would require significant investment (Obaya, Ramos, and Romero, 2022).

CONICET) launched a call for research papers to expand knowledge on trade and the environment in LAC. The call sought to generate relevant data and analyses that would inform decision-making in LAC's public and private sectors.

The call was addressed to individual researchers and research teams from universities, research centers, and think tanks in LAC. The research areas open for proposals were: a) energy transition and international integration; b) trade policy and environmental distortions; and c) environmental goods and services and productive/export supply.

Proposals could cover a specific country, several countries, a subregion (e.g., the Andean Community, MERCOSUR, the Caribbean Community, the Central American Integration System), or LAC as a whole. The call included one area for researchers in general and another for young researchers (those under 35 years of age at the time of submission).

The call was open through February 2023, and 235 proposals were received, six of which were selected by the jury (from the general category).[2] The jury members were Ana Basco (Director, INTAL IDB), Mauricio Mesquita Moreira (Senior Economic Advisor, INT IDB), Andrés López (Director, IIEP UBA/CONICET), Ricardo Rozemberg (Senior Integration and Trade Associate, INTAL IDB), Martina Chidiak (Senior Researcher, IIEP UBA/CONICET), Johan Mulder (Head of International Trade Unit, ECLAC), Rodrigo Rodríguez Tornquist (Climate Change Specialist, UNSAM), and Juliana Almeida (Senior Climate Change Specialist, IDB).

The studies presented here were written between April and September 2023 and aim to contribute to the challenging environmental and trade agendas, providing novel empirical and conceptual evidence on key issues for promoting this agenda at the regional level with a view to generating relevant policy proposals and recommendations for LAC countries.

The team from the University of Chile (Marco Schwartz, Werther Kern, Ricardo Marchant, and Francisco Mango) focused their research on the potential for using fruit residues as inputs for the Chilean food industry. The authors identified exportable food products from the valorization of waste from the apple and olive oil processing industries (pomace and orujo, respectively), notably in cereal-based snacks and yogurts. These products

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2 · The area of the call targeting young researchers was declared deserted.

seem to be the most promising for Chile and have a potential export market of US$236 million. Developing the circular economy in this way would make a small contribution to meeting Chile's environmental commitments.

The project by a team from Fundar Argentina (Jimena Calvo, Victoria Arias Mahiques, María Fernanda Villafañe, Pablo de la Vega, Leonardo Park, Ángeles Sancisi, and Verónica Gutman) analyzes the potential impact of the EU's new Deforestation Regulation on Argentinian exports. The study found that approximately 40%[3] of Argentinian exports of goods to the EU (which represent 5% of the country's total global exports) would be affected by the new regulation. This impact is almost entirely explained by soybean and beef exports. Using a general equilibrium model, the authors estimate that GDP could fall by between 0.15% and 0.26% if the country's exports do not comply with the new regulation. The paper ends with policy recommendations to ensure compliance with the regulation and strengthen Argentina's institutions and international integration.

The paper by the team at the Getúlio Vargas Foundation in Brazil (Rodrigo Fagundes Cezar, Juliana Camargo, and Yixian Sun) analyzes the impact of voluntary environmental certifications derived from trade regulations on gender equality. Specifically, the study focuses on how these have impacted the gender gap in rural property ownership in different Brazilian municipalities since the implementation of private certification schemes for sugar and ethanol production (known as BONSUCRO schemes), which certify compliance with European biofuel standards (Renewable Energy Directive, RED). Based on econometric analysis and using data from agricultural censuses, the results show that environmental/trade certification schemes of this sort do not have a significant impact on gender equality in rural Brazilian property ownership over time. Consequently, the authors propose that this social dimension (gender equality and inclusion) be included in sustainability certificates. They also advocate for the promotion of synergies between regulations that favor family farming and women's participation and voluntary trade regulation schemes.

The study coordinated by researchers at Argentina's National Institute of Agricultural Technology (INTA) (Silvina Cabrini, Estela Cristeche, Ignacio Amaro, Claudia Faverin, Laura Gastaldi, Ignacio Pace Guerrero, Demián Olemberg, Marcela Piperata, Paulo Recavarren, Andrés Said, María Paz Tieri, and Ruy Vidal) evaluates the potential impact of the implementation of an EU border carbon adjustment mechanism on international beef trade, with a special focus on Argentinian exports. To this end, the authors use the CLIMTRADE partial equilibrium model to simulate the impact of these regulations based on different price scenarios and mitigation measures. The results show a reduction of between 8% and 7% in Argentinian beef exports, given the current emissions intensity, which could be reduced by 20% under a mitigation scenario that improves the efficiency of Argentinian livestock systems. The recommendations from this study suggest that, regardless of the current uncertainty around the possible application of these measures to EU beef imports, concrete public–private actions are rapidly needed to promote climate-smart livestock production. These should be based on technologies and practices that improve production efficiency, reduce methane emissions, and promote carbon sequestration, accompanied by evaluation and traceability processes that allow the certification of environmental performance. This will make it possible to monitor the emissions generated by livestock production and promote net emissions calculations, taking into account carbon sequestration in livestock systems.

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[3] · Although the document states that a large proportion of the goods in question originate in provinces where deforestation has not taken place, this would not eliminate the cost of proving the origin of production, which the EU regulation may require. There is also uncertainty as to whether the regulation will consider the country as a whole or will origins to be broken down into subnational units.

The paper by the team from the Autonomous University of Campeche, Mexico (Ileana Canepa Pérez, Salvador Meneses Requena, Ricardo Dzul Caamal, Randall Góngora García, and Adán Martínez Cruz) examines the feasibility of implementing a honey traceability system in Mexico. Specifically, it estimates the price beekeepers in the state of Campeche could expect to receive on the international market if their honey was produced using environmentally sustainable practices and a system that guarantees product traceability. To do so, the authors use a discrete choice experiment that allows them to estimate the price premium that beekeepers would consider sufficient to compensate for changes in production practices. The results indicate that beekeepers in Campeche would be willing to switch to production practices that preserve the environment and biodiversity and participate in a traceability system in exchange for a premium equivalent to slightly less than US$2 per kilogram of honey.

Finally, an article by the INTAL IDB team analyzes the challenges facing LAC in relation to the growing use of carbon leakage mitigation measures in international trade. These include environmental provisions in trade agreements and unilateral measures such as bans on imports from deforested areas and border carbon adjustment mechanisms.

We believe this body of work contributes significantly to the trade and investment policy agenda for sustainable growth in LAC. The region has made progress on policies to reduce greenhouse gas emissions and policies to mitigate the negative impacts of these. All the same, there are major outstanding challenges in this area that relate to further expanding the international integration of these economies. The studies presented here aim to inform the design of policies to improve LAC export performance based on increasingly widespread environmental sustainability criteria. Developing and promoting a positive environmental agenda at the sector level will be fundamental to putting exports at the heart of a modern strategy for sustainable productive development in LAC.

María Priscila Ramos (IIEP UBA/CONICET)
and **Ricardo Rozemberg** (INTAL IDB)

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CHAPTER 1

THE CIRCULAR ECONOMY AND CHILE'S AGROINDUSTRIAL EXPORT POTENTIAL

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ACRONYMS AND ABBREVIATIONS

- **CH$_4$**
methane

- **CO$_2$**
carbon dioxide

- **ECLAC**
Economic Commission for Latin America and the Caribbean

- **FLW**
food losses and waste

- **GDP**
gross domestic product

- **GHGs**
greenhouse gases

- **HS**
Harmonized System

- **IPCC**
Intergovernmental Panel on Climate Change

- **ISIC**
International Standard Industrial Classification

- **LAC**
Latin America and the Caribbean

- **MtCO$_2$eq**
million metric tons of carbon dioxide equivalent

- **NDCs**
nationally determined contributions

- **NRCA**
normalized revealed comparative advantage

- **VSs**
volatile solids

ABSTRACT

This study focuses on the circular economy within agrifood chains and its relationship with sustainable trade. Its objectives are to identify food products suitable for export that derive from the valorization of waste from the apple and olive oil processing industries in Chile (pomace and orujo, respectively), and to estimate the environmental benefits that would come from developing these businesses through reductions in greenhouse gas emissions. A synthetic trade indicator based on the methodology of Hidalgo et al. (2007) is used to identify food products that contain dietary fibers from agroindustrial residues. The application of this tool revealed cereal-based snacks and yogurts to be the most promising such products for Chile, as they may yield a potential market of US$236 million. Developing the circular economy this way would contribute 0.03% to Chile's nationally determined contributions and 0.08% to its methane reduction commitments.

1. INTRODUCTION

Agriculture plays a critical role in Latin America and the Caribbean's (LAC) contribution to climate change. The region accounts for 7% of world GDP and but is responsible for 20% of global agricultural greenhouse gas (GHG) emissions agricultural emissions. This is partly due to LAC's factor endowment, which is intensive in land and natural resources, and to widespread farming practices like deforestation, intensive livestock production, and pesticide use (FAO, 2023).

However, food losses and waste (FLW) also account for a major share in the region's emissions. This term refers to the "the edible part of food that is lost or wasted" along the value chain, including the production, distribution, and consumption stages (Gustavsson, Cederberg, and Sonesson, 2011). **According to data from the FAO (2023), 7.8% of LAC's total GHG emissions in 2019 were due to FLW, three times more than the world average.**

Given this context, the circular economy is a business model that can contribute systemically to reducing LAC's emissions. The circular economy is an entropic system that promotes the recycling, recovery, and reuse of waste in a continuous production cycle, extending the life cycle of materials and promoting sustainable development. Through this model, the large quantities of waste plant matter generated by the LAC agrifood industry—such as grape pomace, orange bagasse, orujo from the olive oil industry, wheat bran, coffee, and pomace—can be reused to obtain fiber. This has high nutritional value and can be consumed directly as a flour or added to other foods such as beverages, dairy or meat products, or baked goods (Waldbauer, McKinnon, and Kopp, 2017).

LAC's productive specialization puts it in the ideal position to leverage global awareness of sustainable development and healthy eating. The region exports 55% of its value-added in food products, generating the largest trade surplus in the world—an annual US$150 billion (UN Comtrade). Businesses that valorize FLW are an opportunity for LAC to start building up green trade circuits.

This chapter aims to identify opportunities for the use and commercialization of wastes from the apple and olive oil processing industries in Chile to promote export diversification and reduce the environmental impact of FLW. To achieve this, we designed a sequential three-stage methodology to estimate the GHG emission reductions that Chile would achieve if it moved in the proposed direction. First, based on Hidalgo et al.'s (2007) concept of proximity, we will try to identify the foods most likely to incorporate fiber from food waste and enter the international trade cycle. Second, we estimate the potential market for the Chilean products that are most likely to enter the circular economy, based on the standardized comparative advantages in Yu, Cai, and Leung (2009). Finally, drawing on existing technical studies on the environmental impact of the reuse of waste from the apple and olive oil processing industries (pomace and orujo), we will estimate the potential reductions in GHG emissions. The few existing studies on this topic are highly susceptible to environmental conditions, so these estimates should be approached cautiously and taken as a preliminary exercise to be expanded in future research.

This chapter is structured as follows. Section 2 introduces the concept of the circular economy and its connections with the environment and trade. Section 3 examines the importance of the Chilean agrifood industry and FLW and looks at recent developments in the sector. Section 4 describes our methodology and results. Sections 5 and 6 contain conclusions and policy recommendations, respectively.

2. THE CIRCULAR ECONOMY

There is no single theoretical framework for the circular economy as a development model. It is based on different interpretations of development objectives that emphasize its contributions to society. The Economic Commission for Latin America and the Caribbean (ECLAC) describes the circular economy as a model that allows economic activity to be decoupled from resource use and waste generation while promoting new business and employment models (CEPAL, 2021). **In short, the circular economy is a production, distribution, and consumption model in which products, materials, and other resources remain valuable for as long as possible and are used sustainably and efficiently, reducing waste generation** (Ghosh, 2020).

The term "circular bioeconomy" is often associated with biowaste, which brings together the concepts of bioeconomy and the circular economy, and seeks to convey an economically, socially, and environmentally sustainable model (Carus and Dammer, 2018). The bioeconomy brings together all economic activities involved in the production, transformation, and direct or indirect use of resources of biological origin to produce and transform biomass into food supply, feed, materials, energy, and services.

The areas of overlap between the bioeconomy and the circular economy include better use of resources, ecoefficiency, reduced carbon footprints, lower demand for fossil fuels, and waste valorization (Carus and Dammer, 2018). Thus conceived**, the circular bioeconomy is a paradigm that orients business decisions toward the reuse of organic waste to create new products** (which may be original or may complement the product whose waste they derive from) that contribute to producing food for a growing population. This approach mitigates the effects of climate change and reduces dependence on hydrocarbons (Manniello et al., 2020).

The circular bioeconomy can be conceived of as a subset of the circular economy. The latter includes not only materials from the agricultural sector but also minerals, metals, and other abiotic

inputs. Under this paradigm, the technological and economic focus is on transforming waste from production processes to mitigate the pollution that these cause (George, Lin, and Chen, 2015). **It is a business model based on the reuse of waste** (Pieroni et al., 2019), **which can contribute to enhancing firms' productivity, competitiveness, and interdisciplinary initiatives**, thus helping them develop cleaner production processes (Beluhova-Uzunova, Shisshkova, and Ivanova, 2019).

The circular economy brings value to related industries through its role in product development and innovation. In this sense, society implicitly values the circular economy since it can help reduce the negative externalities caused by industries that generate environmentally harmful waste (Nuñez-Cacho et al., 2018). This perspective argues for the reduction, recycling, reuse, recovery, and/or valorization of waste. Once processed, this waste is of greater value to society (Sariatli, 2017), which acknowledges its positive effects and economic benefits for industry.

Generally speaking, the noncircular production model generates significant amounts of waste even when it is limited to certain sectors. This is partly true for the agrifood industry, where waste is rife and sustainability has been further aggravated by the industry's effects on climate change and biodiversity loss (Jurgilevich et al., 2016). These factors make it imperative for supply chains to adapt to the circular economy (Esposito et al., 2020). Ju et al. (2017) posit that circular initiatives that decrease food losses increase the population's consumption efficiency. Secondi et al. (2019) argue that it is possible to reuse and thus revalue 85% of the waste from the tomato industry. Meanwhile, other authors propose using waste from the olive oil industry to obtain biogas (Lovrak et al., 2022). Likewise, grape pomace from the winemaking industry can be used by extracting the colorant anthocyanidin and transforming it into a concentrated aqueous extract or microencapsulated powder for use as an ingredient in the food industry (Schwartz, Muñoz, and Sepulveda, 2002).

These examples illustrate that waste from the agrifood industry can be used as raw materials for production processes within the logic of the circular economy (Jurgilevich et al., 2016). Given the above, a high-impact, scalable innovation ecosystem needs to be consolidated, along with mechanisms to promote entrepreneurship and business development in order to implement initiatives to valorize these wastes and transform them into economic and social assets (Donner et al., 2021).

The move toward circular economy models is expected to significantly impact the structure of global value chains and the dynamics of international trade. Yamaguchi (2021) notes that new interrelationships include the creation of circular supply chains for raw materials, intermediate goods, and final products, as well as increased trade in end-of-life goods such as waste and scrap, secondary raw materials, and second-hand goods. Trade in circular services also has a key role to play (Tamminen et al., 2020; Yamaguchi, 2021).

The essence of the concept of "circular trade" is that increased free trade will help to ensure that goods and materials can be moved around the world to be reused, repaired, recycled, or valorized efficiently and viably, especially in the countries that offer the greatest competitive advantages (Mulder and Albaladejo, 2021). In addition to creating and expanding circular trade flows, the transition to this model will likely have broader implications for the global trade system. For example, as countries become more autonomous in terms of resources, the geopolitical landscape will shift. Examples of such changes include the increased risk of trade protectionism in relation to critical raw materials through tariff or nontariff barriers (Kettunen et al., 2019) and the potential for low-income countries to become recipients of toxic and low-value waste (Cotta, 2020; Qu, 2019). At the same time, circular

economy strategies could reduce dependence on raw materials, most of which come from developing countries.

For all of these reasons, **the nexus between international trade and the circular economy is garnering increasing attention in both the academic and global policy spheres**. However, despite the vital role that international trade is said to play in the transition to a circular economy, **there is only limited knowledge on the link between the two** (Hofstetter et al., 2021) **and few comprehensive studies that shed light on these issues** (Yamaguchi, 2021).

Although the circular economy has been introduced into public policy agendas and private activities through public competitive funding, legislation, and incentives for food collection companies to implement recycling and/or recovery activities in several LAC countries and other regions of the world, the link between the circular economy and international trade remains relatively unexplored (Mulder and Albaladejo, 2021).

3. WASTE GENERATION IN THE CHILEAN AGRIFOOD INDUSTRY

Agrifood is one of Chile's largest industries, generating approximately 10% of annual GDP and 1.5 million jobs). The sector also produces large amounts of organic waste, including leftover fruit, vegetables, meat, fish, and other food products.[1] **These wastes are often used as organic crop fertilizers and raw material for composting**, resulting in a stabilized product with heightened potential as a fertilizer and organic soil amendment. **However, if processing conditions are not appropriate, methane, a GHG, is produced.**

FLW is a particularly significant issue in Chile. Its agricultural emissions are lower than those of neighboring countries, as the sector is less oriented toward livestock and extensive farming. However, Chile's share in LAC emissions doubles if FLW is included (figure 1) and triples if composting is also taken into account (FAO, 2023). Another way of illustrating the problem is to compare how many million metric tons of CO_2 equivalent ($MtCO_2eq$) are emitted from FLW for every 100 $MtCO_2eq$ of GHGs generated in the agricultural sector.[2] While in LAC and globally this share is 21 $MtCO_2eq$, in Chile it is 51 $MtCO_2eq$.

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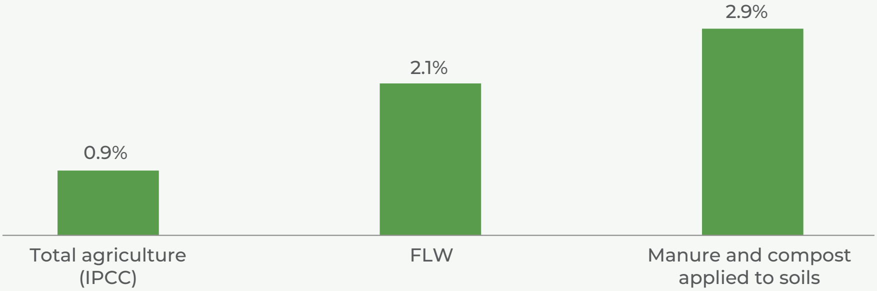
[1] · The Chilean agrifood industry is working to reduce the amount of waste it produces. Some of the measures taken include the development of new technologies for composting and converting organic waste into energy, the use of biodegradable food packaging, the reduction of agrochemical use, and the creation of support programs related to circular economy and clean production practices (CORFO, 2023).

[2] · We arrived at this estimate by taking into account that not all FLW emissions are deemed to be agricultural emissions by the IPCC inventory methodology. For example, emissions generated by transportation, distribution, and consumption are not counted as agricultural emissions. It is thus not possible to calculate the exact share of FLW in Chile's total agricultural emissions.

Figure 1.

Chile's Share (%) in LAC's GHG Emissions, by Source (2019)

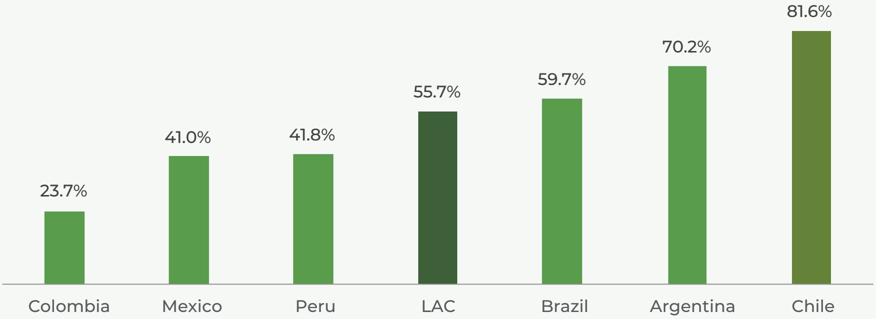


Source: Authors based on the FAOSTAT Agrifood Systems and Land-Related Emissions database.

This situation points to a major challenge: how can **these wastes be valorized according to the principles of the circular economy in order to include them in international trade?** Chile is well-positioned in the new global trade scenario, which is marked by growing restrictions based on environmental footprints ("green protectionism") and concerns about healthy eating. **Among the "big six" economies in LAC, Chile is the most food export–oriented** (81.6% of its value-added in food products is sold abroad). **It also has comparative advantages in fruit and vegetable processing**, including oil and wine production (figure 2). These two industries are the main sources of waste for recovery, due to economies of scale and lower transportation costs for collection. Production units are also smaller and more concentrated than is the case for primary crop residues. Along with Guatemala and Mexico, Chile is one of the three LAC countries with the greatest number of comparative advantages in products from these processing industries (figure 3).

Figure 2.

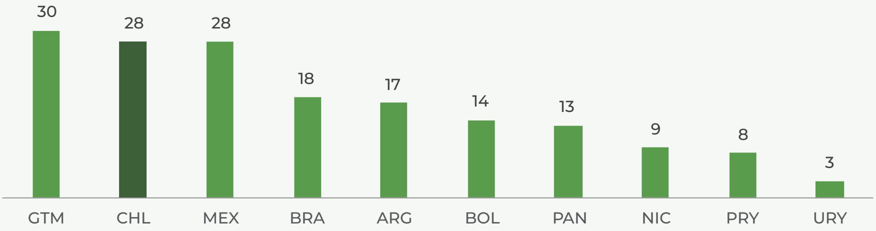
Food Exports as a Share of Value-Added (%) in LAC, Annual Average (2016–2019)



Note: Value-added was estimated as the sum of International Standard Industrial Classification (ISIC) Rev. 3 sectors 01, 02, and 05 from CEPALSTAT and ISIC Rev. 3 sector 15 from the UNIDO Statistics Data Portal. Exports were calculated as the sum of these sectors in the UN Comtrade database.
Source: Authors based on CEPALSTAT, UNIDO Statistics Data Portal, and UN Comtrade.

Figure 3.

Number of Products from the Fruit and Vegetable Processing Industries with Revealed Comparative Advantages by Country (2016–2022)

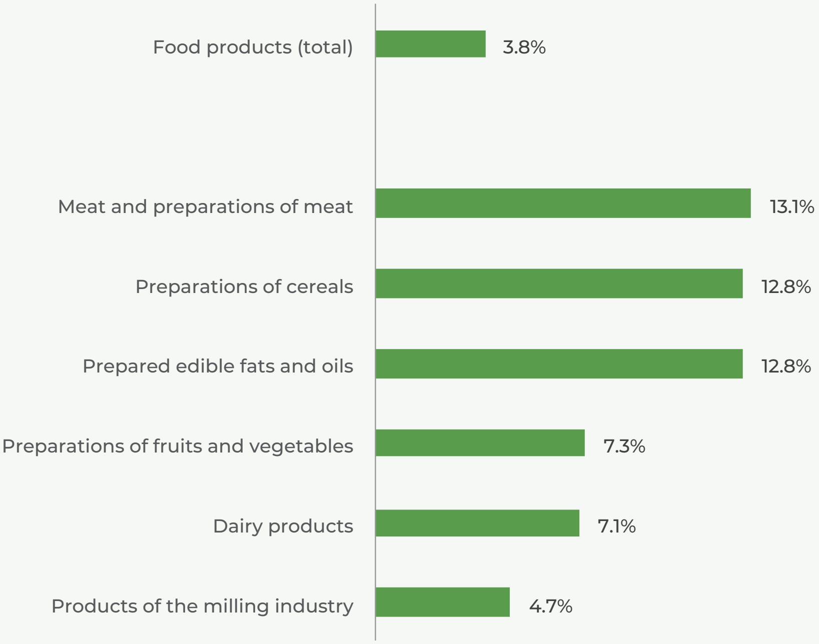


Note: The fruit and vegetable processing sectors were defined as ISIC Rev. 3 classes 1513, 1514, 1542, 1551, and 1552. These products correspond to the subheadings of the 2002 Harmonized System associated with the ISIC classes in question. The calculation includes LAC countries that systematically reported information between 2003 and 2022.
Source: Authors based on UN Comtrade.

The development of the circular economy is an opportunity for Chile to overcome some of its export-related challenges, mainly the diversification of export sectors. Although the country produces a broad range of products, its comparative advantages are currently concentrated in just three sectors. In contrast, the average country in LAC has comparative advantages in 4.5 sectors (UN Comtrade). Valorizing waste from these competitive sectors and including it in new foods as a source of dietary fiber would make it possible to differentiate exports downstream in the chain.

One of the motivations for this research is that a process of this sort may be emerging in Chile. **Agroindustrial exports increased by 3.8% annually between 2017 and 2022, and growth was highest in nontraditional sectors** (figure 4). These were the only sectors with above-average growth and were associated with trends toward the circular economy, as this chapter will explore.

Figure 4.
Compound Annual Growth Rate (%) of Chile's Exports by Sector (2017–2022)

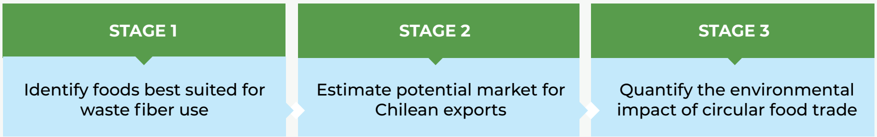


Note: Sectors were defined at the 4-digit ISIC level. Agroindustrial exports do not include the processing and preserving of fish and fish products (ISIC Rev. 3 division 05 and class 1512).
Source: Authors based on UN Comtrade.

Originally, the production and export of agroindustrial by-products in Chile took the form of flours or fibers. One example is the commercialization of (apple) pomace flour and inulin extracted from chicory (*Cichorium intybus L.*). The country acquired comparative advantages in both products in the mid-2000s. **Another related line of business is the use of these residues as a functional ingredient that provides fiber in prepared foods.**

This chapter aims to estimate the market potential for Chilean foods based on the valorization of residues from the fruit and vegetable processing industry and to quantify their impact in terms of reducing GHG emissions. To this end, we propose a three-stage methodology that seeks, first, to identify the edible products with the best export prospects; second, to measure the expected short-term market size for these products; and, finally, to evaluate the contribution of a circular food economy to Chile's climate objectives under the Paris Agreement (figure 5).

Figure 5.
Three-Stage Methodology for Estimating Reductions in GHG Emissions Derived from the Circular Economy.


STAGE 1
Identify foods best suited for waste fiber use
STAGE 2
Estimate potential market for Chilean exports
STAGE 3
Quantify the environmental impact of circular food trade

Source: Authors.

Before presenting our methodology, we should point out that although waste fibers can be obtained from different agroindustrial processes, this study focuses on waste from the apple processing and olive oil industries (pomace and orujo) as a benchmark for calculating this environmental impact. We selected these products for two reasons. The use of apples for dehydration and juice production were historically second only to wine production among Chile's fruit-processing industries. In the mid-2010s, the area used for apple growing decreased to make way for other products that are more profitable on the international market, such as cherries. **However, an industry has developed to revalorize pomace, the main waste from the apple processing industry.** In 2022, apple exports accounted for almost US$700 million, representing 10% of Chile's agrifood exports. There are currently 32,000 hectares of apple trees in Chile, which yield approximately 1.05 million tons of apples. Some 600,000 tons of these apples are exported fresh, while 200,000 tons go to the domestic market, and 250,000 tons are processed to obtain juice and dehydrated and frozen apples. Exports of 68 Brix apple juice concentrate reached US$87 million, while those of dehydrated apple yielded US$28 million (Ministerio de Agricultura de Chile, 2023). Based on the usual technical coefficients for the production of concentrated juice and dehydrated apples, the country's apple processing industry generates an estimated 130,000 tons of FLW (pomace) annually.

Although producing and exporting olive oil is still an emerging economic activity in Chile, the **entrepreneurial environment in the sector is conducive to attracting new stakeholders to create a circular chain around orujo**. In 2022, Chile exported 14,540 tons of olive oil, worth US$70 million, up from just US$26,000 20 years ago (UN Comtrade). Assuming a yield of about 13% from extra virgin oil processing, the country produces about 100,000 tons of orujo per year.

4. A METHODOLOGY FOR ESTIMATING THE ENVIRONMENTAL IMPACT OF CIRCULAR TRADE

As mentioned above, our methodology is a three-stage process that aims to identify how the valorization of pomace and orujo in new food exports might contribute to meeting Chile's climate commitments. **The first stage is to identify which downstream products in the agroindustrial chain would be most likely to include residual fiber as functional, healthy ingredients.** Unfortunately, the international trade classification does not allow us to identify these products directly. The first reason for this is that the classification scheme does not distinguish between foods based on their functional content. Second, several agroindustrial by-products are grouped with other products in "nonspecific" tariff headings, and there are discrepancies between different national customs authorities as to which headings new products should fall under. Finally, the current nomenclature does not specifically identify agricultural residues beyond certain cases with a longer history of valorization, such as cocoa shells, bran, or meal for animal feed.

4.1.

STAGE 1

IDENTIFICATION OF CIRCULAR FOODSTUFFS USING AN ECONOMIC COMPLEXITY ANALYSIS

To overcome the information-related limitations described above, this section proposes a methodology to identify foods that are more appropriate for the circular economy based on the foundational work of Hidalgo et al. (2007). These authors introduced the concept of economic complexity to describe the linkages within a production system and, more importantly, **to predict when a new product will arise within a country's export basket. In short,** Hidalgo et al. (2007) state that if **effective exports of product *i* are associated with effective exports of product *j*, then the two products are said to be connected.** This makes it possible to calculate the probability of country *c* starting to export good *i* given that it already effectively exports a set of connected products *j*.

Formally, this probability—which is known in the literature as "proximity"—is defined as follows:

$$(1)\ \omega_{c,i} = \frac{\Sigma_j CA_{c,j} * \theta_{i,j}}{\Sigma_j \theta_{i,j}}$$

Where θ is the similarity or proximity between product *i* and product *j*, and *CA* is the comparative advantage of country *c* in product *j*.[3] From equation (1), it **follows that the greater the comparative advantage of *c* in the set of goods *j* with the greatest proximity to product *i*, the more likely it would be for *c* to export *i*.**

---

[3] Proximity is normally a measure of conditional probability: that is, the occurrence of simultaneous exports of *i* and *j* over the total or possible instances of *i*. Hidalgo et al. (2007) take the minimum pairwise conditional probability between *i* and *j* and j and *i* to obtain a symmetric measure. Likewise, *CA* is a binary variable: it takes the value of 1 if country *c* has comparative advantages in product *j*, and 0 otherwise.

However, the proximity between *i* and *j* is produced through two transmission channels: either because *j* is an input (*ins*) of *i* or because *j* is a good that shares complementary resources (*comp*) with the production of *i*, be they physical (building structure, machinery, inputs) or intangible (knowledge, information, know-how, etc.). That is:

$$(2)\ \omega_{c,i} = \frac{\Sigma_{inp} CA_{c,inp} * \theta_{i,inp} + \Sigma_{comp} CA_{c,comp} *\ \theta_{i,comp}}{\Sigma_j \theta_{i,inp} + \Sigma_j \theta_{i,comp}}$$

A simple mathematical transformation yields the following:

$$(3)\ \omega_{c,i} = \frac{\Sigma_{inp} CA_{c,ins} * \theta_{i,inp}}{\Sigma_{inp} \theta_{i,inp}} * \frac{\Sigma_{inp} \theta_{i,inp}}{\Sigma_j \theta_{i,j}} + \frac{\Sigma_{comp} CA_{c,comp} * \theta_{i,comp}}{\Sigma_{comp} \theta_{i,comp}} * \frac{\Sigma_{comp} \theta_{i,comp}}{\Sigma_j \theta_{i,j}}$$

if *j* is equal to the sum of inputs (*imp*) and complementary goods (*comp*).

Equation (3) implies that the probability of exporting *i* can be expressed as the density of *c* in inputs and complementary resources weighted by the relative importance of the channels that associate *i* with the rest of the economy. Therefore, **the higher the position of a good in the economy as a demander of inputs and the greater *c*'s input density**—understood as the quantity of inputs needed to produce *i* that it produces competitively—**the more likely it is that *c* will export *i*.**

In this study we use this approach to identify the foods most likely to emerge in Chile's export basket in response to the availability of agroindustrial wastes. To achieve this, we construct an indicator that seeks to capture the main features of the circular economy in the sector, as described above.

This indicator is based on a four-step process. First, given the trade nomenclature limitations mentioned above, the availability of residual fibers in a country is inferred from its comparative advantages in agricultural and agroindustrial production. The assumption is that the greater the comparative advantage, the greater the domestic supply of residual fibers that could be valorized as food industry inputs.

However, since scale and transportation costs limit the industrial valorization of primary production wastes, **this assumption only applies to comparative advantages in agroindustrial processing.** In this sense, in equation (4) we divided *inp* into primary (*prim*) and processed (*pros*) inputs. Assuming that there are no comparative advantages from *prim*, we suppress this fraction in equation (5).

$$(4)\ \omega_{c,i} = \frac{\Sigma_{prim} CA_{c,prim} * \theta_{i,prim}}{\Sigma_{prim}\theta_{i,prim}} * \frac{\Sigma_{prim}\theta_{i,prim}}{\Sigma_j \theta_{i,j}} + \frac{\Sigma_{pros} CA_{c,pros} * \theta_{i,pros}}{\Sigma_{pros}\theta_{i,pros}}$$

$$\frac{\Sigma_{pros}\theta_{i,pros}}{\Sigma_j \theta_{i,j}} + \frac{\Sigma_{comp} CA_{c,comp} * \theta_{i,comp}}{\Sigma_{comp}\theta_{i,comp}} * \frac{\Sigma_{comp}\theta_{i,comp}}{\Sigma_j \theta_{i,j}}$$

$$(5)\ \omega_{c,i} = \frac{\Sigma_{pros} CA_{c,pros} * \theta_{i,pros}}{\Sigma_{pros}\theta_{i,pros}} * \frac{\Sigma_{pros}\theta_{i,pros}}{\Sigma_j \theta_{i,j}} + \frac{\Sigma_{comp} CA_{c,comp} * \theta_{i,comp}}{\Sigma_{comp}\theta_{i,comp}}$$

$$* \frac{\Sigma_{comp}\theta_{i,comp}}{\Sigma_j \theta_{i,j}}$$

provided that

$$\frac{\Sigma_{prim} CA_{c,prim} * \theta_{i,prim}}{\Sigma_{prim}\theta_{i,prim}} = 0$$

Second, residual fibers are a homogeneous input that by-products of various agroindustrial processes can easily substitute. Thus, healthfood *i* could incorporate functional ingredients contained in wastes from the apple, tomato, olive oil, or fish industries. **In this sense, having comparative advantages in one or other of these processes does not confer a particular advantage when producing and exporting *i*.** To capture the substitutability of inputs, fraction $\frac{\Sigma_{pros} VC_{c,pros} * \theta_{i,pros}}{\Sigma_{pros}\theta_{i,pros}}$ in equation (5) should not be calculated based on comparative advantages in a given *pros*, but on the basis of the average global affinity between exports of food *i* and the various *pros*.

$$(6)\ \omega_{c,i} = E\,(\Sigma_{pros}\theta_{i,pros}) * \frac{\Sigma_{pros}\theta_{i,pros}}{\Sigma_j \theta_{i,j}} + \frac{\Sigma_{comp} CA_{c,comp} * \theta_{i,comp}}{\Sigma_{comp}\theta_{i,comp}} * \frac{\Sigma_{comp}\theta_{i,comp}}{\Sigma_j \theta_{i,j}}$$

Third, we seek to assess how far circular food systems are an emerging global trend, including within international trade. We do this by looking at the increasing proximity between trade in processed food and processed agroindustrial products over time. The underlying hypothesis is that **a growing connection between the two segments could be indicative of increased use of waste from the latter in the manufacture of the former.**

$$(7)\quad \left[\frac{(\Sigma_{pros}\,\theta_{i,pros})}{\Sigma_j\,\theta_{i,j}}\right]_{t1} > \left[\frac{(\Sigma_{pros}\,\theta_{i,pros})}{\Sigma_j\,\theta_{i,j}}\right]_{t0}$$

Through this methodology, (7) is added to equation (6) multiplied by a scalar *k*, which allows us to calibrate the value of this component of the indicator based on judgments made during the research process. **As *k*→∞, more importance is attached to the growth (trend) in the connectedness of inputs and outputs than the degree of proximity per se.** This is important when it comes to distinguishing robust, established linkages between a given food and one or more processed inputs (which is very marked in the production of sauces and dressings, for example), on the one hand, and connections whose competitiveness is increasingly tied to the presence of *pros* in the country (which function as a proxy for a domestic supply of residual fiber) as a result of increased environmental and nutritional awareness worldwide.

$$(8)\ \omega_{c,i} = E\,(\Sigma_{pros}\,\theta_{i,pros}) * \frac{\Sigma_{pros}\,\theta_{i,pros}}{\Sigma_j\,\theta_{i,j}} + \frac{\Sigma_{comp}\,CA_{,comp} * \theta_{i,pros}}{\Sigma_{comp}\,\theta_{i,comp}} * \frac{\Sigma_{comp}\,\theta_{i,comp}}{\Sigma_j\,\theta_{i,j}}$$

The fourth step is analyzing the importance of complementary goods. Which food items have the best prospects of appearing in the export basket depends partly on existing capacities. For example, a country that specializes in dairy products will have a better chance of exporting new dairy products. However, entrepreneurship within the circular economy is another important factor. **The circular economy is ripe with opportunities for new players with new resources and ideas. As a result, the products that emerge might differ from those anticipated by the proximity calculation.**[4]

Consequently, a scalar *m* between 0 and 1 is added to the fraction in equation (8) to enable us to reassign $\frac{\Sigma_{comp}\,CA_{c,comp} * \theta_{i,comp}}{\Sigma_{comp}\,\theta_{i,comp}}$ the importance of complementary resources in food export proximity based on value judgments made during the research process. **As the presence of new players in the circular economy increases, complementary resources become relatively less important in the development of new healthfood products, and m→0.**

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[4] · International interest in and demand for fiber encouraged the farming of chicory in Chile for inulin extraction by Belgian company Orafti S.A., which has led to the country becoming the world's third-largest producer. There are also several companies that produce dietary apple fiber from the juice and dehydration industries. These include Revitta, which has vegan certification from Switzerland, Cáscara Foods (whose products are made by Azana, Surfrut, and Agrocepia), and Becactus, which produces and exports nopal fiber. There are numerous companies on the international stage that export fiber from different sources, such as Vitacel (wheat, oats, apple), Van Nuland (sugar cane, wheat, bamboo), Ingredion (corn, corn dextrin), and Beneo (inulin).

In summary, our synthetic indicator is as follows:

$$(9)\ \omega_{c,i} = E\left(\Sigma_{pros}\theta_{i,pros}\right) * \frac{\Sigma_{pros}\theta_{i,pros}}{\Sigma_{j}\theta_{i,j}} + m * \frac{\Sigma_{comp} CA_{c,comp} * \theta_{i,comp}}{\Sigma_{comp}\theta_{i,comp}} * \frac{\Sigma_{comp}\theta_{i,comp}}{\Sigma_{j}\theta_{i,comp}} + k * \delta\left(E\left(\Sigma_{pros}\theta_{i,pros}\right)\right)$$

4.1.1
RESULTS

To calculate the indicator, t_1 was taken as the average for 2016–2022 and t_0 as the average for 2003–2009. We selected these periods because we drew on data from the Harmonized System (HS) 2002 to measure structural variations in input-output linkages. To this end, all existing data between 2003 and 2022 was included and divided into three equal periods, after which only the extremes of the series were retained. To reduce the weight of well-established fixed links and emphasize growing connections, a weight of 2 was assigned to the parameter *k* in equation (9). A weight of 50% was chosen for *m* to give equal weight to existing capabilities and the possibility of new players starting new businesses. The data, sources, and complete methodology for the calculation are included in table A1 of the annex.

Table 1 shows the top ten products best suited to the circular economy according to the proposed indicator. Table A2 in the annex shows the top 20% of the 207 foods identified.

Table 1.

Top Ten Food Products That are Best Suited to the Circular Economy According to the Proposed Indicator

SUBHEADING	PRODUCT
1904.90	Cereals (other than maize (corn)) in grain form, precooked or otherwise prepared, excluding by swelling or roasting
2302.30	Bran, sharps, and other residues of wheat
1602.90	Other prepared or preserved meat, including preparations of blood of any animal
1602.31	Prepared turkey meat, excluding liver
1905.90	Other bread, pastry, cakes, biscuits, and bakers' wares
1904.20	Prepared foods obtained from unroasted cereal flakes or from mixtures of unroasted cereal flakes and roasted cereal flakes or swelled cereals
1106.30	Flour, meal, and fruit powder of edible fruits
0403.10	Yogurt
1905.40	Rusks, toasted bread, and similar toasted products
2202.10	Waters, including mineral waters and aerated waters, containing added sugar or other sweetening matter or flavored

Note: Products correspond to HS 2002 subheading level.
Source: Authors.

Of the top ten products with the closest potential links to the circular economy according to the proposed indicator, four are cereal-based preparations and bakers' wares (1904.20, 1904.90, 1905.40, and 1905.90); two are meat products (mixed meat preparations, 1602.90, and turkey, 1602.31); two are flour and bran (1106.30 and 2302.30); and the remaining two are yogurt (0403.10) and mineral waters and flavored drinks (2202.10). The findings are consistent with those of Dennehy et al. (2016), who found that cereal snacks, yogurt, beverages, and sausages with fiber-substituted fat are all foods that contain ingredients from agroindustrial waste and/or purified waste, meet the marginal conditions of a diet free of or reduced in specific nutrients (salt, saturated fat, and sugars), and contain healthy (functional) compounds. They also converge with the findings of Schwartz et al. (2022), who additionally determined the economic viability of the above products.

Interestingly, the methodology in their paper pinpoints subheading 1106.30 (flours, meals, and powders of the fruits contained in HS Chapter 8, which includes apples). As mentioned, pomace flour is one of the main products that have emerged in the discussion around waste revaluation in Chile. However, highly undifferentiated subheadings are also extremely prevalent in the study (these are usually listed as "unspecified" or "other"). In particular, new foods that emerge from circular economy initiatives by mixing or adding fibers from different farming products may not find a secure place in the nomenclature. For example, meat preparations that replace fat with fiber will likely be placed under subheading 1602.90. In the case of cereal snacks, different products will be placed under headings 1904 and 1905. For example, subheading 1905.40 will be used to identify toast-type snacks, while cereal bars could be placed under either subheadings 1904.90 or 1905.90, depending on how they look and what customs authorities decide. **Unfortunately, this is a limitation of the methodology put forward in this paper.**

4.1.2

THE EVOLUTION OF CHILEAN EXPORTS: A CIRCULAR DYNAMIC?

Figure 4 showed that in recent years, Chile's exports from nontraditional sectors increased at a higher rate than total agrifood exports. There has even been an increase in traditional exports of processed agroindustrial products, which is the main industry generating residual fibers. One of the motivations underlying this study is to analyze **whether Chile is making early progress in the circular economy in a way that contributes to significant, quality growth in exports.** Specifically, one of the main applications of Hidalgo et al.'s (2007) methodology is predicting the appearance of new products in a country's export basket (Hidalgo, 2021).

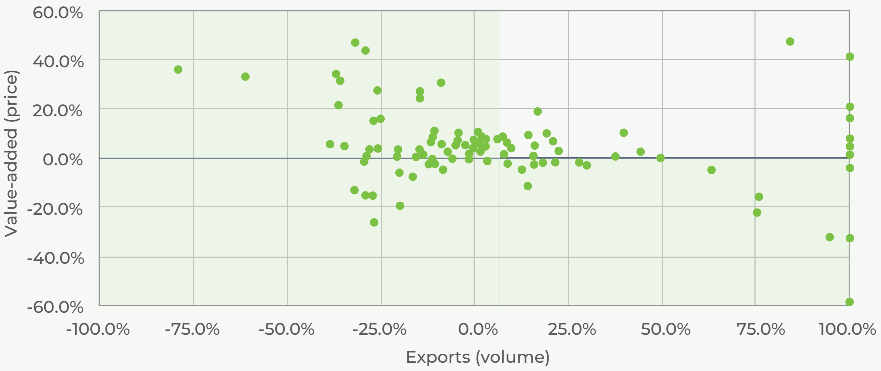
To infer this possibility, we took the subset of "nontraditional" processed foods—that is, products that are downstream of the agroindustrial chain and in which the country does not have comparative advantages. The hypothesis is that if Chile were to develop a circular economy in this sector, qualitative growth would be expected in a certain number of subheadings identified in table 1. **By "qualitative growth" we mean not**

only a faster increase compared to total agroindustrial exports but also an increase in value-added.

Although no information on trade in value-added is available at the subheading level, one way to approximate it is to return to equation (4). This reveals that the production and export of a given food are linked to a group of inputs (*inp*). If the change in the export price of good *i* is greater than the change in the export price of its inputs *inp*, weighted by their proximity to *i*, an increase in value-added can be assumed.

Using information on values and quantities from Chile's customs authority, figure 5 divides the 105 subheadings of nontraditional processed foods exported by Chile in 2002 into four quadrants: those that grew above or below the agroindustrial average of 4.8% per year and those for which the variation in export price was greater or less than the variation in the price of inputs.

Figure 6.
Variation in Nontraditional Processed Food Exports in Chile, Annual Average (2017–2022)



Note: The variation in the volume of total agroindustrial exports in Chile is 4.8%, according to data from the Chilean customs authority. Variations of 100% indicate that the product was exported for the first time in the period under consideration. The variation in value-added is the difference between the annual growth in the export price of the product and the price of inputs. The input price is the average variation in the price of products considered by the methodology to be primary inputs, weighted by their proximity to the exported food product in question.
Source: Authors based on outbound operations reported by the Chilean customs authority.

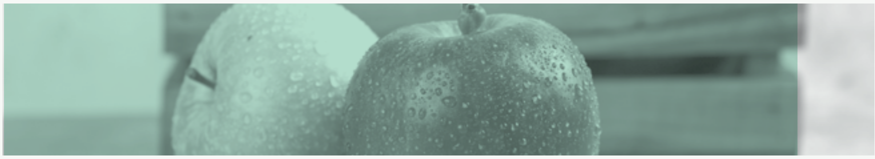
According to figure 4, only 25 of these 105 products have simultaneously registered faster growth in quantities and value-added. Three of these fall under the top ten subheadings for the proposed circular economy indicator: mixed meat preparations (1602.90), other bakers' wares (1905.90), and yogurt (0403.10). Given the limitations of the nomenclature explained above, the first two subheadings have been approximated as meat preparations with fat substituted by vegetable fiber and cereal snacks. If the analysis is extended to the top 10% of the proposed indicator (21 products), there is a fourth food that falls within the 25 experiencing qualitative growth in Chile: buttermilk (0403.90). Between 2017 and 2022, exports of these four foods quadrupled (from US$3.5 million to US$13.6 million), with yogurt and snacks concentrating almost all of the exported value. Table A3 in the annex shows the 25 products in the qualitative growth quadrant.

The evolution of flour exports is also noteworthy. Between 2000 and 2010, Chile acquired comparative advantages in nine food products, two of which are closely linked to the valorization of residual fibers: subheadings 1106.30 (fruit flours) and 1102.90 (other cereal flours). As mentioned above, the first of these products is among the top ten for the proposed circular economy indicator, while the second is in the top 20. However, exports of fruit flour (1106.30) declined between 2017 and 2022, which coincides with the decline in apple growing in Chile. In contrast, the "other cereal flours" subheading (1102.90) is within the quadrant of the 25 foods experiencing qualitative growth. In total, the two products contributed US$12 million to the country's exports.

In summary, this study finds indications that Chile may be making early progress in the valorization of residual fibers from its competitive fruit and vegetable processing industry, such as apple dehydration and juicing, and the growing production of olive oil.

4.2.

STAGE 2: PROJECTION OF THE EXTERNAL MARKET FOR THE CIRCULAR ECONOMY IN CHILE



Although there is evidence of growth in exports of food products related to the circular economy in Chile, this process is still in the early stages. In 2022, the exported values of the products identified in the previous exercise (snacks, yogurts, sausages, and whey) represent only 0.8% of processed food exports (up from 0.2% in 2017). **However, given that the circular economy is a relatively new development worldwide, the point here is to identify the trend.**

How can we project the potential growth of these products? The approach we propose is through Yu, Cai, and Leung's (2009) normalized revealed comparative advantage (NRCA). The index of economic complexity takes a binary approach to comparative advantage: that is, a country either has or does not have a comparative advantage in a given product (Hidalgo et al., 2007). Indeed, much of the problem with analyzing comparative advantage is that there is no direct interpretation of the calculation beyond whether the country has (or does not have) a higher (or lower) international market share in a given good relative to its share of total world exports. Instead, Yu et al. **propose a normalization method that allows cross-country and within-country comparative advantages to be compared over time in terms of world market value.**

Their formula is as follows:

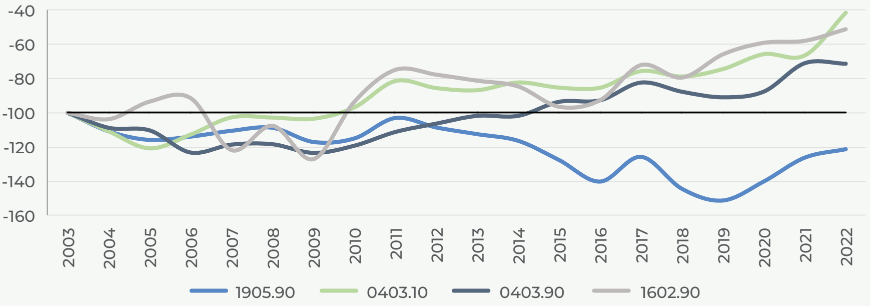
$$(10)\ VCRN_i^c = \frac{E_i^c}{E} - \frac{E_i * E^c}{E^2}$$

where E^c_i is the exports of product i from country c, E^c is the total exports of country c, E_i is the world exports of product i, and E is total world exports. As constructed, the formula indicates the difference between the actual exports of good i from country c and the theoretical value if c had the same share in the world market for i as in its total exports. Since this difference is relativized by the size of the world market E, the range of is between -1 (the country does not export the product today) and 1 (it is the only supplier in the global market). If the index is greater than 0, current exports exceed the theoretical value and the country has a comparative advantage in i.

Yu, Cai, and Leung's (2009) methodology allows comparative advantages to be analyzed dynamically. That is, negative values are not all discarded, as in the binary formula, but relativized over time. The question is thus whether Chile is on the way to becoming a competitive exporter of processed foods with stronger links to residual fibers from agroindustrial processes, in the context of an incipient development of the circular economy.

Figure 7.

Evolution of Chile's Standardized Comparative Advantages by Product, Index (-100 = 2003)



Note: The value -100 indicates the value of Chile's NRCA in product i (negative). A value of 0 in the index would indicate the acquisition of comparative advantage in i.
Source: Authors based on Yu, Cai, and Leung (2009) and UN Comtrade.

Figure 7 shows that the NRCA value is highest for yogurt (0403.10) and prepared or preserved meats (1602.90) in 2022, while the maximum value for buttermilk (0403.90) is for 2021. In 2022, the "other bakers' wares" subheading (1905.90) reached its highest NRCA value since 2014, although the maximum was recorded in 2003. This is yet another demonstration of the limitations of nomenclature in identifying specific foods. It is likely that the actual products exported under this subheading are very different today than they were in the mid-2000s. For example, the price per kg in 2022 was 43% higher than 12 years ago, according to data from the Chilean customs authority.

Interestingly, flours also play an important role here. The "other cereal flours" subheading (1102.90) systematically acquired NRCA status (NRCA > 0) from 2018 onward and reached its maximum value in the entire series in 2022. However, fruit flour (1106.30) reached its maximum NRCA value in 2017 and then declined, as was to be expected given the decline in apple growing. However, it is worth noting that this value has been consistently positive since 2008.

It would therefore be possible to conclude that Chile is developing dynamic comparative advantages in products that we have identified as suitable for the circular economy. The next question concerns future projections of the international market. In the absence of better instruments in the current international context,[5] we propose the methodology of Yu, Cai, and Leung (2009) as **a measure of opportunities for growth in the world market value of products for which a comparative advantage is emerging.** The question, therefore, is what is the value of Chilean exports compatible with the acquisition of a comparative advantage in the products identified above? This value is that of theoretical exports in equation (10).[6]

Table 2.

Potential Exports from Chile in the Event of Achieving Comparative Advantages in Processed Foods with Closer Linkages to the Circular Economy (millions of US$)

SUBHEADINGS	THEORETICAL EXPORTS	2022 EXPORTS	INCREASE
1905.90	236	8.1	227.9
0403.10	15.3	5.4	9.9
0403.90	17.8	0.157	17.6
1602.90	2	0.06	1.9
Total	271.1	13.7	257.4

Source: Authors based on Yu, Cai, and Leung (2009) and UN Comtrade.

By applying equation (10) to Chile's and global trade values, we conclude **that if the upward trend in the comparative advantages of processed foods with more links to the circular economy continues, exports could increase by US$236 million in other bakers' wares, US$15.3 million in yogurt, US$17.8 million in buttermilk, and US$2 million in prepared or preserved meats** (table 2). In terms of total processed food exports in 2022, this represents growth of 11%. There is no question that Chile should take advantage of its agroindustrial competitiveness and the knowledge acquired in the valorization of waste to make progress in the qualitative development of circular exports.

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[5] · For example, the International Trade Center (ITC) recently published a methodology for estimating potential exports at the country-product level (ITC, 2022). However, this publication includes some parameters that might lead to an overestimation of potential market size. The most important of these is the projection based on the extrapolation of the previous five years' growth. In the case of food, this implies carrying forward the 13% jump in agricultural commodity prices that followed Russia's invasion of Ukraine. In September 2023, food prices were 15% lower than the 2022 peak. However, the ITC weights the tariff advantages of countries with a significant number of trade agreements, as is the case for Chile. However, in the case of subheadings that are relatively unspecified within the international nomenclature, the products being compared may not be similar. A more disaggregated examination of national nomenclature levels is necessary to identify such advantages in this case.

[6] · Note that we suppress the exponential component of E, as this is part of the subsequent standardization of the Yu, Cai, and Leung's (2009) methodology.

4.3.

STAGE 3: ENVIRONMENTAL IMPACT OF THE DEVELOPMENT OF THE CIRCULAR ECONOMY IN CHILE

As mentioned at the start of this chapter, the food industry generates large quantities of organic by-products that require appropriate, environmentally friendly disposal. In Chile, by-products are conventionally treated through composting, which releases GHGs that contribute to climate change, such as methane and CO_2 (box 1).

The main problem in measuring the impact is that the magnitude of the production of these gases depends on several factors, such as the composition of the waste, the temperature of the digestion, the acidity of the medium, the duration of the process, the introduction of microbial inoculants, and mixing with other organic wastes, thermal, and ultrasonic pretreatments. The combination of some of these variables is another issue. **As a result, the technical coefficient data reported in different studies varies widely—that is, the volume of biogas per ton of waste or the volume of volatile solids** (VS) (Bong et al., 2018; Jutidamrongphan, 2018; Bautista et al., 2018; Fisgativá, Tremier, and Dabert, 2016; Negri et al., 2016; Tulun and Bilgin, 2018).

Regarding the olive oil industry, it is generally estimated that 330.24 L of methane can be produced per kg VS of orujo, a value that is within the typical range (250–550 L CH_4/kg VS) found in many operations relating to anaerobic solid waste digesters in various countries. It has been estimated that 36.61 L/kg VS are produced for each ton of apple pomace subjected to anaerobic digestion, equivalent to 0.62 kg of CO_2 (Ampese et al., 2022). Gil, Rodríguez et al. (2018) find that one ton of orujo produces 23 L of biogas (methane and CO_2) during dry anaerobic

BOX 1
CHILE AND NATIONALLY DETERMINED CONTRIBUTIONS UNDER THE PARIS AGREEMENT

In 2020, Chile submitted the first update of its nationally determined contributions (NDCs) under the Paris Agreement. In it, Chile committed to annual GHG emissions of 95 $MtCO_2eq$ in 2030 and a total budget for 2020–2030 of 1,100 $MtCO_2eq$ (Ministerio de Medio Ambiente, 2022). This means that the country will have to reduce its GHG emissions by at least 12 $MtCO_2eq$ in the next two to three years compared to the average level between 2015 and 2020 (107 $MtCO_2eq$) to achieve the two targets simultaneously. In addition, at the 2021 UN Climate Change Conference in Glasgow (COP26), Chile pledged to strengthen its NDCs through specific methane-related efforts (Ministerio de Medio Ambiente, 2022). At this summit, 150 countries adhered to the Global Methane Pledge, which sets a global target of -30% of methane emissions by 2030, which could make a potential contribution of -0.2°C to the earth's temperature by 2050 (Climate & Clean Air Coalition Secretariat, 2023).

Although Chile has not set a specific methane reduction target, methane accounts for 14% of the country's total GHG emissions. Should the Chilean government decide to adopt the 30% target set for the global total, this would contribute 12.5% to the country's efforts under its NDCs. Globally, 95% of emissions from FLW are methane gas.

digestion. Meanwhile, Varnero et al. (2014) estimate that an average of 30 L of biogas (13% methane, 60% CO_2 and other gases) is produced under anaerobic conditions per ton of residue made up of 27% orujo, 68% vegetable waste, and 5% guano containing methanogenic bacteria. The total yield during the time in which the anaerobic and then aerobic fermentation was carried out was 700 L.

Czubaszek, Wysocka-Czubaszek, and Tyborowski (2022) determined the methane output for apple pomace after 30 days of digestion and the maximum daily average. Their calculation yielded 232.2 L/kg VS-1 and 26.8 L/kg VS-1 measured under normal conditions.

4.3.1.

THE CIRCULAR ECONOMY AND THE REDUCTION OF METHANE EMISSIONS: ADDING ORUJO AS A SOURCE OF FIBER TO CEREAL SNACKS

In section 4.1, we identified the processed foods that were most suitable for the circular economy in Chile. In section 4.2, we estimated potential demand should the country further develop circular exports based on its competitiveness in agroindustrial processing. In this last section, we calculate the environmental benefits that this shift would bring. Indeed, **the valorization and reuse of agroindustrial waste as dietary fiber in different foods is presented as a zero-emission alternative** to the normal treatment of waste in Chile (aerobic and anaerobic digestion), which is intensive in methane gas emissions. In this way, **the circular economy contributes not only to increasing the value-added of the export basket but also to reducing the national GHG inventory.**

We base our calculation for this potential impact on the use of orujo in cereal-based snacks. Cereal bars were chosen because of the potential market size (US$236 million, according to table 2) and because Chile has a longer history of production in this area (the country has had different comparative advantage peaks in 2003, 2011, and 2022). Likewise, a cereal bar uses more residual fiber per kg than yogurt, resulting in a greater use of waste.[7] The decision to focus on orujo is due, first, to the fact that Chile's nascent olive oil industry has created an entrepreneurial environment that is attracting new players to the sector, as was the case with apples not long ago. Second, cereal bars are estimated to use more orujo than pomace per kg.[8]

Therefore, if Chile specializes and acquires comparative advantages in cereal bars, its total exports could amount to US$236 million, according to the methodology in Yu, Cai, and Leung (2009). At current prices, this represents 56,500 kg of food and demand for 19.8 kg of orujo (350 g per kg).

To quantify the environmental impact, we now need to define the amount of orujo contained in 1 kg of VS waste. Guerrero (2012) carries out the same calculation for biogas production in a 60-liter digester.

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[7] It is estimated that 1 kg of cereal bar can incorporate up to 350 g of alperujo, while 1 kg of yogurt can only use up to 10 g.

[8] 350 g versus 250 g.

As orujo is plant matter with low nitrogen content, mixing it with other waste would be needed to reduce the carbon-to-nitrogen (C:N) ratio from over 50:1 to 30:1. In his calculation, Guerrero (2012) obtains 1.73% VS in 1 kg of orujo. Using this value as a benchmark, orujo generates 5.7 L of methane (CH_4) per kg (0.18 kg of CO_2eq).

Applying this value, the savings in GHG emissions from reusing fiber and achieving international specialization in cereal bars is 0.0036 MtCO_2eq. Table 3 shows the logical steps taken to arrive at this conclusion.

Table 3.

Estimation of the Impact of the Revalorization and Reuse of Waste from the Export-Oriented Olive Oil Industry in Chile on GHG Emissions

Projected exports under subheading 1905.90, "other bakers' wares" (proxy for cereal bars)	56,563 kg(1)
Demand for orujo per kg of product	350 gr
Total projected demand	19,797 kg
Methane gas emissions per kg of composted orujo	5.7 L(2)
	0.18 kg CO_2eq(3)
Emissions saved by revalorizing and reusing orujo in cereal bars for export	19,797 kg. * 0.18 kg CO_2eq = 3,555 kg CO_2eq
Chile's total GHG emission reduction commitment (NDCs)	12,000,000 kg CO_2eq(4)
Theoretical methane reduction commitment (2021 pledge)	4,483,800 kg(5)
Contribution of the food export-related circular economy to Chile's NDCs	0.03% total
	0.08% methane

(1) Assuming that Chile specializes in cereal snacks that include residual fiber and achieves comparative advantage in this subheading. (2) Assuming normal measurement conditions (0°C and 1 atm of pressure) and the use of 1.73% VS per kg of orujo. (3) Conversion coefficient: 1 L CH_4 = 0.031 kg CO_2eq. (4) Authors' approximation based on the targets adopted by Chile under the Paris Agreement (95 MtCO_2eq in 2030 and an accumulated 1,100 MtCO_2eq between 2020 and 2030) and current emissions (107 MtCO_2eq annual average for 2015–2020). (5) Assumes that Chile will reduce current methane emissions by 30%, in line with the global target of the Global Methane Commitment.
Source: Authors.

This simple change makes a highly significant contribution to Chile's climate agenda. Based on the NDCs described in box 1, the **0.0036 MtCO_2eq is equivalent to 0.03% of the target reduction of approximately 12 MtCO_2eq** needed in the next two to three years to simultaneously achieve the two targets submitted by the country to the Intergovernmental Panel on Climate Change (IPCC). **If we compare this with current methane emissions** (15 MtCO_2eq) and a hypothetical target reduction of 4.4 MtCO_2eq in line with the Global Methane Commitment, **the contribution would be 0.08%.**

5. CONCLUSIONES

The valorization of waste materials from the olive oil and apple industries in the form of value-added products is a possible path for improving the circular economy in Chile. In this context, the challenge is how to turn these residues into competitive health foods and functional ingredients. For some time now, circular practices such as composting and the cogeneration of energy have been developed in Chile**. However, the transition to the circular economy should multiply and scale business models and advance toward the gradual application of higher priority strategies, such as valorization.** Considering that there is significant global demand for fiber and that this demand is increasing for technological and health reasons, the production and commercialization of fiber in the Chilean and international markets would contribute to **building the circular economy and achieving Chile's NDCs under the Paris Agreement.**

The international trade classification scheme does not allow us to analyze the development of the circular economy in food products. First, the scheme does not distinguish between foods according to their functional content, such as in terms of the incorporation of residual fibers. Second, several intermediate products that are emerging around waste fall under nonspecific tariff classifications, and there are also discrepancies between national customs authorities as to where to place these new products.

The scalable, potentially impactful value proposition of using orujo (waste from the Chilean olive oil industry) and apple pomace flour would take the form of incorporating orujo into beverages and yogurt, and pomace flour (fiber) into snack bars.

Cereal snacks, yogurt, beverages, and sausages containing dietary fiber as a fat substitute are part of the portfolio of products selected using the methodology applied in this study and that meet the following marginal conditions: they provide a diet free of or reduced in certain critical nutrients (salt, sugar, saturated fat), contain healthy (functional)

compounds, and contribute to the reduction of GHGs because they contain ingredients such as fiber obtained from pomace and orujo.
If Chile were to specialize in cereal-based snacks in the international market, additional exports could amount to a potential market of US$236 million, representing 56,000 kg of product.

The development of these circular businesses in Chile could contribute 0.03% of the country's NDCs in the coming years and 0.08% of its methane commitments.

Waste materials from the agrifood industry can be included in other food products but could also be an export business in their own right. This is particularly true for isolated dietary fiber.

Given that comparative advantages are dynamic and that the circular economy is still an emerging trend, Chile is on its way to becoming a competitive supplier of the foods identified here based on the methodology used in this research.

This study shows that Chile is making early progress in the circular economy, incorporating residual fiber into foods that are new for the country and the global market, contributing to the comprehensive growth of Chilean exports.

6. POLICY RECOMMENDATIONS

Chile has implemented different **public policy measures that are supporting the implementation of the circular economy and helping businesses make effective use of it.** One such example is Law 20920 on Extended Producer Responsibility and the Promotion of Recycling, which establishes that from 2023, Chilean companies must move from a linear model to a circular one in terms of raw materials within 12 years, which includes the reuse of materials, recycling, and waste recovery.

However, **a fundamental aspect in boosting Chile's foreign trade by exporting products from the circular economy is certifications and compliance with standards that can help establish the credibility of exported products.** These standards signal to international buyers that products comply with the principles of the circular economy and are therefore environmentally sustainable. Chile is already involved in these practices: there are certifying companies operating in the country that deal with environmental protection and thus with the circular economy. Some of these global standards used in Chile promote sustainable practices, waste minimization, and the reduction of GHG emissions, thus contributing to the circular economy worldwide. These include the Fair Trade, Waste Management, Zero Waste to Landfill Certification, Carbon Footprint, and B-Corps certifications.

The development of standards and certifications can ensure the circularity of goods, services, and production processes. However, in some cases, the creation of multiple standards has led to mutual incompatibility, requiring considerable effort on the part of producers while yielding few results. To address this problem, equivalent standards or common criteria for circular economy certifications need to be established. In this regard, regional coordination is required in instruments such as environmental labeling.

Other **more general public policy recommendations** could be considered to boost Chile's foreign trade through the export of circular economy products. First, the state should promote the adoption of circular practices in the private sector by developing and strengthening innovation ecosystems for companies, as well as capacity-building, coordination, and knowledge transfer among the different stakeholders. Similarly, the introduction of fiscal or financial incentives could serve as tools to encourage companies to move toward more sustainable models. In conclusion, public policies to promote international trade through the export of circular economy products require a holistic approach, ranging from promoting the adoption of circular practices by companies to educating consumers and working with other stakeholders (e.g., government agencies and business associations). Integrating the circular economy into countries' export strategies could create economic, social, and environmental benefits for both exporting and importing countries, thus contributing to achieving the goals of the Paris Agreement.

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ANNEX.

DATA AND SOURCES

All data was obtained from UN Comtrade, Harmonized System (HS) 2002, at the subheading level. To avoid selection bias, information was taken from all countries that systematically reported trade data for the entire period (2003–2022). We opted for the HS 2002 because we sought to capture structural variations in the proximity between *i* and *j* over time. To this end, we divided the series into three equal periods: 2003–2009, 2010–2015, and 2016–2022. Since not a large number of new subheadings were added to sections I to IV (food) of the HS in subsequent updates, not using more recent versions of the classification scheme does not lead to a significant loss of information.

This approach is based on the universe of agricultural products. To define it, we used all HS subheadings corresponding to ISIC Rev. 3 sectors 01 (crop and animal production) and 15 (manufacture of food products and beverages). There are a total of 573 subheadings, which were subdivided into primary inputs (*prim*), processed inputs (*pros*), and processed food, based on the 4-digit ISIC codes.

The sectors that process fruits, vegetables, and sugar and make alcoholic beverages (sectors 1513, 1514, 1542, 1551, and 1552) are considered producers of "residues" that fall into the *pros* category. In the case of milling (1531 and 1532), these products were placed in the "downstream" food group because residual fiber meals can be both an intermediate product in the manufacture of processed foods and a direct product for human consumption. With regard to alcoholic beverages, beers are included as food in the proposed definition because beermaking produces malt, which is a nutritional drink. All other ISIC 15 subsectors except the processing of fish products contain food for human consumption and are the exportable products targeted by this research (the *i* goods). They also enter into equation (9) as complementary goods (*comp*) of other food products (table 1).

Table A1.

Classification of Sectors and Products Into Primary Agricultural Inputs, Processed Agricultural Inputs, and Processed Food Products

	PRIMARY INPUTS	PROCESSED INPUTS	PROCESSED FOOD PRODUCTS
ISIC sectors	0111, 0112, 0113, 0121, 0122	1513, 1514, 1542, 1551, 1552	1511, 1520, 1531, 1532, 1533, 1541, 1543, 1544, 1549, 1553, 1554
Subheading no.	207	159	207

Note: The fishing sector (sectors 05 and 1512) was not included in this study.
Source: Authors.

Among the main methodological decisions we made were the exclusion of nonfood products (tobacco, rubber, wool, and cotton) from the primary input group and the inclusion of food waste not classified by ISIC (e.g., bran and cocoa shells) in various sectors. The most important decision was the reallocation of HS heading 0813 from ISIC sector 0113 (primary inputs) to sector 1513 (processed inputs). We made this decision because these products are part of the dehydration industry, but ISIC includes them as primary products to make them compatible with fruits, as the HS does not distinguish between fresh and dehydrated fruit in headings 0805 to 0810. This reallocation affects apples and plums, in which Chile has comparative advantages.

Table A2.

Top 20% of Food Products That are Best Suited to the Circular Economy According to the Proposed Indicator

SUBHEADING	PRODUCT
1904.90	Cereals (other than maize (corn)) in grain form, precooked or otherwise prepared, excl. by swelling or roasting
2302.30	Bran, sharps, and other residues of wheat
1602.90	Other prepared or preserved meat, incl. preparations of blood of any animal
1602.31	Prepared turkey meat, excl. liver
1905.90	Other bread, pastry, cakes, biscuits, and bakers' wares
1904.20	Prepared foods obtained from unroasted cereal flakes or from mixtures of unroasted cereal flakes and roasted cereal flakes or swelled cereals
1106.30	Flour, meal, and fruit powder of edible fruits
0403.10	Yogurt
1905.40	Rusks, toasted bread, and similar toasted products
2202.10	Waters, incl. mineral waters and aerated waters, containing added sugar or other sweetening matter or flavored
2301.10	Flours, meals, and pellets of meat or meat offal
0901.90	Coffee husks and skins; coffee substitutes
1601.00	Sausages and similar products of meat, meat offal, or blood; food preparations based on these products
1904.10	Prepared foods obtained by the swelling or roasting of cereals or cereal products
2105.00	Ice cream and other edible ice, whether or not containing cocoa
0403.90	Fermented or acidified milk and cream, excl. yogurt
1905.31	Sweet biscuits
1602.10	Homogenized preparations of meat, meat offal, or blood
0201.10	Meat of bovine animals, fresh or chilled, carcasses and half-carcasses
0201.20	Meat of bovine animals, fresh or chilled, other cuts with bone in
2106.10	Protein concentrates and textured protein substances
1602.42	Pork shoulders and cuts thereof
1602.49	Other pork preparations, incl. mixtures
0401.10	Milk and cream of a fat content, by weight, not exceeding 1%
0401.20	Milk and cream of a fat content, by weight, exceeding 1% but not exceeding 6
1102.90	Cereal flours other than wheat, rye, maize (corn), and rice flours

1602.50	Prepared or preserved meat, meat offal, or blood of bovine animals
1107.20	Toasted malt
1103.13	Maize (corn) groats and meal
1101.00	Wheat or meslin flour
0405.20	Dairy spreads exc. butter
1903.00	Tapioca and substitutes prepared from starch
2302.10	Bran, sharps, and other residues of maize (corn)
1104.23	Pearled or flaked maize (corn)
1806.31	Filled chocolates
2201.10	Mineral waters and aerated waters
1806.20	Other chocolate and other food preparations containing cocoa in containers or immediate packings, of a content exceeding 2 kg
1501.00	Pig fat (incl. lard) and poultry fat
2104.20	Homogenized composite food preparations
0209.00	Pig fat, free of lean meat, and poultry fat, not rendered or otherwise extracted, fresh, chilled, frozen, salted, in brine, dried or smoked

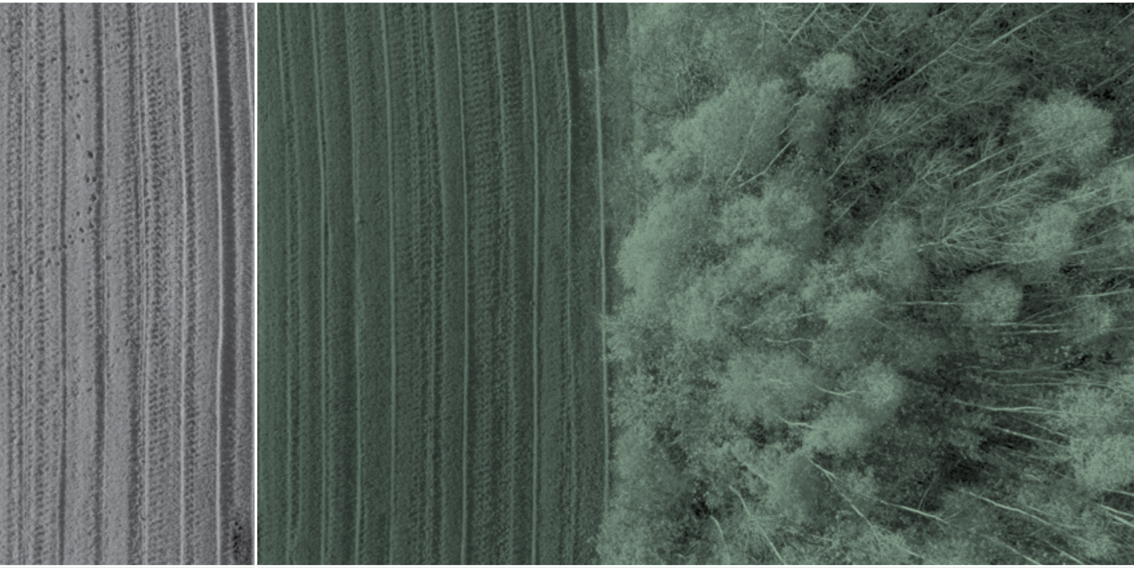
Table A3
Processed Foods with the Greatest Increase in Exported Volume and Value-Added

SUBHEADING	DESCRIPTION
0202.20	Meat of bovine animals, frozen, other cuts with bone in
0202.30	Meat of bovine animals, frozen, boneless
0204.22	Meat of sheep or goats, fresh, chilled, or frozen, other cuts with bone in
0204.23	Meat of sheep or goats, fresh, chilled, or frozen, boneless
0206.22	Livers of bovine animals, frozen
0206.49	Edible offal of swine excl. livers, frozen
0207.26	Cuts and offal of turkeys, fresh or chilled
0210.20	Meat of bovine animals salted, in brine, dried, or smoked
0402.10	Milk and cream in powder, granules, or other solid formss
0402.21	Milk and cream in powder, granules, or other solid forms, of a fat content by weight exceeding 1.5%, not containing added sugar or other sweetening matter
0402.91	Other milk and cream not containing added sugar or other sweetening matter
0403.10	Yogurt
0403.90	Fermented or acidified milk and cream, excl. yogurt
0901.22	Roasted, decaffeinated coffee
1102.90	Flours excl. wheat, rye, maize (corn), and rice flours

1602.90	Other prepared or preserved meat, meat offal, or blood of any animal
1702.11	Lactose and lactose syrup containing by weight 99 % or more lactose, expressed as anhydrous lactose
1702.40	Glucose and glucose syrup, containing in the dry state at least 20% but less than 50% by weight of fructose
1704.90	Other sugar confectionery not containing cocoa, excl. chewing gum
1805.00	Cocoa powder, not containing added sugar or other sweetening matter
1806.31	Filled chocolates
1901.90	Other food preparations
1902.30	Other pasta
1905.90	Other bread, pastry, cakes, biscuits, and bakers' wares
2102.20	Inactive yeasts; other single-cell micro-organisms, dead

CHAPTER 2

THE IMPACT OF THE EUROPEAN GREEN DEAL ON EXPORTS FROM ARGENTINA



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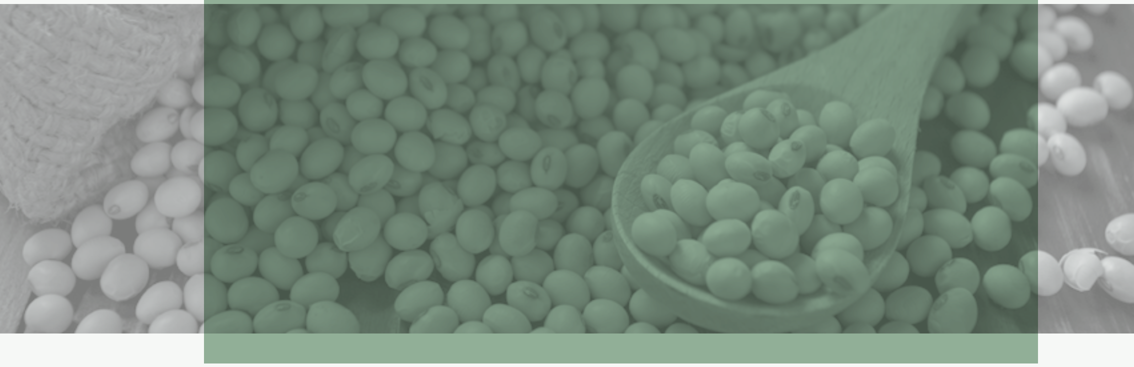
ACRONYMS AND ABBREVIATIONS

- **AFIP**
  Federal Tax Authority of Argentina
- **CAR**
  Rural Environmental Cadaster
- **CBAM**
  EU Carbon Border Adjustment Mechanism
- **CET**
  constant elasticity of transformation
- **CGE**
  computable general equilibrium
- **CIAM**
  Center for Environmental Information
- **CIARA**
  Argentine Edible Oil Association
- **COP26**
  2021 UN Climate Change Conference in Glasgow
- **EU**
  European Union
- **EUDR**
  European Union Deforestation Regulation
- **EWS**
  early warning system
- **FAO**
  Food and Agriculture Organization of the United Nations
- **GDP**
  gross domestic product
- **GHGs**
  greenhouse gases
- **HS**
  Harmonized System
- **SAM**
  social accounting matrix
- **SENASA**
  National Service for Agrifood Health and Quality
- **SINIA**
  Integrated Environmental Information System
- **SNMBN**
  Native Forest Monitoring System of Argentina
- **ViSeC**
  Gran Chaco Monitoring System

ABSTRACT

This chapter **examines the potential economic and trade impacts** that the new European Union (EU) regulation on the importation of **deforestation-free products** (the EU Deforestation Regulation or EUDR) may have in **Argentina**. It examines the regulatory framework that will enter into force in January 2025, which establishes a traceability system for primary products with ties to deforestation, such as soybeans, cattle, cocoa, coffee, oil palm, rubber, and wood. **The chapter assesses deforestation in Argentina** from this perspective, taking into account the fact that the EUDR establishes January 2021 as the cut-off date for deforestation. Approximately 40% of Argentina's exports of goods to the EU will be subject to this regulation, mainly soybeans and cattle. Using a computable general equilibrium model, we estimate that Argentina's gross domestic product could fall between 0.15% and 0.26% if exports do not comply with the EUDR. We conclude by making policy recommendations that seek to improve compliance with the new regulation and promote international trade in sustainable agricultural products.



1. INTRODUCTION

Through the guidelines set out in its Green Deal, the European Union (EU)[1] has developed sector-specific trade strategies to achieve its goal of becoming carbon-neutral[2] by 2050. In light of the scope and current status of the regulatory process, two measures are particularly relevant for Argentina. The first is the Border Carbon Adjustment Mechanism (CBAM), a scheme that will function as a premium to be charged based on the carbon footprint of products imported into the EU. The products covered by the first stage of the CBAM will be iron and steel, cement, fertilizers, aluminum, and electricity. Under the scheme, EU importers will be required to purchase certificates at a price equal to the cost of the goods if they had been produced under the carbon pricing rules in place in the European single market.

The second significant measure is the Regulation on Deforestation-Free Products or the EU Deforestation Regulation (EUDR). The EUDR seeks to establish a traceability system for certain primary products and their supply chains that have been identified as drivers of deforestation because their production is associated with the expansion of the agricultural frontier. These products include soybeans and livestock products. The scheme will take the form of mandatory due diligence statements that prove that goods imported by the EU have not been produced on land deforested after December 31, 2020, combined with a comparative risk assessment system for each country.

According to our estimates, which are based on UN COMTRADE trade data for 2022, only 0.17% of Argentina's total exports to EU countries are covered by the CBAM,[3] or 0.02% of the country's total exports. In contrast, as this article shows, the EUDR would make a significant impact as it may affect approximately 42% of total exports to the EU, which represent more than 5% of Argentina's total goods exports.

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[1] · The EU currently comprises 27 countries: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Poland, Portugal, Romania, Slovenia, Spain, Sweden, and Slovakia.

[2] · Carbon neutrality means achieving a net result of zero greenhouse gas emissions; that is, emitting the same amount of gases into the atmosphere as is absorbed in other ways.

[3] · Of the US$19 million in export value, iron and steel products account for US$17.8 million and aluminum products for US$1.6 million.

Given the greater relative importance of the EUDR, this paper analyzes what economic and trade implications its introduction as a requirement for EU market access will have for Argentina. This chapter is structured as follows. Section 2 describes the EU regulatory framework for the importation of deforestation-free products and Argentina's current situation regarding deforestation and land use. Section 3 then shows which Argentinian exports are affected by the EUDR based on an analysis of international trade data. In section 4, we use a computable general equilibrium model to simulate the impact of the implementation of the new EU regulations on the Argentinian economy and share our findings. Finally, section 5 contains conclusions and policy recommendations based on our analysis and a series of interviews with key informants that we conducted throughout the research process. Specifically, we conducted 14 interviews with key stakeholders from the public, private, and international sectors (listed in annex C) to better understand the EUDR and how it is being interpreted, find out more about progress on implementation, and be able to put forward policy recommendations that address the need for compliance in the coming months.

2. GLOBAL DEFORESTATION THE EU REGULATION ON DEFORESTATION-FREE PRODUCTS AND WHERE ARGENTINA STANDS

Forests[4] cover 4.06 billion hectares (ha), or 31% of the earth's total area. More than half (54%) of the world's forests are located in just five countries: the Russian Federation, Brazil, Canada, the United States, and China (FAO, 2020). Over the last 5,000 years, 1.8 billion ha of forest have been lost, and most of this loss (1.4 billion ha) has occurred in the last 300 years. Driven by the continued expansion of land for agriculture, the world has lost as much forest cover in just over 100 years as in the previous 9,000 (Ritchie and Roser, 2021).

Tackling deforestation is central to the fight against climate change. At the 2021 United Nations Climate Change Conference in Glasgow (COP26), world leaders committed[5] to promoting international and domestic trade and development policies that do not encourage deforestation and forest degradation.

The EU is pioneering this issue. The European Green Deal establishes different initiatives, such as the Farm to Fork Strategy, which aims to make food systems fair, healthy, and environmentally friendly, and the Biodiversity Strategy for 2030. Both programs are being implemented through specific actions. This framework includes the EUDR—Regulation 2023/1115 on "the making available on the EU market and the export from the EU of certain commodities and products associated with deforestation and forest degradation."

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[4] · Forest is "land spanning more than 0.5 hectares with trees higher than 5 meters and a canopy cover of more than 10 percent, or trees able to reach these thresholds in situ. It does not include land that is predominantly under agricultural or urban land use" (FRA, 2015).

[5] · See https://webarchive.nationalarchives.gov.uk/ukgwa/20230418175226/https://ukcop26.org/glasgow-leaders-declaration-on-forests-and-land-use/.

The core of the EUDR establishes conditions for the production of raw materials and products that are considered to be global drivers of deforestation.[6] It places the burden of proof of compliance with these conditions on the importer of these goods into the EU market. Specifically, the raw materials and products in question may only be imported into and traded in the EU if they meet the following conditions:

1 · They have been produced on land **that was not deforested after December 31, 2020.**
2. **They have been produced in accordance with the laws of the producing country,** both in terms of the environment and the protection of human rights and the rights of Indigenous peoples.
3. **They are supported by a due diligence process in the form of a statement** that provides detailed, verifiable information on how the exporter's products meet the requirements of the EUDR.

Although the regulation entered into force in June 2023, the deadline for the implementation of more than half of its provisions is December 2024.[7] These include the entry ban and the due diligence obligation, which are the provisions that are most relevant for third countries, either because they potentially prevent them from importing products into the EU or because of the possible transfer of operators' costs associated with measures to ensure compliance with market access conditions.

The regulation mentions the need for coordination with the exporting countries on several occasions.[8] In this sense, third countries need to be aware of three fundamental aspects, which are discussed in the following paragraphs.

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[6] · Regulation (EU) 2023/1115 concerns importation into, and trade within, and exportation from the EU market of products that contain, are fed to, or are processed using relevant commodities, namely: **cattle, cocoa, coffee, oil palm, rubber, soybeans, and timber** (see annex B for more details).

[7] · See annex A, table A1.

[8] · The regulation includes an entire chapter on benchmarking and cooperation with third countries. It establishes that the European Commission "shall engage in a coordinated approach with producer countries and parts thereof that are concerned by this Regulation, in particular those classified as high-risk." To this end, a comprehensive EU strategic framework will be developed, and cooperation mechanisms will be promoted, focusing on "the conservation, restoration, and sustainable use of forests, deforestation, forest degradation, and the transition to sustainable commodity production, consumption, processing, and trade methods." This cooperation may take different formats, such as structured dialogue, administrative agreements or joint roadmaps, as well as support for legal reform processes in these countries. The regulation also mentions the development of integrated land-use planning processes, the adoption of relevant legislation in producer countries, multilateral processes, and fiscal or trade incentives, among other instruments.

The first issue arises from the distinction between the concepts of **“deforestation-free” and “legal compliance.”** The EU will not admit raw material produced on deforested land even if production complies with local law in the producing country. This aspect entails a potential coordination challenge for countries where a raw material has been produced in a forested area whose conversion to agricultural use has been legally authorized.

The second issue relates to compliance with the producing country’s legislation, which entails verifying **compliance with a broad legal framework**.[9] On this point, the due diligence obligation mainly entails criteria[10] related to institutional[11] and socioenvironmental[12] factors in the producing country. Analyzing how local legislation fits into the traceability mechanisms currently in place for trade with the EU is key to fully understanding the degree of complexity the new regulation introduces.

The EUDR does not stipulate that certifications are required to demonstrate compliance and only mentions that certifications may be submitted as supplementary information to the risk assessment. What this means in practice is that the format of the due diligence system and the compliance mechanisms deemed reliable have not yet been defined. **This could be seen as a risk to the clear implementation of the EUDR but could also be interpreted as an opportunity for producer countries to design strategies that are appropriate to their circumstances.** Two issues are key here: the level of traceability required and attention to the specificities of operations at each stage in the supply chains for the raw materials in question.

Third, **the EUDR establishes a benchmarking system[13] to classify countries according to their risk level**. There are three groups: high-risk, low-risk, and standard risk. The risk level that the EU assigns to each country or area of a country **will determine the complexity and thoroughness of the due diligence process that the operator must comply with.** For example, a simplified due diligence procedure may apply to products from low-risk countries, while more exhaustive checks will be carried out on products from high-risk countries. There is still considerable uncertainty regarding the scope of these control mechanisms, the costs they may eventually generate (in terms of efforts to demonstrate compliance to the importer and the regulator), and their format (i.e., whether there will be a single EU-wide procedure or separate procedures for each EU member country).

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[9] · Article 2, paragraph 40. The issues involved include land-use rights, environmental protection, forest-related rules (including forest management and biodiversity conservation when directly related to wood harvesting); third parties’ rights; labor rights; human rights; the principle of free, prior, and informed consent; tax, anticorruption, trade, and customs regulations.

[10] · See annex A, table A2.

[11] · For example, concerns about corruption, the falsification of documents and data, lack of law enforcement, human rights violations, and others.

[12] · In terms of the information required by the EUDR, the use of geolocation is not mandatory nor does the regulation refer to a specific forestation database or platform. Consequently, the potential use of existing geolocated maps in producing countries and the EU and whether these are compatible with the due diligence system are relevant issues.

[13] · The EUDR establishes three risk tiers: high-risk (countries or parts of countries in which the assessment detected commodities that do not comply with the deforestation-free conditions of the regulation), low-risk (countries or parts of countries for which the assessment concludes that there is sufficient assurance that instances of production that are not deforestation-free are exceptions), and standard (countries or parts of countries that do not fall into either of the above categories).

Argentina has more than 53 million ha of native forest.[14,15] **This represents 19.2% of the country's continental area, 6.5% of the native forest in South America, and 1.4% of the world's native forest** (MAyDS, 2020). **The provinces with the largest forested areas are Santiago del Estero, Salta, Chaco, and Formosa** (the Parque Chaqueño region).[16]

According to the Ministry of Environment and Sustainable Development (MAyDS, 2021), **between 1998 and 2018, around 6.5 million ha of native forest were lost in Argentina** (21% of the total area of native forest in the country).[17] **The expansion of the agricultural frontier in the Parque Chaqueño region represents the second-largest instance of deforestation in South America after the Amazon. Globally, Argentina is among the ten countries to have experienced the greatest net loss**[18] **of forest cover between 2000 and 2015 (Ritchie and Roser, 2021) and among the ten countries with the greatest increase in loss of primary forest**[19] **in 2020–2022 (Global Forest Review, 2022).**[20]

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14 · Native forests are natural forest ecosystems composed mainly of mature native tree species, with various associated species of flora and fauna, together with the surrounding environment—soil, subsoil, atmosphere, climate, water resources. Together, these form an interdependent network with specific features and multiple functions. A dynamic equilibrium emerges when the forest system is in its natural state, providing society with multiple environmental services, in addition to various natural resources that can be used for economic purposes. The definition includes primary native forests, in which there has been no human intervention, and secondary forests, which arose after deforestation, and those resulting from voluntary replanting or restoration (Law No. 26.331, section 2).

15 · Native forests provide services to society such as water regulation; soil formation and conservation; biodiversity conservation; carbon sequestration; the provision of food, water, energy sources, construction materials, or medicines; and the preservation and defense of cultural identity, among others.

16 · This is largest area of native forest in Argentina and includes humid zones in the east and dry zones in the west. It is an extremely diverse environment, which translates into high levels of biodiversity. Its surface area is distributed over the provinces of Catamarca, Chaco, Córdoba, Corrientes, Formosa, Jujuy, La Rioja, Salta, San Juan, San Luis, Santa Fe, Santiago del Estero, and Tucumán (MAyDS, 2022).

17 · According to data from the First National Inventory of Native Forest (2005), which reported 31,443,873 ha.

18 · Net forest cover change measures any increase in forest cover through either the growth of natural forests or afforestation through tree planting, minus deforestation (Ritchie and Roser, 2021).

19 · These are areas of mature rainforest that are particularly important for biodiversity, carbon storage, and the regulation of regional and local climate impacts (Global Forest Review, 2022).

20 · The top ten countries were determined by comparing the average primary forest loss in 2015–2017 with the average loss in 2020–2022. Includes countries with at least 1 Mha of tropical primary forest in 2001 (Global Forest Review, 2022).

In 2007, Argentina's Forest Act was passed: Law 26.331 on Minimum Public Financing for the Environmental Protection of Native Forests. The law sets minimum environmental protection budgets for the enrichment, restoration, conservation, use, and sustainable management of native forests and the environmental services they provide.

It introduced two key instruments: (i) a mandate for Argentina's provinces to implement a Territorial Ordinance on Native Forests (each province is the primary owner of its natural resources, including the native forest ecosystem), and (ii) the creation of the National Fund for the Enrichment and Conservation of Native Forests, which was designed to strengthen local enforcement authorities and the national enforcement authority, and to provide financial support for forest owners to implement sustainable forest management schemes. This law also laid the groundwork for an institutional framework for environmental services in Argentina.[21]

According to the latest report on compliance with the Forest Act, all 23 provincial jurisdictions have laws on land-use planning for native forests, and the area of native forests managed or under conservation in the country has gradually increased. Currently, under Law 26.331, areas that are classified as green in a province's Territorial Ordinance on Native Forests can be deforested, subject to the presentation of a deforestation plan that must be approved by the competent authority in each province.[22] The deforestation of areas classified yellow and red is prohibited by law. **The problem is that if deforestation plans are authorized now in green areas, anything subsequently produced there cannot be exported to the EU, as explained above.**

The authorities responsible for monitoring these activities are overseen by the Ministry of Environment and Sustainable Development and include the Center for Environmental Information (CIAM), the Integrated Environmental Information System (SiNiA), the Native Forest Monitoring System of Argentina (SNMBN), which includes an early warning system[23] (EWS) to prevent illegal deforestation and audit authorized deforestation. Specifically, the EWS monitors the loss of native forests through continuous semiautomated processes based on satellite imagery, with the aim of strengthening the oversight and monitoring work of local authorities in the enforcement of Law 26.331.

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[21] · These services include water regulation, biodiversity conservation, the maintenance of soil and water quality, the capture of GHG emissions, the diversity and beauty of the landscape, and the preservation of cultural identity (art. 5).

[22] · Before issuing a permit for the clearing or sustainable use of land, the enforcement authority in each jurisdiction must carry out an environmental impact assessment. This process is mandatory in the case of land clearing and when the sustainable management project could potentially cause significant environmental impacts.

In relation to product traceability, soybean producers have developed a private-sector initiative,[24] the Gran Chaco Monitoring System (ViSeC) initiative, a space promoted by The Nature Conservancy, Tropical Forest Alliance, and the Argentine Edible Oil Association (CIARA). Launched in 2019, the initiative aims to bring together different stakeholders in the soybean value chain to reduce the negative environmental impacts of soybean production. ViSeC has developed a monitoring, reporting, and verification system to trace soybeans from the farm where they are grown to the port from which they are shipped. The system covers industrial processing such as storage, crushing, and so on. Processing companies are responsible for reporting on what enters and leaves their plants, thus guaranteeing the coherence and consistency of the entire traceability process. At this stage, it may be feasible for certification bodies to become involved in the process.

There are public sector information systems for trade in soybeans run by the Federal Tax Authority of Argentina (AFIP) and for livestock run by the National Service for Agrifood Health and Quality (SENASA), which could be integrated.

The Ministry of Foreign Affairs and Worship has created the Economic Relations and Environment Unit to monitor regulatory processes.[25] This unit focuses specifically on the challenges posed by the new environmental trade rules and is playing a crucial role in following up on the guidelines being developed to ensure that Argentinian exports comply with the EUDR. It is also important in bilateral negotiations with the EU member countries where the main ports of entry for Argentinian products are located since each member state is responsible for enforcing the EUDR in their territories.

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23 · This system was initially implemented in the Parque Chaqueño region. Expansion to the Tucumán-Bolivia Jungle and Misiones Jungle regions is planned, which will help reduce and control deforestation in these other regions (MAyDS, 2021).
24 · For more information, see https://www.visec.com.ar/#nosotros.
25 · Created through Resolution no. 169/2022.

One final factor that should be considered is the entry into force of the EUDR and future changes or revisions to it, given forecasts that in 2025 its scope may be extended to other commodities and products (such as corn or biofuels) and to other ecosystems and lands with high carbon stocks and high biodiversity value, such as grasslands, peatlands, and wetlands.

3. TAILORING ARGENTINIAN EXPORTS TO COMPLY WITH THE EUDR

The EU is the second-largest destination for Argentinian goods exports.[26] In 2022, Argentina posted total goods exports of US$88.45 billion, 12.3% of which went to the EU. The only economy ahead of the EU was Brazil, Argentina's top export destination, which accounted for 14.3% of total foreign sales. The other countries in Argentina's top five export markets in 2022 were China (9%), the United States (7.6%), and Chile (5.7%). In 2012–2022, Brazil and the EU were the first- and second-largest export destinations for Argentinian products, respectively.

Given this state of affairs, Argentina needs to determine what percentage of its exports to the EU would potentially be affected by the implementation of the EUDR. To answer this question, we analyzed the following: i) annual imports from Argentina reported by the EU, disaggregated by product,[27] and ii) the list of products covered by the EUDR (see annex B). By cross-checking these two datasets, we quantified the volume of Argentinian exports to the EU affected by the EUDR.[28]

The data reveals that if Argentina does not comply with the EUDR, it could have a substantial impact on the country's exports to the EU. As figure 1 shows, in 2022, US$4.7 billion of products now covered by the EUDR were sold to the EU, **representing more than 40% of Argentina's goods exports to the bloc or about 5% of its total goods exports in 2022.**[29] In 2012–2022, on average, US$4 out of every US$10 of goods exported to the EU were products that are now covered by the EUDR.

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[26] · Since the EUDR does not cover service exports, the following section focuses only on Argentina's goods exports

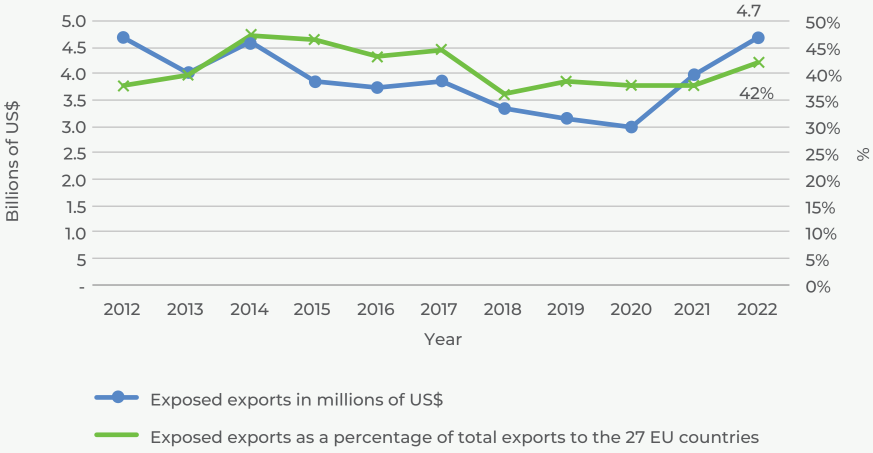
[27] · This database is disaggregated by product at the Harmonized System (HS) 6-digit level. It is available from UN COMTRADE.

[28] · We did not use export data by product as reported by Argentina because these databases have entailed measurement problems since 2018 that ultimately conceal the product type of a significant number of exports. For this reason, we opted to draw on the "mirror" data for Argentinian exports—that is, imports from Argentina as posted by the country's trading partners.

[29] · We did not use export data by product as reported by Argentina because these databases have entailed measurement problems since 2018 that ultimately conceal the product type of a significant number of exports. For this reason, we opted to draw on the "mirror" data for Argentinian exports—that is, imports from Argentina as posted by the country's trading partners.

Figure 1.

Argentinian Exports Impacted by the EUDR (2012–2022) (Billions of US$ and as a percentage of exports to the EU)



Note: Import data from the 27 countries that currently comprise the EU (excluding the United Kingdom).
Source: Authors based on UN COMTRADE.

A closer look at the exports affected by the EUDR shows that the main Argentinian export products at stake are soybeans and cattle. In fact, of the nearly US$4.7 billion in exports to the EU in 2022 that would be affected by the measure, soybeans account for 80% and cattle for 15%.[30] Oil palm products account for nearly 5%, and the remaining 0.4% is made up of wood, rubber, cocoa, and coffee. Table 1 shows the top five products at the HS 6-digit level affected by the EUDR. The first of these, accounting for 32% of total exports to the EU, is oilcake and other solid residues resulting from the extraction of soybean oil, such as ground soybeans or pellets (HS code 2304.00). The second is boneless beef, fresh or chilled (HS code 0201.30), which accounts for 5% of exports to the EU.

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30 · Most of Argentina's soybean exports in 2022 went to Spain (US$918 million), Poland (US$820 million), and Italy (US$661 million). The largest importers of Argentinian beef were Germany (US$335 million), the Netherlands (US$130 million), and Italy (US$70 million).

Table 1.
Top Five Products Covered by the EUDR Exported by Argentina to the EU, at the HS 6-digit Level (2022)

HS CODE	DESCRIPTION	PRODUCT TYPE	EXPORTS FROM ARGENTINA TO THE EU, 2022 (MILLIONS OF US$)	% OF EXPORTS TO THE EU
2304.00	Oilcake and other solid residues, whether or not ground or in the form of pellets, resulting from the extraction of soybean oil	Soybeans	3,559	31.9%
0201.30	Boneless meat of bovine animals, fresh or chilled	Cattle	560	5.0%
3823.19	Industrial monocarboxylic fatty acids; acid oils from refining; industrial fatty alcohols (excl. stearic acid, oleic acid, and tall oil fatty acids)	Palm oil	220	2.0%
4104.41	Tanned or crust hides and skins of bovine (including buffalo) or equine animals, without hair on, in the dry state (crust), full grains, unsplit; grain splits	Cattle	84	0.8%
1507.90	Soybean oil and its fractions, whether or not refined, but not chemically modified	Soybeans	67	0.6%

Note: Products are classified according to HS Nomenclature 2022 Edition.
Source: Authors based on UN COMTRADE.

Argentina is one of the main suppliers of soybeans and domestic cattle to the EU: in 2022, the bloc imported US$18.73 billion in the former and US$4.35 billion in the latter. Looking more closely at the origin of these imports, **Argentina is the second-largest supplier of soybeans and the third-largest supplier of cattle to the EU**, accounting for about 20% and 16% of imports of these products, respectively. The number one supplier of both products is Brazil, which accounted for nearly 46% of soybean imports and 21% of cattle imports into the EU in 2022. Other countries competing with Argentina and Brazil in the soybean market include the United States (which in 2022 accounted for 17% of EU soybean imports), Ukraine (5%), and Canada (3%). Other major origins of cattle shipped to the EU are the United Kingdom (17%), the United States (9%), and Uruguay (8%).

Figure 2.

Major Exporters of Soybeans (Figure 2a) and Cattle (Figure 2b) to the EU (2022)

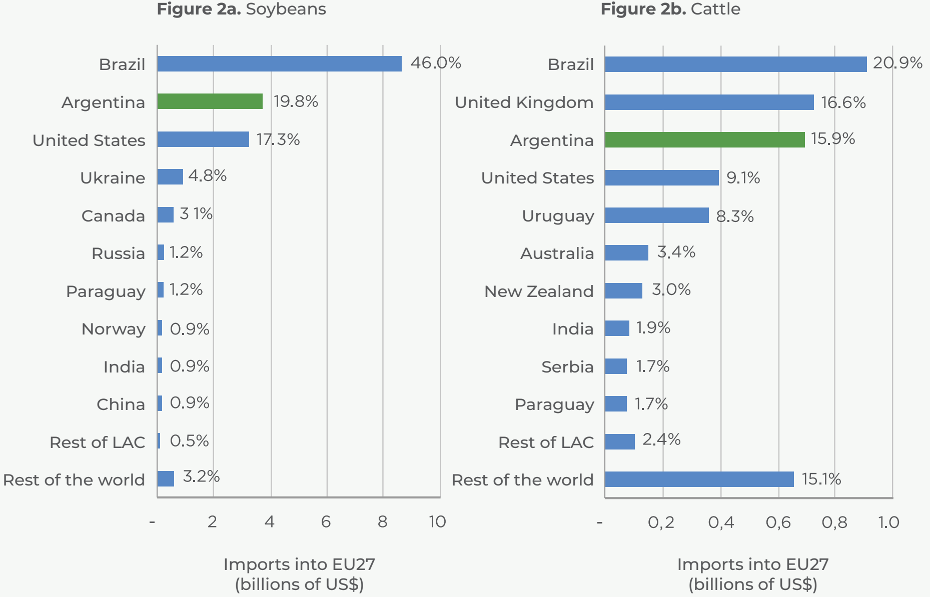

Figure 2a. Soybeans

Figure 2b. Cattle

Note: Import data from the 27 countries that currently make up the EU (excluding the United Kingdom). The percentages refer to each country's share in total EU imports of each product.
Source: Authors based on UN COMTRADE.

In production terms, the EU has a low level of self-sufficiency in soybeans but is self-sufficient in beef. According to the latest estimates, for the 2022/2023 trade year, only 3% of the soybean meal used as a protein source for animal feed in the EU was produced within the bloc, while the remaining 97% was imported[31,32]. In contrast, it is estimated that the EU will have produced 5% more domestic beef in 2023 than it was projected to consume that year.[33] **This data reveals the EU to be highly dependent on soybean imports but not on beef imports.**

So far, we have focused on the impact of the EUDR on Argentina as a whole. However, not all the country's provinces will be affected by the new regulation to the same extent. In fact, **the provinces that account for the largest shares of Argentina's exports of products covered by the EUDR have deforested the least number of hectares in recent years** (see figure 3).

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31 · Directorate-General for Agriculture and Rural Development, European Commission. 2022. "Crops market observatory." https://agriculture.ec.europa.eu/data-and-analysis/markets/overviews/market-observatories/crops_en.

32 · If other oilseed meals used as feed protein (including rapeseed and sunflower) are included, the EU produces 24% of what it needs to feed its livestock sector, a figure that is still low.

33 · Directorate-General for Agriculture and Rural Development, European Commission. 2023. "Meat products short-term outlook." https://agridata.ec.europa.eu/extensions/DashboardSTO/STO_Meat.html.

Between 2017 and 2019, three provinces accounted for 93% of the country's exports of the products in question: Santa Fe (53%), Buenos Aires (25%), and Córdoba (15%).[34] These provinces have deforested the least hectares: in 2021, they accounted for less than 5% of the total deforested area in Argentina.[35] In contrast, Santiago del Estero, Formosa, Chaco, and Salta accounted for 64% of deforested hectares but only 2% of the exports impacted by the EUDR.[36] These differences in deforestation between provinces point to the need for a subnational approach to EUDR adaptation efforts.

Figure 3.
Deforestation in Argentina and Impact of the EUDR by Province

Figure 3a. Loss of Forest Land and Other Land, by Province (hectares, 2021)
Figure 3b. Province-Specific Exports of Products Covered by the EUDR (billions of US$, annual average 2017–2019)

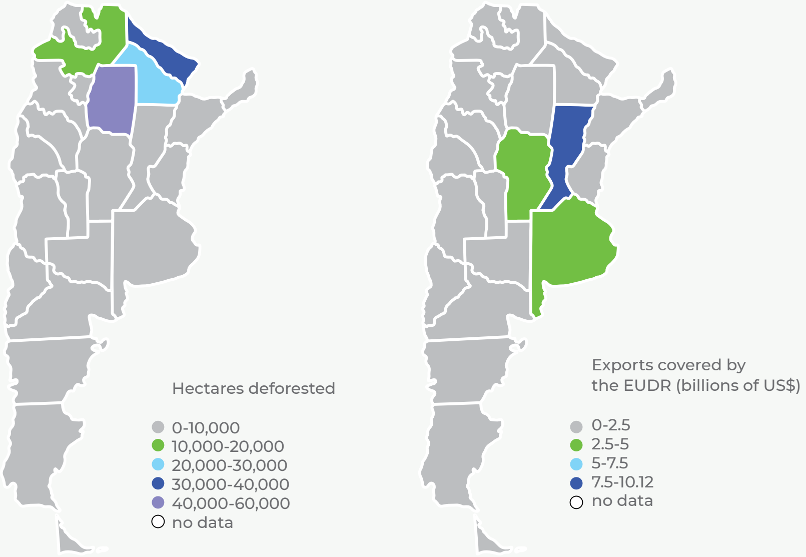
Figure 3a
Figure 3b

Source: Authors based on CEP XXI (Centro de Estudios para la Producción) and the Native Forest Monitoring System of Argentina.

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[34] Province-specific export data was obtained from CEP XXI (Centro de Estudios para la Producción). Unfortunately, at the time of writing, it was not possible to access this data disaggregated by destination country. The data presented here is for province-specific exports of products covered by the EUDR to the world (and thus not necessarily to the EU). Furthermore, the province listed as the origin of these exports is not necessarily where the raw material was produced, but instead refers to the location of the establishment where it was manufactured.

[35] Deforestation is defined here as the loss of forest and other nonforest land according to data from the Argentinian Inventory of Native Forests.

[36] Although the provinces with the highest deforestation rates do not account for a significant share of the national exports covered by the EUDR, the new regulation could still have a major impact on the total exports of some provinces. The EUDR affects 22% of exports from Santiago del Estero, the province with the highest levels of deforestation during this period. This share is 17% in Chaco, 8% in Salta, and 6% in Formosa.

The share of Argentinian exports affected by the EUDR is significant: more than 40% of the country's exports will need to be accompanied by the necessary documentation for the European importer to submit due diligence in the destination country from January 2025 onward. However, given the specific provinces that these products come from in Argentina and the low rate of deforestation there, the risk of being denied entry to the EU market could be reduced.[37] It is therefore also important to understand how the risk associated with each exported will be assessed by the EU (or each EU country): will an average risk level be calculated (weighted or otherwise) for the country, or will risk be area-specific?

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37 · Although the document states that much of the universe of goods covered by the EUDR comes from provinces where deforestation has not taken place, this would not eliminate the costs of proving the origin of production that may be required by the EUDR. There is also uncertainty as to whether the EUDR will consider the country as a whole or will allow product origins to be broken down into subnational units.

4. SIMULATION OF THE IMPACT OF THE EUDR WITHIN A CGE MODEL FOR ARGENTINA

To quantify the potential impact of the introduction of the EUDR on the Argentinian economy, we simulate a maximum scenario for this regulation within a computable general equilibrium (CGE) model.

Given the multiplicity of direct and indirect effects associated with this shock, general equilibrium models need to be used to analyze them. CGE approaches make it easier to capture the interdependencies between productive sectors, the interaction between economic agents, the role of prices in decision-making, and the macroeconomic constraints under which an economy normally operates. These models are also a simulation tool: once calibrated using real data from the economy in question, they allow us to simulate the effects of policy changes (among other shocks), as other authors have done in relation to international trade (Scollay and Gilbert, 2000; Gilbert and Wahl, 2002; Robinson and Thierfelder, 2002; Lloyd and McLaren, 2004).

The instrument used in this paper is a static real CGE model, which has features in common with the International Food Policy Research Institute standard (Lofgren et al., 2002).[38] Generally speaking, the model considers a small open economy in which producers and consumers maximize profits and utility, respectively, in competitive markets. One important feature of our approach is that the equations related to the balance of payments are disaggregated by trading partner to better account for the introduction of export restrictions (see annex B for a further description of the model.[39]

The main source of data used to calibrate a CGE model is the social accounting matrix (SAM), an accounting record of all economic transactions that have taken place in an economy during a given period. For this study, we use the SAM in Banerjee and Cicowiez (2021a), which is for 2018 and disaggregates trade flows by trading partner using data from UN Comtrade. The disaggregation of products and activities focuses specifically on those affected by the EUDR, such as agricultural activities and related manufacturing.

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[38] · The model was also used in De la Vega (2020), which describes the mathematical approach and clarifies further methodological issues.

[39] · The model requires closure rules to be specified for three macroeconomic balances: government, saving and investment, and balance of payments. All simulations assume that: i) government borrowing is balanced by changes in real domestic financing; ii) private investment is endogenously determined by the level of savings; and iii) savings from the rest of the world are exogenous (measured in the currency of the rest of the world), so that the real exchange rate varies endogenously to balance foreign exchange inflows and outflows. The numeraire for the model is the consumer price index.

The SAM data is complemented with elasticities and other behavioral parameters that are obtained from the literature and/or econometric estimations. For this project, these inputs are obtained from the publicly available resources of the Inter-American Development Bank's Open IEEM project (Banerjee and Cicowiez, 2021a and 2021b). Specifically, the model used in this research requires values for: i) the wage elasticity of unemployment (-0.1), and ii) the elasticities of substitution between domestic purchases and imports (Armington, 1969) and of transformation between domestic sales and exports (constant elasticity of transformation—CET), which range from 0.9 to 2.

This model is used to assess the isolated effect of the introduction of the EUDR. The shocks are introduced by exogenously changing the quantities of the affected products exported to the EU. The magnitude of the shocks is related to the share of the products affected by the EUDR in the value of exports to the EU, as shown in column 1 of table 2. For example, 1% of the total volume of product 1 that is exported to the EU is affected by the EUDR,[40] which means that this is the percentage included in the categories covered by the new regulation. The exercise thus provides an upper bound for estimating the potential reduction in exports of the affected products, assuming that 100% of these products do not comply with the new regulations and therefore cannot be exported.

In addition, columns 2 and 3 of table 2 show the share of exports to the EU in total exports (column 2) and the share of exports in the total demand for each sector (column 3). Intuitively, the higher the shares in the three columns of the table, the greater the impact of the shock. For example, 100% of the exports of product 20 (paper and paperboard products) to the EU would be affected by the EUDR. However, only 1% of exports of this product actually go to the bloc, which in turn represents just 0.11% of sales in this sector. The direct impact of the shock on the sector is thus expected to be small. On the other hand, 97% of exports of product 11 (leather goods) to the EU would be affected by the EUDR since the bloc accounts for 22% of exports and 43% of sales in the sector. The direct impact (shock) on this industry is expected to be significant. Therefore, the sectors that could be most affected include meat and meat products (product 6), residues from the extraction of vegetable oils (8), leather goods (11), and chemical products (22). However, the final impact also depends on the share of each item in intermediate and final demand, the evolution of activity levels, agents' incomes, and the role of prices.

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[40] The introduction of the EUDR was also simulated for other destination markets such as the USA and China, in addition to the EU. The results are qualitatively similar to those presented here, although they are of course quantitatively higher when the share of exports covered by the measure is higher.

The sensitivity of the results is assessed by making different assumptions about the magnitude of the key elasticities in our analysis, such as the transformation elasticities between domestic sales and exports. Since the range estimated in the literature for these elasticities is between 0.9 and 2, the simulations are iterated by modifying the value of the elasticities of the products affected by the EUDR within this range. Obviously, the lower the value of this elasticity, the larger the impact is expected to be (i.e., the higher the adjustment costs) since it will be more difficult to "convert" exports to a given destination into domestic sales or sales to other destinations.

Table 2.
Magnitude of the Impact of the EUDR as a Share of Exports and Sales in Each Sector

PRODUCT	IMPACT: % OF TOTAL EXPORTS TO THE EU AFFECTED BY THE EUDR	% EU IN TOTAL EXPORTED	% EXPORTS IN DEMAND
Product 1: Oilseeds and oleaginous fruit	1.12	35.74	12.19
Product 2: Beverage crops and spices	5.58	12.39	1.60
Product 3: Live animals	-	9.27	0.33
Product 4: Other animal products	7.38	28.31	5.91
Product 5: Wood in the rough and other forestry products	1.69	52.55	2.19
Product 6: Meat and meat products	87.55	23.04	14.30
Product 7: Animal and vegetable oils and fats	7.78	2.55	25.40
Product 8: Residues resulting from the extraction of vegetable fats or oils	97.21	29.92	67.60
Product 9: Cocoa, chocolate, and sugar confectionery	81.78	1.39	3.96
Product 10: Other food products n.e.c.	0.31	1.55	6.55
Product 11: Leather goods	96.87	21.84	43.28
Product 12: Sawn or sliced wood; wood in the rough	100.00	0.75	6.15
Product 13: Builders' joinery and carpentry of wood	100.00	1.6	0.16
Product 14: Wooden packing cases, crates, and boxes; casks and barrels	100.00	8.22	0.32
Product 15: Other articles of wood or cork	76.86	15.60	0.58
Product 16: Paper pulp, paper, and cardboard	99.90	0.49	5.57
Product 17: Books, maps, and printed scores	100.00	10.79	2.28
Product 18: Newspapers, journals, magazines, and periodicals	100.00	1.14	0.40
Product 19: Other printed items	100.00	3.21	0.81
Product 20: Articles or paper or paperboard	100.00	0.99	0.11
Product 21: Basic organic chemicals	62.53	23.30	8.22
Product 22: Miscellaneous basic chemicals	30.45	16.13	54.06
Product 23: Tires and other rubber products	95.38	2.35	7.81
Product 24: Furniture	98.65	2.06	0.87
Product 25: Prefabricated buildings	100.00	0.01	5.94
Product 27: Industrial wastes	0.30	46.43	28.94

Source: Authors based on Banerjee and Cicowiez (2021), COMTRADE, and WTO.

5. RESULTS OF THE EUDR SIMULATIONS USING THE CGE MODEL

The introduction of the EUDR exogenously reduces external demand for the products directly affected by the measure. This reduction may be offset by an increase in sales to other destinations or to the domestic market. When the values of the elasticities are reasonable, the first of these effects (the reduction in demand) is expected to dominate, such that a fall in production would be seen.

In fact, as figure 4, panel A shows, real output declines for virtually all products except 1, 2, 5, and 28 (oilseeds, edible plants and spices, unprocessed wood, and other agricultural products of agricultural activities), demand for which is offset by increases in exports (figure 4, panel B) and domestic sales (figure 4, panel A). However, higher demand for investment does not offset the decline in manufactured goods (product 26). The performance of exports, meanwhile, is mixed (figure 4, panel B). While some products are jeopardized directly by the introduction of quantitative restrictions associated with the EUDR (meat products, leather, and wood products), others benefit indirectly from real depreciation (services and other manufactured goods).[41] Nevertheless, total exports are falling. In line with the drop in production and as a consequence of the increase in the real exchange rate, imports in most sectors decline (figure 4, panel C). Domestic sales of most products fall for most products (figure 4, panel D).

These sector-specific impacts have consequences on factor demand and remunerations and the income of the owners of these factors, which are households. Table 3 presents the impacts on the labor market, GDP and its components, the government budget, and welfare. Three columns show the minimum, median, and maximum values for the elasticities (CET), as explained in the previous section.

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41 · Products below 0 are disadvantaged and the opposite is true for those above 0.

With respect to the labor market, although the unemployment rate remains practically unchanged, between 7,000 and 13,000 jobs are lost, and real wages fall by between 0.22% and 0.39%. As a result, disposable household income is reduced, as evidenced, for example, by lower private consumption and a decline in direct tax revenues.

Looking to macroeconomic aggregates, absorption declines in all the simulations due to private consumption, despite a slight increase in investment, as there is an increase in external financing to the private sector in local currency as a result of the real depreciation that is confirmed in all the simulations. This appreciation of the real exchange rate is necessary to close the balance of payments, which deteriorates in response to a growing trade deficit. Overall, GDP falls by between 0.15% and 0.26%. As far as the government accounts are concerned, the decline in economic activity, household income, and demand for imports translates into reductions in revenue collection and total income. Since public spending remains unchanged, public debt increases.

Finally, since the model is static, changes in welfare can be measured through variations in prices and income based on the definition of equivalent variation.[42] This refers to the additional monetary income needed for the household to enjoy the same welfare at initial prices after the shock has taken place. Welfare declines in all the simulations. The results depend on the model (the design of the simulation scenario) and the data used (base year and calibration parameters). For example, the lower the value of the CET elasticities, the higher the adjustment costs. However, the main results are robust, particularly in terms of their qualitative interpretation, such that they take into account the direction and range of magnitude of the effects, which are necessary to inform policy decisions aimed at mitigating negative effects and exploiting the positive externalities of regulations of this sort. Future research could refine these estimates once countries exporting to the EU have started gaining experience in implementing the new regulation (e.g., in relation to the costs of any associated certifications) and once the operational consequences of the risk levels assigned to each country are known.

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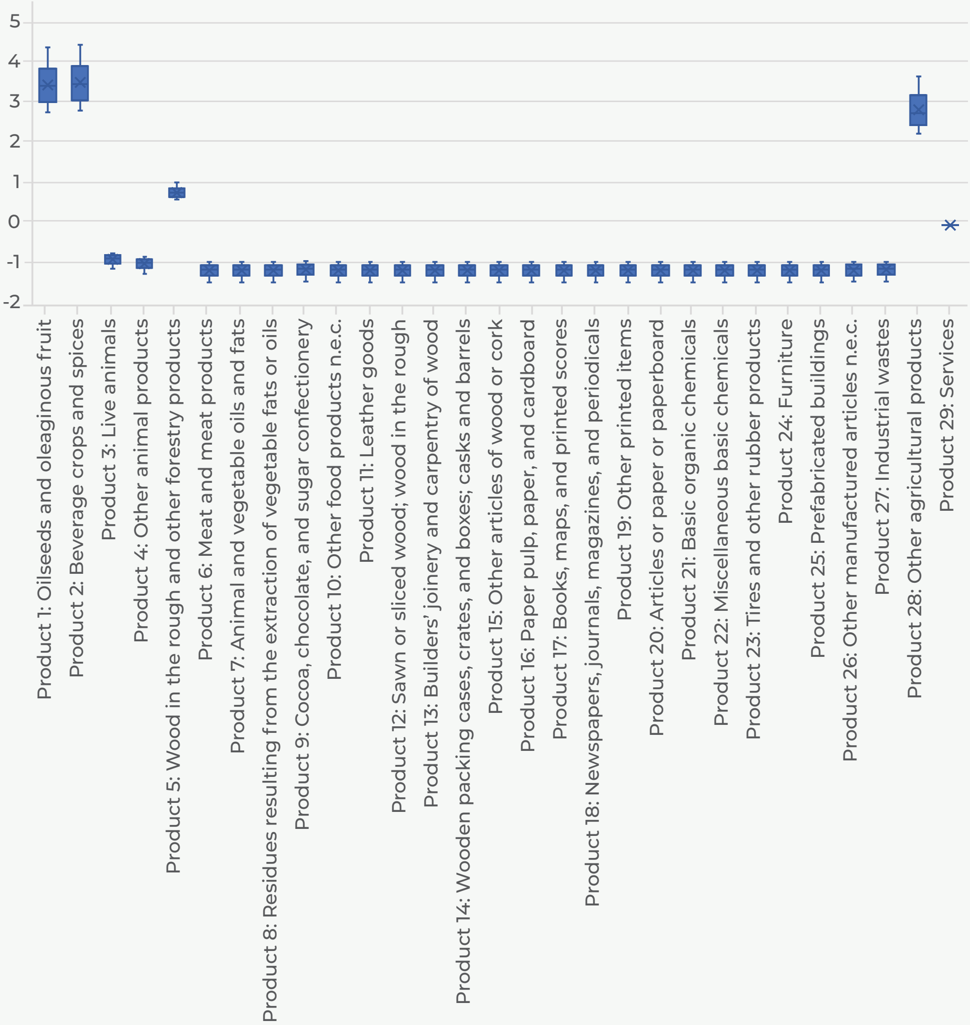
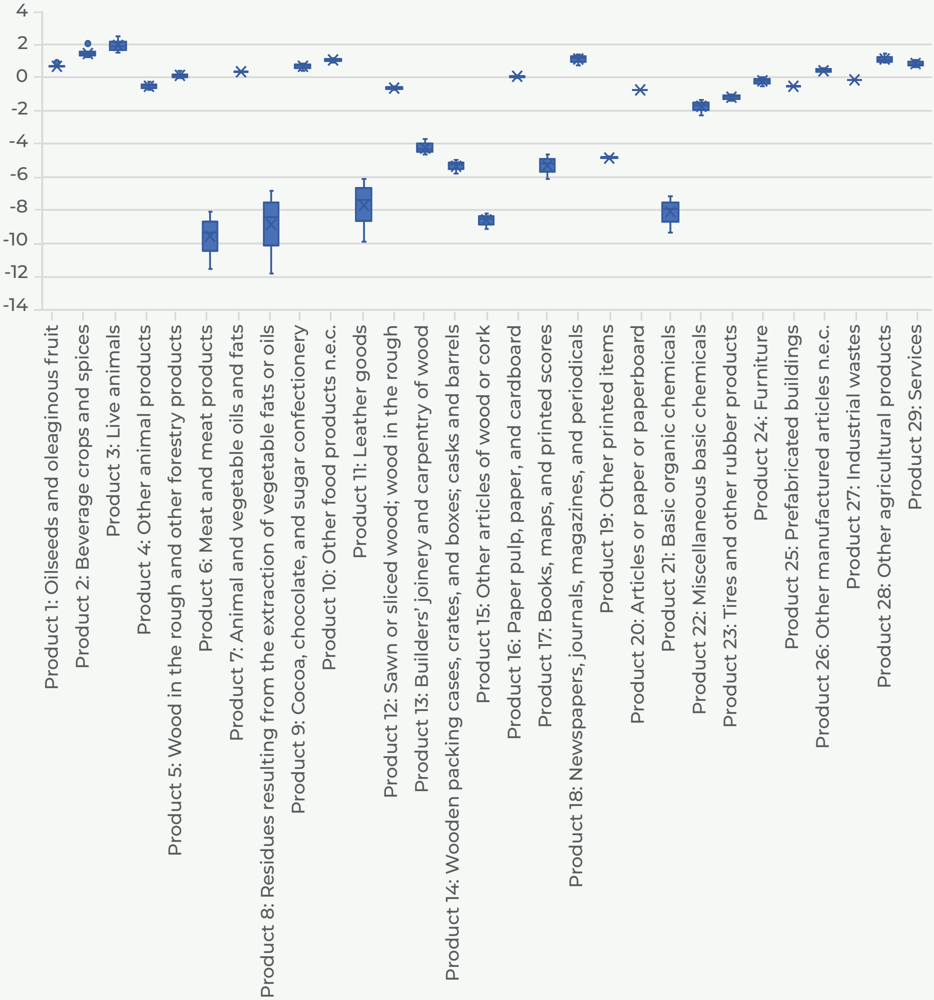
42 · In a dynamic model, the assumption of endogenous investment means that equivalent change is not a good indicator of changes in welfare, since changes in investment lead to changes in the capital stock in the following period.

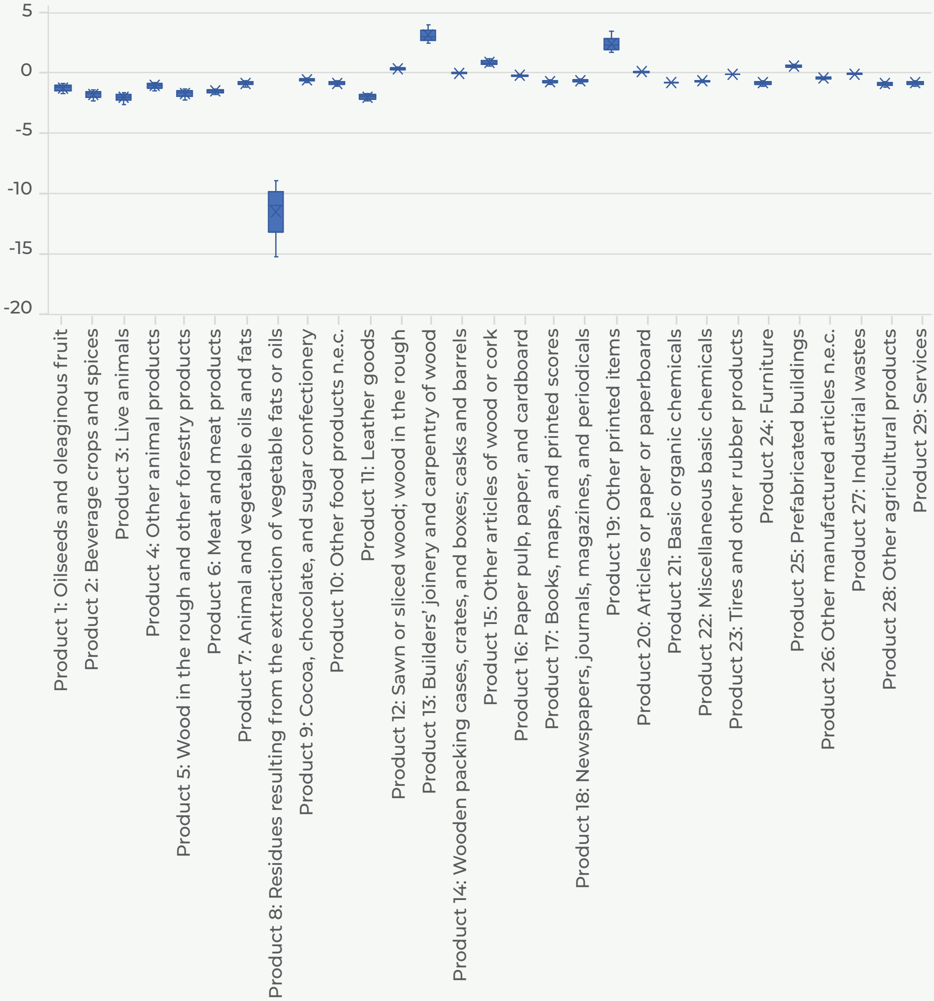
Figure 4.
Sector-Specific Effects
Panel A: Real Output (% change from baseline scenario)



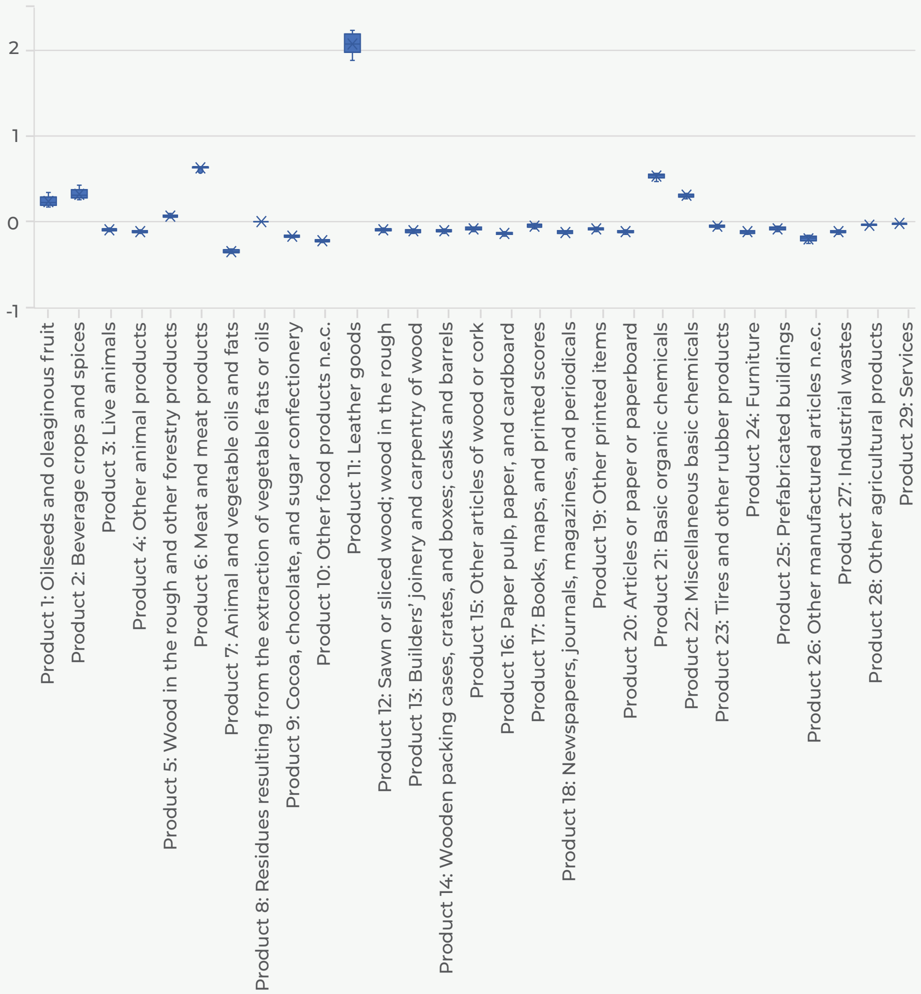


Panel B: Exports (% change from baseline scenario)

Panel C: Exports (% change from baseline scenario)



Panel D: Domestic Sales (% change from baseline scenario)



Note: The figure shows the distribution of the scores for each sector. The solid line within each box represents the median.
Source: Authors.

Table 3.
Effects on Macro Aggregates and the Labor Market

	BASE	MIN	P50	MAX
Macro results (actual amounts) (% change from baseline)				
Absorption	15.02	-0.20	-0.14	-0.11
Private consumption	10.24	-0.31	-0.22	-0.18
Government consumption	2.33	0.00	0.00	0.00
Investment	2.25	0.04	0.05	0.07
Various stocks	0.20	0.00	0.00	0.00
Exports	2.14	-1.31	-0.98	-0.79
EU	0.39	-24.38	-24.29	-24.12
United States	0.27	0.75	0.86	1.05
China	0.12	3.91	4.57	5.01
Rest	1.36	4.40	4.94	5.26
Imports	2.42	-0.48	-0.60	-0.80
GDPmp	14.74	-0.26	-0.19	-0.15
Indirect taxes	2.07	-0.23	-0.17	-0.13
GDPfc	12.67	0.00	0.00	0.00
Real exchange rate	1.00	0.62	0.77	1.03
Labor market				
Employment in millions (% change from baseline)	20.,57	-0.06	-0.05	-0.04
Unemployment rate (p.p. change)	0.14	0.00	0.00	0.00
Wages (% change)	0.26	-0.39	-0.28	-0.22
Government budget (% change from baseline)				
Direct household taxes	0.94	-0.34	-0.24	-0.19
Direct taxes on factors	1.25	-0.43	-0.31	-0.25
Taxes on activities	-0.19	-0.17	-0.22	-0.31
Taxes on products	2.16	-0.36	-0.26	-0.20
Taxes on imports	0.11	0.15	0.19	0.25
Transf. RoW	0.05	0.54	0.67	0.88
Total income	4.31	-0.35	-0.25	-0.20
Consumption	2.33	-0.28	-0.20	-0.16
Domestic transf.	2.36	0.00	0.00	0.00
Transf. RoW	0.01	0.54	0.67	0.88
Public savings	-0.38	2.25	1.61	1.25
Total expenditure	4.31	-0.35	-0.25	-0.20
Welfare				
GDP per capita (% change from baseline)	0.33	-0.26	-0.19	-0.15
Equivalent variation		-0.03	-0.02	-0.02

Note: Unless otherwise stated in the table, the unit of measurement is billions of ARS. Three columns show the minimum, median, and maximum results, considering different values for the CET elasticities ranging from 0.9 to 2.
Source: Authors.

6. CONCLUSIONS AND POLICY RECOMMENDATIONS

Argentina's trade pattern is such that about 40% of the country's exports to the EU will be affected by the implementation of the EUDR. This exposure is almost entirely explained by sales of soybeans and cattle to the EU. The EUDR simulations that we conducted within a CGE model for Argentina show that GDP could fall between 0.15% and 0.26% if the country's exports do not comply with the new regulation. Indeed, total exports would fall between 0.79% and 1.31%. Reductions in employment and real wages would translate into lower household income and, therefore, lower private consumption (between 0.18% and 0.31% lower). The drop in activity and household income would also lead to a fall in tax revenues of between 0.20% and 0.35%.

In response to these findings, we propose forward a series of policy recommendations that aim to address three fundamental issues: i) operational compliance with the EUDR, ii) strengthening Argentina's institutional agenda and its integration into international trade, and iii) improving collaboration with the private sector.

I) Recommendations regarding operational compliance with the EUDR.

Traceability systems will need to be in place for raw materials and products to be exported to the EU, and products will need to be tracked from the field to the point of shipment. To this end, it is essential to **identify all public and private information systems in place in the respective value chains and to further analyze whether data from the AFIP and SENASA systems is compatible with the EUDR's requirements.** One example of a private-sector information system is the ViSeC initiative mentioned above. It would also be advisable to analyze the feasibility of extending this system in the future to include to other raw materials and products that may be included in the EUDR in the coming years or in the regulations of other export markets.

Traceability systems for exportable products should take the following aspects into account:

- **Integrating geolocation systems.** Geolocations of plots, parcels, and/or production units will be part of the information to be included in the due diligence. In this regard, **it would be advisable to examine the data in existing government systems in greater detail in order to work toward integrating these systems.** The geolocation system must accurately reflect the status of every production unit involved since data errors (due to either timing or spatial representation issues) may lead to the production unit in question being excluded from trade with the EU.

- **In this sense, it would be desirable to implement an integrated data system that provides information on production units and deforested plots.** Given the existence of the CIAM, SiNiA, and SNMBN[43] data systems (mentioned in section 2), we recommend creating a platform that allows access to different levels or layers of information by entering a certain coordinate (for example, the province-specific Territorial Ordinance on Native Forests category to which the plot belongs, whether it has been deforested since December 31, 2020, whether a land-use management plan has been submitted, who owns the land, etc.). It w**ould be useful to create a registry of geolocated rural properties** using a model like Brazil's Rural Environmental Cadaster (CAR).[44] This is a mandatory national online public registry of all rural properties that combines information on agricultural establishments with environmental information.

II) Recommendations to strengthen Argentina's national and subnational institutional agendas. The potential impact of the EUDR points to the need to analyze Argentina's national and subnational institutional capacities in relation to both the main objective of the EUDR (reducing deforestation in supply chains) and the instrument itself (a trade regulation with environmental objectives). The following suggested initiatives respond to the first of these points:

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[43] · https://www.argentina.gob.ar/ambiente/bosques/monitoreo-bosques-nativos.
[44] · https://www.gov.br/pt-br/servicos/inscrever-imovel-rural-no-cadastro-ambiental-rural-car.

- **Implement land-use planning schemes. The state should promote integrated land-use planning that takes into account the different biomes** (forests, wetlands, etc.) as well as economic factors. This should identify agricultural areas, conservation areas, forest areas, and urban areas based on environmental impact analysis and strategic environmental assessments based on a federal agreement in which the private sector, civil society, local communities, and Indigenous peoples are involved.

- **Strengthen monitoring mechanisms.** Argentina was among the ten countries with the greatest net loss of forested area in 2000–2015 (Ritchie and Roser, 2021), pointing to the need to develop better mechanisms for monitoring and overseeing deforestation. This requires strengthening local information systems.

- **Provide awareness-raising and training for provincial authorities.** Provide training on the implications of authorizing land clearing in green areas in relation to trade with the EU.

- **Promote and develop economic instruments. There is a need to design better incentives for forest owners not to deforest by considering payment schemes for ecosystem services.** Standing forests could be valued, for example, by assessing the merits of issuing carbon credits on international voluntary markets or by creating a national cap-and-trade system that sets maximum GHG emission levels for specific sectors (such as energy, transportation, or construction) and allows the regulated sectors to offset their emissions through payments to forest owners for the ecosystem services their forests provide. There is also a need to promote the development of a “green” bank that will finance projects to promote environmental sustainability, especially projects related to developing the technologies needed to guarantee the traceability and verification of products exported to the EU that will be affected by the EUDR.

With regard to the state’s capacity to regulate environmental trade, **a common, coordinated national strategy needs to be developed to address environmental, climate, trade, and sector-specific aspects**

in an integrated manner. The following suggestions address this need:

- **Value institutional progress.** The dialogue with the EU on risk assessment should ideally take into account the progress made on strengthening protection mechanisms (the Forest Act, the spread of management and conservation plans, and the development of forest monitoring tools).

- **Include a public-private approach to the issue in the National Climate Change Cabinet's agenda.** Given that the fight against deforestation is closely tied to reducing emissions, we recommend making the most of the opportunities for coordination provided by the National Climate Change Cabinet. One of its objectives is to generate coordinated responses to support vulnerable sectors in adapting to the impacts of climate change. The National Climate Change Cabinet includes various mechanisms (ministerial meetings, focal point meetings, interprovince coordination meetings, and wider meetings) that could be used to discuss and design comprehensive policy responses.[45]

- **Monitor developments relating to the EUDR.** Particular attention should be paid to whether regulations are extended to cover new feedstocks and products (such as corn or biofuels) or other ecosystems and areas with high carbon stocks and biodiversity value (grasslands, peatlands, and wetlands), as the EUDR advises may happen.

III) Improving collaboration with the private sector.

- **Capitalize on progress on existing initiatives** (such as ViSeC) and lessons learned (forestry, biodiesel, Hilton Quota). **Establish a joint working group with the relevant government sectors** (Ministry of Economy, Secretariat of Agriculture, Ministry of Environment, AFIP, SENASA, Ministry of Foreign Affairs, among others).

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[45] The National Climate Change Cabinet is Argentina's governing body for the design of climate change adaptation and mitigation policies, created by Law 27.520 on Minimum Public Financing for Adaptation to and the Mitigation of Global Climate Change.

- **Strengthen existing networks to continue promoting sustainable practices in agricultural production** through support and incentives for adopting these.

- **Raise awareness and provide training to agricultural production units (especially smaller ones).** Provide training on the implications of deforestation in green areas for future trade opportunities with the EU.

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ANNEX A.

KEY ASPECTS OF EU REGULATION 2023/1115 IMPLEMENTING THE EUDR.

Table A.
Scope of Application and Requirements

Activities and products covered	Importation,[1] and placing on the EU market and export from the EU of relevant products containing, fed to, or processed using cattle, cocoa, coffee, oil palm, rubber, soybeans, and wood.[2]
To enter the EU, products should be:	Deforestation-free: includes deforestation (the conversion of forests into cropland)[3] and forest degradation (structural changes in forest cover)[4].
	Produced in accordance with the legislation of the producing country.
	Be accompanied by a due diligence statement, the system for which includes three steps: 1) information requirements,[5] 2) measures to assess the risk of noncompliance, and 3) risk mitigation measures.
Components of the benchmarking system	Classification of countries using three-tier risk scheme (low-risk, standard, high-risk)[6].
	Risk level classification based on: · rate of deforestation and forest degradation; · rate of expansion of agricultural frontier associated with the relevant raw materials; · production trends for relevant raw materials and products.[7]
	The risk level for the country or region determines the complexity and thoroughness of the due diligence the operator must carry out.

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1 · Although the regulations use the term “introduction,” we use the term “importation” instead for ease of understanding.
2 · Article 1.
3 · Article 2, paragraph 3. Broadly speaking, this definition is in line with that of the Food and Agriculture Organization of the United Nations (FAO).
4 · While deforestation implies the elimination of the forest to allow the land to be used for another purpose, forest degradation refers not to the reduction of the forest surface area but to the deterioration of soil function and the loss of flora and fauna.
5 · The due diligence submission must conclusively demonstrate compliance with the requirements of deforestation-free production that complies with local legislation in verifiable manner. It must list descriptions and quantities of products entered, the country of production, the geolocation of all plots of land on which the raw materials were produced, and the date or time of production, among other things.
6 · When the EUDR enters into force, all countries will initially be placed in the standard category. The Commission is expected to draw up the lists of high- and low-risk countries by December 2024.
7 · The assessment may also take into account information submitted by the producing country, regional authorities, operators, and/or NGOs or third parties on agreements and other instruments concluded between the producing country and the EU and/or its Member States to address deforestation and forest degradation; the existence of national or subnational legislation in the producing country; the adoption of measures to address deforestation and forest degradation; and the transparent disclosure of relevant data, among other things.

Table A2.
Criteria for Assessing Risk of Noncompliance

FACTOR	CRITERIA
Institutional factors in the producing country	i) Risk level assigned to all or part of the producing country; ii) consultation and cooperation with indigenous peoples in all or part of the producing country; iii) existence of duly substantiated claims by the indigenous peoples regarding the use or ownership of the area used to obtain the raw materials, based on objective, verifiable information; iv) concerns regarding all or part of the producing country/country of origin, such as corruption levels, the prevalence of falsification of documents and data, lack of law enforcement, human rights violations, etc. Information source, reliability, and validity.
Socioenvironmental factors in the producing country	i) Presence of forests in all or part of the producing country; ii) presence of Indigenous peoples in all or part of the producing country; iii) prevalence of deforestation or forest degradation in all or part of the producing country.
Trade-related factors	i) Complexity of the supply chain in question and the level of processing of the relevant products; ii) risk of circumvention or mixing with relevant products of unknown origin or produced in areas where deforestation or forest degradation has occurred or is occurring.
Institutional factors in the EU	i) Conclusions of the Commission’s expert group meetings justifying the application of the EUDR.
Operator governance	i) Substantiated concerns raised and information on the history of noncompliance with the EUDR by operators or traders along the relevant supply chain; (ii) additional information on compliance with the EUDR, which may include information provided by certification or other third-party verification schemes, including voluntary schemes recognized by the Commission.

Source: Authors based on article 10, paragraph 2 of the EUDR.

ANNEX B.

This chapter uses a static real CGE model, which has features in common with the International Food Policy Research Institute standard (Lofgren et al., 2002). The model was also used in De la Vega (2020), which describes the mathematical approach and clarifies further methodological issues. Generally speaking, it considers a small open economy in which producers and consumers maximize profits and utility, respectively, in competitive markets.

Each production sector is represented by a profit-maximizing activity, subject to the production technology. The value-added production technology is assumed to be a Cobb-Douglas function combining capital and labor, while intermediate inputs are used in fixed proportions. With respect to factor markets, capital is assumed to be fully employed and sector-specific, while labor supply is assumed to be exogenous and perfectly mobile across sectors. In addition, rigidities are introduced through a wage curve, which allows for the negative empirical relationship between the wage level and the unemployment rate (Blanchflower and Oswald, 1994). Equilibrium in the labor market is then determined by the intersection of labor demand and the wage curve.

In terms of institutions, a single representative household is modeled that receives income from the productive factors it owns and from transfers from the government and the rest of the world. This income is used to pay direct taxes, save, make transfers to other institutions, and consume goods. Private consumption demand is derived from the maximization of a Cobb-Douglas utility function. The government collects revenues through taxes on households, factors, activities, domestic sales, and foreign trade, and receives transfers from the rest of the world. It then uses these revenues to purchase goods for consumption, invest, make transfers to households, and save.

As noted above, a small economy is modeled so that international prices are given. As is usual in the literature, and following Armington (1969), we assume imperfect substitution between goods that differ by origin, so that the demand for imports follows a constant elasticity of substitution function that combines domestic and imported goods. Meanwhile, the supply of exports is modeled symmetrically by a constant elasticity of transformation function, which reflects the fact that producers decide to allocate their output to the domestic market or export, depending on relative prices. One important feature is that the equations related to the balance of payments are disaggregated by trading partner to better account for the introduction of export restrictions by trading partners.

Finally, the model requires closure rules to be specified for three macroeconomic balances: government, saving and investment, and the balance of payments. All simulations assume that: i) government borrowing is balanced by changes in real domestic financing; ii) private investment is endogenously determined by the level of savings; and iii) savings from the rest of the world are exogenous (measured in the currency of the rest of the world), so that the real exchange rate varies endogenously to balance foreign exchange inflows and outflows. The numeraire for the model is the consumer price index.

ANNEX C.

RELEVANT RAW MATERIALS AND PRODUCTS

RELEVANT RAW MATERIAL	RELEVANT PRODUCTS[1]
Cattle	0102 21, 0102 29 Live cattle ex 0201 Meat of bovine animals, fresh or chilled ex 0202 Meat of bovine animals, frozen ex 0206 10 Edible offal of bovine animals, fresh or chilled ex 0206 22 Edible livers of bovine animals, frozen ex 0206 29 Edible offal of bovine animals (excluding tongues and livers), frozen ex 1602 50 Other prepared or preserved meat, meat offal, or blood of bovine animals ex 4101 Raw hides and skins of bovine animals (fresh, salted, dried, limed, pickled, or otherwise preserved, but not tanned, parchment-dressed or further prepared), whether or not dehaired or split ex 4104 Tanned or crust hides and skins of bovine animals, without hair on, whether or not split, but not further prepared ex 4107 Leather further prepared after tanning or crusting, including parchment-dressed leather, of bovine animals, without hair on, whether or not split, other than leather of heading 4114.
Cocoa	1801 Cocoa beans, whole or broken, raw or roasted 1802 Cocoa shells, husks, skins, and other cocoa waste 1803 Cocoa paste, whether or not defatted 1804 Cocoa butter, fat, and oil 1805 0 Cocoa powder, not containing added sugar or other sweetening matter 1806 Chocolate and other food preparations containing cocoa
Coffee	0901 Coffee, whether or not roasted or decaffeinated; coffee husks and skins; coffee substitutes containing coffee in any proportion

Oil palm	1207 10 Palm nuts and kernels 1511 Palm oil and its fractions, whether or not refined, but not chemically modified 1513 21 Palm kernel or babassu kernel oils and fractions thereof, whether or not refined, but not chemically modified. 1513 29 Palm kernel or babassu kernel oils and fractions thereof, whether or not refined, but not chemically modified (excluding crude oils) 2306 60 Oilcake and other solid residues, whether or not ground or in the form of pellets, resulting from the extraction of palm nuts or kernels ex 2905 45 Glycerol, at least 95% pure (calculated on the dry weight) 2915 70 Palmitic acid, stearic acid, their salts and esters 2915 90 Saturated acyclic monocarboxylic acids and their anhydrides, halides, peroxides, and peroxyacids; their halogenated, sulfonated, nitrated, or nitrosated derivatives (excluding formic acid; acetic acid; mono-, di-, or trichloroacetic acids; propionic acid; butanoic acids; pentatoic acids; and palmitic acid, stearic acid, their salts and esters; and acetic anhydride) 3823 11 Industrial stearic acid 3823 12 Industrial oleic acid, 3823 19 Industrial monocarboxylic fatty acids; acid oils from refining; industrial fatty alcohols (excluding stearic acid, oleic acid, and tall oil fatty acids) 3823 70 Industrial fatty alcohols
Rubber	ex 4005 Compounded rubber, unvulcanized, in primary forms or in plates, sheets or strip ex 4006 Other forms (for example, bars, rods, tubes, and profile shapes) and articles (for example, discs and rings) of unvulcanized rubber ex 4007 Vulcanized rubber thread and cord ex 4008 Plates, sheets, strip, rods, and profile shapes, of vulcanized rubber other than hard rubber ex 4010 Conveyor or transmission belts or belting, of vulcanized rubber ex 4011 New pneumatic tires of rubber ex 4012 Retreaded or used pneumatic tires, of rubber; solid or cushion tires, tire treads and tire flaps, of rubber ex 4013 Rubber inner tubes for tires (pneumatic tires) ex 4015 Articles of apparel and clothing accessories (including gloves, mittens, and mitts), for all purposes, of vulcanized rubber other than hard rubber ex 4016 Other articles of vulcanized rubber other than hard rubber, not specified in Chapter 40 ex 4017 Hard rubber (e.g., ebonite) in all forms, including waste and scrap; articles of hard rubber
Soybeans	1201 Soybeans, whether or not broken 1208 10 Soybean flour and meal 1507 Soybean oil and its fractions, whether or not refined, but not chemically modified 2304 Oilcake and other solid residues, whether or not ground or in the form of pellets, resulting from the extraction of soybean oil

Wood	4401 Fuel wood; wood in chips or particles; sawdust and wood waste and scrap, whether or not agglomerated in logs, briquettes, pellets, or similar forms
	4402 Wood charcoal (including shell or nut charcoal), whether or not agglomerated
	4403 Wood in the rough, whether or not stripped of bark or sapwood, or roughly squared
	4404 Hoopwood; split poles; piles, pickets, and stakes of wood, pointed but not sawn lengthwise; wooden sticks, roughly trimmed but not turned, bent, or otherwise worked, suitable for the manufacture of walking-sticks, umbrellas, tool handles, or the like; chipwood and the like
	4405 Wood wool; wood flour
	4406 Railway or tramway sleepers (crossties) of wood
	4407 Wood sawn or chipped lengthwise, sliced, or peeled, whether or not planed, sanded, or end-jointed, of a thickness exceeding 6 mm
	4408 Sheets for veneering (including those obtained by slicing laminated wood), for plywood or for similar laminated wood and other wood, sawn lengthwise, sliced, or peeled, whether or not planed, sanded, spliced, or end-jointed, of a thickness not exceeding 6 mm
	4409 Wood (including strips and friezes for parquet flooring, not assembled) continuously shaped (tongued, grooved, rebated, chamfered, V-jointed, beaded, molded, rounded, or the like) along any of its edges, ends, or faces, whether or not planed, sanded, or end-jointed.
	4410 Particle board, oriented strand board (OSB) and similar board (for example, waferboard), of wood or other ligneous materials, whether or not agglomerated with resins or other organic binding substances
	4411 Fiberboard of wood or other ligneous materials, whether or not agglomerated with resins or other organic substances
	4412 Plywood, veneered wood, and similar laminated wood 4413 Densified wood in blocks, plates, sheets, boards, or profile shapes
	4414 Wooden frames for paintings, photographs, mirrors, or similar objects
	4415 Packing cases, boxes, crates, drums, and similar packings, of wood; cable-drums of wood; pallets, box pallets and other load boards, of wood; pallet collars of wood (excluding packing material used exclusively to support, protect, or transport another product placed on the market)
	4416 Casks, barrels, vats, tubs, and other coopers' products and parts thereof, of wood (including staves)
	4417 Tools, tool bodies, tool handles, broom, or brush bodies and handles, of wood; boot or shoe lasts and trees, of wood
	4418 Builders' joinery and carpentry of wood, including cellular wood panels, assembled flooring panels, shingles, and shakes
	4419 Tableware and kitchenware, of wood
	4420 Wood marquetry and inlaid wood; caskets and cases for jewelry or cutlery, and similar articles, of wood; statuettes and other ornaments, of wood; wooden articles of furniture not falling in Chapter 94
	4421 Other articles of wood
	Wood pulp and paper of Combined Nomenclature chapters 47 and 48, except bamboo-based products and products for recycling (waste and scrap)
	ex 49 Printed books, newspapers, pictures, and other products of the printing industry; manuscripts, typescripts, and plans.
	ex 9401 Seats (other than those of heading 9402), whether or not convertible into beds, and parts thereof, of wood
	9403 30, 9403 40, 9403 50, 9403 60, and 9403 91 Wooden furniture, and parts thereof
	9406 10 Prefabricated wooden buildings

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1 · The list of relevant products is defined according to the Combined Nomenclature, which is the nomenclature used by the EU to classify goods for customs purposes. The maximum level of disaggregation of this nomenclature is eight digits, and the first six digits correspond to the Harmonized System nomenclature. Since the table lists products with a maximum disaggregation of six digits, the codes presented here also correspond to the Harmonized System.

ANNEX D.

STAKEHOLDER INTERVIEWS

INSTITUTION	AREA
EU Delegation in Argentina	Economic and Trade Section, Cooperation Section
Ministry of Foreign Affairs, International Trade, and Worship of the Argentine Republic	Center for International Economics Undersecretariat for Economic Negotiations Economic Relations and Environment Unit
Ministry of the Environment and Sustainable Development	National Forestry Authority
Ministry of Production, Industry, and Employment	Provincial Forest Authority
Gran Chaco Monitoring System (ViSeC)	
National Institute of Agricultural Technology	Forestry
World Chambers Federation	Global Task Force for Sustainability
Genneia	Corporate Affairs
Consultant specializing in climate change and agriculture	Former Secretary of Climate Change
Argentinian Institute of Standardization and Certification (IRAM)	
Argentinian Forest Certification System (CERFOAR)	Program for the Endorsement of Forest Certification Argentina
Secretariat of Agriculture, Livestock, and Fisheries	Undersecretariat for Policy Coordination

CHAPTER 3

SUSTAINABLE TRADE REGULATIONS AND GENDER IN RURAL BRAZIL

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ACRONYMS AND ABBREVIATIONS

- **CBAM**
  Carbon Border Adjustment Mechanism
- **CEPESP-FGV**
  Center for Public Sector Policy and Economics, Getúlio Vargas Foundation
- **CIBE**
  CIBE
  International Confederation of European Beet Growers
- **COPA-COGECA**
  Committee of Professional Agricultural Organizations, General Confederation of Agricultural Cooperatives
- **DID**
  difference-in-differences
- **EU**
  European Union
- **EUDR**
  EU Deforestation Regulation
- **GAPKI**
  Indonesian Palm Oil Association
- **GHGs**
  greenhouse gases
- **IBGE**
  Brazilian Institute of Geography and Statistics
- **MBA**
  Malaysian Biodiesel Association
- **MTK**
  Central Union of Agricultural Producers and Forest Owners of Finland
- **PAA**
  Food Acquisition Program (Brazil)
- **PSM**
  propensity score matching
- **PT**
  Workers’ Party
- **RED**
  EU Renewable Energy Directive
- **SDGs**
  Sustainable Development Goals
- **UNICA**
  Sugarcane and Bioenergy Industry Association of Brazil

ABSTRACT

Voluntary certification is fast becoming a tool for ensuring compliance with binding sustainable trade regulations. This chapter examines whether voluntary certification helps promote gender equality among rural landowners. Focusing on Bonsucro, a private certification for sugar and ethanol production that complies with the European Union biofuels regulations (the Renewable Energy Directive—RED), we analyze the gender gap in rural land ownership in Brazil at the municipal level. To do so, we use a difference-in-difference econometric analysis, drawing on data from the Brazilian Agricultural Census. The results show that Bonsucro certification does not significantly affect the gender gap in ownership over time. This is interesting because Bonsucro certification generally seems to have significant positive effects on community empowerment, regardless of compliance with the RED. However, no such effects were found when we looked exclusively at producers complying with the RED-specific requirements. After considering the limitations of our analysis, we discuss the implications of our findings regarding the effects of private certification schemes and policies on the sustainability of international trade.

1. INTRODUCTION

The European Union (EU) has been promoting sustainability regulations that affect developing countries' exports to the bloc. The Border Carbon Adjustment Mechanism (BCAM) and the EU Deforestation Regulation (EUDR) are well-known examples of such regulations. Another is the EU Renewable Energy Directive (RED), first introduced in 2009. Some of these regulations are or have become binding and rely on private stakeholders to help ensure compliance. The combination of private certification processes and public regulations should theoretically make sustainability policies more effective by leveraging the monitoring capacity of the former and the strictness of the latter (Marx, Depoorter, and Vanhaecht, 2022; Abbott and Snidal, 2010). **However, there is still no evidence on whether combining private and public trade regulation promotes local socioeconomic development, which is a crucial part of a holistic approach to sustainable development.**

This chapter begins to **bridge this gap by assessing the gender implications of sustainability certifications that comply with EU import regulations.** Specifically, it looks at the Bonsucro standard, a key multilateral initiative that certifies producers participating responsibly in the sugar and ethanol supply chain in several countries, including Brazil. Hereafter, we use "Bonsucro-RED" to refer to the specific version of the Bonsucro standard that meets the EU sustainability requirements established in the RED. Our analysis combines data from Bonsucro-RED-certified producers and Brazilian agricultural censuses using difference-in-differences (DID) and propensity-score-matching (PSM) models to measure trends in women's empowerment.

Why look at gender and certifications? Gender equality became the United Nations (UN) Sustainable Development Goal (SDG) 5 in 2015 but remains elusive. Recent data from the UN (2022) suggests that the progress made toward this goal by 2030 will be limited. Women face challenges in relation to leadership and ownership, particularly in the Global South. In sectors such as agriculture, these factors are exacerbated by climate change and the COVID-19 pandemic. In 2020, for example, women accounted for 39% of global employment but 54% of job losses.[1] At the same time, most owners of rural land in Brazil are still men. As a major sugarcane and ethanol producer, the country could benefit significantly from sustainability certifications if these generate positive externalities in the form of gender equality among rural farm owners. **This chapter anticipates that the effects of Bonsucro certification on land access and community empowerment dynamics will help explain its impact on the gender gap. We expect the driving force behind these dynamics to be the EU RED.**

The central arguments of this paper derive from two opposing expectations in the literature: (1) the Bonsucro certification promotes gender equality by facilitating women's access to formal land titles and improving their economic opportunities, as since 2022 it has included gender-specific requirements, such as providing training to empower women in their community and promote women in management and skilled positions; (2) the Bonsucro certification undermines gender equality by displacing family farms, where the gender gap is narrower. After controlling for variables that could bias the results, we find evidence that Bonsucro-RED certification does not significantly affect the gender gap in land ownership or the prevalence of family farms in Brazil. However, Bonsucro-RED appears to have a negative effect on community empowerment, in contrast to versions of the certification that do not comply with the RED.

Our findings help further understandings of the relationship between sustainable trade regulation, certification, and local socioeconomic development. Several studies have examined the socioeconomic and environmental impacts of major certification programs, but the results are inconsistent, suggesting that voluntary regulations do not always deliver on their social and environmental promises in producing countries (Bartley, 2018; DeFries et al., 2017; Schleifer and Sun, 2020; van der Ven and Cashore, 2018). Other research has pointed to trade-offs between socioeconomic and environmental outcomes (Grabs, 2020; Vanderhaegen et al., 2018). These negative perspectives have led some scholars to argue that while certification may address some sustainability challenges, it exacerbates others (Auld et al., 2008;

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[1] · Available at: https://www.mckinsey.com/featured-insights/future-of-work/covid-19-and-gender-equality-countering-the-regressive-effects.

Cashore and Bernstein, 2022). This chapter contributes to the body of knowledge in this academic field and identifies some important policy implications.

The findings presented here suggest that even when private certifications are supported by public sustainable trade regulations, they may not adequately address fundamental local socioeconomic issues, which may undermine a comprehensive approach to sustainable development through trade. Several policy recommendations emerge as a result of these findings: (1) public regulators should push private regulators to incorporate the community-based empowerment of women and smallholders into their schemes and adopt a holistic sustainability perspective; (2) public regulations need to refine and expand mechanisms for regulatory updates based on experiential learning, ensuring inclusive stakeholder participation; and (3) state and federal authorities in Brazil and other Latin American countries should promote synergies between public regulations that favor family farming and women's participation in prevailing international voluntary standards.

This paper is structured as follows. Section 2 reviews the relevant literature. The relationship between Bonsucro, the RED, and the gender gap in Brazil is examined in detail in section 3. Section 4 presents the data and methods used in our analysis, and sections 5 and 6 contain our findings and conclusions on policy implications and consider possible paths for future research.

2. CAN CERTIFICATION SCHEMES HELP REDUCE GENDER GAPS IN COMBINATION WITH SUSTAINABLE TRADE REGULATIONS?

As gender equality is an important aspect of sustainable development, the sustainability standards of several voluntary certification programs have included requirements to reduce gender inequality and empower women. This trend has become increasingly relevant since the creation of the 2030 Agenda, as various private governance organizations have integrated the SDGs into their work programs as a way of including a holistic approach to complement global public institutions in terms of sustainable development. Recent research shows that the stated policy goals of voluntary sustainability standards cover SDG 5 (gender equality) particularly well (Schleifer et al., 2022). These findings reveal that gender issues have been acknowledged by most certification standards, even if they were not their foremost concern in the past. This shift is particularly important in the agricultural sector, as millions of women's livelihoods depend on it. To help producers comply with gender standards, different agricultural certification programs in developing countries have organized awareness-raising campaigns, special training programs, and capacity-building activities to promote gender equality within farming communities (Smith et al., 2018). If the standards in question are designed carefully and implemented fully, they should increase women's influence on decision-making at the household and community level and thus increase their economic opportunities.

How can sustainability certification improve gender equality? Lyon et al. (2010) **point to three mechanisms for empowering women: increasing their influence in household and community decision-making, supporting their acquisition of formal land titles or tenure, and improving their economic opportunities.** First, in terms of women's influence on decision-making, the standards of some certification programs, such as Fairtrade, explicitly encourage women to participate in producer organizations. This hypothesis has been supported by certain empirical evaluations. For example, Elder et al. (2012) showed that women's participation in cooperative decision-making increased after the introduction of certification, but this did not necessarily translate into women being promoted to leadership positions. Second, certification programs can help women obtain formal land titles. At the same time, however, the requirements of some of these standards may have unintended consequences for women, especially in countries where they cannot legally hold such titles (Oosterveer et al., 2014). **It is therefore important to conduct an empirical assessment of how far private certification changes women's rights to legal access to land.** Third, as the mission statements of various programs state, certification promises to increase women's incomes and thus their economic opportunities. In this regard, numerous researchers have examined the relationship between certification and women's incomes or asset ownership. Some studies have found evidence of positive effects in this regard (Bolwig et al., 2007; Chiputwa et al., 2015; Meemken and Qaim, 2018). However, as one meta-analysis of the impact of the Fairtrade certification shows, its positive effects on women's income and household position appear to be highly context-dependent, in that they often vary in relation to other covariates like age, education, and wealth (Smith, 2013). Given the proactive stance on gender issues and community participation taken by several certification programs, it would be plausible to expect their standards to improve equity among certified producers.

At the same time, several academic studies have cast doubt on whether voluntary sustainability certification has a positive impact on gender issues. This is partly because there are large gaps in the research on the relationship between SDGs and private certification and a lack of empirical evidence on what impact standards have by comparing certified and noncertified producers (Rubio-Jovel, 2022). More importantly, however, several studies have shown certification to have negative and sometimes unintended impacts on smallholder farmers, a group in which women tend to play a prominent role and remain vulnerable. For example, a study on palm oil certification in Indonesia shows that the spread of voluntary standards to improve environmental sustainability has undermined the socioeconomic status of smallholder farmers by excluding them from global value chains when foreign customers demand certified products only (Brandi, 2017). Similarly, the findings of several impact assessments on the effects of certification on smallholder coffee farmers ran counter to expectations that private sustainability standards would empower smallholders: instead, these standards have actually disadvantaged small producers, who lack the capacity and resources to obtain certification (Akoyi and Maertens, 2018; DeFries et al., 2017). The prevalence of women in small-scale production makes them more likely to be affected by such measures.

Traditionally, women have faced structural socioeconomic barriers to participation in international trade (Barafani and Verna, 2020), and market-based initiatives alone may have difficulty addressing such constraints. Moreover, the market-based nature of private certification can often lead to a downward spiral in regulatory effectiveness, as cost factors may make sustainability regulations weaker and weaker. **Public-sector trade regulations could help to address some of these gaps.** As a guarantee of the lowest common denominator of sustainability, binding laws and the presence of the state can help limit downward spirals of this sort and make private regulation schemes more effective (Dietz et al., 2022; Naiki, 2021). For example, the state can set minimum requirements for gender parity, thereby helping to reduce the trend toward greater gender inequality in rural land ownership. More efficient certification schemes could thus arise from partnerships between the state (a guarantor of minimum standards) and private regulators (key players with deep first-hand experience in sustainable production).

To realize this potential, however, a holistic approach to sustainable development that links environmental protection and local socioeconomic development must be at the core of public regulations that support private certification. The literature has clearly demonstrated that capacity and size are strong predictors of a firm's ability to participate in the politics and economics of international trade (Osgood, 2021; Madeira, 2016). Consequently, even when stricter import requirements focus on environmental protection, they may further limit smallholder farmers' access to foreign markets due to high compliance and monitoring costs (Elamin and Córdoba, 2020). Regulatory costs are largely fixed, allowing wealthier producers a higher marginal return and enabling them to take advantage of economies of scale. In this sense, larger producers are more likely to meet stricter quality standards (Oya, Schaefer, and Skalidou, 2018; Meemken, 2021), meaning they can comply more easily with more stringent environmental standards. **Unless private and public regulations address sustainable trade holistically, additional requirements and higher compliance costs could further limit the role of smallholders and women in agricultural production.**

3. BONSUCRO, EUROPEAN UNION POLICIES, AND THE GENDER GAP IN BRAZIL

3.1

HOW MIGHT BONSUCRO CERTIFICATION AFFECT THE GENDER GAP IN RURAL BRAZIL?

In 2021, Brazil accounted for 36.7% of the world's sugar supply, twice as much as the second-largest supplier.[2] Although only 2.42% of Brazilian sugar exports[3] went to the EU, this was true of 6.57% of ethanol exports. The EU is the third-largest importer of Brazilian ethanol after South Korea (37.03%) and the United States (25.22%). Given the importance of the sugar and ethanol sectors in the Brazilian export matrix, what role could certifications in the sugar value chain play in promoting gender equality among producers?

Bonsucro is a major multilateral certification initiative that aims to reduce the social and environmental impacts of the sugarcane value chain (see box 1). **However, Bonsucro has only recently included gender-specific requirements in its sustainability indicators.** In its 2022 production standards, the organization added a new evaluation criterion: "the operator should promote gender inclusion in management and skilled positions and should conduct training to empower women in the community" (Bonsucro, 2022). Prior to 2022, Bonsucro only mentioned women in indicators 2.1.2 (absence of discrimination) and 2.1.3 (absence of forced or compulsory labor).

On the one hand, this lack of an explicit mention of gender equality before 2022 might reduce the impact of Bonsucro certifications on this

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2 · Available at: https://www.gov.br/en/government-of-brazil/latest-news/2022/brazil-leads-sugar-production-and-exports.
3 · Available at: https://unicadata.com.br/.

issue in Brazil, given the short time that has passed since the scheme began. **On the other hand, Bonsucro might have an indirect impact on gender equality in the country through its impact on smallholder empowerment.** This is because the gender gap in rural land ownership has generally been narrower among smaller production units. In 2017, the year of Brazil's most recent agricultural census, 81% of farms were owned or managed by men. However, the gender gap is also smaller for family farming than for nonfamily farming (figure 2). In Brazil, family farming[4] accounts for about 77% of rural properties and more than 23% of the total land area occupied by rural properties in the country. It is the main economic activity for more than 90% of Brazilian municipalities with less than 20,000 inhabitants and employs more than 66% of the rural workforce in Brazil.[5] Women play a central role in these production units. Indeed, the proportion of female workers involved in the Brazilian Food Procurement Program (PAA) was as high as 80% in 2019, partly in response to government policies that promote women's participation in such programs, such as Regulation 44/2011 (CONAB, 2021). Overall, women's participation in family farming increased by 38% between the 2006 and 2017 censuses (figure 1).[6]

Regardless of the recent inclusion of gender-specific criteria in its production standards, Bonsucro's earlier efforts to promote community-based empowerment may also have indirectly benefited women landowners. Community-based empowerment can create greater awareness of government incentives to support small-scale agriculture and women's entrepreneurship. As a result, women living and working near certified farms and enterprises may be better positioned to seek opportunities through government incentives, such as rural credit schemes. Bearing this in mind, **we expect the relationship between certification and the gender gap in rural land ownership in Brazil to be influenced by the effect of Bonsucro certification on smallholder farmers.**

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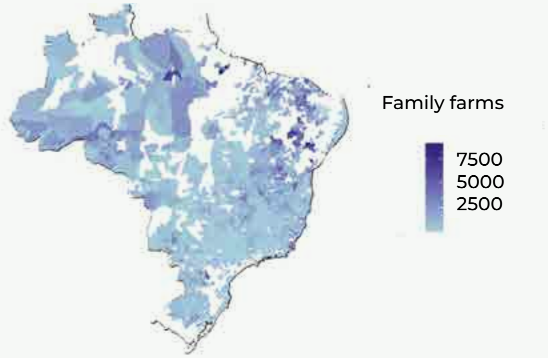
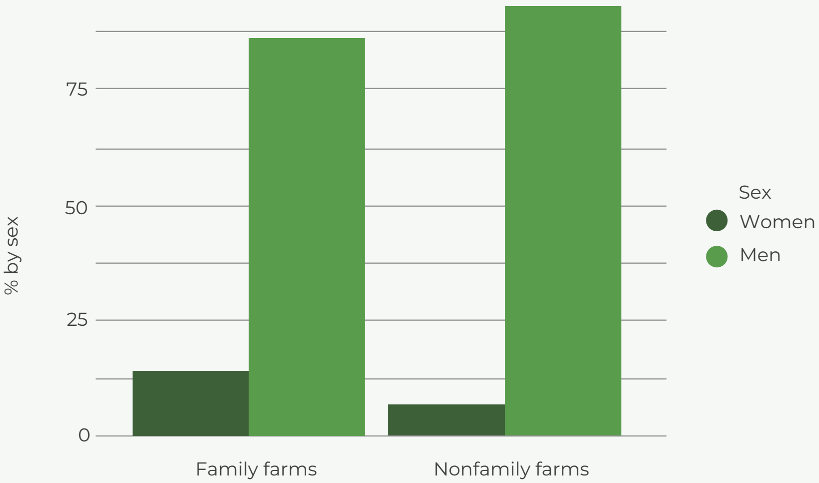
[4] · Defined by law 11326/2006 as an economic activity carried out within a small rural property that predominantly employs family labor and provides the owner's family with most of its income. A small property is defined as one consisting of up to four "fiscal modules"—the name for the minimum area necessary for farming activity in Brazil.
[5] · Available at: https://www.ibge.gov.br/apps/atlasrural/pdfs/11_00_Texto.pdf.
[6] · Available at: https://www.cnabrasil.org.br/noticias/mulheres-na-agricultura-representatividade-crescente-e-muito-trabalho.

Figure 1.
Women Landowners per Municipality



Source: Authors based on IBGE.

Figure 2.
Gender Gap in Rural Brazil. Proportion of Women and Men Running Family and Nonfamily Farms (2017)



Source: Authors based on IBGE.

BOX 1
WHAT IS BONSUCRO CERTIFICATION?

Bonsucro is a leading multilateral initiative that seeks to reduce the social and environmental impacts of the sugarcane supply chain. It operates in over fifty countries and is governed by a seven-member board of directors. The board receives advice, opinions, and recommendations from the membership council, which is made up of Bonsucro members. To obtain Bonsucro certification, producers and retailers have to demonstrate compliance with six key principles. There are specific criteria and indicators for each principle that are audited by several private certification companies, such as Control Union, FoodChain ID, and SCS Global Services. The six principles are:

1. Respect the law.
2. Respect labor rights and standards.
3. Manage input, production, and processing efficiencies to enhance sustainability.
4. Actively manage biodiversity and ecosystem services.
5. Continuously improve other key areas of the business.
6. Additional mandatory requirement for biofuels under the EU RED.

There are currently 239 Bonsucro-certified establishments worldwide, the majority of which (102 or 42.7%) are in Brazil. Some 78 of these Brazilian operations are certified as complying with the EU RED. Bonsucro certification is expanding significantly: the scheme now covers more than 1.5 million hectares, and over 10,000 smallholders are certified. No conservation land has been converted to agricultural land on these certified farms.[7]
Given Brazil's prominence in ethanol and sugar production and exports, Bonsucro's presence in the country is key to achieving its global mission.

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7 · Available at: https://www.cnabrasil.org.br/noticias/mulheres-na-agricultura-representatividade-crescente-e-muito-trabalho.

3.2

HOW MIGHT EUROPEAN UNION REGULATIONS IMPACT THE GENDER GAP IN RURAL BRAZIL THROUGH BONSUCRO CERTIFICATION?

The EU is in the midst of an unprecedented expansion of its regulations on imports of products that may cause deforestation. This process is part of the bloc's increasingly ambitious goals for sustainable and renewable energy production. As part of this process, the RED defines a set of sustainability criteria for imports of products in the biofuel supply chain, including sugar, palm oil, and ethanol. Initially introduced in 2009,[8] the RED became legally binding in June 2021 to help the EU meet its target of using 32% renewable energy by 2030. The RED establishes a "common framework for the promotion of energy from renewable sources," binding national energy targets, and rules and criteria regarding what are considered sustainable biofuels and bioliquids.

Article 18 of the first (2008) version of the RED establishes that "Where the Community has concluded agreements containing provisions relating to matters covered by the sustainability criteria set out in Article 17(2) to (5), the Commission may decide that those agreements demonstrate that biofuels and bioliquids produced from raw materials cultivated in those countries comply with the sustainability criteria in question." In other words, **the draft implementing decision leaves room for private certification schemes to play a monitoring role and ensure compliance with EU regulations.** By the end of June 2023, there were 15 accredited voluntary certifications (box 2), and Bonsucro was among them.

By interacting with private regulators, the EU can improve the legitimacy and effectiveness of its regulations (Marx, Brando, and Lein, 2017). Likewise, the linkages and guidance provided by EU regulations can help address some of the limitations of private certification, such as the difficulty of ensuring compliance with voluntary standards

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[8] · Available at: https://www.cnabrasil.org.br/noticias/mulheres-na-agricultura-representatividade-crescente-e-muito-trabalho.

(Dietz et al., 2022). Ideally, EU RED certification would help to reinforce the positive effects of Bonsucro certification. It would do so in two main ways: (1) by making compliance with Bonsucro standards a valid path for accessing EU markets, the RED may incentivize more establishments to seek certification, which would extend the reach of Bonsucro's community engagement practices, and (2) the RED would also set higher sustainability standards and incentivize greater compliance with voluntary schemes, given that these facilitate entry into EU markets.

However, the interplay between private and public regulations can also lead to unintended consequences (Oosterveer et al., 2014; Heilmayr, Carlson, and Benedict, 2020), impacting gender equality in rural Brazil. As certification requirements become more demanding, they simultaneously become more costly, especially for smallholder farmers (Pyk and Abu Hatab, 2018). Indeed, RED-compliant Bonsucro certification entails meeting additional requirements that impose higher costs on smallholder farmers (box 2). This process has two main consequences: (1) smallholders may experience difficulties in accessing Bonsucro certification; (2) the increased cost of certification may have the side effect of increasing demand for land and raising prices, potentially leading to greater concentration of land ownership in rural Brazil.

Ethanol and sugar production in Brazil are primarily concentrated among larger producers. In other words, the logic of sugarcane production in Brazil differs considerably from other countries participating in the Bonsucro initiative, such as Pakistan, where smallholder farmers account for more than 64% of production.[9] In São Paulo state, the number of biofuel production companies with more than 100 employees in 2020 was higher than that of companies with up to four employees.[10]

Looking specifically at ethanol production, Brazilian municipalities exporting to the EU had a maximum of five companies in the sector, and most were large companies such as Raízen, which has more than 30,000 employees. For sugarcane production, the Brazilian

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9 · Available at: https://bonsucro.com/wp-content/uploads/Bonsucro-outcome-Report-2021.pdf.

10 · In this chapter, the terms "company" or "firm" are used here as synonyms for what the IBGE classifies as "local units," namely a physical location where economic activities are carried out and that is listed with Brazilian National Registry of Legal Entities (CNPJ).

municipality exporting to the EU with the most production units was Itápolis, which had more than 1,200. Nevertheless, according to data from the Brazilian Institute of Geography and Statistics (IBGE), one in three rural farms in Brazilian municipalities producing and exporting sugarcane/ethanol to the EU is larger than 2,500 hectares, on average. **Bonsucro-RED certification could incentivize the expansion of larger properties and increase the cost of land in the vicinity of these farms as a result of the higher initial investments and privileged access to EU markets that the certification entails, combined with a pre-existing trend toward the concentration of land ownership (Gilberto de Souza 2010).** Given that the availability of affordable land is paramount to the expansion of smallholder agriculture, Bonsucro-RED certification could have unintended consequences on women's rural land ownership.

I**n summary, the interplay between private and public trade and sustainability regulations can help promote greater efficiency, but it can also lead to unintended consequences or negative side effects.** Given that RED requirements are a political issue as well as a technical one (box 3), these negative consequences are significant as they reflect the preferences of EU countries and may or may not prioritize local socioeconomic development in third countries. This could result in EU sustainable trade initiatives either promoting or hindering the positive impact of Bonsucro's community empowerment initiatives on women producers.

BOX 2.
RED REQUIREMENTS AND RECOGNIZED CERTIFICATION PROGRAMS AS OF JUNE 2023

To obtain RED-compliant Bonsucro certification, stakeholders in the biofuel value chain must meet certain key criteria:

1. **Land use criterion:** ethanol producers are assessed to determine whether land with high biodiversity or carbon capture has been converted to farmland since 2008. The criterion also assesses whether information on land use is communicated to the next economic operator in the value chain. This is important for the last economic operator (the

stakeholder in the value chain that sells the product to the first retailer) as it enables them to calculate aggregate data on the sustainability impact of a product's value chain.

2. **Greenhouse gas (GHG) criterion**: this includes a calculation of GHG emissions from production and transportation as well as emissions associated with any change in land use that has occurred since 2008. It also includes an assessment of whether information on GHG emissions is communicated to the next economic operator in the value chain.
3. **Chain of custody criterion:** a set of requirements and information aimed at guaranteeing the traceability of the good to ensure that it meets sustainability criteria throughout the value chain. To provide certification, certifying organizations verify that the material produced has been correctly identified and audit data on product volumes.

As of June 2023, the following private regulators are recognized by the RED:

- Biomass Biofuels Voluntary Scheme
- Better Biomass
- Bonsucro EU
- International Sustainability and Carbon Certification
- KZR INiG System
- REDcert
- Red Tractor Farm Assurance Combinable Crops and Sugar Beet Scheme (Red Tractor)
- Roundtable of Sustainable Biofuels EU RED
- Round Table on Responsible Soy EU RED
- Scottish Quality Farm Assured Combinable Crops
- Trade Assurance Scheme for Combinable Crops
- Universal Feed Assurance Scheme
- Sustainable Resources Voluntary Scheme
- Sustainable Biomass Program
- Austrian Agricultural Certification Scheme

Source: Eur-LEX.

BOX 3.
THE POLITICAL TUG-OF-WAR BEHIND THE EU RED

The EU RED did not emerge in a political vacuum: it reflects the growing demand for sustainable imports in the EU among both the public and businesses. As evidence of the public importance of this issue, a 2019 consultation on updating the RED received more than 37,000 comments from citizens across the EU. A significant proportion of respondents argued that the EU's renewable energy strategy is particularly important for achieving its Green Deal targets. Three major political forces in the EU have stakes in the debate on renewable fuels: NGOs and other public interest organizations, domestic sugar producers, and producers of both fossil fuels and renewable energy. Agricultural producer organizations—such as the International Confederation of European Beet Growers (CIBE), the Central Union of Agricultural Producers and Forest Owners of Finland (MTK), and most notably the Committee of Professional Agricultural Organizations of the General Confederation of Agricultural Cooperatives (COPA-COGECA)—argued that the RED should be as rigorous as possible to harmonize imports and Green Deal obligations. These organizations were joined by public interest organizations and think tanks, including the Environmental Justice Foundation, Biofuelwatch, the Global Forest Coalition, the Corporate Europe Observatory, and Greenpeace. These groups generally faced some opposition from foreign biofuel value chain associations, such as the Malaysian Biodiesel Association (MBA) and the Indonesian Palm Oil Association (GAPKI), which noted their fear of discrimination against smallholder exports.[11]

Fuente: Eur-LEX.

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[11] · Available at: https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/12553-Revision-of-the-Renewable-Energy-Directive-EU-2018-2001/public-consultation.'

4. DATA AND ECONOMETRIC MODEL

The main source of data for this chapter was the Brazilian Agricultural Census for 2006 and 2017, which provides comprehensive information on rural family farms, including profiles, production practices, and socioeconomic aspects. Municipal-level variables were collected from this source, which contains data for every municipality in Brazil on the size of sugarcane plantations in hectares, local sugarcane and ethanol production units, the proportion of family and nonfamily farms, and the proportion of women and men managing each of the production units (see annex). Only municipalities with sugarcane plantations occupying areas greater than 0 hectares were included in our database. Data was also collected from the Brazilian Ministry of Agriculture on agricultural loans received by each municipality before 2006 and between 2006 and 2017. Data on political dynamics was obtained from the Center for Public Sector Policy and Economics at the Getúlio Vargas Foundation (CEPESP-FGV). This included whether the mayor of each municipality is a member of the Workers' Party (PT) and the proportion of PT politicians in local legislative chambers. Additional data on private certification was also collected from Bonsucro's official sources. In total, 42% of all Bonsucro-certified companies worldwide are located in Brazil, and 23% of these companies are certified under the Bonsucro-RED scheme.

The main dependent variable in this study is the gender gap between male and female owners of rural establishments at the municipal level ("gender_gap"). This variable was derived from data collected by the Brazilian Agricultural Census and is expressed as the proportion of rural establishments owned by men minus the proportion of rural establishments owned by women. Prior to 2022, Bonsucro did not include a specific gender criterion in its standards. Consequently, the effects of Bonsucro-RED are expected to be mediated by the proportion of family farms in a given municipality ("prop_af"). This is because Bonsucro certification may promote women's empowerment indirectly by promoting community empowerment, which has long been a criterion for obtaining certification. The RED-compliant Bonsucro certification might then reinforce this process by setting minimum standards.

In this chapter, the independent variable is the presence of a company holding Bonsucro and Bonsucro-RED certification in a given municipality at a given time. The full sample includes all Brazilian municipalities producing sugarcane in the 2006 and 2017 censuses (N=6,841). However, the sample is highly unbalanced because the number of municipalities with Bonsucro-certified companies is very small compared to the total number of sugarcane-producing municipalities. Two statistical techniques are combined to correct for this.

The propensity-score-matching method (PSM) is used in conjunction with the difference-in-differences method (DID) to assess the impact of export-oriented certification on gender equality in land ownership. DID has previously been used to measure the impact of trade agreements on foreign direct investment (Baltagi, Egger, and Pfaffermayr, 2008) and gender equality (López Mourelo and Samaan, 2018) and serves as a solid starting point for our analysis. In this chapter, impact is defined as the difference between the situation of the group that received an intervention after having access to it (the treatment group) and the situation this group would have been in had the intervention not taken place. The impact of the intervention is defined by the contrast between two situations: the actual situation (i.e., that of the group that benefited from the intervention after it had taken place) and a hypothetical one (i.e., the situation the same group would have been in had the intervention not taken place).

In the DID analysis, the treatment variable is binary: it takes a value of 1 if there is at least one Bonsucro- or Bonsucro-RED-certified company in a municipality, and 0 otherwise. The treatment is the same for municipalities with more than one Bonsucro-certified company. To create a control group with characteristics that are comparable to the treatment group, units were matched based on propensity scores (the similarity score between units). A DID analysis was then implemented to estimate the average treatment effect. The treatment variable in the PSM exercise was a binary indicator that takes a value of 1 if a municipality has at least one Bonsucro-certified enterprise and 0 otherwise. The propensity score represents the predicted probability of a unit receiving the treatment, given a set of observed variables. A logistic regression model was estimated to obtain the propensity scores. The formula for estimating the propensity score is as follows:

$$\textit{Propensity Score}\ (x) = Pr\ (\textit{Treatment} = 1 / X)$$

In this study, three variables were included in the propensity score calculation to create the control group. We derived these variables from the theory, selecting those we found to be the most useful for comparing the control and treatment groups. The three variables we chose to balance the number of variables in the PSM, the number of observations, and bias were as follows:

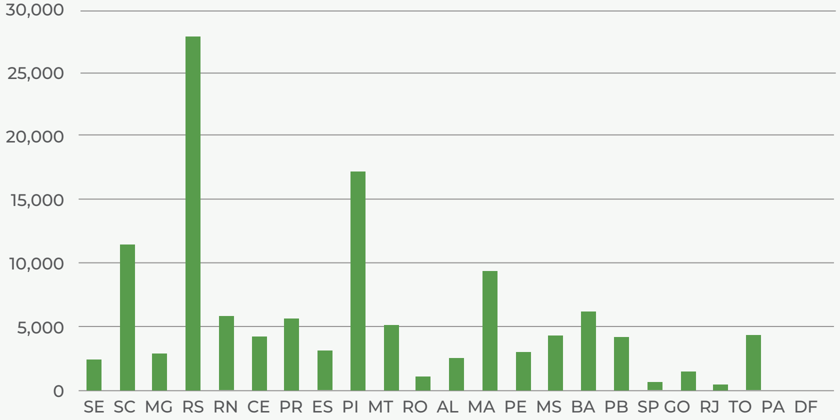
1) Number of local sugarcane- and ethanol-producing units.

2) Total sugarcane plantation area in hectares. Andersson (2019) shows that the size of a country's agricultural sector is important when it comes to understanding the impact of certification on exports. In addition, the number of local production units and total production volumes are likely to affect the ability of smallholders and women to access land: when land is in high demand due to the expansion of monocultures, land prices tend to increase over the years. With this in mind, we compiled data on total ethanol and sugar production based on IBGE data.

3) Total agricultural credit. Smallholder production and women's access to land are also likely to be strongly affected by government policies (Saraiva and Borges, 2021), as well as by the presence or absence of large sugar and ethanol producers. This factor should thus be included in the control group. We used the variation in access to government-subsidized agricultural credit as a proxy for access to government policies to support small farmers and vulnerable groups (figure 3), drawing on data from the Brazilian Ministry of Agriculture website.

Figure 3.

Individuals Benefiting from Government-Subsidized Agricultural Credit (2000–2022)



Notes: SE: Sergipe; SC: Santa Catarina; MG: Minas Gerais; RS: Rio Grande do Sul; RN: Rio Grande do Norte; CE: Ceará; PR: Paraná; ES: Espírito Santo; PI: Piauí; MT: Mato Grosso; RO: Roraima; AL: Alagoas; MA: Maranhão; PE: Pernambuco; MS: Mato Grosso do Sul; BA: Bahia; PB: Paraíba; SP: São Paulo; GO: Goiás; RJ: Rio de Janeiro; TO: Tocantins; PA: Pará; and DF: Distrito Federal.
Source: Authors based on the Ministry of Agriculture of Brazil.

To match units, we used the nearest neighbor criterion, a technique that matches each treated unit to the most similar control unit based on their propensity scores. We used a one-to-one correspondence ratio, meaning that each treated unit was matched with one control unit. This explains why there are only 188 total observations (94 in the treated group and 94 in the control group). **Although the choice of a one-to-one matching ratio limits the number of observations, it allows the most appropriate matches to be selected, which helps to ensure that the treatment and control groups are fully comparable.**

It is important to note that the PSM exercise was conducted without replacement, meaning that once a control unit had been selected as a matching partner for a treated unit, it could not be reused as a partner for another treated unit. This approach ensures that each control unit is only used once in the matching process. The sample size after the PSM is shown in table 1. The density plots for all covariates included in the propensity score for Bonsucro and Bonsucro-RED before and after matching (see annex) **show that the distribution of all covariates included in the logistic regression after matching is balanced between treated and control units, meaning that the two groups are comparable.**

Table 1.
Sample Size Before and After PSM

	BEFORE	AFTER
TOTAL	6,747	94
Matched (total Bonsucro)	94	94
Unmatched	6,653	0
Discarded	0	0

Source: Authors.

The next step is estimating the average treatment effect using the DID method. This strategy is based on exploiting two main sources of variation: variation over time and within-treatment-group variation. The treatment conditions used in this chapter were two time periods (2006 and 2017) and Bonsucro (all Bonsucro-certified units) and Bonsucro-RED (only RED-certified units). The basic idea was to compare the changes in the outcome variable before and after treatment between the treatment and control groups. The functional formula that was estimated is given by:

$$Y_{t,m} = \alpha + \gamma_1 Treatment_m + \gamma_2 Time_t + \gamma_3 Treatment_m {*} Time_t + Z'_{m,t}\,\zeta + \varepsilon_{m,t}$$

Where $Y_{m,t}$ is the outcome variable (percentage of smallholders or the gender gap), $Treatment_m$ is an indicator variable that takes a value of 1 if the municipality has Bonsucro or Bonsucro-RED certifications, $Time_t$ is an indicator variable that takes a value of 0 when the treatment has not yet occurred (2006) and 1 otherwise (2017), and $Z'_{m,t}$ is a vector of control variables. α is the intercept and $\varepsilon_{m,t}$ represents the error term.

In addition to using PSM, **we controlled for certain political variables that could potentially affect outcomes.** Specifically, to capture possible political party differences across municipalities, we included the proportion of municipal councilors belonging to the PT in each municipality and a dummy variable indicating whether the mayor of the municipality in question belonged to the PT.

Table 2.
Descriptive Statistics for the Sample

	TREATED (N=94)	CONTROL (N=6,747)	TOTAL (N=6,841)
Gender gap			
Mean	0.758 (0.117)	0.751 (0.129)	0.751 (0.129)
Median [min, max]	0.777 [0.349, 1.00]	0.780 [-0.0421, 1.00]	0.780 [-0.0421, 1.00]
Proportion of establishments that are family farms			
Mean	0.601 (0.146)	0.751 (0.149)	0.749 (0.150)
Median [min, max]	0.611 [0.125, 0.879]	0.775 [0.0521, 0.996]	0.773 [0.0521, 0.996]
Hectares of sugarcane plantations			
Mean	21,800 (16,200)	2,130 (6,130)	2,400 (6,770)
Median [min, max]	18,900 [0, 75,200]	62.0 [1.00, 99,000]	67.0 [0, 99,000]
Mayor belongs to PT			
Mean	Mean 0.0652 (0.248)	0.0620 (0.241)	0.0620 (0.241)
Median [min, max]	[0, 1.00]	0 [0, 1.00]	[0, 1.00]
Missing	2 (2.1%)	37 (0.5%)	39 (0.6%)
Share of municipal councilors belonging to PT			
Mean	5.76 (8.61)	6.18 (9.40)	6.18 (9.39)
Median [min, max]	0 [0, 33.3]	0 [0, 66.7]	0 [0, 66.7]
Missing	0 (0%)	3 (0.0%)	3 (0.0%)
Companies in the ethanol production sector			
Mean	0.883 (1.87)	0.0870 (0.368)	0.0979 (0.436)
Median [min, max]	0 [0, 15.0]	0 [0, 5.00]	[0, 15.0]
Companies in the sugarcane sector			
Mean	27.4 (61.6)	3.31 (26.0)	3.64 (26.9)
Median [min, max]	8.00 [0, 478]	0 [0, 1,250]	0 [0, 1,250]

Source: Authors.

5. RESULTS OF THE ECONOMETRIC ANALYSIS

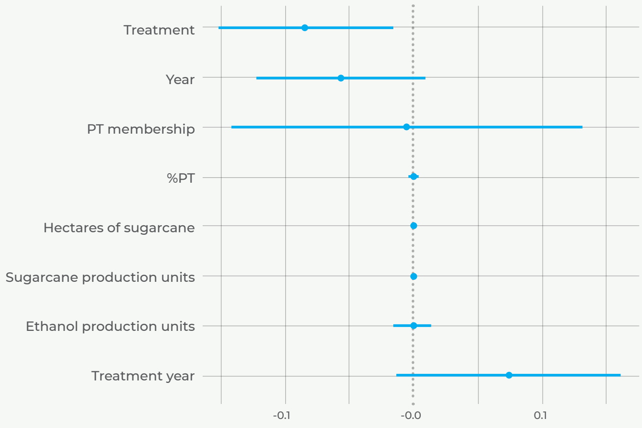
In this section, we show the correlation between Bonsucro and Bonsucro-RED certification and the gender gap, using the potential impact of certification on land ownership as an intervening variable. **The results are consistent across different model specifications: there is no significant correlation between the adoption of the Bonsucro-RED standard and the gender gap in rural Brazil.** This suggests that the Bonsucro-RED certification, which aims to promote more sustainable trade, does not have a significant impact on land ownership and gender inclusion in the areas near certified establishments. In other words, EU sustainability regulations do not generate any significant additional positive externalities for gender equality in the areas near the Brazilian municipalities that are the main exporters of sugarcane and ethanol to the bloc.

Figures 4 and 5 present the results of the regression analyses. Each line represents a different covariate included in the model. The figures show a point estimate for each covariate—this is the coefficient estimate, which indicates the average change in the dependent variable associated with changes in the covariate when all other variables are held constant. The horizontal lines extending from the point estimate represent 95% confidence intervals. These intervals provide a range of plausible values for the true effect of the covariate. When the confidence interval crosses the vertical line representing no effect (the line at 0), it indicates that the effect of the covariate is not statistically significant. **The treatment group does not perform significantly better than the control group on the main outcome variables ("treated_dyear" and "red_dyear") at the 95% confidence level.**

interestingly, however, when the confidence level is set at 90%, the effect of Bonsucro certification on smallholders is statistically significant and positive, albeit marginally so. This means that being in the treatment group increases the likelihood of there being a higher number of family farms compared to the control group. In other words, although the direct effect of Bonsucro certification on the gender gap is not statistically significant, its effect in terms of the mediating variable is (CI = 90%). The results suggest that in municipalities where there are Bonsucro-certified companies that do not export to the EU, certification has a positive, significant effect on relevant local socioeconomic variables. This positive, significant impact on community empowerment is not observed in regions that comply with the Bonsucro-RED standard. These results should be treated with caution, however, as any community with at least one certified enterprise/farm is included in the treated group, regardless of the density of such certified establishments, no data on which was available. All the same, the results raise the alarm about the impact (or lack thereof) of extraterritorial regulations on local socioeconomic development.

Figure 4.

Effects of Standard and RED-Compliant Bonsucro Certification on Smallholders (Left) and the Gender Gap (Right) at CI = 95%



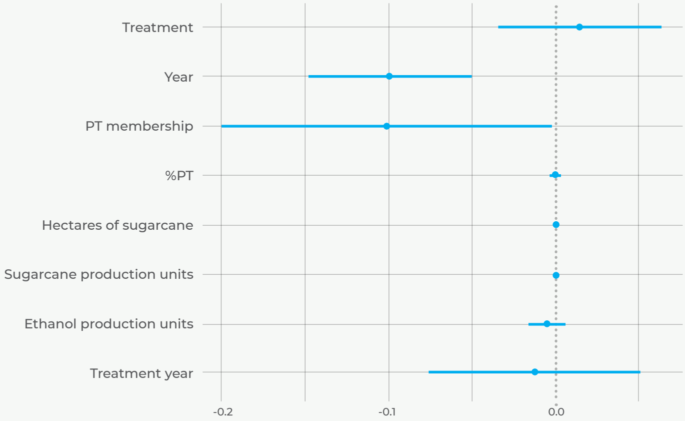
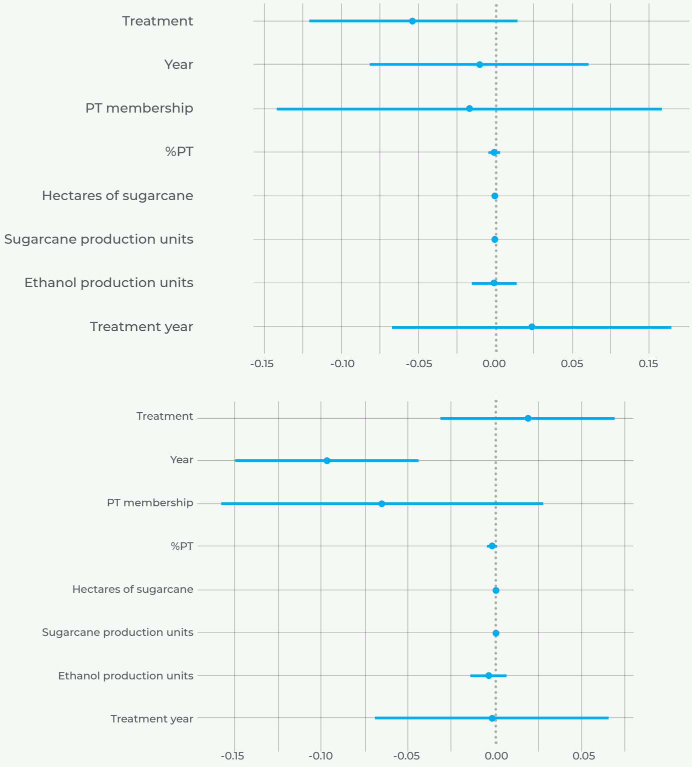


Figure 5.
Effects of Bonsucro Certification on Smallholders (Left) and the Gender Gap (Right) at CI = 95%



Source: Authors.

6. CONCLUSIONS AND POLICY RECOMMENDATIONS

Do private certifications that comply with sustainable trade regulations contribute to gender equality? This chapter has attempted to answer this question using Bonsucro certification and the EU RED as a case study. Bonsucro is one of the accredited organizations that monitors the requirements of the EU RED. This directive seeks to contribute to the EU's transition to renewable fuels by establishing rules to reduce GHG emissions from biofuel imports. Under a holistic approach to sustainability, the RED's goal of promoting greener imports should also contribute to generating positive socioeconomic externalities. However, the potential for negative side effects and the conflicting findings on certifications and their actual impact warrant a closer examination of the Bonsucro-RED standard.

In this chapter, we have argued that the Bonsucro-RED certification may bring about two opposing effects: (1) it may empower women, including by providing access to information on government resources that could facilitate their integration into the economy as landowners; or (2) it may contribute to increased land concentration, thus reducing the number of women landowners in rural areas. **The DID analysis did not find a significant relationship between Bonsucro-RED and an increase or decrease in the gender gap in land ownership in Brazil.** However, it is interesting to note that the effect on smallholders is positive and significant (CI=90%) in municipalities where there are Bonsucro-certified companies, but not for municipalities where companies hold the RED-compliant version of the Bonsucro certification. However, these results should be treated with caution due to potential limitations associated with the small number of observations included in this study. The decision to select alternative matching variables may also affect our results. Consequently, the findings from this study can offer no more than suggestions toward understanding the relationship between certification and sustainable trade policies.

Our results confirm the findings of the existing literature on the limited impact of large-scale agriculture on neighboring smallholder farmers (Ali, Deininger, and Harris, 2019). **At best, the findings present a mixed picture of the joint effect of private certification and sustainable trade regulations such as those promoted by the EU.** The Bonsucro-RED standard does not seem to exacerbate the gender gap in rural land ownership in Brazil, but this is far from reassuring news, given that the sector is still characterized by few opportunities for women, despite some recent improvements. Moreover, the results obtained by comparing companies with standard Bonsucro certification and the RED-compliant version suggest that regulations aimed at promoting sustainable exports may work poorly if both private and public regulatory bodies do not adequately address the concentration of land ownership and unequal access to export opportunities.

Our results also add nuance to the existing literature on the potential impacts of international trade. Some studies have found that exporting to partners with higher social and environmental standards (e.g., the EU) can lead to increased adoption of such standards locally: the “trading-up effect” or “California effect” (Vogel, 1997). **Our findings suggest that the trading-up effect of exporting to the EU may potentially be limited by negative externalities or policies that are not fully designed to address sustainability in a holistic fashion.** Given the proliferation of unilateral EU sustainability regulations with extraterritorial effects, including the CBAM and RED, the results of this report serve as a warning: policymakers need to address environmental protection without losing sight of local socioeconomic development, especially in developing countries.

A series of policy recommendations can be derived from these conclusions:

1. **Public regulators need to encourage private regulators to fully integrate community-based empowerment practices targeting smallholders and women into their schemes and adopt a holistic approach to sustainability.** This means that the goal of limiting carbon emissions must go hand-in-hand with a realistic assessment of local socioeconomic conditions in less developed countries. In particular, we recommend that local socioeconomic development be clearly and explicitly integrated into the production standards of private certification schemes (e.g., through training programs). Bonsucro has

started to do this through version 5.1 of its production standards, which were launched in January 2022. It is important for private standard setters to strengthen the participation of women and communities in their organizations' decision-making processes and to engage in deeper forms of social auditing. These might include conducting assessments, reporting, and making improvements based on social performance and ethical behavior, especially through dialogue with stakeholders (Ebrahim, 2003).

For EU policymakers, the issue of financial support mechanisms to mitigate adjustment costs in third countries is likely to be key to ensuring the legitimacy of the bloc's policies in the future. This includes increased funding for existing schemes such as the Amazon Fund, as well as the facilitation and institutionalization of science-based impact assessment mechanisms for evaluating funded projects in terms of local socioeconomic development.

For Brazil to gain the support of EU policymakers and ensure that sustainable trade policies generate positive externalities, several conditions must be in place. The first step toward this entails Brazil establishing a clear set of sustainability objectives and scientific evidence of potential negative side effects. The country needs to do this in close collaboration with stakeholders from civil society, including those most active in Brussels, in order to make its position more convincing and ensure that clear gender and sustainability demands are included in bilateral discussions, including the EU-Mercosur trade agreement.

2. **Given the potential impact of public regulation on private certification schemes, policymakers should institutionalize and expand existing mechanisms to update and/or modify existing regulations, ensuring broad stakeholder participation.** For autonomous/unilateral regulations with impacts outside the EU, as is the case with the RED, policymakers should ensure that stakeholders from the countries most affected by these regulations participate in their design. These participation mechanisms should not only take the form of "passive" consultation opportunities, where stakeholders

present their positions to the EU, but should involve the active facilitation of foreign stakeholder participation (e.g., through open events organized and funded by EU consulates and embassies in affected countries). If such mechanisms are not implemented, the unintended local effects of sustainability rules and regulations could create feedback effects that might limit public support for these standards and regulations.

In this regard, the Brazilian government should work with the EU to raise awareness that unilateral measures such as the EUDR and RED II are not enough to solve the problem of gender inequality on the ground and may even exclude smaller producers from the market. We also suggest that the EU should provide more financial and technical support to producers in Brazil and the Global South in general to achieve its sustainability policy goals.

3. **In Brazil and possibly other Latin American and Caribbean countries, state and federal governments should strive to build stronger ties between existing voluntary certification schemes and public regulations promoting family farming/women's participation.** Given that sustainability governance seems to be moving toward increasingly robust public-private partnerships (Marx, Depoorter, and Vanhaecht, 2022), governments in developing countries can act as "meta-regulators" (Naiki, 2021; Dietz et al., 2022) to mitigate the negative externalities of private certification schemes while amplifying their positive effects. Specifically, these positive effects include using the knowledge and networks of private stakeholders to fill the gap in enforcement left by public environmental regulations, a situation that is especially true in Brazil (Coelho-Junior et al., 2022).

The first step toward achieving this entails conducting baseline studies on the effective use of private certification schemes to complement the state's capacity to enforce public regulations. These studies should identify areas in which the state lacks sufficient enforcement capacity and in which a market-based mechanism could scale up and thrive. They should also identify the voluntary certification schemes that are currently available and rank them from most to least likely to bridge the enforcement gaps that have been identified.

The second step would involve developing communication strategies to inform policy makers in Latin America and the Caribbean of policies and results through joint conferences, courses, and workshops.

These actions would play a key role in the process of developing or modifying public regulations that allow private regulators to monitor compliance, while creating clear and binding minimum social, economic, and environmental standards. For example, Brazil plans to develop a new program known as *Selo Verde* [Green Label]. The goal is to create a voluntary label scheme certifying that Brazilian products are socially and environmentally responsible. Rather than starting from scratch, this program should build on the lessons learned from existing certification schemes.

Finally, the Brazilian government could use public procurement as a sustainability tool by establishing specific sustainability requirements backed by existing certification schemes as a way to purchase services and products. This would include existing public initiatives, such as the Food Procurement Program (PAA), which could include gender requirements.

There are some interesting avenues that could be explored to build on this chapter and establish stronger ties between international certification schemes, sustainable trade, and local socioeconomic impacts. One possibility is studying the impact of certification on smallholdings, regardless of whether they are family farms. This information is available annually for Brazil. It would also be important to extend this research and evaluate the impact of Bonsucro certification in other countries in Latin America and the Caribbean to establish a baseline for comparisons for the results and conclusions of this study. These steps will enable researchers to establish a more solid basis for evaluating the real impact of certifications in Latin American and Caribbean countries and how they may or may not improve the region's green exports while generating positive socioeconomic externalities. Finally, it is important to emphasize that the data used in this report does not allow us to test for parallel trends, as only two time periods are available from the Brazilian censuses. Future research could address this issue by focusing on property size as an outcome variable, as such data is available annually in Brazil. New studies could also go further by using alternative methods for PSM in addition to nearest neighbor, such as kernel matching.

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ANNEX A

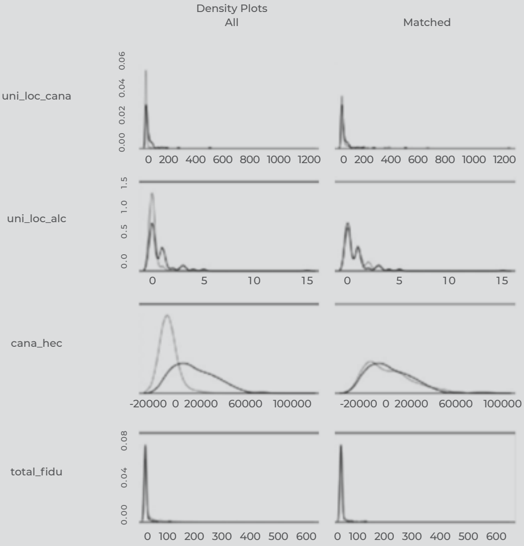
Table A1.
Variable Codes.

VARIABLE	DESCRIPTION	SOURCE
Nome	Name of municipality	IBGE
id_municipio	IBGE municipality code	IBGE
sigla_uf	State the municipality is in	IBGE
year	Year	IBGE
total_saf	Total nonfamily farms	IBGE
total_caf	Total family farms	IBGE
total_saf_h	Total nonfamily farms owned by men	IBGE
total_saf_m	Total nonfamily farms owned by women	IBGE
total_caf_h	Total family farms owned by men	IBGE
total_caf_m	Total family farms owned by women	IBGE
po_cana	People employed in the sugarcane sector	IBGE
po_alc	People employed in ethanol production	IBGE
cana_hec	Hectares of sugarcane plantations	IBGE
uni_loc_cana	Companies in the sugarcane sector	IBGE
uni_loc_alc	Companies in the ethanol production sector	IBGE
share_pt	Share of municipal councilors belonging to the PT	CEPESP-FGV
sexo_pref	Sex of mayor	CEPESP-FGV
prop_caf_h	Proportion of men who manage family farms	IBGE
prop_caf_m	Proportion of women who manage family farms	IBGE
prop_saf_h	Proportion of men who manage nonfamily farms	IBGE
prop_saf_m	Proportion of women who manage nonfamily farms	IBGE
partido_pt	Dummy indicating whether the mayor belongs to the PT	CEPESP-FGV
total_estab_m	Number of women working on farms	IBGE
total_fidu	Total agricultural credit by municipality	Ministry of Agriculture
total_estab	Total farms	IBGE
prop_af	Proportion of establishments that are family farms	IBGE
prop_saf	Proportion of establishments that are nonfamily farms	IBGE

Source: Authors.

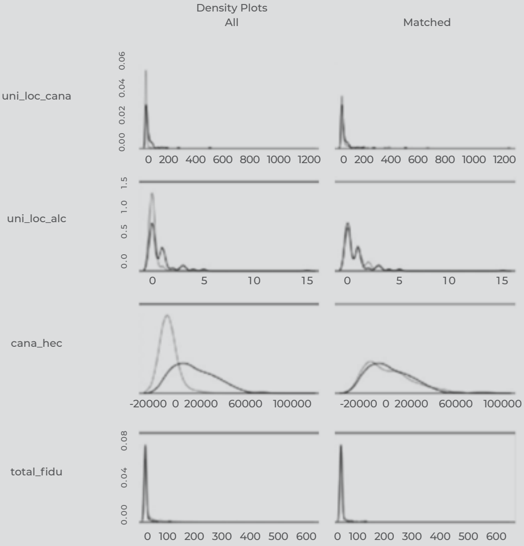
Figure 1 shows the density plots of all covariates included in the propensity score before and after matching. After matching, the distribution of all covariates included in the logistic regression is balanced between treated and control units.

Figure A1.
PSM Covariate Density Plots (Bonsucro and Bonsucro-RED Certification)



Source: Authors.

Figure A2.
PSM Covariate Density Plots (Bonsucro-RED)



Source: Authors.

Table A2.
Regression Results with PSM Including Agricultural Credit

	DEPENDENT VARIABLE: % OF SMALLHOLDER FARMERS	
	PSM based on Bonsucro certification	PSM based on Bonsucro-RED certification
Bonsucro	-0.084**	
	(0.034)	
Bonsucro-RED		-0.055
		(0.034)
Time dummy	-0.057*	-0.016
	(0.033)	(0.036)
Mayor PT	-0.005	-0.021
	(0.069)	(0.063)
% PT	-0.0002	-0.001
	(0.002)	(0.002)
Sugarcane plantations (hectares)	-0.00000***	-0.00000***
	(0.00000)	(0.00000)
Sugarcane plantations	0.00005	-0.00003
	(0.0001)	(0.0001)
Local ethanol production units	-0.001	-0.0004
	(0.007)	(0.007)
Bonsucro * Time dummy	0.074*	
	(0.044)	
Bonsucro-RED * Time dummy		0.034
		(0.046)
Intercept	0.730***	0.716***
	(0.030)	(0.030)
Observations	186	159
R^2	0.108	0.128
Notes:		$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

Source: Authors.

Table A3.
Descriptive Statistics for Bonsucro Post-PSM

	2006	
	Treated (N=47)	Control (N=31)
Gender gap		
Mean	0.800 (0.0878)	0.800 (0.0936)
Median [min, max]	0.812 [0.489, 0.956]	0.827 [0.525, 0.906]
Proportion of establishments that are family farms		
Mean	0.594 (0.177)	0.697 (0.167)
Median [min, max]	0.640 [0.125, 0.879]	0.677 [0.367, 0.969]
Hectares of sugarcane plantations		
Mean	18,000 (14,400)	14,900 (11,800)
Median [min, max]	16,000 [0, 54,000]	12,800 [1.00, 46,000]
Mayor belongs to PT		
Mean	0.109 (0.315)	0.0323 (0.180)
Median [min, max]	0 [0, 1.00]	0 [0, 1.00]
Missing	1 (2.1%)	0 (0%)
Share of municipal councilors belonging to PT		
Mean	8.54 (9.82)	7.11 (8.31)
Median [min, max]	9.09 [0, 33.3]	0 [0, 22.2]
Companies in the ethanol production sector		
Mean	1.21 (2.40)	0.968 (1.30)
Median [min, max]	1.00 [0, 15.0]	0 [0, 5.00]
Companies in the sugarcane sector		
Mean	12.4 (36.1)	16.0 (58.0)
Median [min, max]	4.00 [0, 244]	2.00 [0, 325]

	2017	
	Treated (N=47)	Control (N=63)
Gender gap		
Mean	0.716 (0.129)	0.715 (0.111)
Median [min, max]	0.740 [0.349, 1.00]	0.724 [0.365, 0.909]
Proportion of establishments that are family farms		
Mean	0.608 (0.107)	0.620 (0.152)
Median [min, max]	0.595 [0.406, 0.873]	0.638 [0.200, 0.942]
Hectares of sugarcane plantations		
Mean	25,700 (17,100)	25,700 (22,300)
Median [min, max]	24,000 [0, 75,200]	22,400 [1.00, 99,000]
Mayor belongs to PT		
Mean	0.0217 (0.147)	0 (0)
Median [min, max]	0 [0, 1.00]	0 [0, 0]
Missing	1 (2.1%)	0 (0%)
Share of municipal councilors belonging to the PT		
Mean	2.98 (6.14)	2.84 (5.57)
Median [min, max]	0 [0, 22.2]	0 [0, 33.3]
Companies in the ethanol production sector		
Mean	0.553 (1.06)	0.714 (0.906)
Median [min, max]	0 [0, 5.00]	0 [0, 3.00]
Companies in the sugarcane sector		
Mean	42.5 (76.9)	69.0 (186)
Median [min, max]	20.0 [0, 478]	4.00 [0, 1,250]

	TOTAL	
	Treated (N=94)	Control (N=94)
Gender gap		
Mean	0.758 (0.117)	0.743 (0.112)
Median [min, max]	0.777 [0.349, 1.00]	0.769 [0.365, 0.909]
Proportion of establishments that are family farms		
Mean	0.601 (0.146)	0.645 (0.160)
Median [min, max]	0.611 [0.125, 0.879]	0.645 [0.200, 0.969]
Hectares of sugarcane plantations		
Mean	21,800 (16,200)	22,200 (20,000)
Median [min, max]	18,900 [0, 75,200]	18,700 [1.00, 99,000]
Mayor belongs to PT		
Mean	0.0652 (0.248)	0.0106 (0.103)
Median [min, max]	0 [0, 1.00]	0 [0, 1.00]
Missing	2 (2.1%)	0 (0%)
Share of municipal councilors belonging to PT		
Mean	5.76 (8.61)	4.25 (6.86)
Median [min, max]	0 [0, 33.3]	0 [0, 33.3]
Companies in the ethanol production sector		
Mean	0.883 (1.87)	0.798 (1.05)
Median [min, max]	0 [0, 15.0]	0 [0, 5.00]
Companies in the sugarcane sector		
Mean	27.4 (61.6)	51.5 (158)
Median [min, max]	8.00 [0, 478]	3.00 [0, 1,250]

Source: Authors.

Table A4.
Descriptive Statistics for Bonsucro-RED Post-PSM.

	2006	
	Treated (N=40)	Control (N=31)
Gender gap		
Mean	0.801 (0.0879)	0.779 (0.0998)
Median [min, max]	0.808 [0.489, 0.956]	0.782 [0.501, 0.930]
Proportion of establishments that are family farms		
Mean	0.592 (0.176)	0.668 (0.199)
Median [min, max]	0.634 [0.125, 0.852]	0.684 [0.190, 0.980]
Hectares of sugarcane plantations		
Mean	18.000 (14.900)	14.800 (18.200)
Median [min, max]	16,100 [0, 54,000]	8,840 [1.00, 93,000]
Mayor belongs to PT		
Media	0,103 (0,307)	0,129 (0,341)
Median [min, max]	0 [0, 1,00]	0 [0, 1,00]
Missing	1 (2,5%)	0 (0%)
Share of municipal counsellors belonging to PT		
Mean	8.92 (9.40)	8.78 (10.9)
Median [min, max]	10.0 [0, 33.3]	0 [0, 33.3]
Companies in the ethanol production sector		
Mean	1.43 (2.54)	1.13 (1.50)
Median [min, max]	1.00 [0, 15.0]	1.00 [0, 5.00]
Companies in the sugarcane sector		
Mean	14.1 (38.9)	11.4 (25.8)
Median [min, max]	4.00 [0, 244]	2.00 [0, 116]

	2017	
	Treated (N=40)	Control (N=49)
Gender gap		
Mean	0.727 (0.118)	0.717 (0.117)
Median [min, max]	0.743 [0.447, 1.00]	0.725 [0.365, 0.895]
Proportion of establishments that are family farms		
Mean	0.608 (0.104)	0.628 (0.126)
Median [min, max]	0.595 [0.406, 0.870]	0.615 [0.320, 0.942]
Hectares of sugarcane plantations		
Mean	26,300 (17,700)	27,100 (19,200)
Median [min, max]	24,500 [0, 75,200]	26,000 [1.00, 74,000]
Mayor belongs to PT		
Mean	0.0250 (0.158)	0 (0)
Median [min, max]	0 [0, 1.00]	[0, 0.0]
Missing	0 (0%)	0 (0%)
Share of municipal counsellors belonging to PT		
Mean	3.34 (6.53)	1.57 (3.40)
Median [min, max]	0 [0, 22.2]	0 [0, 11.1]
Companies in the ethanol production sector		
Mean	0.650 (1.12)	0.735 (0.884)
Median [min, max]	0 [0, 5.00]	1.00 [0, 3.00]
Companies in the sugarcane sector		
Mean	42.0 (80.6)	84.2 (199)
Median [min, max]	19.0 [0, 478]	12.0 [0, 1,250]

	TOTAL	
	Treated (N=40)	Control (N=80)
Gender gap		
Mean	0.764 (0.110)	0.741 (0.114)
Median [min, max]	0.777 [0.447, 1.00]	0.763 [0.365, 0.930]
Proportion of establishments that are family farms		
Mean	0.600 (0.144)	0.643 (0.158)
Median [min, max]	0.612 [0.125, 0.870]	0.630 [0.190, 0.980]
Hectares of sugarcane plantations		
Mean	22.200 (16.800)	22.300 (19.600)
Median [min, max]	18,900 [0, 75,200]	17,300 [1.00, 93,000]
Mayor belongs to PT		
Mean	0.0633 (0.245)	0.0500 (0.219)
Median [min, max]	0 [0, 1.00]	0 [0, 1.00]
Missing	1 (1.3%)	0 (0%)
Share of municipal counsellors belonging to PT		
Mean	6.13 (8.52)	4.36 (8.01)
Median[min, max]	0 [0, 33.3]	0 [0, 33.3]
Companies in the ethanol production sector		
Mean	1.04 (1.99)	0.888 (1.17)
Median [min, max]	0.500 [0, 15.0]	1.00 [0, 5.00]
Companies in the sugarcane sector		
Mean	28.0 (64.4)	56.0 (160)
Median [min, max]	8.00 [0, 478]	5.00 [0, 1,250]

Source: Authors.

CHAPTER 4

CATTLE FARMING IN ARGENTINA

THE IMPACT OF CARBON BORDER TAXES

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* This chapter is dedicated to the memory of Amílcar Arzubi, who was part of the original research proposal and who devoted his professional life to studying these issues.

ACRONYMS AND ABBREVIATIONS

- **CERDI**
  Center for Studies and Research on International Development
- **CO$_2$**
  carbon dioxide
- **CREAs**
  Regional Consortia for Agricultural Experimentation
- **ETS**
  Emissions Trading System
- **EU**
  European Union
- **FAO**
  Food and Agriculture Organization of the United Nations
- **FPMA**
  Food Price Monitoring and Analysis
- **gCO$_2$**
  grams of carbon dioxide
- **GHGs**
  greenhouse gases
- **GSIM**
  global simulation model
- **INDEC**
  National Institute of Statistics and Censuses, Argentina
- **INTA**
  National Institute of Agricultural Technology, Argentina
- **IPCC**
  Intergovernmental Panel on Climate Change
- **MtCO$_2$eq**
  million metric tons of carbon dioxide equivalent
- **NDVI**
  Normalized Difference Vegetation Index
- **nETS**
  ETS for sectors not covered by the EU ETS
- **OECD**
  Organisation for Economic Co-operation and Development.
- **RoW**
  rest of the world
- **SENASA**
  National Service for Agrifood Health and Quality, Argentina
- **tCO$_2$eq**
  Metric tons of carbon dioxide equivalent
- **UN**
  United Nations
- **UNCTAD TRAINS**
  United Nations Conference on Trade and Development Trade Analysis and Information System
- **US**
  United States
- **US$**
  United States dollars
- **USDA**
  United States Department of Agriculture
- **WTO**
  World Trade Organization

ABSTRACT

There is currently considerable debate around the inclusion of the agricultural sector into formal carbon pricing mechanisms in the countries of the Global North, led by the EU. There is also a growing trend toward implementing measures to prevent carbon leakage. In this scenario, the trading partners of these countries could face challenges in terms of competitiveness and market access.

This chapter analyzes how the implementation of a carbon adjustment mechanism at the EU border might impact international beef trade, with a focus on Argentinian exports. To conduct this analysis, we used the CLIMTRADE partial equilibrium model to perform impact simulations. Through this methodology, we estimated that Argentinian beef exports would drop by 8% in the current emissions intensity scenario and by 7% in a mitigation scenario (which entails a 20% reduction in emissions intensity due to improvements in the efficiency of Argentinian cattle-farming systems).

It is critical for Argentina to make progress on monitoring the emissions generated by cattle farming and promote the evaluation of its net emissions, taking carbon sequestration in cattle-farming systems into account.

1. INTRODUCTION

Climate change poses complex challenges for agrifood systems. On the one hand, these systems are particularly vulnerable to the effects of climate change, and due to concerns about food security, the focus to date has been on adaptation strategies. On the other hand, agricultural production is a major source of greenhouse gas (GHG) emissions, accounting for 23% of these globally (IPCC, 2020). **Although the agricultural sector has so far faced fewer demands than other sectors in terms of emission reductions, these are expected to increase soon** (Stepanyan et al., 2023; OECD, 2022; European Court of Auditors, 2021). Emissions from the agrifood sector need to be reduced significantly to limit the global temperature rise to below 2°C, as set out in the Paris Agreement.[1] Recent reports even state that the agriculture sector must reduce its emissions while adapting to climate change (Isermeyer, Heidecke, and Osterburg, 2019; OECD, 2022).

The agriculture and livestock sector participates in voluntary carbon credit markets, which have seen strong growth in recent years (World Bank, 2022). All the same, these activities have not yet been included in formal emissions trading systems (ETSs) in the European Union (EU). **The inclusion of the agricultural sector in formal carbon pricing mechanisms now seems imminent** (Isermeyer, Heidecke, and Osterburg, 2019). In the EU, where these markets play a key role in controlling emissions as part of the bloc's Green Deal,[2] alternative approaches to reducing emissions from agricultural production through emissions trading are being discussed (Stepanyan et al., 2023). Agricultural and forestry products are sold internationally on a large scale. As a consequence, carbon pricing in Europe may lead to the relocation of emissions-intensive production activities to third countries (carbon leakage). However, carbon border adjustment mechanisms would help to reduce leakages of this sort to regions where agriculture and forestry are not regulated (Hufbauer et al., 2022).

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1 · At the United Nations Climate Change Conference in Paris in December 2015 (COP21), world leaders laid the foundations for the Paris Agreement, which entered into force in November 2016. The Paris Agreement is a legally binding international treaty signed to date by 193 countries and the EU. It contains commitments by signatory countries to reduce emissions and work together to adapt to the impacts of climate change, as well as a call for signatory countries to increase their commitments over time (United Nations, 2015).

2 · A set of initiatives promoted by the European Commission to adapt EU climate, energy, transportation, and taxation policies with the aim of reducing net GHG emissions by at least 55% by 2030, compared to 1990 levels.

Other key regulations for trade in agricultural products include the initiatives in the EU and the United States (US) regarding compliance with deforestation-free certification requirements. These regulations require exporters to provide geolocated documentation proving that products or the ingredients used in them do not come from land that has been deforested since December 2020. These obligations apply from December 2024, and the documentation process will be stricter in regions with a higher risk of deforestation (Conte Grand, Schulz-Antipa, and Rozenberg, 2023). Despite criticism from major agricultural exporting countries, these regulations are moving forward.

Livestock production, crop fertilization, and land use changes are the main channels through which agrifood systems contribute to GHG emissions (European Court of Auditors, 2021; González Fischer and Bilenca, 2020). **Beef production is likely to be among the industries most affected by carbon pricing initiatives.**

Agricultural exporting countries such as Argentina may be severely affected by changes in the rules of the game involving a carbon tax on beef exports. Argentina has 53 million head of cattle and produces 3 million tons of meat, making it one of the world's top five beef producers and exporters. The country accounts for 5% of global beef production and 7% of global beef exports (USDA, 2023). Beef plays a crucial role in Argentina's economy: it is one of the country's main value chains (the third largest, according to Lódola and Picón, 2023), is the fifth-largest export item (INDEC, 2023), and beef production takes place in most areas of Argentina. However, the industry poses some major environmental challenges, as it accounts for 18% of national emissions (64.9 $MtCO_2eq$ of the total 366 $MtCO_2eq$) (Ministerio de Ambiente y Desarrollo Sostenible de Argentina, 2021) and represents the largest share of emissions within the country's agricultural exports (Marquardt, Gonzales-Zuñiga, and Röser, 2022). A recent simulation-based study found that **different EU border carbon tariffs had significant impacts on Argentinian beef exports to destinations purchasing high-value cuts** (Marquardt, Gonzales-Zuñiga, and Röser, 2022).

In Argentina, beef is produced using a wide variety of production systems in many different parts of the country. Although grains have come to account for a larger share of cattle diets in recent decades, **cattle production in the country remains predominantly grass-fed.** The rearing and rearing phases take place almost exclusively on pasture or rangeland. Rearing generally uses land that is not suitable for agriculture. Likewise, less than 20% of steers are fattened exclusively in feedlots.

The average efficiency of reproduction indices for the rearing phase is low: the average national weaning rate is 65%, although this varies enormously across regions and systems. **The wintering phase typically entails low slaughter weights** due to the domestic market's preference for light animals and producers' need to recover working capital quickly. Consequently, the average weight per animal in Argentina is 230 kg, compared to 376 kg in the United States, 308 kg in Australia, 260 kg in Brazil, and 257 kg in Uruguay.

This chapter examines how the emissions intensity of beef production in Argentina would be impacted by a series of transformation scenarios associated with improvements to production efficiency indicators. **We use a partial equilibrium model to simulate international trade in beef and evaluate whether mitigation scenarios reduce the negative economic impact of EU border carbon taxes** on the Argentinian beef export sector.

The rest of the chapter is organized as follows: section 2 describes the methodology, the trade model and its application, and the design of the different mitigation scenarios. Section 3 discusses the results of the simulation. Finally, section 4 looks at the implications of these results for policies targeting the sector.

2. METHODOLOGY

The CLIMTRADE partial equilibrium model was used to simulate international trade in beef. Through it, we evaluated how far certain emissions mitigation scenarios for Argentinian beef production could reduce the negative economic impacts of an EU border carbon tax. To design the mitigation scenarios used in the simulation, we surveyed lifestock farming experts on the feasibility of improvements to the productivity levels of rearing and wintering systems in Argentina over a 10-year period, assuming sustained that international demand for beef remains stable but entails stricter requirements in terms of GHG emissions intensity per kg of product. We used the national GHG inventory methodology to calculate each mitigation scenario's potential for reducing emissions. Detailed information on the data and methods used is presented in the following sections.

2.1.
CLIMTRADE model

CLIMTRADE[3] (Marquardt, Gonzales-Zuñiga, and Röser, 2022) allows researchers to model how the carbon tariffs and mitigation measures adopted in different countries impact trade flows. CLIMTRADE is based on the global simulation model (GSIM) (Francois and Hall, 2002). The GSIM is a partial equilibrium representation of the global market that considers trade between several countries at the industry or product level, assuming product differentiation by origin (imports from different origins are imperfect substitutes). The model's input variables include an initial bilateral trade matrix at world prices and an initial bilateral ad valorem tariff matrix. The elasticities of substitution, import demand, and export supply are defined exogenously.

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[3] · Developed by the New Climate Institute: https://newclimate.org/

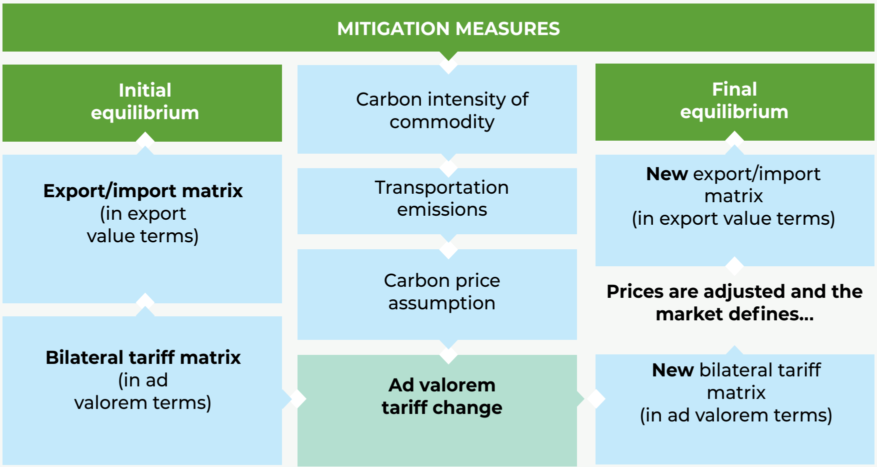
CLIMTRADE expands on the original GSIM approach by incorporating the modeling of the trade effects of carbon tariffs and mitigation measures, expressed in terms of lower GHG emissions intensity per unit of product relative to a baseline scenario. This study uses CLIMTRADE to model how the EU's introduction of a tariff proportional to the emissions intensity of the traded product would impact trade in beef. The magnitude of the tariff increase is a function of (country-specific) beef emission intensities, international transportation emissions, and the carbon price underlying the carbon tariff.

The model estimates a new equilibrium through a price adjustment that determines the absence of excess supply and/or demand in the world market, thus obtaining new equilibrium prices and the corresponding imported and exported quantities of a given product for each country. The total aggregate impact is the sum of the impact on exports and imports for each country and the impact on domestic trade.

We used the CLIMTRADE to evaluate five GHG mitigation scenarios for Argentinian beef production. The baseline scenario was the benchmark to which the mitigation scenarios were compared (more on the definition of the baseline and mitigation scenarios later in the article). Figure 1 provides an overview of the modeling logic that we used.

Figure 1.

Representation of the CLIMTRADE Model for Estimating the Trade Impacts of Mitigation Actions in the Context of a Border Carbon Adjustment Mechanism


MITIGATION MEASURES
Initial equilibrium
Export/import matrix (in export value terms)
Bilateral tariff matrix (in ad valorem terms)
Carbon intensity of commodity
Transportation emissions
Carbon price assumption
Ad valorem tariff change
Final equilibrium
New export/import matrix (in export value terms)
Prices are adjusted and the market defines...
New bilateral tariff matrix (in ad valorem terms)

Source: Authors.

2.1.1.
ASSUMPTIONS OF THE PARTIAL EQUILIBRIUM ANALYSIS

- Our analysis used the default elasticities in Holzner (2004) in combination with others from more recent empirical studies (Cicowiez and Lofgren, 2017) and sector-specific publications (Kawashima and Puspito Sari, 2010; Sbarai and Galvão de Miranda, 2012), namely:
    - Price elasticity of export supply: 1.5
    - Price elasticity of demand for imports: 1.25
    - Substitution elasticity: 2
- Initial tariffs are expressed as bilateral most-favored-nation tariff lines. Preferential tariff lines subject to restricted volume quotas are not counted.
- Nontariff barriers, export taxes, and subsidies are held constant. Existing export taxes in Argentina are assumed to be constant.
- Only EU countries have a border carbon tax that affects the rest of the countries involved in the global beef trade.

- The GHG emissions intensity indicator for beef used in the model does not consider possible sources of carbon sequestration associated with grazing.[4]

2.1.2.
CLIMTRADE DATA SOURCES

The model considers Argentina, a set of 23 countries, and the rest of the world (RoW). The countries that were included in the model are either significant players in the international beef market (either because they are large producers or consumers) or important for Argentina's foreign trade (table 1). The EU countries in the model are the ones that are introducing the carbon tax.

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[4] The potential limitations of using the CLIMTRADE model in our analysis are described in annex A.

Table 1.
Production, Total Exports, Emissions Intensity, and Export Value for Beef in Argentina (2019)

REGION	CODE	PRODUCTION	EXPORTS	EMISSIONS INTENSITY	VALUE OF EXPORTS FROM ARGENTINA TO EACH DESTINATION
		bone-in beef (tons)	product (tons)	kg CO_2eq/kg bone-in beef	US$1,000
Argentina	ARG	3,135,908	562,281	34.05	-
Australia	AUS	2,351,793	1,275,779	23.44	-
Brazil	BRA	10,200,000	1,569,525	38.96	49,268
Canada	CAN	1,393,831	383,698	14.96	-
Chile	CHL	208,313	19,153	24.58	169,368
China	CHN	5,988,070	218	15.38	2,396,639
China (Hong Kong SAR)	HKG	6,632	6,632	0.42	7,387
France	FRA	1,428,460	199,339	22.66	646
Germany	DEU	1,107,000	273,630	15.08	145,460
India	IND	915,640	915,640	140.71	-
Ireland	IRL	619,800	387,827	18.13	-
Israel	ISR	141,138	-	4.18	134,673
Italy	ITA	779,820	117,824	12.84	41,630
Japan	JPN	470,918	3,707	11.40	-
Mexico	MEX	2,027,634	246,395	32.21	-
Netherlands	NLD	424,300	424,300	10.99	81,119
Poland	POL	560,500	377,786	16.07	-
Republic of Korea	KOR	286,000	98	21.56	-
Russia	RUS	1,625,163	5,979	15.54	92,624
Spain	ESP	695,170	191,295	18.69	6,410
United Kingdom	GBR	914,000	135,781	18.85	4,012
United States	USA	12,348,749	899,930	13.71	12,085
Uruguay	URY	580,320	326,617	36.71	-
Uzbekistan	UZB	944,724	-	16.95	-
Rest of the World	RoW	18,761,741	1,568,328	36.32	18,701

Source: Authors based on UN COMTRADE (exports of chilled and frozen beef) and FAOSTAT (beef production, fresh and chilled bone-in beef, emissions intensity).

All data used in the model is for 2019. If this data is not available, the closest available year is taken and reported. We opted to use data from 2019 because it is the most recent data available outside the years affected by the Covid-19 pandemic, which were excluded to avoid trade distortions that could be a consequence of the health crisis).

Data sources are listed below:

- Trade flows between countries: COMTRADE. The categories included are 201 (fresh or chilled beef) and 202 (frozen beef).
- Total beef production: FAOSTAT.
- World beef price: Food Price Monitoring and Analysis (FPMA), Food and Agriculture Organization of the United Nations (FAO). We took the annual average for 2019 of international beef export prices for Brazil and the US.
- Emissions intensity: FAOSTAT. The baseline scenario beef emissions intensity for Argentina and the rest of the countries included in the model is taken from the FAOSTAT database, which follows the tier 1 calculation methodology of the IPCC 2006 guidelines.[5] The emissions categories considered include enteric fermentation, manure management, and manure on pastures, which were the only variables on which comparable data was available for the 23 countries included in the CLIMTRADE model. However, it should be clarified that some of these countries, including Argentina, have validated tier 2 or 3 calculations for their national GHG inventory reports. Argentina (tier 2) estimates an emissions intensity value that is 30% lower than that shown in table 1. Other countries' measurements are similarly affected. The results of the model could thus be adjusted once emissions intensity data calculated to a higher level of complexity becomes available.

To calculate the reduction in the emissions intensity of the mitigation scenarios, the percentage change resulting from the estimation of the baseline and mitigation scenarios was applied to the baseline value using the methodology of the national GHG inventory for the beef cattle sector (a tier 2 calculation, according to IPCC 2006 guidelines).[6]

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[5] · The calculation tier represents the level of methodological complexity of emissions estimates. Tier 1 is the most basic level in terms of complexity and data requirements, tier 2 is intermediate, and tier 3 is the most demanding. Tiers 2 and 3 are considered more accurate as long as adequate local data is available with which to implement the higher level methods, evaluate them, and apply them.

[6] · This point is further discussed in more detail later in the chapter.

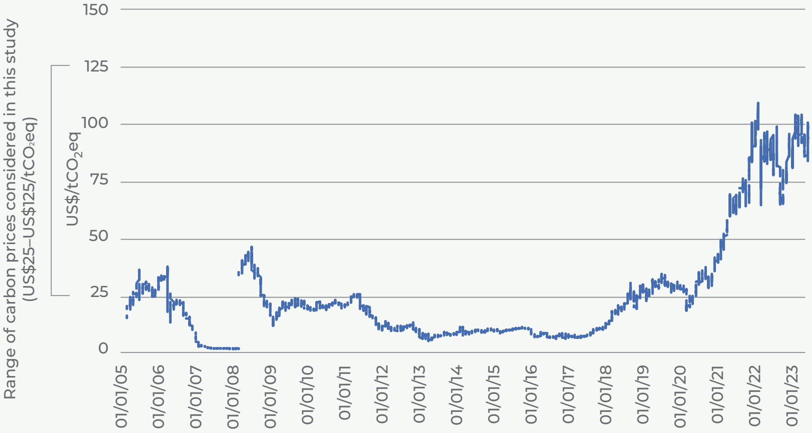
- Distances: sea distance between ports (in km) for all countries according to the CERDI-seadistance database. An average distance is assumed for trade with the rest of the world.
- Transportation emissions intensities: Calculated in gCO_2/ton of product/km. Source: European Chemical Transport Association.
- Tariffs: UNCTAD TRAINS.

2.1.3.

CARBON PRICE

To estimate the impact of border carbon tariffs, a range of carbon prices needs to be defined. In real markets, these price levels are determined by restrictions on emissions (“caps”) and the abatement costs of the activity in question. Both of these factors are difficult to estimate for the agricultural sector. Stepanyan et al. (2023) propose using €100/tCO_2eq to simulate the effects of nETS (ETS for sectors not covered by the EU ETS), which is the maximum EU ETS price reached in 2022 (figure 2), based on the possibility that ETS and nETS markets will converge to these values soon. Other authors (Isermeyer et al., 2019) use wider price ranges of between €25 and €180/tCO_2eq. Based on these studies, this chapter uses a carbon price range of US$25–US$125/tCO_2eq. Trade flow results are reported for a median value in the range of US$75/$tCO_2eq$. At this carbon price, the ad valorem carbon equivalent tariff for Argentinian exports to the EU is 64% in the baseline scenario.

Figure 2.
Evolution of Prices in the EU ETS Carbon Market (US$)



Source: Authors using data from the International Carbon Action Partnership (ICAP).

2.2.

Construction of GHG mitigation scenarios for cattle farming

2.2.1.

SURVEY OF EXPERTS ON FEASIBLE TRANSFORMATIONS TO ARGENTINIAN CATTLE FARMING SYSTEMS

Between June and July 2023, we surveyed experts on the feasibility of improving the productivity of Argentina's rearing and wintering systems over a 10-year period, assuming a context of sustained demand for beef in international markets but with stricter requirements around reducing the intensity of GHG emissions. Based on this information, we developed transformation scenarios for beef production to assess their impact on Argentinian exports and international trade flows in the event of an EU border carbon adjustment mechanism being applied.

We interviewed five professionals with extensive experience in the sector regarding such transformations at the national level. The information from these interviews was complemented by an online survey of 27 experts/consultants from different regions in the country with ties to the National Institute of Agricultural Technology (INTA), Regional Consortia for Agricultural Experimentation (CREA), and the private sector. There were between three and five respondents for each of the country's cattle-farming regions (the questionnaires used in the interviews and online surveys are included in annexes B and C, respectively).

These informants answered questions on the maximum potential levels of technical change in the proposed context for five key indicators that determine the efficiency of beef production and, therefore, emissions intensity: 1) weaning rate; 2) average weaning weight; 3) average slaughter weight; 4) length of the fattening cycle; and 5) age at first calving.[7] Table 2 shows the baseline values, which were calculated based on the modal systems of the national GHG inventory and the range of improvement for the rearing and wintering indicators that emerged as maximum transformation values over a 10-year horizon in the experts' survey responses. The nature of the information gathered was such that a statistical analysis was not possible.

Generally speaking, the experts interviewed pointed to weaning rate and average slaughter weight as having the greatest impact on production efficiency and thus on the intensity of GHG emissions per kg of meat produced. The first variable is associated with the reduction of unproductive categories and the second with the increase in the volume of product per animal slaughtered.

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7 · The five national-level expert respondents gave direct answers on the maximum achievable values for these five indicators. The specialists from each region of the country were asked about changes in the proportion of breeding and wintering systems. We then calculated changes in the indicators based on these. The main systems in place in each production region are used to represent the diversity of cattle-farming systems in the calculation of the national GHG inventory. Cattle-farming systems are classified into low, medium, and high productivity levels for each region. A total of 47 wintering systems are also described, which combine characteristics of the rearing and finishing phases and are differentiated through feed, confinement period, the duration of fattening, and slaughter weight.

Table 2.
Prospects for Transformation of the Rearing and Wintering Phases of Cattle Farming in Argentina by 2033

VARIABLE	UNIT OF MEASUREMENT	BASELINE VALUES	RANGE OF IMPROVEMENT	
			from	to
Age at first calving	months	27	25	20
Weaning rate	%	65	67	75
Average weaning weight	kg	170	170	200
Duration of fattening cycle	months	14	15	12
Average slaughter weight	kg live weight	395	430	460

Note: The data in the table was obtained from a survey of 32 livestock specialists on possible improvements in the productivity levels of rearing and wintering systems in 10 years' time, assuming a favorable export scenario and sustained demand for beef on international markets but with stricter requirements in terms of GHG emissions intensity per kg of product.

Source: Authors.

We used an open-ended questionnaire to gather reflections and observations to support the views on the proposed transformations. The interviewees were also asked about market conditions and the political/institutional factors needed to implement these changes. Finally, we asked the respondents to identify other strategies for reducing emissions from beef production that are not directly related to improved production efficiency. The qualitative information they provided informed our discussion of the transformation initiatives needed to reduce emissions. Annex D provides a detailed description of the changes proposed by the experts consulted for the different cattle-farming regions.

One interesting question is how the improvements in cattle-farming systems identified as feasible by the experts could be achieved and what role government and markets might play in these transformations. The experts and consultants interviewed provided information in this regard. For example, **they identified rearing systems as the most inefficient aspect of the cycle, making it the area with the greatest potential for improvement.** Likewise, the main factors constraining this transformation are shortfalls in the quality and quantity of feed, overgrazing, and management problems.

The experts agree that changing the weaning rate would be difficult but not impossible, "despite the weight of tradition." They also noted that "breeders think about things differently, and outsiders have not been able to call that logic into question." Specifically, rearing activities are managed with less of a focus on output than wintering or agricultural activities, even though they take place on the same farms. A**ccording to the interviewees, changes could be achieved by providing better information about existing technologies. They also emphasized the role of the state in implementing initiatives to strengthen producer groups** (so-called rural transformation initiatives, cooperative groups, etc.) and consultants, who play a key role in facilitating transformations.

As far as wintering is concerned, the experts point out that t**he main short-term limitation is domestic consumers' preference for meat from lighter animals, which remains the main destination of Argentinian beef productio**n despite the continuing decline in the country's per capita consumption. The introduction of feedlot systems has also led to a decrease in slaughter weight compared to grazing systems. This relates to fast capital turnover in a market that penalizes higher slaughter weights.

There are more concrete possibilities for change in response to market incentives in wintering systems. Although higher prices for lower-weight animals on the domestic market are currently the main determinant of low slaughter weights, there are opportunities for a rapid response if these incentives change. Alternative incentives come from export markets and would be accentuated if emissions reduction requirements became stricter. Policies are thus needed to support the sector's export profile in response to this new scenario and provide long-term predictability.

The experts we interviewed acknowledged that the rearing process played a fundamental role in increasing slaughter weight. They emphasized that improving these systems would be one way to achieve the desired outcome and envisioned two strategies for achieving this objective: increasing individual daily weight gain through initiatives to improve the production and quality of feed during the rearing stage, as well as grazing management; or extending the length of the rearing stage while maintaining current daily weight gain.

They also mentioned the need for the state to facilitate access to financing for investment in productive structures. Regarding the new rules of the game, the respondents argued that it is the public sector's responsibility to inform producers about these changes, which are now unavoidable, given current demands around national and global climate action. **The interviewees viewed international environmental requirements as a positive force for change that will prompt a leap in competitiveness and ensure the survival of cattle-farming systems that have historically maintained low productivity levels.**

Finally, they emphasized that both the state and the private sector are responsible for complying with Argentina’s Forestry Act to ensure that deforestation regulations do not jeopardize the beef trade. **The current increase in environmental trade restrictions points to a greater need for traceability mechanisms.** In this sense, Argentina’s National Service for Agrifood Health and Quality (SENASA) has a central role to play in the digitization of livestock farming.

2.2.2.
TRANSFORMATION SCENARIOS FOR CATTLE FARMING THROUGH 2033

Based on the results of our interviews, we developed a series of transformation scenarios for cattle farming through 2033, which are presented in table 3. Each scenario is defined by values for the weaning rate, age at first service, and average weaning and slaughter weights. For the productivity improvement scenario, the average values of the performance variables reported by the experts were used. Since the weaning percentage indicator is of particular interest, two values are included in the scenarios: the average value and the maximum value of the range reported by the interviewees.

Table 3 provides information on diet digestibility in the different scenarios. The specialists were not asked directly about possible changes in this variable, so the values used are derived from the research team’s knowledge, with the aim of identifying the magnitudes of changes in digestibility (due to increased use of pasture, greenfeed, and supplements) that are indirectly associated with daily weight gain, increases in weaning percentage, age at first calving, and increases in weaning and slaughter weights.

Table 3.
Description of the Baseline and Transformation Scenarios for the Argentinian Beef Industry Through 2033

ID	STAGE AT WHICH CHANGES ARE MODELED	WEANING	AGE AT FIRST SERVICE	SLAUGHTER WEIGHT	DIGESTIBILITY	CHANGE IN EMISSIONS INTENSITY*	CHANGE IN TOTAL EMISSIONS*	CHANGE IN MEAT PRODUCTION*
						%	%	%
0	BASELINE	65%	27 months	livestock system inventory	livestock system inventory	--	--	--
1	WINTERING	65%	27 months	heifers+5%, steers+10%	steers and heifers +5%	-5.4	0.41	6.19
2	REARING	75%	24 months	livestock system inventory	rearing cows +10%	-15.4	-3.22	14.35
3	REARING	72%	24 months	20 kg increase in weaning weight, with no improvement in weight gains in the fattening phase	breeding cows +10%	-14.3	-4.77	11.09
4	REARING + WINTERING	75%	24 months	heifers+5%, steers +10%	breeding cows +10%, steers and heifers +5%	-20.3	-2.77	21.98
5	REARING + WINTERING	75%	24 months	Heifers +5%, steers +10%, 20kg+ increase in weaning weight, improvements in fattening efficiency: higher daily gain and shorter duration	breeding cows +10%, steers and heifers +5%	-21	-2.32	23.61

Source: Authors.
Note: *changes in the respective indicators (GHG emissions intensity per kg of meat, total emissions, and total meat production) relative to baseline values (calculated from information on rearing and wintering systems from the 2018 national GHG inventory).

SCENARIO 1—WINTERING

For this scenario, the slaughter weight of the wintering categories is increased by 5% in females and 10% in males. This is achieved through higher daily weight gain due to better feeding (greater use of pastures and greenfeed and greater use of strategic supplements). The length of the fattening stage is not modified. This scenario contemplates a 5% improvement in the digestibility of wintering cattle's diets.

SCENARIO 2—REARING

The combined effects of two improvements to physical parameters during the rearing stage were modeled: increasing the weaning rate from the baseline of 65% to 75%, and reducing the age at first service in heifers from 27 to 24 months. This scenario contemplates a 10% improvement in the digestibility of cattle's diets.[8]

SCENARIO 3—REARING

This scenario entails a slight increase in the weaning rate (+10%) and the weaning weight (+20 kg) and uses the same adjustments to age at first service and diet digestibility as scenario 1.

SCENARIO 4—REARING + WINTERING

Combines scenarios 1 and 2: improvements to rearing stage parameters (weaning rate, age at first service, and diet digestibility) and the increase in finishing weights through a higher daily weight gain that derives from better feeding (greater use of pastures, greenfeed, and strategic supplementation).

SCENARIO 5—REARING + WINTERING

This is a combination of scenario 4—which includes an increase in the weaning rate, the decrease in age at first service, and the increase in finishing weights (due to higher daily weight gain during wintering)—and a 20 kg increase in weaning weight.

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[8] In scenario 2—breeding, approximately 70% of the improvement in emissions intensity is due to increasing the weaning rate, while lowering the age at first service accounts for the remaining 30%.

2.3.

Calculation of GHG emissions intensity reduction for mitigation scenarios

To calculate the emissions intensity and emissions reduction potential of the mitigation scenarios, we used the methodological definitions included in the *Inventario Nacional de Gases de Efecto Invernadero de Argentina* [National Greenhouse Gas Inventory for Argentina] (Ministerio de Ambiente y Desarrollo Sostenible, 2021). We made calculations for a set of rearing and wintering systems for each cattle-farming region and then weighted these according to each system's share in total national beef production. We used the tier 2 calculation method from the 2006 Intergovernmental Panel on Climate Change (IPCC) guidelines for national GHG inventories (IPCC, 2006). To estimate beef production, we based our calculations on a stable herd size and the number of animals at slaughter associated with a given weaning rate and age at first service, while keeping the number of breeding cows constant. We calculated total bone-in beef production based on the live slaughter weights defined in the models and the corresponding yields. The total emissions and meat production values were used to calculate the emissions intensity of each mitigation scenario.

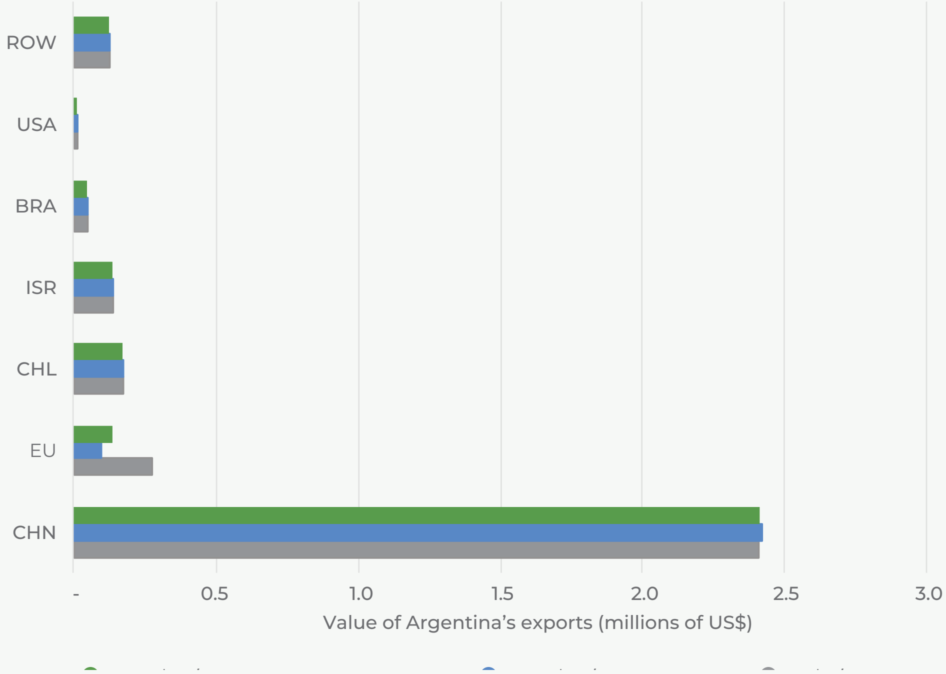
3. RESULTS

The results of running the CLIMTRADE model show relative changes in the flow of international trade, with a focus on beef exports from Argentina. The mitigation scenarios are compared with the baseline scenario for the range of carbon prices contemplated in the analysis.

Figure 3 shows how implementing a border carbon adjustment mechanism would impact Argentinian beef exports. We present beef export values by destination in three sets of circumstances: 1) no border adjustment mechanism; 2) an EU border adjustment mechanism of US\$75/tCO_2eq for the emissions intensity values of the baseline scenario, and 3) a border adjustment mechanism of US\$75/tCO_2eq and the maximum mitigation scenario (scenario 5—rearing + wintering).

The circumstances represented in figures 3 to 5 correspond to the maximum possible mitigation scenario modeled in this chapter, in which emissions intensity is reduced by 21% compared to the baseline (see table 3). The results for the remaining mitigation scenarios follow similar patterns, but their magnitude lies between the baseline and maximum mitigation scenarios (the comparison across scenarios is shown in table 4 and figure 7).

Figure 3.
Impact of the Border Carbon Adjustment Mechanism on the Value of Argentinian Beef Exports by Destination

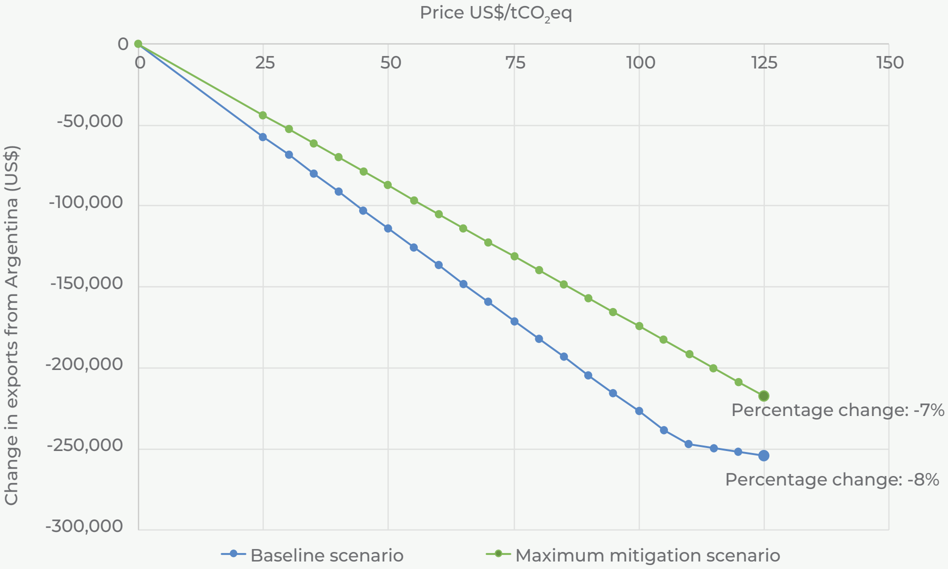


Note: **P=0**: No border carbon adjustment mechanism. **P=US$75/tCO$_2$eq baseline**: EU border adjustment priced at US$75 combined with the emissions intensity values of the baseline scenario. **P=US$75/tCO$_2$eq maximum mitigation scenario**: EU border adjustment priced at US$75 combined with emissions intensity values 21% below the baseline scenario. This is the scenario with maximum improvements to productivity (scenario 5).
Source: Authors.

The implementation of the EU border adjustment mechanism leads to a reduction in Argentinian beef exports to the bloc. In the baseline scenario for Argentinian cattle-farming (no mitigation measures), the value of exports to the EU declines by approximately 65% in response to this measure. In scenario 5 (mitigation), they decline by 50%. This drop in exports to the EU is only partially offset by exports to other destinations.

Figure 4.

Changes in the Value of Argentinian Beef Exports for Different Border Carbon Price Levels: Baseline and Maximum Mitigation Scenario (5)

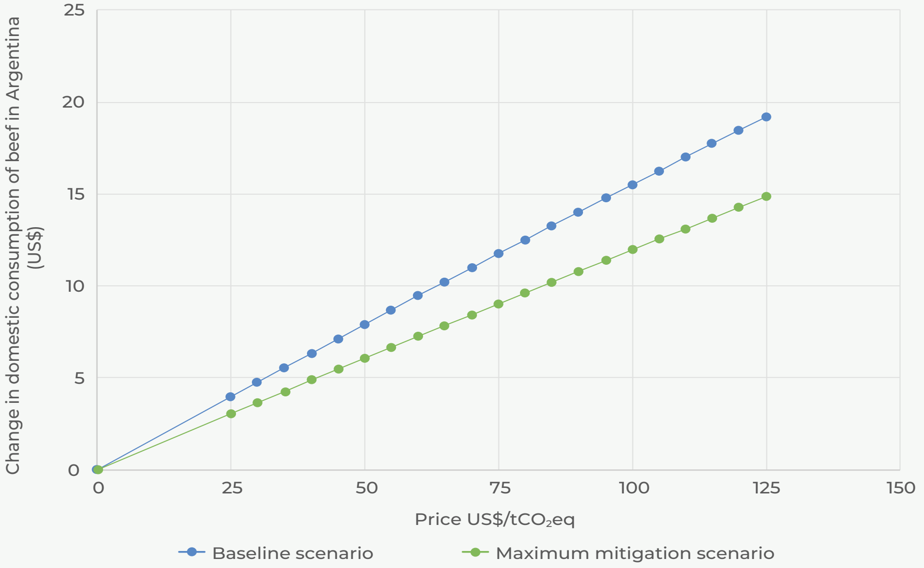


Source: Authors.

Figure 4 shows how exports decline more when the border carbon tax is applied (within the range of prices analyzed), while the emissions mitigation scenario attenuates this effect. This is because the EU border carbon tax triggers an increase in ad valorem equivalent tariffs for the entry of certain products (in this case, beef) into the EU market because beef produced in Argentina is more emissions-intense than beef produced in EU member countries. This leads to a reconfiguration of international equilibrium prices and thus to a redistribution of trade. In the baseline scenario (i.e., in the absence of mitigation strategies and using the original emissions intensities), the new EU carbon tariffs discourage exports to the bloc, discriminating against exporting countries with higher emissions intensities. The higher the carbon tax, the greater this effect. Some of this trade is diverted to the domestic market and other destinations, but the net effect is negative, as discussed below. Conversely, in the maximum mitigation scenario, the emissions intensity of Argentinian beef is lower than the baseline. As a consequence, the EU carbon tariff increases less, such that the effect described above is mitigated. However, despite attenuating the fall in exports, this reduction far from compensates for the loss of trade resulting from the carbon tax.

Figure 5.

Changes in the Value of Argentinian Beef Production Destined for the Domestic Market for Different Levels of Border Carbon Prices: Baseline and Maximum Mitigation Scenario



Source: Authors.

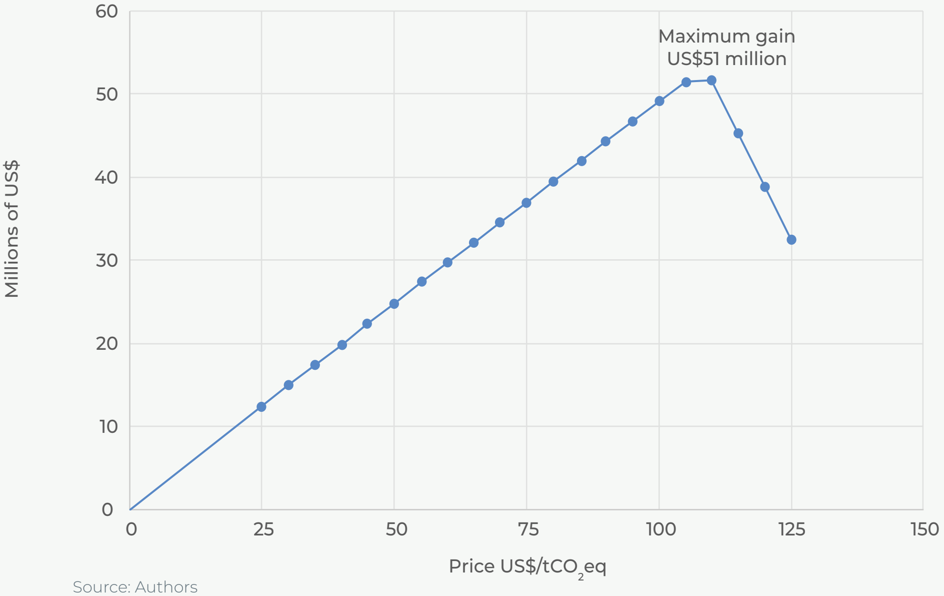
Similarly, as mentioned above, emissions mitigation strategies impact the volume of production destined for domestic consumption, which increases with the carbon price that is implemented. The changes in domestic consumption are much smaller than the changes in exports shown in figure 4. In this case, for the range of prices studied, reducing emissions keeps domestic market growth below that of the baseline scenario since the reduction in exports that needs to be redirected is smaller (figure 5).

Generally speaking, the implementation of the EU border carbon tax causes a reduction in exports to EU countries. These are redirected to the rest of the world, including the exporting countries' own domestic markets. At the same time, EU countries increase their own production and intrabloc trade. For example, for a tax of US$75/$tCO_2$eq in the baseline scenario, where no mitigation strategies are considered, the following changes to trade patterns are observed: a 4% increase in domestic consumption in EU countries; a 7% increase in intra-EU trade; and a 2% decrease in EU exports to non-EU countries and a 79% decrease in imports from non-EU countries.

The final effect of mitigation on the beef sector in Argentina entails two changes: the first is caused by changes in foreign trade, and the second by changes in domestic consumption. The net effect of each mitigation strategy on the beef trade can be expressed as the sum of these two changes, which we call the mitigation gain. The mitigation gain is an additional outcome when comparing different strategies, and also a parameter to consider when evaluating the costs of these. Figure 6 shows the mitigation gain over total sector trade relative to the value of trade for the baseline scenario for each carbon price.

Figure 6.

Mitigation Gain of the Maximum Mitigation Scenario Compared to the Baseline, for Different Border Carbon Prices



Source: Authors

As can be seen in figure 6, the mitigation gain initially increases with the carbon price, but this strategy attenuates the drop in exports more than it restricts the redirection of production to the domestic market. For the last tranche of the range considered (prices above US$110/t$CO_2$eq), the margin of decline in exports is exhausted as Argentina's trade with the EU disappears altogether, such that mitigation has a diminishing effect. At prices above US$110/t$CO_2$eq, the model assigns a value of zero for exports to the EU, even for the maximum mitigation scenario. The maximum net benefit occurs at a carbon price of around US$110/t$CO_2$eq, which is higher than a likely hypothetical value of applying a carbon tariff in the short term under current and recent conditions, according to the literature we reviewed.

As noted above, the results obtained for the different mitigation scenarios are similar in terms of the directions of changes in trade flows, although there is some variation in the magnitude of the effects on the outcome variables. To measure the differences between the scenarios and the particular contributions of the practices modeled, a quantitative comparison is shown below (table 4 and figure 7).

Table 4.

Mitigation Gains for Improvement Scenarios for Cattle-Farming Systems Based on Different Border Carbon Prices

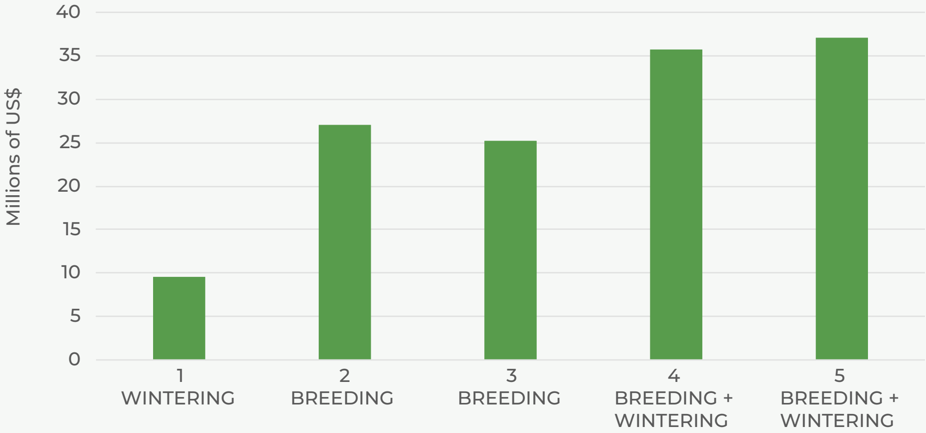
	VARIATION IN EMISSIONS INTENSITY	MITIGATION GAIN (MILLIONS OF US$)		
SCENARIO		P=50	P=75	P=100
1 WINTERING	-5.42	6.40	9.55	12.66
2 REARING	-15.34	18.12	27.04	35.87
3 REARING	-14.28	16.87	25.18	33.40
4 REARING + WINTERING	-20.27	23.96	35.76	47.45
5 REARING + WINTERING	-20.98	24.80	37.01	49.10

Source: Authors.

Table 4 shows all the scenarios we considered, the resulting relative change in emissions intensity, and, as a synthetic outcome measure, the mitigation gain for three benchmark carbon prices: US$50, US$75, and US$100/t$CO_2$eq. Figure 7 plots the mitigation gain for the US$75/t$CO_2$eq pricing scenarios.

Figure 7.

Mitigation Gains for the Different Improvement Scenarios for Cattle-Farming Systems at a Border Carbon Price of US$75/$tCO_2$eq



Note: The mitigation gain is the reduction in the economic loss caused by a border carbon adjustment mechanism in the EU under the mitigation scenario as compared to the baseline scenario.
Source: Authors

Finally, we analyzed how the application of the border carbon adjustment mechanism impacts total GHG emissions from beef production. Based on the results of our simulations for different border carbon prices for both the baseline and maximum mitigation scenarios for Argentina, we calculated total CO_2eq emissions for world production (applying each country's emissions intensity to its total production), for Argentinian production (including exports and the domestic market), and, in particular, for consumption in the EU countries included in the model (calculated by applying each country's emissions intensity to the amount imported, including each country's domestic market).

A border carbon adjustment mechanism can indeed trigger a decrease in total GHG emissions that increases with the carbon price used. The resulting values are quantitatively significant for the emissions associated with EU consumption: for prices between US$25/$tCO_2$eq and US$100/tCO_2eq, emissions are reduced by between 12% and 18%. However, the values are lower for Argentinian and global production—below 1% for the range of carbon prices considered.

4. DISCUSSION AND CONCLUSIONS

Led by the EU with its Green Deal, the countries of the Global North are implementing a range of policies to reduce their GHG emissions. Currently, their main concerns include decarbonizing international trade and avoiding the carbon leakage that may result from increased imports of products from countries with more lax environmental policies. This chapter examined the possible impact of a border carbon adjustment mechanism applied by the EU to international trade in beef, with a particular focus on Argentinian beef exports. In it, we designed different emissions mitigation scenarios for the Argentinian beef sector associated with improvements in production indicators for the rearing and wintering stages. We then evaluated whether these improvements could reduce trade losses if the border carbon mechanism was implemented.

The results show that the implementation of a border carbon adjustment mechanism is expected to have significant impacts on trade with the EU. However, the overall effects on Argentina's trade in beef are small. Much larger impacts could be expected if major buyers in the world beef market also implemented measures of this sort, although calculating such effects was beyond the scope of this study.

The results show that an EU border carbon adjustment mechanism could reduce global trade in Argentinian beef, most notably by generating a sharp decrease in trade with the EU market. Likewise, we observed that **mitigation strategies based on improvements in production efficiency over a 10-year horizon would reduce emissions intensity levels by up to 21% relative to current values, which is associated with a significant reduction in the loss of export value** in response to the implementation of a carbon price similar to those currently applied to other economic activities.

Improving production efficiency indicators is essential to increasing the competitiveness of Argentinian beef in markets that price carbon emissions based on an emissions intensity metric, such as the one discussed in this chapter. However, it should be noted that the nature of the indicator used to estimate emissions per unit of product puts Argentina at a disadvantage compared to EU countries, which produce animals with higher slaughter weights, even if they use more external inputs, consume more natural resources and energy, and do not make a significant contribution to carbon sequestration. **This indicator does not fully consider other benefits associated with cattle farming's potential for sequestering carbon through pasture and rangeland management, the inclusion of forestry in livestock systems, or forest management with integrated livestock production.**

To date, agriculture has been excluded from ETSs due to the potential loss of domestic competitiveness and the risk of carbon leakage. However, the EU is engaged in a public discussion on the inclusion of agriculture in its ETS, while New Zealand will allegedly be including the sector in its ETS starting with farm-level emissions reporting in 2024 and pricing in 2025 (Arvanitopoulos et al., 2021). The potential inclusion of agriculture in the EU ETS would likely be accompanied by a border carbon adjustment mechanism in the medium term. In this respect, the impact assessment documents for the EU border carbon adjustment mechanism initially only cover direct emissions (those resulting from the production process and over which the producer has direct control) and indirect emissions (those resulting from the generation of the electricity consumed during some processes). The assessment process for steel products, electricity, cement, and fertilizers began in late 2023. In the longer term, as information on the carbon content of products becomes more readily available and carbon pricing policies become more comparable across countries, emissions assessments could be expanded and may move toward being complete carbon footprints (European Commission, 2021). However, implementing a measure of this sort may entail a significant administrative burden in terms of measurement, reporting, and verification, which may ultimately lead to bottlenecks, such that **the use of default emissions intensities cannot be ruled out. This would allow exporters of products to the EU to demonstrate that the actual emissions levels of their products are below the default values and can therefore be subject to a lower border adjustment rate.**

According to Isermeyer et al. (2019), **the correct way to assess livestock systems from a theoretical point of view is to estimate the GHG balance at the farm level, taking carbon sequestration into account.** However, in line with what is proposed by the European Commission (2021), these authors warn of the difficulties in determining the necessary information requirements, given the heterogeneity of the variables that characterize each farm and affect GHG emissions and sequestration. One alternative to these challenges is to work on developing ratings and fixed rates. However, the less precise these ratings are, the less they will do justice to how each establishment operates (Isermeyer et al., 2019).

In their analysis of the potential impact of new climate regulations on trade in Latin American countries (border carbon adjustment mechanisms and deforestation-free production requirements), Conte Grand, Schulz-Antipa, and Rozenberg (2023) point out that if WTO disputes are ruled out, there are two ways of adapting to these measures. These entail making headway in two areas. First, assessing the price of carbon that could be paid at the domestic level (either explicitly or through regulations or carbon taxes). And second, estimating the carbon content of products.

When analyzing border carbon taxes in countries without effective carbon pricing systems, one fundamental question relates to the effectiveness of these measures and the uncertainty surrounding their emissions calculations (IPCVA, 2021). Metrics play a critical role in determining product-related emissions. However, various methodological approaches with different objectives are currently in use. These include using GHG estimates at the production chain level using life cycle analysis methodology or conducting a sector-specific assessment following IPCC guidelines (Faverin and Tieri, 2023). These methodologies may produce different values for GHG emission and absorption indicators, depending on their approaches. This leads to uncertainty around which indicator(s) to choose, especially regarding which are most appropriate for assessing emissions and absorption in different contexts and countries. It is therefore essential to address this uncertainty by analyzing and selecting indicators carefully, supported by a strong scientific evidence base. In this regard, **it is encouraging to note that there are national-level efforts underway in Argentina to collect data on activity and emissions factors.**

As a consequence, public and private institutions representing the cattle-farming sector in Argentina need to continue analyzing grazing and silvopasture systems in the country's different regions. **Such efforts could help establish an "Argentina brand" for grass-fed beef with lower net emissions, no deforestation, conservation of biodiversity, low use of pesticides, and attention to animal welfare.**

In conclusion, the results of this chapter contribute to the discussion on how the agricultural emissions reduction policies promoted by developed countries may impact Argentinian beef production and their implications for international trade. However, there is still uncertainty around the implementation of these policies. All the same, this new outlook requires rapid, concrete action by alliances between the public and private sectors to promote climate-smart beef production. Such joint action should take into account heterogeneity within and between regions and be based on the adoption of technologies and management practices that increase production efficiency, reduce emissions, and promote carbon sequestration, supported by the development of scientific knowledge and assessment and traceability processes that allow environmental performance to be certified.

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ANNEX A.

LIMITATIONS OF THE MODEL

- Domestic trade flows are represented as commodity volumes and converted to values using world prices. Since the data is obtained from different sources and combined, potential inconsistencies between the estimates reached may limit accuracy.

- Standard elasticities are used. The assumption of uniform elasticities does not reflect differences in specific supply and demand elasticities between economies of significantly different sizes.

- Nontariff barriers, export taxes, and subsidies are held constant. Nontariff barriers such as certification requirements are not included. These are likely to become increasingly important as countries impose stricter domestic environmental requirements in addition to carbon pricing schemes.

- Economic impacts are modeled at the commodity level, and the results for all other commodities are held constant. A carbon tariff is likely to change production patterns in exporting countries. It is also possible that importers will substitute between products in response to a carbon tariff. These feedback mechanisms or interactions are not taken into account when aggregating impacts at the product level.

- The CLIMTRADE model assumes that a border adjustment mechanism is implemented in the adopting countries, while the remaining countries are deemed to be affected by this. The adopting countries (in this case, the EU countries) do not implement a carbon tax domestically or within the bloc. In other words, the measure only increases the price of imports from outside the region (i.e., imports from the affected countries).

ANNEX B.

SUGGESTED QUESTIONS FOR INTERVIEWS WITH EXPERTS ON ARGENTINIAN CATTLE FARMING

INTRODUCTION

We greatly appreciate your taking part in this interview/consultation. Your opinion is very important to the research we are currently undertaking. The idea is to have a 30–60-minute conversation regarding possible changes to cattle farming in Argentina. We would like to start by giving you some background on our work.

As part of a call for proposals from the IDB on the new green trade agenda in Latin America and the Caribbean,[1] a team of technicians from different areas of INTA and the Secretariat of Agriculture, Livestock, and Fisheries presented a project entitled "The International Beef Trade: The Impacts of Proposals to Transform Beef Production in Argentina in Response to Different Border Carbon Tax Scenarios." Ours was one of the six proposals selected from the 235 put forward by research teams throughout the region.

The design for our project takes into account the negotiations taking place at the United Nations Framework Convention on Climate Change and the growing trend among countries of the Global North to mainstream a climate perspective in public policies, including trade policies, given the evidence of how trade impacts the climate, ecosystems, and human well-being.

In this context, border carbon adjustment mechanisms have been promoted as a compelling solution to the increase in greenhouse gas (GHG) emissions from consumption. Indeed, a pilot of this mechanism was launched in the EU in 2023 for steel and chemical products. The second phase of the mechanism is expected to be extended to agrifood products.

In this context, there is a need for more research into the relationship between strengthening climate action and sustainable development, promoting the transition to low-carbon economies that are sustainable, competitive, and climate-resilient.

Our project seeks to quantify the impact of the main proposals for the transition to climate-smart beef production on the value of Argentinian beef exports and international trade flows in response to different EU border carbon tax scenarios. To this end, we would like your opinion on the feasibility of improving the productivity of cattle rearing and wintering systems in Argentina in the long term (10 years from now). This will allow us to design and assess alternative approaches to transforming the Argentinian cattle-farming sector, assuming that there will be sustained international demand for beef but with stricter requirements in terms of the GHG emissions intensity per kg of product.

Do you have any questions about the project or the purpose of this conversation?
We would like to ask for your permission to record this meeting to help us with the

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[1] · "Going Green: The New Trade Agenda in Latin America and the Caribbean" was a call for proposals organized by the IDB's Institute for the Integration of Latin America and the Caribbean (INTAL) and IIEP UBA, CONICET. The initiative aims to select research projects that analyze different aspects of international trade and environmental issues in Latin America and the Caribbean and put forward specific policy proposals.

note-taking process and ensure we record your comments accurately. Interviewees will be anonymized when we publish the results of this study. Do you have any objections, or can we start recording?

Questionnaire

1. Generally speaking, what possible transformations do you foresee for the Argentinian beef industry under the circumstances described above, focusing on the international beef market and assuming that domestic beef consumption remains stable. In other words, if there is sustained international demand for beef, but requirements become stricter in terms of GHG emissions intensity per kg of product, and soil carbon is not included in the emissions intensity calculation.

2. Which variables do you think may entail significant room for improvement during the rearing and wintering stages, without taking into account policy factors that may or may not be favorable to such changes? Thinking in terms of maximum values, how far do you think these improvements could affect values at the country level? Please complete the following table considering current values for the variables in question and 10 years from now.

	COUNTRY-LEVEL VALUES	
VARIABLE	NOW	IN 10 YEARS
Weaning rate (%)		
Average weaning weight (kg)		
Average slaughter weight (kg)		
Length of fattening cycle (months)		
Stocking rate (head/ha)		
Age at first calving (months)		
Any indicators you would like to add		

3. Do you see differences in the potential for transformation of these variables at the regional level (with an emphasis on the Pampas region, northeastern Argentina, and northwestern Argentina)?

VARIABLE	MAJOR AREAS											
	SOUTHEASTERN AND SOUTHWESTERN PAMPAS		NORTHERN AND WESTERN PAMPAS		NORTHEASTERN ARGENTINA		NORTHWESTERN ARGENTINA		SEMIARID ZONE		PATAGONIA	
	NOW	IN 10 YEARS	NOW	IN 10 YEARS	NOW	IN 10 YEARS	NOW	IN 10 YEARS	NOW	IN 10 YEARS	NOW	IN 10 YEARS
Weaning rate (%)												
Average weaning weight (kg)												
Average slaughter weight (kg)												
Length of fattening cycle (months)												
Stocking rate (head/ ha)												
Age at first calving (months)												
Any indicators you would like to add												

4. What conditions do you think would be needed for these transformations to be viable (national and international policies, climate factors, etc.)?

....

....

ANNEX C.

ONLINE QUESTIONNAIRE FOR SURVEYING SPECIALISTS FROM SPECIFIC REGIONS (FOR EXAMPLE, FOR THE NORTHERN PAMPAS REGION)

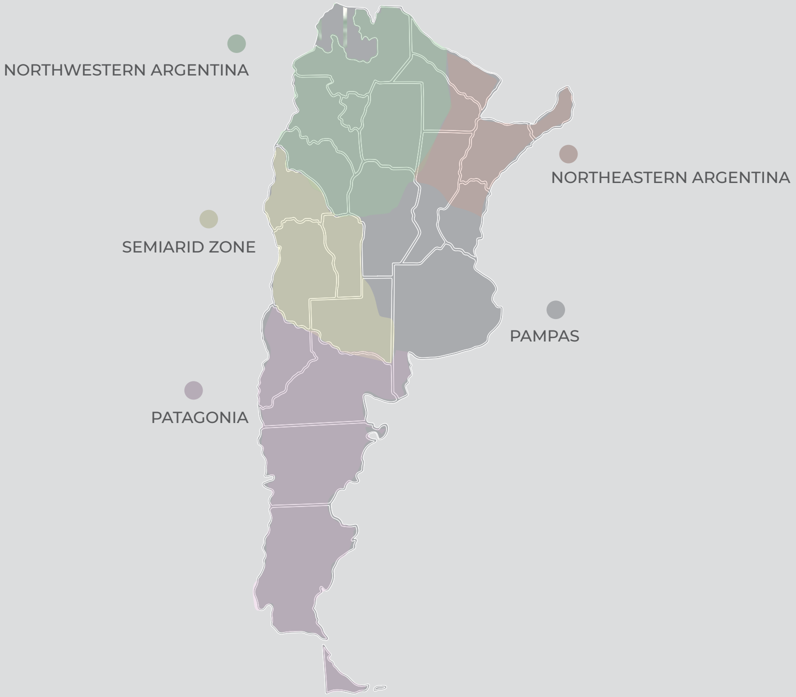
The objective of this survey is to ascertain your opinion on the feasibility of improvements to the productivity levels and economic outcomes of cattle rearing and wintering systems in Argentina in the medium to long term, assuming that international demand for beef remains stable but requirements become stricter regarding GHG emissions intensity per kg of product.

The beginning of each section describes the current rearing and wintering systems in the northern Pampas region, which is followed by questions on the likelihood of change over the next 10 years.

The purpose of this survey is to identify feasible alternative approaches that would reduce the emissions intensity of beef production and thus mitigate their potential impact on Argentina's trade in beef in response to different border carbon tax scenarios.

Thank you for taking part in this survey. Please feel free to contact us with any questions.

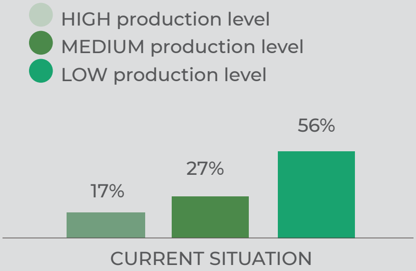
MAP OF CATTLE-FARMING REGIONS



CATTLE REARING IN THE NORTHERN PAMPAS

The rearing systems currently in use in the region are described below. The proportion of breeding cows in each system and the average regional values for the indicators are also shown.

REARING SYSTEMS: NORTHERN PAMPAS			
Productivity level	High	Medium	Low
Rate (cows/ha)	1.00	0.60	0.70
Meat production (kg/ha)	158	81	67
Weaning (%)	74%	62%	38%
Age at first service (months)	15	22	27
Mortality %	2%	4%	5%
Pasture and greenfeed %	10%	5%	0%
Percentage of breeding cows in each production system			
CURRENT SITUATION	17%	27%	56%



REGIONAL AVERAGES	CURRENT SITUATION
Stocking rate (breeding cows/ha)	0.70
≠	82
Weaning (%)	50%
Age at first service: 15 months	17%
Age at first service: 22 months	27%
Age at first service: 27 months	56%
Pasture and greenfeed (ha)	3%

Do you think that these models accurately represent WINTERING systems in the NORTHERN PAMPAS region?

.... Yes (1)
.... No (2)

MITIGATION PROPOSAL

Given that higher technology levels imply lower GHG emissions intensity, please indicate what percentage of breeding cows you expect to be part of low-, medium-, and high-technology systems in 10 years' time (please refer to the table presented in question 1 and your suggestions for change in the descriptions of each technology level).

LOW () ○ ○ ○ ○ ● ○ ○ ○ ○ ○
MEDIUM () ○ ○ ○ ○ ● ○ ○ ○ ○ ○
HIGH () ○ ○ ○ ○ ● ○ ○ ○ ○ ○

Please briefly explain what changes you would introduce to rearing systems to mitigate GHG emissions.

....

....

WINTERING STEERS IN THE NORTHERN PAMPAS

The wintering systems currently in use in the region are described below, based on our interviews with experts on cattle farming. The percentages of cattle in each system are also shown.

WINTERING SYSTEMS FOR STEERS IN THE NORTHERN PAMPAS						
	Island + pen	Grazing + short wintering pen	Grazing + long wintering pen	Short winter grazing	Long winter grazing	Pen
Finishing (days)	270	270	480	330	690	210
Weight gain (kg/day)	0.59	0.89	0.48	0.76	0.45	0.90
Initial weight	170	160	170	170	150	160
Final weight	330	400	400	420	460	350
Feed						
Rangeland	70%	0%	70%	0%	0%	0%
Pasture/greenfeed	0%	50%	0%	90%	85%	0%
Grain	24%	35%	20%	10%	15%	80%
Other (silage, by-products, hay)	6%	15%	10%	0%	0%	20%
Percentage of heads in each type of production system						
CURRENT SITUATION	10%	5%	5%	35%	20%	25%

Do you think that these models accurately represent WINTERING systems in the NORTHERN PAMPAS region?

.... Yes (1)
.... No (2)

If your answer was NO, what parameters would you suggest modifying?

MITIGATION PROPOSAL
Please indicate the percentages of steers that could be fattened in each production system in 10 years' time to lower the carbon footprint of beef production.

Island + pen	○ ○ ○ ○ ● ○ ○ ○ ○ ○
Light grazing + pen	○ ○ ○ ○ ● ○ ○ ○ ○ ○
(short wintering)	○ ○ ○ ○ ● ○ ○ ○ ○ ○
Grazing + pen (long wintering)	○ ○ ○ ○ ● ○ ○ ○ ○ ○
Grazing (short wintering)	○ ○ ○ ○ ● ○ ○ ○ ○ ○
Grazing (long wintering)	○ ○ ○ ○ ● ○ ○ ○ ○ ○
Pen	○ ○ ○ ○ ● ○ ○ ○ ○ ○

Please briefly explain what changes you would introduce to wintering systems to mitigate GHG emissions.

If you are in favor of changing the ratio of animals in each system, please indicate what public- and private-sector initiatives you think are needed to bring about these changes.

Are there any other changes that could be made to cattle farming in the NORTHERN PAMPAS that could help mitigate GHG emissions?

.... Suggestion (1)
.... Suggestion (2)
.... Suggestion (3)
.... Suggestion (4)
.... Suggestion (5)

Please include your name below.

Thank you for taking part in this study.

PROPUESTA DE MITIGACIÓN
Por favor indique cuáles son los porcentajes de novillos que podrían engordarse en cada

ANNEX D.

IMPROVEMENTS TO CATTLE-FARMING SYSTEMS IN ARGENTINA'S DIFFERENT FARMING REGIONS. RESULTS OF THE SURVEY OF COUNTRY-AND REGIONAL-LEVEL EXPERTS (MAY–JUNE 2023).

The specialists we interviewed put forward broadly similar suggestions for improvement for the different regions of Argentina. However, there were some differences between these, especially in the proposals associated with feed management (pastures, rangeland, and the use of supplements). These are due to the specific environmental conditions in each region (soil and climate).

NORTHWESTERN ARGENTINA

The key factors in this region are increasing the productivity of rearing systems, maintaining the grazing system, and managing woodland more efficiently, including strategic supplementation by category and short cycles of feedlot finishing. Another issue that needs to be addressed is sanitary diagnostic exercises to improve reproductive and productive indices.

Training and financial assistance are essential to driving productivity increases and reductions in emissions intensity. Areas in which training could be provided include production planning, early weaning strategies, strategic fertilization to increase the forage plant density, feed management to achieve higher daily weight gains, and marketing systems. Key areas for investment include wire fences to create paddocks for rotational grazing and seasonal servicing and water management systems to improve the quality and quantity of water the cattle consume.

NORTHEASTERN ARGENTINA

Silvopasture systems or the rational use and care of fields with native vegetation are seen as the most viable alternatives for this region. Other key factors include managing the diet of replacement heifers and lactation (early weaning).

The use of pastures (including tropical pastures) and greenfeeds should be increased during the wintering and rearing stages to improve forage supply. The region has a severe deficit in the quantity and quality of the planted areas of perennial tropical forage crops. Strategic supplementation, particularly protein supplements, could help mitigate this issue and shorten the wintering time for steers.

NORTHERN PAMPAS

The key factor for transformation in this production zone is improving process technologies such as reducing the age at first service and the length of service; weaning according to body size and availability of feed; supplementation with concentrates; planting and management of pastures according to soil type and environment; and preparation of forage reserves. The aim is to achieve higher meat productivity through higher weaning rates and weights and higher stocking levels per hectare. Promoting outreach and training programs that encourage these changes is essential to this process.

SOUTHWEST

The most significant factors for change are increased weaning rates, improved forage supply (by planting pasture and making better use of pasture and supplements), and control of venereal diseases. There needs to be greater use of systems that are more efficient and intensive, and thus shorter. Increasing slaughter weights is also important: steers need to be heavier when entering the finishing stage so that the current length of this stage and weight gains during it remain the same.

The use of additives (prebiotics, probiotics, synthetic molecules, algae, tannins, etc.) could potentially help reduce GHG emissions. Field trials would be needed to validate these technologies. Other areas that need to be improved are carbon sequestration in perennial pastures and green pastures and conversion efficiency in energy supplementation.

Public initiatives to support these chances include financing through credit and training. Private initiatives include the strengthening of producer groups (CREA, Cambio Rural).

WEST

Increase forage supply and efficient supplementation (silage, dry grain, wet grain, by-products, and agroindustrial residues) to improve the status of breeding stock and stocking rates. The percentages of alfalfa-based perennial pastures (pure alfalfa and mixed plantings) need to be increased through better planting and appropriate management to ensure durability. Efforts should be made to increase the productivity of annual forage through early attention to soil fertility and/or fertilization (the use of manure as a possible approach to replenishing soil nutrients) and the inclusion of mixtures with annual legumes.

The use of digital identification systems (digital herd management) is also important for obtaining individual information and detecting productivity indicators that can be inherited through genetic improvement.

The public-private initiatives that could be implemented include multirisk insurance to cover risks affecting the feed chain (pasture and crops).

SOUTHWEST

The most influential factors are increasing the weaning rate, improving forage supply, and controlling venereal diseases. Emphasis was also placed on building the capacities of the human resources involved (producers, consultants or extension workers, and farm workers), as their support will be needed to bring about major change. There is also a need to promote the use of the futures market to provide more predictability. If prices are uncertain because they are highly volatile or impacted by trade measures (export duties and/or quotas), high tax rates, etc., incentives to invest or undertake improvements are lost, which affects final productivity.

Supporting programs such as Cambio Rural is also important. The public and private sectors need to promote good communication and training programs that draw attention to the quality and benefits of animal protein. Mention was also made of projects that measure GHG indices using approaches that are tailored to the region. Voisin-style rational grazing was discussed as an environmentally sustainable management strategy, as was the need to plan grazing, build reserves, and design strategic supplementation systems to attain optimal digestibility levels. The use of Normalized Difference Vegetation Index (NDVI) images to design forage chains was also mentioned.

SEMIARID ZONE

Fundamental changes to cattle ranching systems in this area should be oriented toward improving procreation rates and increasing reproductive rates in general. The priority issues include seasonal servicing, genetics, paddock management and access to water, rotational grazing, sanitary planning, and managing systems based on environmental conditions (e.g., stocking rate).

Between 15% and 30% of the arable land in this region is made up of patches of land in the Caldén forest. At least 50% of these fragile areas should be used for perennial tropical pastures. The remainder should be used as part of a rotational grazing scheme that includes perennial pastures. In other words, cattle-farming systems should be tailored to the environment they operate in.

Regarding the impact of cattle-farming decisions on emissions, carbon sequestration, and the carbon balance, awareness needs to be raised among producers by providing examples of productive improvements that also benefit the carbon balance and are often easy and inexpensive to implement. This will require hard work on communication and outreach activities to provide producers with the information they need.

PATAGONIA

The top priority is managing forage resources appropriately, based on improvements in productive infrastructure (watering places and subdivisions), strategic interventions in feeding (fodder type, quantity, and time of year); herd management issues (health, reproduction); and the mechanization/automation of some processes. Other important issues include herd management for seasonal servicing and strategies such as early weaning, especially during droughts.

Another factor that was emphasized was working to professionalize wintering through fodder resource management and feedlot rations in order to reduce grazing time and send animals to feedlots in winter to ensure higher weight gains. Access to credit is another important issue. This should take the form of long-term financing tied to product value to be used for infrastructure investments (pasture, water, facilities); the maintenance and purchase of breeding cattle; machinery; and planting perennial pasture.

CHAPTER 5

SUSTAINABILITY AND INTERNATIONAL TRADE

THE IMPACT OF TRACEABILITY SYSTEMS IN CAMPECHE, MEXICO

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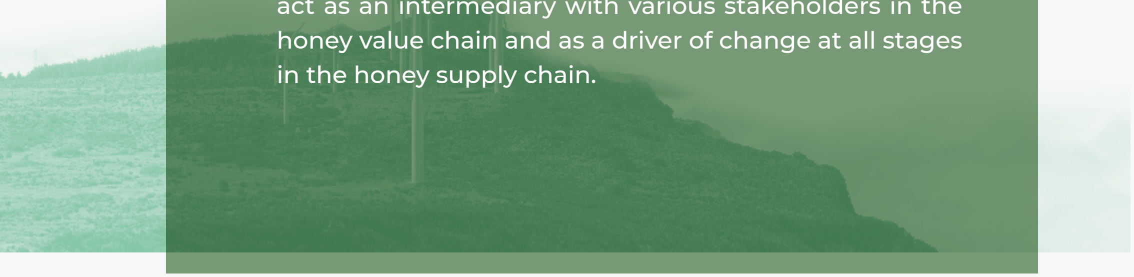
We are grateful to the beekeepers who took the time to answer the surveys analyzed in this chapter. Any errors are the responsibility of the authors.

ACRONYMS AND ABBREVIATIONS

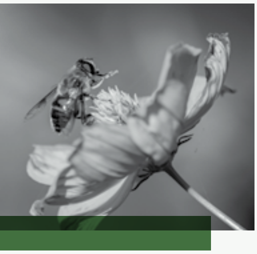
- **BCAM**
border carbon adjustment mechanism
- **DCE**
discrete choice experiment
- **EU**
European Union
- **SDGs**
Sustainable Development Goals
- **SENASICA**
National Service of Health, Food Safety, and Food Quality, Mexico

ABSTRACT

This study analyzes beekeepers' preferences regarding traceability systems in honey production in the state of Campeche, Mexico. We conducted a discrete choice experiment (DCE) in which the beekeepers surveyed were given a choice between three honey production scenarios. The production practices included in the DCE directly impact the health of the bees and their ecosystem services. Some 90% of the honey produced in Campeche is exported. The sale abroad of honey produced in line with the practices considered in this chapter is a channel through which international trade could trigger local growth and environmental conservation. Honey produced according to the practices studied would have a better chance of complying with sanitary regulations in importing countries. Implementing such measures, along with a traceability system, would allow Campeche and Mexico as a whole to position themselves as reliable suppliers. Our findings show that beekeepers would be willing to switch to production practices that conserve biodiversity and participate in a traceability system. In return, they would expect a bonus of just under US$2 per kilogram of honey. This figure is at the lower end of the range reported in previous willingness-to-pay studies for food traceability, suggesting that there appears to be potential to incentivize the production of traceable honey that conserves biodiversity in Campeche. These findings are an opportunity for the state government to act as an intermediary with various stakeholders in the honey value chain and as a driver of change at all stages in the honey supply chain.



1. INTRODUCTION



According to the 2030 Agenda for Sustainable Development, international trade could potentially become a driver for inclusive economic growth and poverty reduction, thus contributing to sustainable development (United Nations, 2015). As a direct consequence of this expectation, trade policymakers in developed and developing countries have initiated measures to seek a global green transition supported by international trade.

On the one hand, several regional trade agreements have been renegotiated to promote the spread of environmentally friendly goods and technologies by reducing or removing trade barriers (Gisselman and Merkus, 2023). On the other hand, trade rules have been amended to discourage imports of goods that contribute to deforestation. Indeed, the European Union (EU) has implemented a regulation on supply chains that seeks to stop member countries from contributing to deforestation in countries outside the bloc (López Bejarano, 2022).

At the same time, trade officials in each country have been forced to rethink conventional tariff schemes—the traditional objective of which is to protect domestic goods and services, balance the current trade account, and/or increase government revenues—to meet the dual objective of promoting domestic economic growth while "saving the planet" (Moreira and Dolabella, 2023). In this sense, several measures that form part of the European Green Deal are examples of new designs for tariff schemes to promote cleaner industries in countries outside the EU (European Commission, 2022). One such measure is the Border Carbon Adjustment Mechanism (BCAM), which seeks to set prices that reflect the carbon emitted during the entire production process of goods imported by EU member countries.

Traceability systems can help international trade go green. These systems systematically document the production, sale, and distribution of a given product (Beltrán and Coronado, 2021). They allow value chains to be monitored and checked, enabling consumers to trace the steps taken to manufacture the good or service they are about to purchase right back to the point of origin. Traceability systems can thus provide support for the monitoring of producers' environmental and biodiversity conservation initiatives. This is particularly important for goods that are traded internationally and therefore cannot be closely monitored by potential end consumers. At the same time, traceability systems can enhance the role of international trade in promoting inclusive economic development and poverty reduction. This role is important when producers are part of poor or vulnerable groups whom consumers may potentially be willing to compensate for using environmentally friendly practices.

Traceability mechanisms reduce information asymmetries throughout the production process, meaning that they can potentially promote differential payments to producers for adopting sustainable practices. As a consequence, more environmentally friendly goods can be placed on the international market. This study focuses on honey produced in the state of Campeche, Mexico, whose position on the international market could potentially be improved. Between 2009 and 2019, Mexico consolidated its position as one of the world's top 10 honey exporters. Luis-Rojas et al. (2022) interpret this as evidence that the European and US markets have revealed their preferences for honey perceived as being less likely to contain chemical components that are not approved by importing countries' regulations, as opposed to honey from China (the leading international producer), much of which is deemed to be of substandard quality (Jones Ritten et al., 2019; Luis-Rojas et al., 2022; Moore et al., 2012). Mexican honey is analyzed in laboratories in importing countries and is not allowed to enter their markets when the concentration of chemicals in it exceeds regulation ceilings, as happened in August 2012, when honey from Campeche was found to contain sulfates (Vázquez Martínez, 2022). According to European standards (FAO/WHO Codex Alimentarius Commission, 2023), Mexican honey is analyzed for pesticide residues and heavy metals, which are common components of fertilizers, pesticides, and paints.

Even though honey from Mexico is perceived as less likely to contain high concentrations of regulated chemicals, beekeepers engage in practices that expose bees to chemicals that can affect pollination services and biodiversity conservation. For example, to use hive boxes for longer, beekeepers paint them with products containing chemicals that affect pollination because they decrease the bees' ability to fly (Balbuena et al., 2015) and modify their sleep/rest patterns (Vázquez et al., 2020). Another practice that may affect pollination is the distance between apiaries and farmland. The greater the distance between these, the less likely it is that bees will come into contact with pesticides and fertilizers. While Mexico's National Service of Health, Food Safety, and Food Quality, Mexico (SENASICA) recommends there should be at least 3 km between the apiary and crops or farmland (SENASICA, 2022, p. 5), beekeepers in the country do not take measures to ensure this distance is maintained.

Given these findings, adopting a honey traceability system in Mexico could make beekeepers' practices transparent, notably by implementing changes that would improve bees' pollinating capacity, which would then have an impact on biodiversity. This would allow Mexican honey to strengthen its position in international trade. Indeed, there is already evidence of consumer segments that are interested in paying a premium for foods that can be tracked through traceability systems, including for honey (e.g., Mora and Menozzi, 2008; Cosmina, 2016; Jonas Ritten, 2019) and other foods like meats and grains. **By establishing a honey traceability system, Mexico has the chance to achieve two things at once: first, to consolidate itself as a reliable honey exporter by helping its beekeepers to produce honey that would earn a premium price; and second, to contribute to international trade through sustainable economic development by combating the sale of fraudulent honey, and thus countering the environmental impacts of this.**

The initial obstacle to implementing a traceability system in Mexico is the unwillingness of beekeepers in the country to do so. There is currently a traceability system for Mexican honey (SENASICA, 2022) but it depends on beekeepers keeping logbooks in which they report on their production practices. Mexican honey is thus not traceable in practice because only a tiny percentage of beekeepers keep logs—for example, according to figures reported in section 4, only 19% of beekeepers in Campeche do so. However, producers are likely unaware of the potential premium that consumers are willing to pay for traceable honey.

In this study, we begin to explore the feasibility of implementing a honey traceability system in Mexico. Specifically, we estimate the price beekeepers in the state of Campeche could expect to receive on the international market if their honey were produced using environmentally sustainable practices and a system that guarantees product traceability. To do so, we use a discrete choice experiment (DCE) to estimate the price premium that beekeepers would consider sufficient to compensate for changes in production practices. To address the possibility of beekeepers not being aware of the potential premium associated with traceability, the DCE is combined with a split-sample approach. This methodology works as follows: half of the respondents are explicitly told that the government is exploring the implementation of a traceability system that would allow them to receive higher prices for their honey on the international market, while the other half are told that the government is exploring a scheme that would allow them to receive higher prices but without providing any specifics.

On the international demand side, we present a range of prices that consumers would be willing to pay for traceable honey from Campeche, based on a review of estimates made in previous studies. We also discuss the potential for Mexican honey to access a higher-value segment of the international market. The DCE was presented to a representative sample of beekeepers in Campeche. These beekeepers were chosen because (i) Campeche is the second-largest honey-producing state in Mexico (SIAP, 2022) and (ii) the state authorities responsible for agricultural and beekeeping policies expressed interest in using the recommendations of this study to inform beekeeping policies that would enhance beekeepers' welfare. This interest translated into the authorities providing direct input into the design of the DCE implemented in this study.

The rest of this document is divided into five sections. Section 2 describes the beekeeping sector, focusing on (i) the international structure of the honey market; (ii) beekeeping production in Mexico; and (iii) the production, harvesting, and sale of honey in Campeche, with an emphasis on the components of this process that represent a challenge for the implementation of a traceability system. Section 3 describes food traceability. Section 4 provides a detailed description of the DCE that is implemented and analyzed in this study. Section 5 reports the results of the study. Finally, section 6 discusses the implications of these results and makes some recommendations.

2. THE HONEY MARKET

I) International structure

In 2022, global exports of natural honey[1] increased by an average of 17% compared to 2018, reaching US$2.27 billion. China was the world's leading honey exporter, accounting for US$277.7 million (10.5% of the total market). The figures also reveal that New Zealand, Argentina, India, and Ukraine contributed significantly to this market: together, they accounted for 43.1% of international sales of natural honey in 2022 (Workman, 2023).

Recent research has documented that China makes fraudulent claims regarding the quality and origin of its honey. This includes shipping honey to other Asian nations where the packaging is changed and the product is falsely labeled, such that it appears to originate in a country other than China (Ahmad and Khairatun, 2021; Ritten et al., 2019). These tactics allow Chinese producers to avoid (temporary) import restrictions in the United States and Europe. Production practices in China also include the use of unauthorized antibiotics (Trotta, 2013). One practice that is particularly detrimental to the environment and human health is the dilution of honey to conceal the presence of pesticide, herbicide, fertilizer, and paint residues (Bottemiller, 2013; Johnson, 2014). In terms of economic value, research conducted in 2011 revealed that the value of honey imported fraudulently into the United States from China is equivalent to US$80 million annually (Leeder, 2011).

China is not the only exporter that engages in fraudulent practices: these have been identified in other countries and have become a shared concern in terms of honey quality and environmental impacts, including on biodiversity. For example, García and Phipps (2018) note that the increase in global honey exports in 2015–2017 does not square with the relatively constant numbers of hives and the decrease in productivity per hive. This inconsistency suggests that fraudulent honey flooded the market during this period. The authors suggest that

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[1] · The honey market also includes by-products such as pollen, propolis, beeswax, and apitoxin or bee venom. Details on these by-products can be found in Dussart (2007).

India potentially also traded fraudulent honey as both China and India appear to experience a decline in hive numbers and productivity per hive, but a substantial increase in exports. They further emphasize that drops in productivity are an indicator of environmental degradation in the ecosystems where honey is produced.

Broadly speaking, fraudulent food represents an obstacle to achieving the Sustainable Development Goals (SDGs) (Chandan et al., 2023). It is particularly problematic when measuring progress in responsible consumption and production (SDG 12). Trade in fraudulent food implies that the international food supply chain still lacks transparency, which is an indispensable part of moving toward the SDGs. Transparent supply chains allow monitoring and verification mechanisms to be established, which are essential for implementing policies in the event of emergencies relating to foodborne diseases or contamination. Transparency is also a vital part of assigning responsibility through a monitoring system that both consumers and producers can trust. Such systems are also needed to verify the environmental outcomes of agreements reached under the EU Green Deal.

II) Production, harvesting, and sale of honey products in Campeche

Mexico ranks ninth globally in terms of its honey exports, which are mainly purchased by Germany and the United States (SIAP, 2022). The country's hive productivity (measured in kilograms of honey per hive) is high, outstripped only by Argentina and China[2] (Magaña Magaña et al., 2016).

The states of Yucatán, Campeche, and Quintana Roo produce a third of the country's total supply (SIAP, 2022). This region, known as the Yucatán Peninsula, has a long tradition of beekeeping that is important to its economy but is also socially and culturally significant (Güemes-Ricalde et al., 2003).

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[2] · The estimated productivity for China includes fraudulent honey because the calculations made by Magaña Magaña et al. (2016) are based on aggregated country-level data available from FAOSTAT.

In Campeche, the second-largest honey-producing state in Mexico, there are around 7,500 honey producers. In 2021, honey production and sales in Campeche reached 8,951 tons, with an average price per ton of US$2,600. Around 90% of Campeche's honey reaches export markets, most notably Germany and Saudi Arabia (SIAP, 2022; Vázquez Martínez, 2022).

The honey harvesting and sales process in Campeche is shown in figure 1, which is divided into three panels that correspond to the main three stakeholders: beekeepers, collection centers, and sales companies. Each panel shows the activities carried out by each actor. The arrows across the panels show the activities in which these stakeholders are involved.

Generally speaking, the process starts with the beekeeper collecting, storing, and transporting the honey harvest. This can then be sold directly to the collection center or to a third party who acts as an intermediary. Once the honey is at the collection center, it is checked for moisture content, which is a determining factor in the price beekeepers receive—optimum levels are between 17% and 18%.[3] Once purchased, the honey is stored and homogenized by the collection center and then transferred to the premises of the sales company. This firm receives and stores the honey in batches that it sells to the international or domestic market. If a batch might be sold on the foreign market, a sample is taken and sent to laboratories approved by European buyers. Sale on the international market depends on the results of laboratory tests indicating that the chemicals present in the honey are below the thresholds established in European regulations (e.g., the Codex Alimentarius).

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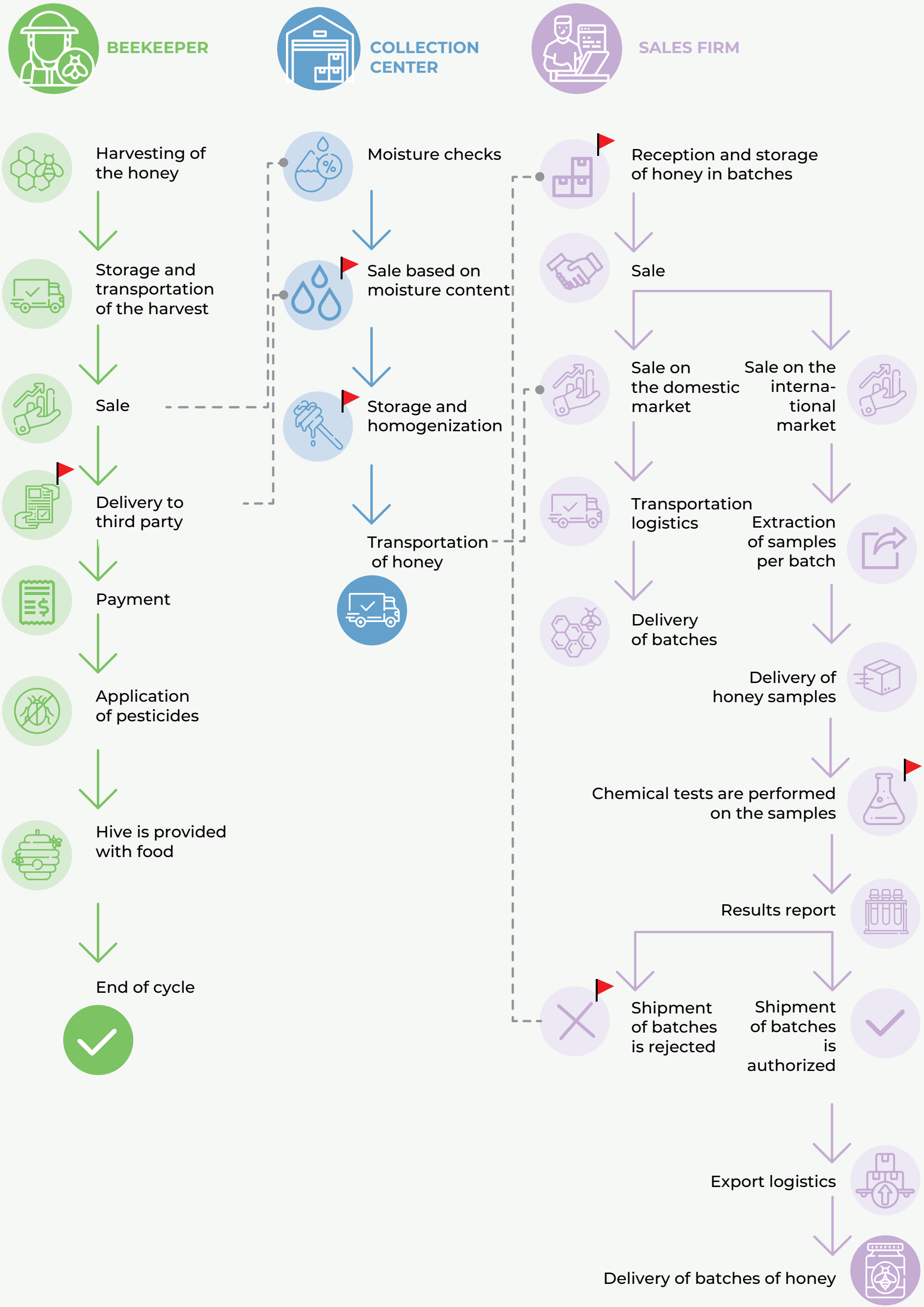
3 · The higher the moisture content of the honey, the lower the price, because moisture accelerates the fermentation process.

Figure 1.
Description of the Honey Harvesting and Sales Process in Campeche


BEEKEEPER
COLLECTION CENTER
SALES FIRM
Harvesting of the honey
Storage and transportation of the harvest
Sale
Delivery to third party
Payment
Application of pesticides
Hive is provided with food
End of cycle
Moisture checks
Sale based on moisture content
Storage and homogenization
Transportation of honey
Reception and storage of honey in batches
Sale
Sale on the domestic market
Sale on the interna-tional market
Transportation logistics
Extraction of samples per batch
Delivery of batches
Delivery of honey samples
Chemical tests are performed on the samples
Results report
Shipment of batches is rejected
Shipment of batches is authorized
Export logistics
Delivery of batches of honey

Source: Authors.

However, there are critical points in the harvesting and sales process that reveal the advantages of a traceability system. These points are marked with red flags in figure 1. The first relates to who beekeepers decide to sell their honey to a collection center or a third party that functions as an intermediary. These intermediaries tend to be illegal honey traders known locally as "coyotes,"[4] who offer higher prices but are not officially authorized as collection centers.

The challenge associated with sales to coyotes is that they often do not test honey sufficiently to assess its quality. Three production practices can particularly affect honey quality at this point: painting hive boxes, the distance between apiaries and crops or farmland, and hive-cleaning methods. Although the purpose of painting hive boxes is to keep them in use for longer, the paint used contains chemicals that directly affect both the bees living in them and the honey they produce. The distance between the apiary and crops or farmland is a factor in honey contamination due to agricultural pesticides and fertilizers: the greater the distance between the apiary and crops, the less likely such chemicals are to reach and affect bees and their honey. Cleaning hives is important, but beekeepers tend to do this using methods involving chemicals such as gasoline.

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[4] · "Coyotes" are intermediaries who buy honey from producers and resell it to collection centers or traders. The defining feature of these intermediaries is that they are not regulated, which prevents their activities from being monitored.

The second critical point is when the collection center decides to purchase the honey in question. As figure 1 shows, the collection center evaluates honey moisture levels to decide what price to offer but does not test for chemicals. The advisability of chemical testing relates to the third key moment in the process, when the collection center mixes the honey received from individual beekeepers and intermediaries (i.e., homogenization). If the chemical profile of the honey is unknown, contaminated honey and honey that passes health and safety standards may be homogenized into the same batch.

The fourth critical point is the chemical testing that the trader performs on the honey being sold by the collection center. When the trader receives the honey, before deciding whether or not it can be exported, it stores it in batches without any rigorous origin checks. This is a direct consequence of the way in which homogenization is carried out since it is impossible to trace the origin of the honey at this stage. Challenges can subsequently arise during the export process if chemicals are detected that were not identified at earlier stages. Generally speaking, honey that is not successfully exported is sold on the domestic market, which has implications for the average quality of local honey.

3. FOOD TRACEABILITY

A traceability system is the systematic documentation of the production, sales, and distribution operations that enable a product to be made available to the final consumer (Beltrán and Coronado, 2021). In the food market, it is a tool for managing food hazards (Hobbs, 2004; Pouliot and Sumner, 2008). Traceability systems are extremely useful when handling contamination scares since they can be used to identify sources of risk and enable the affected products to be rapidly removed from circulation.

These systems operate reactively and preventively, as they allow potential sources of contamination to be identified early on and help countries and producers proactively adopt measures to mitigate future risks (Olsen and Borit, 2018). From the producer's point of view, a traceability system is a monitoring and oversight tool that allows data to be measured and collected to determine efficient resource allocations at each stage of production (Hualpa and Rangel, 2023).[5]

The use of a traceability system in the Mexican beekeeping sector **would allow the early detection of chemicals in honey and the identification of stakeholders engaging in questionable practices (beekeepers, collection centers, and/or sales companies),** particularly those active at the critical points described in the previous section. There is evidence that some consumer segments are interested in honey produced in ways that respect the environment and biodiversity (e.g., avoiding the use of pesticides and chemical fertilizers). A traceability system would thus allow Mexican beekeepers to reach those market segments. The sale of environmentally friendly honey would have positive effects on the environment and biodiversity in producing regions (as described in the next section), which would help international trade play its part in the transition to sustainable development. The next section explains how our DCE addresses the relationship between a traceability system and environmentally friendly practices within Mexican beekeeping.

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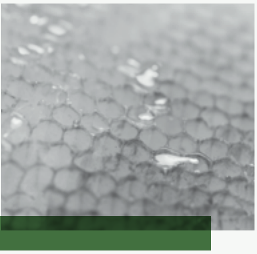
[5] Technology plays a vital role in the effective implementation of traceability systems. Barcodes, radio frequency identification (RFID) tags, and blockchain are some examples of such innovations (Qian et al., 2020).

4. DISCRETE CHOICE EXPERIMENT



I) Earlier studies

This study focuses on documenting the preferences of honey producers in a Latin American country in relation to adopting a traceability system and practices that would reduce environmental impacts. However, most previous research addressing these issues focuses on the preferences of consumers in European countries and the United States for organic and/or locally produced honey.[6]

Cosmina et al. (2016) and Vapa-Tankosić et al. (2020) assess willingness to pay for organic, local honey in Italy and Serbia, respectively. Using a DCE, Cosmina et al. (2016) find that Italian consumers are more willing to pay for honey produced in Italy than organic honey—that is, local production is more important to them than organic production. Vapa-Tankosić et al. (2020) find the reverse preference among consumers in Serbia, who are willing to pay more for organic honey than locally produced honey.
Meanwhile, Arvanitoyannis and Krystallis (2006) and Ureña et al. (2008) explore market segments in Romania and Spain, respectively.

Arvanitoyannis and Krystallis (2006) use cluster analysis to identify three groups of honey consumers in Romania, which they label traditional consumers, enthusiastic consumers, and indifferent consumers.
Enthusiastic consumers would pay a premium for organically produced honey; indifferent consumers are not particularly interested in honey (regardless of whether it is conventional or organic); and traditional consumers are mainly interested in price changes (i.e., they buy cheaper conventional honey and most likely would not buy organic honey because of the premium it entails). The authors emphasize that consumers in Romania do not tend to be interested in labeling systems denoting country of origin or certifications. Ureña et al. (2008) divide their segments into regular consumers, occasional consumers, and potential consumers, reaching fairly similar conclusions to Arvanitoyannis and

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[6] · A group of studies that are related to ours but not included in this review explore how interested Latin American farmers or ranchers are in productive practices that have positive environmental impacts. For example, Colin Castillo et al. (2022) report on the interest of farmers in central Mexico in organic fertilizers and pesticides. Likewise, Ortiz et al. (2023) look at whether cattle farmers in Ecuador are interested in production practices that conserve water.

Krystallis (2006) regarding willingness to consume organic honey. They also observe that men are more willing than women to pay for organic honey in all three groups.

Two exceptions to the focus on organic honey are the studies by Ritten et al. (2019) and Wu et al. (2015). Both of these publications are more closely related to this study in that their starting point is the presence and increase in fraudulent honey and the resulting increase in risks to human health.

Ritten et al. (2019) have conducted laboratory experiments to explore the implications of informing Australian consumers about the negative effects of fraudulent honey on human health. Specifically, they find that providing this information leads to a 27% increase in the premium that consumers report being willing to pay for honey that is produced locally and using approved practices. Wu et al. (2015) document US consumers' interest in locally produced honey but leave out the organic focus that other studies have included. By conducting laboratory experiments, they report that consumers are more willing to pay a premium for locally produced honey, particularly when information is provided about the negative effects that fraudulent (internationally produced) honey can have on human health.

We have only identified two previous studies focusing on honey producers, both in Ethiopia: Girma and Gardebroek (2015) and Tarekegn et al. (2017). Focusing on southwestern Ethiopia, Girma and Gardebroek document an organic honey success story. Small producers in the region experienced an increase in their sales income after signing contracts with a company that sells organic honey internationally. Tarekegn et al. (2017) also find that producers in Chena district mainly sell their honey to cooperatives, which is a successful strategy for maximizing and stabilizing their sales income in the long term. In other words, even though they could sell to unregulated intermediaries offering higher prices (much like the Mexican "coyotes" described in section 2), honey producers in Chena decide to go through cooperatives, even though they offer lower average prices. They do so because cooperatives represent an alternative way of stabilizing incomes by providing access to international markets and offering both formal and informal benefits as part of membership (e.g., training, building social networks, etc.).

II) Atrributes

The DCE described in this chapter consisted of asking the beekeepers surveyed to choose one of three alternatives regarding different honey production practices. These options were presented via choice cards. Figure 2 shows the format of the DCE choice cards, which presented respondents with three alternatives. Each was described in terms of five attributes relating to production practices and one attribute relating to compensation, namely the price that the international market should pay, from the beekeeper's point of view, to compensate them for adopting the beekeeping practices presented in the DCE.

The circumstances shown in the last column of figure 2 represent the status quo—in other words, the conditions that currently prevail for beekeeping in Campeche. At present, beekeepers paint hive boxes, keep apiaries less than 1 km from crops or farmland, apply chemical pesticides to these crops, use nonrecommended chemical treatments in the hive, and do not keep field diaries.[7]

The honey production practices included in this DCE have a direct impact on the health of the bees and the ecosystem services they provide. The recent report by Vides Borrell et al. (2023) describes the effect of these practices. The authors document bee poisoning in Hopelchén, Campeche, which was reported on March 22, 2023. Their findings illustrate that bee poisoning occurs not only due to direct contact with chemicals—via paint or hive-cleaning strategies—but also through the use of pesticides in crop fields. Vides Borrell et al. (2023) document that the wind carried an insecticide used in a field at least 1.6 km away, affecting apiaries located in an area covering 11,304 ha. The authors calculate that the insecticide affected 110 apiaries belonging to 80 beekeepers and containing 3,365 hives. In economic terms, they estimate that 13,200 days of rural employment and US$741,000 were lost, including losses in honey production, swarm replacement, and pollination services.

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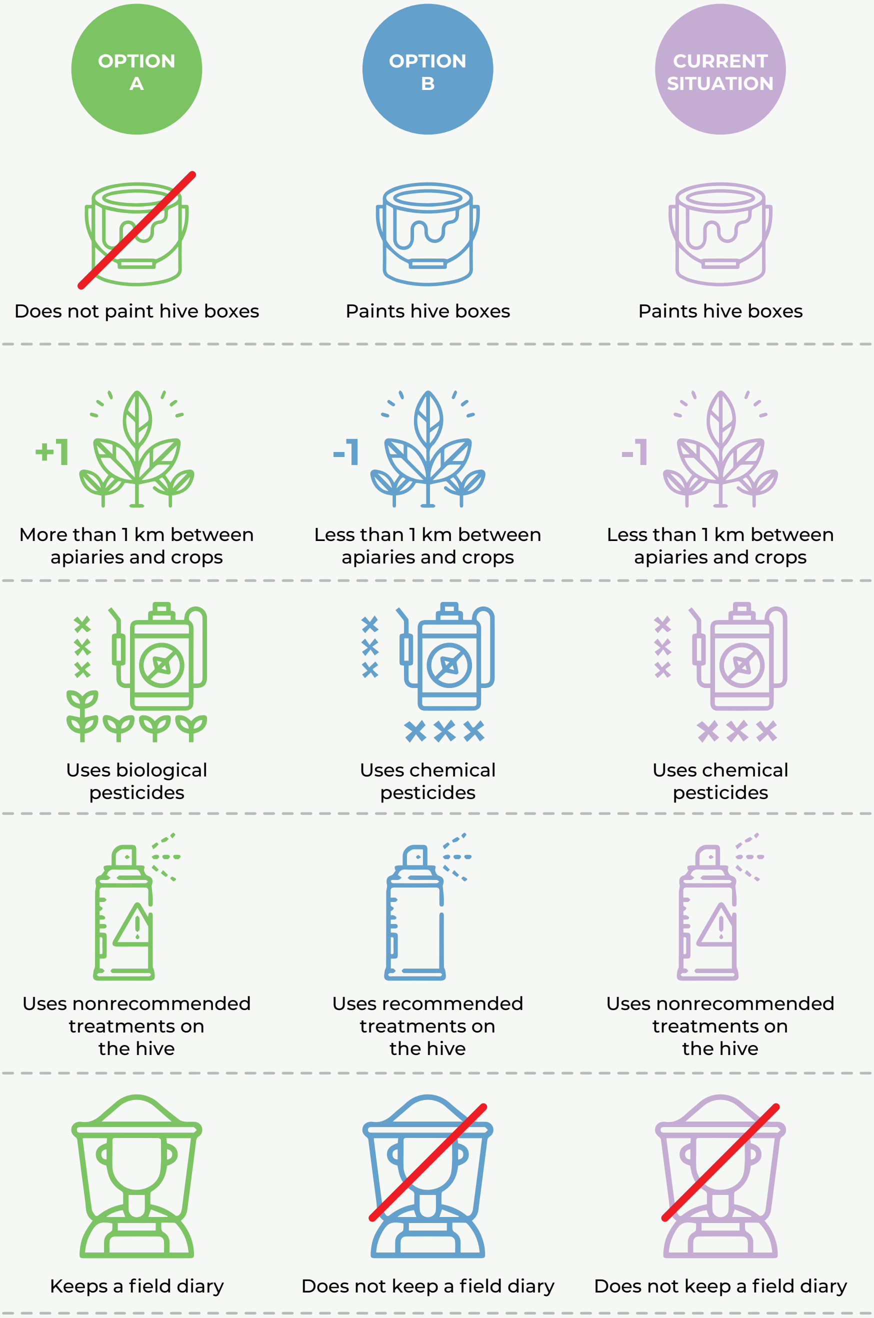
[7] · Other production practices are also part of the current modus operandi. These include how often queen bees are replaced or whether beekeeping is the respondent's main activity. However, our DCE does not include these or other practices for two reasons. The first reason is that we have prioritized practices that may have an impact on reducing the presence of chemicals in honey. The second is that the cognitive demand of the exercise increases with the number of attributes included. The number of attributes included in this study is consistent with the state of the art—less than 20% of the studies reviewed by Martínez-Cruz (2015) use seven or more attributes.

Pollination services are the most significant in terms of the impacts of current beekeeping practices on the environment and biodiversity. Globally, it is estimated that bees account for 80% of pollination (Gill et al., 2012). Bees' pollination capacity is not only jeopardized by exposure to chemicals that cause their death, as in the case documented by Vides Borrell et al. (2023). Gill et al. (2012) also argue that chronic exposure to chemicals affects bees' performance and feeding patterns even if it does not cause mortality—for example, bees that are chronically exposed to chemicals collect pollen less efficiently. This decrease in pollination efficiency is due to effects on both their flight ability (Balbuena et al., 2015) and sleep/rest patterns (Vázquez et al., 2020).

Figure 2.
Choice Card Presented to Respondents.



OPTION A	OPTION B	CURRENT SITUATION
Does not paint hive boxes	Paints hive boxes	Paints hive boxes
+1 More than 1 km between apiaries and crops	-1 Less than 1 km between apiaries and crops	-1 Less than 1 km between apiaries and crops
Uses biological pesticides	Uses chemical pesticides	Uses chemical pesticides
Uses nonrecommended treatments on the hive	Uses recommended treatments on the hive	Uses nonrecommended treatments on the hive
Keeps a field diary	Does not keep a field diary	Does not keep a field diary

Source: Authors.

Regarding the positive externalities of bees in productive activities, Garibaldi et al. (2016) document the potential for what they call ecological intensification—that is, increasing crop yields by enhancing biodiversity. Specifically, they observe a pattern that holds in several regions of the world and for a variety of crops: in agricultural fields of 2 ha or less, pollinator density and variety are positively associated with crop yields.

Production practices that help to reduce bees' exposure to chemicals thus have a positive impact on their pollination capacity, with corresponding effects on biodiversity, without jeopardizing the positive externalities for agricultural production. **Given these findings, the DCE seeks to explore whether beekeepers are interested in moving toward more environmentally friendly practices and, if so, how much they would require as compensation for doing so.** One additional advantage of modifying the practices examined in this study is that honey produced by beekeepers in Campeche would stand a better chance of meeting importing country regulations. This, when combined with a traceability system, would help the state position itself as a reliable global honey supplier.

Table 1 shows the six attributes of the DCE and the options for each, which are known as levels. These levels reflect potential changes in production practices compared to the current situation. The first attribute in table 1 is the practice of painting hive boxes to keep them in use for longer. Since paint contains chemicals that contaminate the honey, it is recommended that hive boxes be left unpainted. There are thus two levels to this attribute: painting the hive box (current situation) and not painting the hive box. Abandoning this practice would align with the recommendations listed in the official Mexican organic beekeeping manual, which states that hive boxes should "be made of natural materials that do not present contamination risks for bees" (SENASICA, 2022, p. 5).

The second attribute is the distance between apiaries and agricultural crops. The greater this distance, the less likely it is that the honey produced in apiaries will be contaminated by agricultural pesticides and fertilizers. The organic beekeeping manual suggests that this distance should be at least 3 km (SENASICA, 2022, p. 5), which is beyond the possibilities of most beekeepers in Campeche.[8] As a result, we have proposed a more achievable distance in this DCE of at least 1 km.

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[8] · Pérez Canepa and Pérez Akaki (2017) report that most beekeepers in Campeche are small-scale in terms of both the number of hives they manage (less than 25, on average) and the area of land under their control (usually less than 2 ha). Given these circumstances, they cannot comply with the recommendation to keep their apiaries at least 3 km from crops.

The third attribute is the type of pesticide used on crops. Because the recommended distance between apiaries and crops is not realistic for most beekeepers, using biological pesticides is a complementary measure that will help reduce the probability of honey contamination due to proximity to crops. The transition to biological pesticides aligns with the recommendations of Mexico's official organic beekeeping manual (SENASICA, 2022, p. 5).

The fourth attribute concerns measures used to clean hives, which have a direct impact on hive health. Beekeepers in Campeche generally clean their hives using liquids containing chemicals that contaminate honey (e.g., gasoline). There are thus two levels for this attribute: nonrecommended treatments (current situation) and recommended treatments, which focus on avoiding the use of chemicals. The aim of the recommended treatments is for "workers who have direct or indirect contact with hives not to represent a risk of contamination" (SENASICA, 2022, p. 15).

The fifth attribute is the keeping of a field logbook, which would be the first step in establishing a traceability system that would allow consumers to trace honey to its origin. Keeping a logbook is necessary regardless of the technology used to implement traceability schemes. Logbooks are currently being promoted by SENASICA through the "certification program for producers interested in reducing the risks of honey contamination" (SENASICA, 2022, p. 12). To obtain certification, beekeepers must document the stages of production in their logbook.

The sixth attribute is financial, which was described to the respondent as the price they would receive on the international market per kilogram of honey produced using the practices mentioned in each alternative on the choice cards. There are five levels for this attribute. The current situation is MOP36, which is the average price received per kilogram of honey in 2022. The other four levels are MOP40, MOP43, MOP47, and MOP50.

Table 1.
Attributes and Levels Used in the Discrete Choice Experiment

ATTRIBUTE	DESCRIPTION	LEVELS
Painting hive boxes	Beekeepers paint hive boxes to keep them in use for longer, but the paint contains chemicals that contaminate the honey. It is therefore recommended that hive boxes should not be painted.	Yes (current situation); No.
Distance between apiaries and crops	The greater this distance, the less likely it is that the honey will be contaminated by agricultural pesticides and fertilizers.	Less than 1 km (current situation); more than 1 km.
Type of pesticide used on crops	The probability of honey contamination is higher for chemical pesticides compared to biological ones.	Chemical pesticides (current situation); biological pesticides.
Hive cleaning	Beekeepers usually maintain and clean their hives using treatments that contain chemicals that can contaminate the honey.	Nonrecommended treatment (current status); recommended treatments.
Keeping a field logbook (diary)	Field logbooks are the first step toward establishing a traceability system. They allow producers to make their production process more efficient.	No (current situation); Yes.
Price per kilogram of honey (in MOP)	Price that the beekeeper would receive per kilogram of honey produced using the practices described in each alternative scenario.	MOP36 (current situation); MOP40; MOP43; MOP47; MOP50.

Source: Authors

This DCE allows us to infer the willingness of beekeepers to accept monetary compensation for implementing the production practices described above. These practices—whether implemented together or separately—reduce the probability of honey being contaminated with chemicals, which has positive implications for Mexico's environment and biodiversity, and increase the probability of it complying with importing country standards.

Specifically, the DCE allows us to infer how much beekeepers would need to be compensated for using logbooks, the first step in establishing a traceability system. This would also entail several other steps, such as coordination between beekeepers and honey collection and sales companies. If this sort of coordination were simple, it would already be happening. The government has a part to play in this process, namely by designing and promoting a traceability system and facilitating and coordinating communication between beekeepers and collection and sales companies. However, beekeepers do not necessarily associate the traceability system with an opportunity to access better international prices.

We thus used a split-sample strategy to explore whether beekeepers are interested in a traceability system that would be designed and implemented by the state government and would allow them to access better international prices. The split-sample strategy consisted of introducing half of the respondents to the DCE by broadly describing the government's efforts without explicitly mentioning the traceability system. This is referred to in the following section as the "standard scenario":

> *The state government is considering implementing a program to enable honey producers in Campeche to obtain higher prices on the international market. In exchange, honey producers would be required to adapt their beekeeping practices.*
>
> *The Ministry of Agricultural Development has commissioned researchers from the University of Campeche and colleagues of theirs from the Swedish University of Agricultural Sciences to explore what changes you are willing to make to your beekeeping practices. The results of this study will be presented to authorities at the ministry, who will decide whether to implement the program based on this information.*

The remaining respondents were introduced to the DCE by being told about the government's efforts, making explicit mention of the traceability system. This is described in the following section as the "traceability scenario":

> *EThe state government is considering implementing a traceability program to enable honey producers in Campeche to obtain higher prices on the international market. Traceability would allow honey producers to document the health conditions of the honey they sell. This would imply honey producers adapting their beekeeping practices.*
>
> *The Ministry of Agricultural Development has commissioned researchers from the University of Campeche and colleagues of theirs from the Swedish University of Agricultural Sciences to explore what changes you are willing to make to your beekeeping practices. The results of this study will be presented to authorities at the ministry, who will decide whether to implement the program based on this information.*

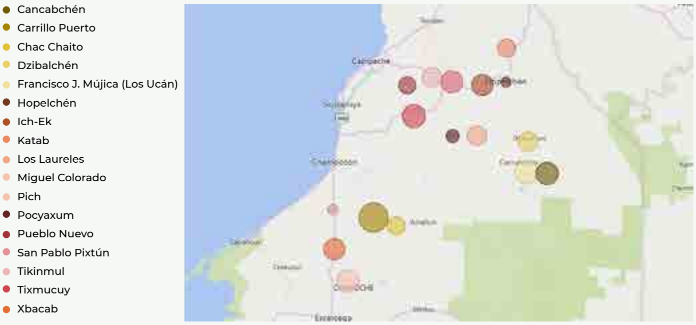
This split-sample design enables us to infer whether beekeepers express more interest (measured in willingness to accept compensation) in changing their production practices when the possibility of implementing a traceability system is explicitly mentioned compared to when no such possibility is mentioned.

III) Application and descriptive statistics

In June and July 2023, we visited 17 areas in the three honey-producing municipalities of the state of Campeche (Hopelchén, Champotón, and Campeche) and surveyed 196 beekeepers: 66 in Hopelchén municipality, 66 in Champotón municipality, and 64 in Campeche municipality. Figure 3 shows the locations visited. To select the areas to be surveyed, we used simple random sampling on the list of beekeepers for 2018 provided by the Ministry of Agriculture and Rural Development.

Figure 3.

Locations Where the Discrete Choice Experiment Was Conducted


● Cancabchén
● Carrillo Puerto
● Chac Chaito
● Dzibalchén
● Francisco J. Mújica (Los Ucán)
● Hopelchén
● Ich-Ek
● Katab
● Los Laureles
● Miguel Colorado
● Pich
● Pocyaxum
● Pueblo Nuevo
● San Pablo Pixtún
● Tikinmul
● Tixmucuy
● Xbacab
Campeche
Champotón

Source: Authors.

Table 2 provides socioeconomic information of the beekeepers who responded to the DCE: 84% are men, with an average age of 47; 11% are women, with an average age of 40; and the 5% who preferred not to state their gender have an average age of 56.

Table 2.
Socioeconomic Information for Beekeepers Surveyed (n=196)

SEX	#	AVERAGE AGE	AVERAGE # OF APIARIES	AVERAGE PRODUCTION (KG/HIVE)	PEOPLE WHO ARE ECONOMICALLY DEPENDENT ON PRODUCTION	PERCENTAGE SPEAKING AN INDIGENOUS LANGUAGE
Men	165	47	3	1,207	2	46%
Women	22	40	3	941	2	36%
Not specified	9	56	2	405	2	44%

Source: Authors.

Table 2 reveals that beekeepers in the state of Campeche are small-scale producers. Regardless of their gender, they each work three apiaries, on average. The average production reported by female beekeepers represents just 78% of the level reported by male beekeepers (941 kg/hive versus 1,207 kg/hive). Regardless of the gender of the beekeeper, an average of two other people are financially dependent on them. With regard to the significance of beekeeping in the region's culture, 46% of the male beekeepers and 36% of the female beekeepers state that they speak an indigenous language.[9]

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[9] · See annex tables A1, A2, and A3 for a more detailed description of respondents by municipality of residence.

5. INTEREST OF BEEKEEPERS IN CAMPECHE IN A TRACEABILITY SYSTEM

Table 3 shows the results of the conditional logit specifications carried out on the beekeepers' responses in each split sample and in the total sample. Participants in the sample presented with the standard scenario answered exactly the same choice cards as those in the group presented with the traceability scenario. As explained in the previous section, the difference between these two groups is that the latter were explicitly told that the state government is considering implementing a traceability system to enable beekeepers to access better international prices. **The logic behind this split-sample design is to explore whether beekeepers are more interested in changing their production practices when they are explicitly informed that the instrument being considered is a traceability system, as opposed to simply providing a very general description of the government's intention to implement a program to enable beekeepers to access better international prices (the generic scenario).**

Although we collected information from 196 participants (98 in each split sample), the econometric specifications reported in table 3 are based on 5,292 observations (2,646 in each split sample). The number of observations is the result of each respondent answering nine choice cards,[10] which each present three alternatives. That is, each observation reported in table 3 describes whether or not the participant selected each of the alternatives that were presented to them.

Keeping in mind that the coefficients reported in table 3 cannot yet be interpreted as measures of willingness to accept compensation—which are presented in table 4—these coefficients suggest the overall direction of preferences for the corresponding attributes or levels. That is, a positive coefficient indicates that respondents are interested in the attribute or level in question.

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[10] · Using this number of choice cards is consistent with the state of the art. For example, 53% of the studies reviewed by Martínez-Cruz (2015) use six to eight cards. The lowest number of cards used is four, and the highest number is 16.

The first coefficient in table 3 reflects the participants' preferences for the current situation, which refers jointly to the practices that are in practice at present (the third alternative for each choice card, as illustrated in figure 2): painting hive boxes, keeping their apiaries at less than 1 km from crops, using chemical pesticides on the crop, applying nonrecommended chemical treatments to the hive, and not keeping a field logbook.[11] The positive sign of the coefficient implies that, from the outset, beekeepers prefer the current situation. In other words, they would rather not change their production practices. This preference is observed in both of the split samples and thus in the whole sample.

The following five coefficients in table 3 denote each production practice as it currently stands: painting the hive box, a distance of less than 1 km between apiaries and crops, the use of chemical pesticides, not following hive-cleaning recommendations, and not keeping a field logbook. The aim of including these factors in the econometric specification is to assess whether there are specific preferences for one or more practices, on top of the inclination for these practices as a whole, as revealed by the coefficient associated with the current situation. There are two practices that respondents report having a particular preference for: chemical pesticide use and the current approach to hive cleaning. To forgo these activities, beekeepers in Campeche would require higher compensation (i.e., more than they earn in the current situation). These preferences are observed in both of the split samples and thus in the whole sample.

The last coefficient in table 3 reveals a highly significant preference for higher payment, which is consistent with both human intuition and classical economic theory: beekeepers prefer to be paid more rather than less. The coefficients reported in table 3 could be used to estimate beekeepers' willingness to accept monetary compensation to implement changes in their production practices via price increases on the international market.[12] Table 4 shows estimates of beekeepers' willingness to accept compensation that were obtained on the basis of both the full sample and each split sample. Given the 95% confidence intervals, average values do not differ between the split samples, indicating that the explicit mention of the traceability system does not change beekeepers' average preferences. In the remainder of this paper, we refer to the estimates obtained from the full sample. These estimates are expressed in MOP/kg.

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[11] · To capture preferences for the current situation, the econometric specification includes a binary variable that takes value of 1 if the alternative under consideration describes the current situation, and 0 if the alternative is one of the two options that represent changes to production practices.

[12] · Willingness to accept compensation for changing production practices is the result of dividing the corresponding coefficient by the coefficient associated with the monetary factor. In practice, this calculation yields a marginal rate of substitution between money and practice, which is equivalent to marginal willingness to accept compensation. See Ortiz et. al (2023) for a brief description of the theoretical and economic underpinnings of these calculations.

The first row in table 4 shows the amounts required for beekeepers in Campeche to consider changing their practices. On average, they would require MOP25.4/kg to do so. However, beekeepers are not willing to change all the practices considered in the DCE unless they are compensated for specific changes—namely refraining from using chemical pesticides and changing their hive-cleaning practices. They are willing to accept compensation of MOP3.6/kg for using biological pesticides and MOP3.9 for following hive-cleaning recommendations. In other words, beekeepers require a minimum of MOP33/kg in compensation (the sum of MOP25.4, MOP3.6, and MOP3.9). This is equivalent to approximately US$1.89 and represents 92% of the average price they received in 2022. **In other words, they would require their sale price to increase from $36 to $69.**

These results are estimates made from the full sample. Annex tables A4 through A9 report coefficients from the econometric estimates and estimates of willingness to accept compensation for the samples obtained within each of the three municipalities that produce honey in Campeche (Holpechén, Champotón, and Campeche). The results by municipality yield similar estimates, although there are some differences between them.[13]

How do the compensation-related aspirations of Campeche's beekeepers compare with foreign consumers' willingness to pay a premium for certain products? This comparison will allow us to ascertain whether traceability systems could potentially become the starting point for international trade to contribute to sustainable economic development in Campeche, allowing access to market segments that would pay a premium for production practices that conserve the environment and biodiversity in Campeche.

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13 · Although discussing differences in preferences between municipalities would be instructive, we do not enter into such a discussion because we used simple random sampling, in line with our objective of obtaining a representative sample of beekeepers at the state level in Campeche. That is, our samples by municipality do not necessarily reflect the preferences of beekeepers in the corresponding municipality.

Table 3.
Conditional Logit Specifications (respondents=196, choice cards=9, alternatives on each card=3)

	COMPLETE SAMPLE	STANDARD SCENARIO	TRACEABILITY SCENARIO
Current situation	1.961*** (0.402)	2.210*** (0.574)	1.735** (0.565)
1 if hive boxes are painted	0.0281 (0.0546)	0.00324 (0.0777)	0.0537 (0.0770)
1 if the distance between apiaries and crops is greater than 1 km	0.0298 (0.0528)	0.0638 (0.0753)	-0.00391 (0.0744)
1 if using chemical pesticide	0.282*** (0.0549)	0.207** (0.0780)	0.357*** (0.0778)
1 if hive-cleaning recommendations are not followed	0.308*** (0.0529)	0.374*** (0.0752)	0.244** (0.0748)
1 if no field logbook is maintained	0.00982 (0.0540)	0.0787 (0.0767)	-0.0595 (0.0763)
Price per kilogram of honey (Mexican pesos)	0.0773*** (0.00826)	0.0817*** (0.0118)	0.0733*** (0.0116)
Observations	5,292	2,646	2,646
Respondents	196	98	98
Pseudo-R2	0.235	0.239	0.235
Log-likelihood	-1481.6	-737.4	-741.6
AIC	2,977.3	1,488.7	1,497.1
BIC	3,023.3	1,529.9	1,538.3

Source: Authors.

Table 4.
Willingness of Beekeepers to Accept Compensation for Changing Their Practices (estimates from specifications reported in table 2, CI=95%)

WILLINGNESS TO ACCEPT (MEXICAN PESOS/KILOGRAM)	COMPLETE SAMPLE	STANDARD SCENARIO	TRACEABILITY SCENARIO
For changing current situation	**25.4**	**27.0**	**23.7**
Lower limit	19.1	18.5	12.8
Upper limit	29.7	32.3	29.9
For not painting hive boxes	0.4	0.0	0.7
Lower limit	-1.1	-1.9	-1.4
Upper limit	1.78	1.9	2.9
For increasing the distance between apiaries and crops	0.,4	**0.8**	0.0
Lower limit	-0.9	0.9	-2.12
Upper limit	1.7	2.6	1.85
For using biological pesticides	**3.6**	**2.5**	**4.8**
Lower limit	2.2	0.7	2.69
Upper limit	5.3	4.7	7.7
For following hive-cleaning recommendations	**3.9**	**4.6**	**3.3**
Lower limit	2.5	2.6	1.3
Upper limit	5.9	7.5	6.4
For keeping a field logbook	0.1	0.9	0.8
Lower limit	-1.2	-0.8	-2.89
Upper limit	-1.5	2.9	1.31

Source: Authors.

Since this study does not focus on consumers, and in the absence of studies estimating consumers' willingness to pay more for traceable honey, we drew on the estimates reported by previous studies that looked at consumers' willingness to pay a premium for traceability in other foods. Table 5 lists the studies we drew on.

Table 5.
Studies Reporting Willingness to Pay for Food Traceability

COUNTRY	YEAR	BONUS	BONUS (US$ 2023)	QUANTITY	PRODUCT	STUDY
Norway	2003	NOK12	6	kg	Beef	(Alfnes and Rickertsen, 2003)
France	2003	US$9.26	7.5	kg	Beef	(Lusk, Roosen, and Fox, 2003)
Germany	2003	US$7.31	6.1	kg	Beef	(Lusk, Roosen, and Fox, 2003)
United Kingdom	2003	US$8.1	6.5	kg	Beef	(Lusk, Roosen, and Fox, 2003)
Germany	2004	€0.47	1.04	kg	Pork	(Enneking, 2004)
Netherlands	2007	€4.18	9.09	kg	Pork	(Meuwissen, Van Der Lans, and Huirne, 2007)

Source: Authors.

The estimates reported by each study have been converted into 2023 US$. **The compensation-related aspirations of Campeche beekeepers for changing their production practices and taking part in a traceability program is equivalent to just under US$2.** In terms of the average price per kilogram of honey, this figure represents an increase in value of almost double relative to the amount they are currently being paid. **However, in terms of willingness to pay for a traceability system, table 5 reveals that European consumers have reported values ranging from US$1 (in Germany) to US$9 (in the Netherlands).** In other words, the aspirations of Campeche beekeepers seem to be within a reasonable range.

6. CONCLUSIONS AND RECOMMENDATIONS

This study has focused on the supply of honey subject to a traceability system and has explored the preferences of beekeepers in Campeche, Mexico, in relation to this. Two tasks remain in analyzing the feasibility of a honey traceability system in the country. The first would be to study the preferences of a representative sample of beekeepers in Mexico. Although the state of Campeche ranks second in total production, it represents only 13% of the national total. It is therefore appropriate to view this study as a pilot that provides an initial understanding of beekeepers' preferences, on the understanding that **further research needs to be carried out to reach conclusions regarding beekeepers at the national level. The second pending task would be to explore demand for honey subject to a traceability system.** There are indications that consumers of honey and other food products are willing to pay a premium for the existence of a traceability system. However, to provide useful recommendations, we would need to know whether the willingness to accept compensation estimated in this study is matched by a willingness to pay among potential consumers. This demand study should focus on consumers in Europe and the United States (the two main markets for Mexican honey).

With these limitations in mind, the message of this study is clear: beekeepers in Campeche would be willing to move toward production practices that preserve the environment and biodiversity of their state and to participate in a traceability system that allows international consumers to verify production practices. In return, beekeepers would expect to receive additional payment of just under US$2/kg of honey. This aspiration seems reasonable because it is at the lower end of the range reported in studies of willingness to pay for food traceability—US$1 to US$9. As noted in section 2, honey **collection centers and sales companies are key players in the success of a**

traceability system Government initiatives should thus include these stakeholders in any discussions.

Developing a mechanism that allows collection centers to measure the presence of chemicals in honey would reduce contamination during the homogenization process. This will make it possible to establish financing schemes that benefit all stakeholders in the chain and thus to implement the traceability scheme.

In other words, since there are economic benefits for all stakeholders in the supply chain, the cost of investing in infrastructure and training to implement a traceability system throughout the supply chain do not have to be covered entirely by public funds.

However, subnational agricultural authorities have two immediate opportunities for action to promote traceability: (i) by facilitating discussions among the various stakeholders in the chain and (ii) through their beekeeping extension program to promote best practices, such as providing information on alternative techniques and materials for painting boxes (vegetable oils and ecological paints) and appropriate oversight of hive health. In the medium term, work needs to be done in partnership with the economic development and foreign trade departments and the academic and private sectors to calculate and determine the premium that the international market is willing to pay for traceable honey. This will make it possible to set up financing schemes that will benefit all stakeholders in the chain and allow the traceability system to be implemented.

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ANNEX

Tables documenting descriptive statistics and econometric results by municipality.

Table A1

Socioeconomic Information for Beekeepers Surveyed in Hopelchén Municipality (respondents=66.

SEX	#	AVERAGE AGE	AVERAGE # OF APIARIES	AVERAGE PRODUCTION (KG/HIVE)	PEOPLE WHO ARE ECONOMICALLY DEPENDENT ON PRODUCTION	PERCENTAGE SPEAKING AN INDIGENOUS LANGUAGE
Men	56	44	3	1,390	3	61%
Women	7	42	2	850	2	43%
Not specified	3	71	2	283	1	33%

Source: Authors.

Table A2

Socioeconomic Information for Beekeepers Surveyed in Champotón Municipality (respondents=66)

SEX	#	AVERAGE AGE	AVERAGE # OF APIARIES	AVERAGE PRODUCTION (KG/HIVE)	PEOPLE WHO ARE ECONOMICALLY DEPENDENT ON PRODUCTION	PERCENTAGE SPEAKING AN INDIGENOUS LANGUAGE
Men	63	47	3	1,078	2	44%
Women	3	43	7	2,770	1	67%

Source: Authors.

Table A3

Socioeconomic Information for Beekeepers Surveyed in Campeche Municipality (respondents=64)

SEX	#	AVERAGE AGE	AVERAGE # OF APIARIES	AVERAGE PRODUCTION (KG/HIVE)	PEOPLE WHO ARE ECONOMICALLY DEPENDENT ON PRODUCTION	PERCENTAGE SPEAKING AN INDIGENOUS LANGUAGE
Men	46	51	4	1,160	2	30%
Women	12	38	2	456	2	25%
Not specified	6	54	2	467	2	50%

Source: Authors.

Table A4

Conditional Logit Specifications for Beekeepers' Responses in Hopelchén Municipality (respondents=66, choice cards=9, alternatives on each card=3)

	COMPLETE SAMPLE	STANDARD SCENARIO	TRACEABILITY SCENARIO
Current situation	2.081** (0.689)	1.880** (0.958)	2.375* (1.003)
1 if hive boxes are painted	-0.341*** (0.0977)	-0.256* (0.134)	-0.446** (0.145)
1 the distance between apiaries and crops is less than 1 km	0.109 (0.0942)	0.203 (0.131)	0.00623 (0.138)
1 if using chemical pesticide	0.315** (0.0978)	0.204 (0.136)	0.440** (0.143)
1 if hive-cleaning recommendations are not followed	0.298** (0.0941)	0.360** (0.131)	0.244* (0.137)
1 if no field logbook is maintained	0.109 (0.0962)	-0.0374 (0.133)	0.274* (0,142)
Price per kilogram of honey	0.0675*** (0.0144)	0.0635** (0.0200)	0.0732*** (0.0209)
Observations	1,782	918	864
Respondents	66	34	32
Pseudo-R2	0.163	0.159	0.175
Log-likelihood	-546.5	-282.7	-260.9
AIC	1,106.9	579.4	535.8
BIC	1,145.3	613.1	569.1

Table A5

Willingness of Beekeepers in Hopelchén to Accept Compensation for Changing Practices (estimated from specifications reported in Table A.4, 95% confidence interval)

WILLINGNESS TO ACCEPT COMPENSATION (MEXICAN PESOS)	COMPLETE SAMPLE	STANDARD SCENARIO	TRACEABILITY SCENARIO
For changing current situation	**30.8**	**29.6**	**32.5**
Lower limit	19.0	1.5	13.2
Upper limit	36.7	37.8	39.7
For refraining from painting hive boxes	**-5.0**	-4.0	**-6.1**
Lower limit	-10.3	-12.9	-15.1
Upper limit	-2.2	0.14	-2.1
For increasing the distance between apiaries and crops	1.6	3.2	0.1
Límite inferior	-0.9	-0.7	-4.1
Límite superior	4.7	9.6	4.2
For using biological pesticides	**4.6**	3,2	**6.0**
Lower limit	1.8	-0.9	2.1
Upper limit	8.7	9.6	13.1
For following hive-cleaning recommendations	**4.4**	**5.6**	3.3
Lower limit	1.6	1.4	-0.3
Upper limit	9.6	18.1	10.4
For keeping a field logbook	1.6	-0.6	**3.7**
Lower limit	-1.1	-5.7	0.1
Upper limit	5.1	4.4	10.4

Source: Authors.

Table A6

Conditional Logit Specifications for Beekeepers' Responses in Champotón Municipality (respondents=67, choice cards=9, alternatives on each card=3)

	COMPLETE SAMPLE	STANDARD SCENARIO	TRACEABILITY SCENARIO
Current situation	2,003** (0.708)	4,097*** (1.113)	0.517 (0.956)
1 if hive boxes are painted	0.0543 (0.0922)	-0.0866 0.140	0,191 (0,129)
1 if the distance between apiaries and crops is greater than 1 km	0.0188 (0.0893)	0.208 (0.137)	-0,143 (0,123)
1 if using chemical pesticide	0.294** (0.0934)	0,254* (0,141)	0,380** (0,132)
1 if hive-cleaning recommendations are not followed	0.158* (0.0898)	0,324** (0,134)	0,0239 (0,126)
1 if no field logbook is maintained	0.0168 (0.0911)	0.398* (0.139)	-0,322** (0,129)
Price per kilogram of honey	0.0944*** (0.0142)	0.134*** (0.0220)	0,0656*** (0,0194)
Observations	0.129	891	918
Respondents	0.0188	33	34
Pseudo-R2	(0.0893)	0.208	0.297
Log-likelihood	(0.137)	-0.143	-236.5
AIC	(0.123)	435.1	487.0
BIC	0.294**	468.6	520.7

Source: Authors.

Table A7

Willingness of Beekeepers in Champotón to Accept Compensation for Changing Their Practices (estimated from specifications reported in Table A.6, 95% confidence interval)

WILLINGNESS TO ACCEPT COMPENSATION (MEXICAN PESOS)	COMPLETE SAMPLE	STANDARD SCENARIO	TRACEABILITY SCENARIO
For changing current situation Lower limit Upper limit	**21.2** 9.6 28.2	**30.6** 20,7 36,5	7.9 -44.2 23.3
For refraining from painting hive boxes Lower limit Upper limit	0.6 -1.5 2.6	-0.6 -3.1 1.4	2.9 -1.2 9.1
For increasing the distance between apiaries and crops Lower limit Upper limit	0.,2 -1.6 1.9	1.5 -0.3 3.6	-2.1 -7.9 1.36
For using biological pesticides Lower limit Upper limit	**3.1** 1.1 5.4	1.9 -0.1 4.1	**5.6** 1.7 13.7
For following hive-cleaning recommendations Lower limit Upper limit	1.7 -0.1 4.1	**2.4** 0.5 5.2	0.6 -3.2 6.2
For keeping a field logbook Lower limit Upper limit	0.2 -1.6 2.2	**2.9** 1.1 5.6	**-4.9** -12.9 -0.8

Source: Authors.

Table A8

Conditional Logit Specifications in Beekeepers' Responses in Campeche Municipality (respondents=63, choice cards=9, alternatives on each card=3)

	COMPLETE SAMPLE	STANDARD SCENARIO	TRACEABILITY SCENARIO
Current situation	1.580** (0.737)	0.640 (1.021)	2.617** (1.079)
1 if hive boxes are painted	0,377*** (0.0991)	0.370** (0.139)	0.393** (0.142)
1 if the distance between apiaries and crops is greater than 1 km	-0.0352 (0.0943)	-0.205 (0.133)	0.141 (0.136)
1 if using chemical pesticide	0.253** (0.0985)	0.198 (0.140)	0.310** (0.140)
1 if hive-cleaning recommendations are not followed	0.505*** (0.0955)	.479*** (0.135)	0.540*** (0.137)
1 if no field logbook is maintained	-0.104 (0.0975)	-0.119 (0.139)	-0.0842 (0.138)
Price per kilogram of honey	0.0733*** (0.0149)	0.0556** (0.0208)	0.0927*** (0.0217)
Observations	1.701	837	864
Respondents	63	31	32
Pseudo-R2	0.299	0.289	0.316
Log-likelihood	-436.7	-217.9	-216.3
AIC	887.3	449.8	446.5
BIC	925.4	482.9	479.9

Source: Authors.

Table A9

Willingness of Beekeepers in Campeche to Accept Compensation for Changing Practices (estimated from specifications reported in Table A.8, 95% confidence interval)

WILLINGNESS TO ACCEPT COMPENSATION (MEXICAN PESOS)	COMPLETE SAMPLE	STANDARD SCENARIO	TRACEABILITY SCENARIO
For changing current situation	**21.5**	11.5	**2.2**
Lower limit	3.6	-77.4	11.1
Upper limit	30.1	28.,5	36.3
For refraining from painting hive boxes	**5.1**	**6.6**	**4.2**
Lower limit	2.5	1.7	1.1
Upper limit	9.3	24.9	8.8
For increasing the distance between apiaries and crops	-0.5	-3.7	1.5
Lower limit	-3.2	-16.8	-1.3
Upper limit	1.9	0.9	4.8
For using biological pesticides	**3.4**	-3.6	3.3
Lower limit	0.7	-16.8	0.4
Upper limit	6.9	0.9	7.4
For following hive-cleaning recommendations	**6.,9**	**8.6**	**5.8**
Lower limit	3.9	3.1	2.7
Upper limit	12.5	34.0	12.0
For keeping a field logbook	-1.4	-2.,1	-0.9
Lower limit	-4.5	-11.5	-4.4
Upper limit	1.3	3.5	2.3

Source: Authors.

CHAPTER 6

THE CHALLENGE OF GREEN TRADE BARRIERS: HOW SHOULD LATIN AMERICA AND THE CARIBBEAN RESPOND?

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ACRONYMS AND ABBREVIATIONS

- **CBAM**
  Carbon Border Adjustment Mechanism
- **CBDR2**
  Common But Differentiated Responsibilities
- **CO_2**
  Carbon dioxide
- **EITE**
  emissions-intensive trade-exposed
- **EPs**
  environmental provisions
- **EU ETS**
  EU Emissions Trading Scheme
- **EU**
  European Union
- **EUDR**
  EU Regulation on Deforestation-Free Products
- **FAO**
  Food and Agriculture Organization of the United Nations
- **GATT**
  General Agreement on Tariffs and Trade
- **GHGs**
  greenhouse gases
- **MERCOSUR**
  Southern Common Market (MERCOSUR)
- **HS**
  Harmonized System
- IDB
  Inter-American Development Bank
- **LAC**
  Latin America and the Caribbean
- **MFN**
  most-favored nation
- **Mha**
  millions of hectares
- **NDCs**
  Nationally Determined Contributions
- **SDGs**
  Sustainable Development Goals
- **SMEs**
  small and medium-sized enterprises
- **tn/Ha**
  metric tons per hectare
- **UNCTAD**
  United Nations Conference on Trade and Development
- **USMCA**
  United States-Mexico-Canada Agreement
- **WTO**
  World Trade Organization

ABSTRACT

Despite the global nature of environmental problems, targets to address these vary widely from country to country. This gap can lead to so-called carbon leakage—an increase in emissions in one country due to stricter environmental regulations in another. Given the difficulty of achieving coordinated global action, the strategies attempting to respond to this risk include measures relating to international trade. Among these initiatives are trade agreements and unilateral measures such as bans on imports of goods from deforested areas and border adjustment mechanisms based on carbon content. The proliferation of these measures will have a significant impact on international trade flows, particularly in certain countries and sectors in LAC. This chapter analyzes where countries in the region stand in relation to such measures and discusses the main challenges they will face in mitigating their negative impacts and adapting to the new rules of the game.

1. INTRODUCTION

The countries of Latin America and the Caribbean (LAC) are among the lowest emitters of greenhouse gases (GHGs) in the world. China accounted for about 30% of total global GHG emissions in 2022, while the European Union (EU) contributed around 7% and the United States 11%. LAC countries contribute approximately 7% of global emissions, with Brazil and Mexico being the largest emitters in the region (EDGAR, 2023). All the same, LAC's emissions have increased significantly over the past 30 years. The main sources of GHG emissions in the region are land-use change and agriculture, unlike the situation globally, which is dominated by energy production (World Bank, 2022).

Signatory countries to the 2015 Paris Agreement collectively committed to capping the average global temperature increase at below 2°C and to pursue efforts to limit global warming to 1.5°C above pre-industrial levels. To achieve this goal, GHG emissions must be reduced by 43% by 2030 compared to 2019 levels (UNFCCC, 2023). Despite the global nature of the problem, each country is free to choose specific measures and policies to meet its individual emission reduction targets, which are set out in the Nationally Determined Contributions (NDCs). To date, 195 countries have submitted NDC pledges, including all 33 LAC countries.[1]

The differences in the scope of the policies implemented in each country encourage what is known as "carbon leakage"—that is, when a sector or company moves its operations to a country with lower carbon prices or laxer regulations (Mehling et al., 2019; Al Hussein and Khan, 2023). As a result of carbon leakage, reductions in domestic emissions following the implementation of policies at the national level may not translate into reductions in global emissions and may even increase these (UNCTAD, 2022).

This is because there is a difference between the environmental impacts within a country's territory (production impacts) and those caused by producing and transporting the goods and services that a

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[1] · Data from https://unfccc.int/NDCREG, accessed November 1, 2023.

country consumes (consumption impacts). This gap is known as the environmental impact of trade. **About 30% of global CO_2 emissions are embodied in exports of goods and services (WTO, 2022).** External trade flows account for between 19.8% and 26.5% of total emissions in LAC, depending on whether household emissions and land use are included in the calculation (Mesquita Moreira and Dolabella, 2022).

In response to this situation, mechanisms to compensate for carbon leakage are being sought. The optimal solution would be to introduce a globally coordinated carbon-pricing mechanism, but this poses a number of design-related challenges (Cramton et al., 2017). Implementing international carbon pricing and calculating the carbon content of products and services require detailed, up-to-date information that may be lacking in some countries or sectors. A global carbon-pricing mechanism also requires considerable coordination between countries. Financial and technological transfers may also be necessary, which could further complicate negotiations (WTO, 2022).

There are a growing number of environmental provisions in trade agreements and unilateral trade measures that are strictly environmental in nature, which could be an alternative solution to this problem. Against this backdrop, the purpose of this chapter is to assess some of the challenges faced by LAC economies in applying these rules and to analyze alternatives at both the national and regional levels with the aim of mitigating their negative effects and adapting to the new rules of the game.

This introduction is followed by an analysis of the channels through which trade is linked to climate change. Section 3 then analyzes the evolution and scope of provisions included in global trade agreements, with a focus on agreements involving LAC countries. Sections 4 and 5 look at two unilateral trade measures that the EU has recently implemented and that are also on the agenda for some of LAC's other major trade partners: the ban on imports of goods originating in deforested areas and border taxes based on carbon content. In these sections, we analyze the potential impact of these two mechanisms on LAC's foreign trade. Finally, we draw some conclusions regarding the trade challenges posed by implementing these environmental regulations.

2. THE CONNECTION BETWEEN TRADE AND THE ENVIRONMENT

To better understand the environmental impacts of international trade, a conceptual framework has been developed in the economic literature that **identifies the channels through which international trade may contribute to increases or decreases in GHG emissions.** Specifically, it breaks down the impact of major trade flows into three distinct effects: scale, composition, and technique (Grossman and Krüger, 1993; Antweiler et al., 2001).

The scale effect is the increase in production, transportation flows, and the consumption of goods and services as a result of trade. All other factors being equal, increases in economic activity will lead to higher energy consumption, which will bring about a corresponding increase in GHG emissions.

The composition effect relates to the impact of trade on a country's production structure by reallocating economic resources to sectors with comparative advantages. Although international trade helps to improve economic efficiency, its effect on GHG emissions will depend on which sectors have comparative advantages. The composition effect triggers an increase (decrease) in emissions if a country's comparative advantage (disadvantage) is in carbon-intensive sectors. International trade may lead countries to specialize in industries that pollute more or less, depending on the availability of their factors of production. If a country's comparative advantage is in natural resource-intensive or carbon-intensive sectors such as fossil fuel extraction, specialization may lead to an increase in production for more polluting industries. In LAC, the expansion of the agricultural frontier is the main driver of deforestation, which is strongly linked to products with high export content.

Food production is the economic activity with the largest global water footprint—the total volume of freshwater used directly or indirectly to produce a product. A country's water footprint can increase its carbon footprint and lead to environmental problems such as soil degradation and water and air pollution. Conversely, a country can develop cleaner industries if its comparative advantage is in low-emission, energy-efficient sectors such as information technology, renewable energy, or services.

Finally, international trade can help improve production processes and techniques, thereby reducing emissions per unit of product. This mechanism is often referred to as the technology effect and can occur in two ways. First, trade facilitates the availability of and access to a greater variety of clean technologies. Furthermore, the growth in income resulting from trade liberalization will shift consumer preferences toward a greater appreciation of the environment. Greater social demands for environmental standards can lead to the implementation of stricter climate policies, thus promoting the green transition. Similarly, globalization and international trade also influence societies' consumption patterns, promoting lifestyles and practices that may be more environmentally sustainable or less so. In markets where consumers are aware of the importance of protecting the environment and the environmental impacts of consumption patterns, they can put pressure on companies and governments to adopt more sustainable practices.

Governments could direct the latter two **channels of transmission between trade and the environment—the composition and technical effects**—through environmental regulations and standards, trade policies, and international agreements to specifically design the measures governing this relationship to promote sustainable environmental practices that encourage the exchange of environmentally friendly goods and services. In this chapter, we address this point in specific relation to three policy instruments that could influence the composition and technical effects: trade agreements, import bans on deforestation-intensive goods, and border adjustments.

3. ENVIRONMENTAL PROVISIONS IN TRADE AGREEMENTS

Rising levels of concern about the impact of trade on the environment have prompted a growing public interest in analyzing **the role of environmental provisions (EPs) in trade agreements.** Countries include EPs through specific articles, the "greening" of chapters on other issues (including chapters devoted entirely to environmental matters), or side agreements. The motives often associated with including EPs in agreements include sustainable development and the global environmental agenda, promoting trade in environmental goods and services, and leveling the playing field among signatories to agreements (OECD, 2023; George and Yamaguchi, 2018).

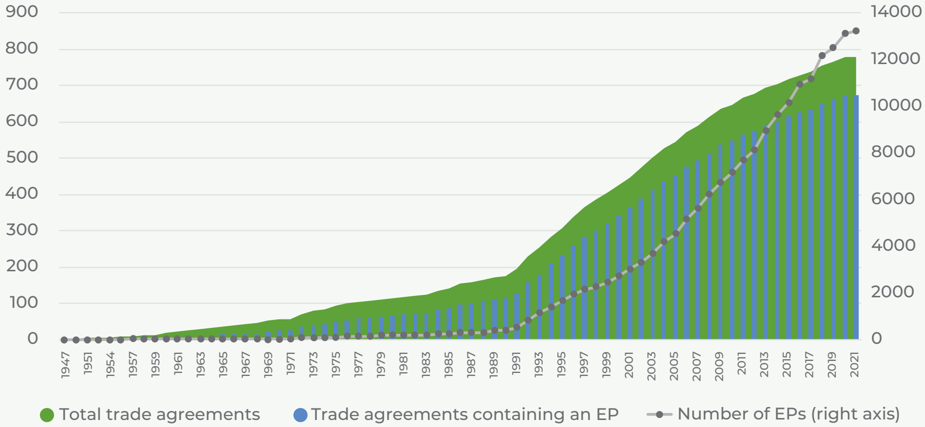
Recent studies have found that the inclusion of EPs in integration agreements contributes to environmental sustainability: countries that have ratified agreements with such provisions have lower pollution levels (Martínez-Zarzozo and Oueslati, 2016), lower GHG emissions, more limited deforestation, and less habitat destruction (Abman et al., 2021). Furthermore, there is also some evidence of ex ante and ex post effects as countries adapt during the negotiation process (Weber, 2021; Bstiaens and Postnikov, 2017). In any case, the results may mask reverse causality, possibly because countries that reduce emissions may export their domestic standards as a result of larger market size and bargaining power.

Weber (2021) underlines that the legal enforceability of environmental agreements means that these have the potential to trigger changes in domestic laws to strengthen environmental regulations; they may also provide legal mechanisms for private stakeholders and civil society organizations to seek environmental rights not covered by local regulations.

According to the Trade and Environment Database (TREND) developed by Morin, Dür, and Lechner (2018),[2] **the number of agreements with at least one EP has increased.** Before the 1990s, an average of 65% of trade agreements contained at least one EP. Since 1991, this share has risen to more than 90% (figure 1). However, Weber (2021) points out that the mere inclusion of one or a handful of EPs in an agreement does not have an impact on GHG emissions, but countries that include numerous clauses in their trade agreements tend to reduce their CO_2 emissions per capita more than those without such clauses. Although fewer agreements were signed per year between 2009 and 2021 than between 1991 and 2008, the average number of EPs per agreement has increased substantially, going from an average of 14 to an average of 47 in recent years.

Figure 1.
Evolution of Agreements Containing Environmental Provisions

Cumulative number of total trade agreements, trade agreements with at least one environmental provision, and cumulative number of provisions.



Source: Authors based on the TREND database. The right axis shows the number of EPs.

Of the 778 trade agreements signed between 1947 and 2021, only 104 do not contain EPs. The agreements that contain EPs of some sort include regional agreements, multilateral agreements such as the General Agreement on Tariffs and Trade (GATT)[3] and the 1994 World Trade Organization (WTO)[4] Agreement, and the 2014 Trade Facilitation Agreement and the Plurilateral Agreement on Government Procurement, all of which fall within the remit of the WTO.

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[2] · The database was updated in 2022 and includes all agreements signed up to 2021.
[3] · The article on exceptions to obligations under the GATT contains just two EPs.
[4] · The WTO Agreement includes 22 EPs, most of which relate to exceptions and coherence between international institutions.

The most common measures relate to trade in goods, which feature in 81% of the agreements in the TREND database (figure 2). Among the most common of these are exceptions to trade in goods that are connected to the conservation of natural resources or flora and fauna. Another is the right to develop or apply technical barriers to trade when these relate to the (promotion or protection of) the environment. The ability to overturn procedures for the adoption of technical barriers during an emergency is one more example.

Relations with other international institutions are the second-most-common type of EP, as they are included in 59% of agreements, including the Paris Agreement and the Kyoto Protocol. The most common of these EPs were mentions of other institutions or specific environmental agreements, the implementation of other agreements on environmental issues, and the preemption of other environmental agreements should these prove incompatible.

Third, cooperation clauses are included in 38% of the agreements. The enforceability of these clauses is limited, as the most common are vague obligations for countries to cooperate and exchange information on environmental matters or protective measures.

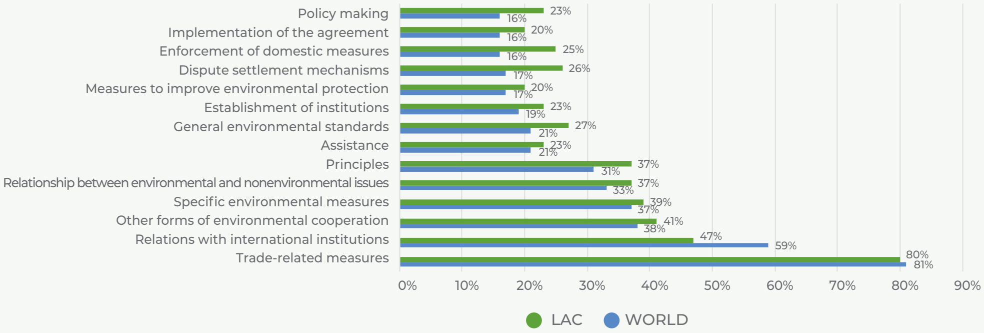
Fourth, specific environmental measures are included in 37% of the agreements signed during the period in question: the most common EPs relate to hazardous waste and biodiversity. Weber (2021) finds that the environmental impact is greater when specific clauses are included.

Fifth, measures on coherence between environmental and nonenvironmental issues are included in 33% of the agreements, mainly those seeking coherence with domestic trade or investment policies.

Figure 2.

Types of Environmental Provisions Included in Trade Agreements

Percentage of agreements signed between 1947 and 2021 containing at least one environmental provision, by group



Source: Authors based on the TREND database.

EPs relating to principles are found in almost a third of the agreements. The most common of these is a mention of the environment in the preamble to the agreements, one of the weakest and least enforceable types of EP. However, some important EPs fall within this group, such as the precautionary principle included in the Peru–Colombia agreement, based on which insufficient scientific evidence cannot be used to reject measures to protect the environment.

Assistance measures are included in 21% of global agreements, the most common of which is technical assistance or training provided by another signatory.

Another group of standards that are also found in 21% of the agreements are general environmental standards that stress it is inappropriate to relax environmental measures to attract investment.

The remaining measures are included in less than 20% of the agreements. The least common categories include obligations such as enforcement of domestic measures, dispute settlement mechanisms, or implementation of the agreement.

According to Weber (2021), strong enforcement mechanisms have an even greater impact on CO_2 emissions than specific environmental measures included in the agreements. Only 16% of the agreements include EPs related to implementation, most of which seek to establish a contact point for environmental issues, provisions for public participation in implementing these, and communication measures on the actions taken to implement the agreement.

The structure of measures in LAC is similar to these global patterns. However, there is a larger share of mechanisms for dispute resolution, enforcement of domestic measures (mainly binding obligations), and policy-making (mainly public participation in the adoption of environmental measures).

The agreements with the highest number of EPs involving an LAC country are the United States–Mexico–Canada Agreement (USMCA) and the agreements between the United States and Peru, Panama, Colombia, and Chile. Also noteworthy are the agreements between the EU and LAC countries (Central America, Colombia, Peru, Cariforum). Indeed, the EU–Cariforum agreement marked the start of the inclusion of EPs in EU agreements.

One of the countries whose agreements most often contain EPs is Chile, whose 2004 agreement with Singapore was the first in the region to include an environmental chapter. Chile is also unusual in having signed agreements with enforceability clauses (Weber, 2021). Its agreements with other LAC countries are among those with the highest number of EPs, particularly those with Brazil, Argentina, Ecuador, Uruguay, and Colombia.

LAC's agreements with Canada include stronger enforcement clauses than those with the EU. **Canada's agreements with LAC countries have some of the highest numbers of EPs, particularly those with Panama, Honduras, Colombia, and Peru.**

4. MEASURES TO REDUCE DEFORESTATION

Forests are an essential resource for human livelihoods and a critical factor for sustainable development. They provide global public goods and are a core component of the fight against climate change, as (i) they are the planet's main CO_2 sinks, absorbing 2.6 gigatons per year, which is equivalent to a third of the anthropogenic gases released by burning fossil fuels; (ii) they provide habitat for a large share of land-based biodiversity while harboring high levels of genetic diversity; and (iii) they provide humanity with a wide range of environmental services, such as regulating water flows, providing drinking water, protecting soils, and providing a large amount of food and raw materials (FAO, 2022a; Friedlingstein et al., 2020; IPCC, 2019; World Bank, 2016).

However, **the last 30 years have seen a significant increase in global tree cover loss,** bringing the issue to the forefront of public debate. The world is estimated to have lost approximately 420 million hectares (Mha) of forest to deforestation between 1990 and 2020.[5] This figure represents 10% of the world's total forest area (FAO, 2020). All the same, the data shows that the rate of global forest loss is slowing.

In fact, the annual deforestation rate has been declining since 1990–2000, such that 6 Mha less forest was cut per year in 2015–2020 than the annual average for the 1990s (figure 3). Although this trend seems encouraging, deforestation remains a global challenge. Even if this rate of reduction is maintained, it will take about another 25 years to reach Sustainable Development Goal (SDG) 15 ("Life on Land"), which focuses on halting global deforestation (FAO, 2022a).

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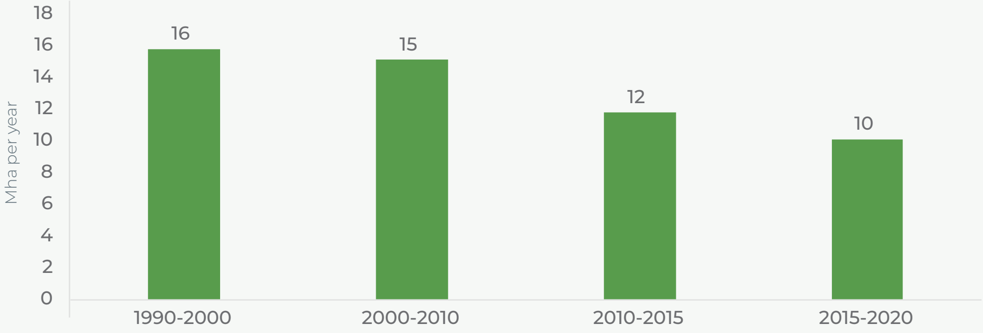
5 · There is currently no consensus on the definition of deforestation. There are two conflicting criteria: "loss of tree cover" and "land use." Global Forest Watch (Hansen et al., 2013) publishes based on satellite imagery applying the loss-of-tree-cover criterion, which only considers the biophysical properties of forests (minimum height thresholds, canopy cover, and tree cover). In this case, the methodology used does not distinguish if the logging occurred in rotation forests where the tree surface will grow back and deforestation is not permanent. The second methodology, which is used in FAO calculations, defines deforestation as the conversion of forests to other uses, such as agriculture and infrastructure. Unlike the previous methodology, this approach does not consider loss of tree cover to be deforestation if it is temporary and is regenerated naturally or through forestry activities. This article uses FAO data, as this is the benchmark in the international debate on global deforestation and for IPCC assessments of the impact of deforestation on climate change (Brown and Zarin, 2013).

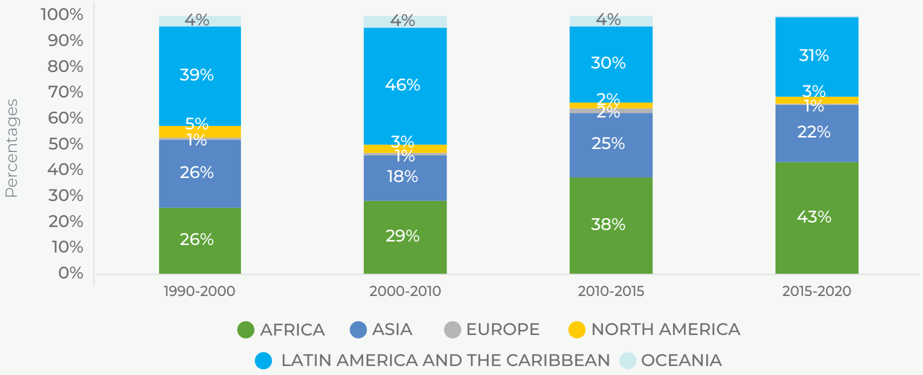
Between 2010 and 2015, **Africa was the continent that lost the largest area to deforestation, overtaking LAC, which had previously topped the list.** This shift is due to a significant decline in the average rate of deforestation in LAC, which fell from 6.1 Mha in the 1990s to half that in 2015–2020. The data for 2015–2020 show that more than 40% of global deforestation occurred in Africa. LAC is next in order of importance, accounting for a third of the total, while Asia ranks third with more than 20% of forest cover loss (figure 4). These three regions account for more than 90% of global deforestation.

Figure 3.
Global Deforestation Rate



Source: INTAL (IDB) based on FAO (2020).

Figure 4.
Deforestation Rate by Region



Source: INTAL (IDB) based on FAO (2020).

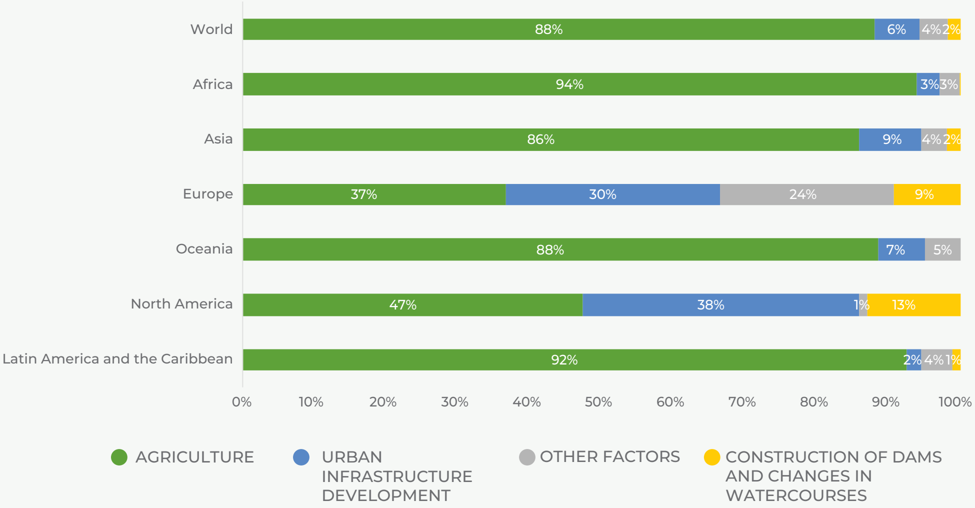
The empirical evidence suggests that **carbon sequestration is highest in subregions where tropical forests are abundant**—these are where the main global deforestation hotspots are currently located (Xu et al., 2021; Schimel, Stephens, and Fisher, 2014; Lewis et al., 2009). Tropical forests in Central Africa, South America, and Southeast Asia concentrate carbon stocks of up to 130 t/ha, while Northern Hemisphere forests retain only a third of this value (Farrokhi et al., 2023).

The most significant driver of global deforestation appears to be the expansion of the agricultural frontier, be it for commercial or subsistence agriculture (Pendrill et al., 2022; FAO, 2022b; Jayathilake et al., 2021; Seymour et al., 2019; Laurance et al., 2014; Lewis, Edwards, and Galbraith, 2015; Hosonuma et al., 2012; Kissinger et al., 2012; Gibbs et al., 2010; Geist and Lambin, 2002). FAO (2022b) estimates that about 90% of global deforestation in 2000–2018 was associated with growth in the agricultural sector (including livestock). This driver is particularly strong in Africa and LAC, where agricultural activity accounts for 95% and 94% of deforestation, respectively. It also accounts for 86% and 88% in Asia and Oceania, respectively, while in North America and Europe, the main drivers of deforestation are other factors that are less significant globally (figure 5).

Figure 5.

Drivers of Deforestation by Region, 2000–2018

Percentages



Source: INTAL (IDB) based on FAO.

Deforestation-intensive products are concentrated in a handful of commodities that are closely linked to agricultural and forestry activities. Their characteristics vary by country and continent. According to calculations by Pendrill et al. (2019) for 2005–2013, regional trends for Latin America show that beef has the highest “deforestation content”[6] (more than 60% of deforested land is used for beef production). However, other products also stand out in specific countries, including oilseeds and cereals, especially in Argentina (47%) and Paraguay (49%), and coffee in Honduras (17%), Ecuador (10%), and Peru (7%). In Asia, palm oil, forestry products, and rubber are the main drivers of deforestation, while in Africa it is beef, although coffee and cocoa are also relatively important.

Although deforestation is mainly driven by domestic demand, **a considerable share of the products associated with forest loss are major export products.** This factor is particularly significant given that global demand for these commodities is constantly increasing, putting more pressure on land-use change (Pendrill et al., 2019; Wood et al., 2018; Mayfroidt et al., 2013; Yu et al., 2013; Weinzettel et al., 2013). International trade thus creates a link between supply and demand for land use between different countries (Farrokhi et al., 2023). Countries with a deficit in the domestic supply of land-intensive products must thus rely on the international market for their supply, creating incentives for production to be increased in countries where land is a relatively abundant factor. If international demand cannot be met by increasing land productivity, one alternative is to expand the agricultural frontier through deforestation.

In response to this global problem and in the absence of a binding multilateral agreement[7] to halt the destruction of forests, particularly tropical forests, **some developed countries are using trade policy as a unilateral tool to restrict imports of deforestation-intensive goods** to make consumption and production patterns more sustainable and environmentally friendly. One such precedent has emerged in the United States, which has temporarily blocked imports of Peruvian forest products as part of a sector-specific arrangement within the

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7 · Following the methodology of Pendrill et al. (2019), the total amount of deforestation embodied in the production of a given commodity in a specific year is calculated as the total deforestation attributed to the land use that produced that commodity in the previous T years, divided by T (where T is the amortization time).

7 · The most far-reaching international agreement on this matter is the Glasgow Declaration on Forests and Land Use, signed at the 2021 UN Climate Change Conference (COP26) by more than 140 countries. In it, the signatories expressed their willingness to work collectively to “halt and reverse forest loss and land degradation by 2030,” but the declaration contains no binding clause to that effect (Abdenur, 2022).

free trade agreement between the two countries on the grounds that they are made from illegally harvested timber.[8,9] In the same vein, the United States is currently considering implementing trade measures to combat the international deforestation associated with the processing of agricultural commodities (US Department of State, 2023). Meanwhile, the United Kingdom included provisions in schedule 17 of the Environment Act 2021[10] that prohibit the importation of any commodity with a risk of deforestation if the processing of the product violates local law in the exporting country. This standard requires importers to follow a due diligence process to ensure that products are deforestation-free. However, the regulations specifying the requirements for compliance with this process and the universe of goods covered by the measure have yet to be formalized (Cosbey et al., 2023). Finally, through state-owned enterprises such as COFCO International, the country's largest food retailer, the People's Republic of China has recently proved itself willing to curb deforestation driven by commodity exports. Indeed, in November 2023, it sealed a deal worth more than US$30 million to purchase soybeans from Brazil that includes a clause requiring the oilseed to be deforestation-free, the first time a provision of this sort has been specified.[11]

The most concrete example of such initiatives is the EU's Regulation on Deforestation-Free Products (EUDR),[12] which will come into force in June 2023 as part of the strategy to combat climate change embodied in the EU's Green Deal.[13] The main objective of the EUDR is to minimize the consumption and domestic production of goods from supply chains with ties to deforestation or forest degradation. To this end, the EUDR prohibits European exports to third markets and the sale of goods in the EU market that do not comply with a set of procedures and measures designed to ensure that goods are deforestation-free. It covers a list of commodities that are consumed

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8 · For more information, see the following USTR press release (2023): https://ustr.gov/aboutus/policy-offices/press-office/press-releases/2023/october/ustr-announces-enforcement-action-block-illegal-timber-imports-peru.

9 · In 2021, the US FOREST Act was introduced in the US Congress, which aims to restrict the entry of goods made from illegally deforested land in the country of origin. It is currently stalled in the Senate Finance Committee (IDB, 2023).

10 · Environment Act 2021, United Kingdom. Available at: https://www.legislation.gov.uk/ukpga/2021/30/enacted#:~:text=An%20Act%20to%20make%20provision,that%20fail%20to%20meet%20environmental.

11 · https://www.reuters.com/sustainability/land-use-biodiversity/chinas-cofco-modern-farming-group-sign-deforestation-free-soybean-deal-2023-11-08/.

12 · https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32023R1115.

13 · This consists of a package of initiatives whose main objective is to make the EU carbon neutral by 2050. It focuses on making the EU economy environmentally sustainable, addressing climate change mitigation, adaptation to new climate conditions, biodiversity conservation, and the promotion of a just transition to a circular, low-carbon economy. The Green Deal seeks to integrate environmental considerations into all policies and sectors and to stimulate innovation and green investment to achieve climate and sustainability goals.

at high rates in the EU and are closely linked to the expansion of the global agricultural frontier, such as beef, timber, palm oil, soy, coffee, rubber, cocoa, and products that derive from these industries, such as leather, chocolate, cellulose, and furniture.[14] Unless a company can prove through the due diligence process established by the law (box 1) that the products covered by the EUDR have been produced on farms or plantations or farms that have not been deforested since December 31, 2020, importing these into the EU market and exporting them to third countries will be prohibited.

As noted by Calvo et al. in chapter 2, although the EUDR will enter into force in June 2023, most of the obligations established by the due diligence process will not apply until December 2024. This includes the geolocation of the land where the goods to be traded were grown or reared and the traceability process for the final products and/or the inputs used to produce them.[15] The EUDR thus poses significant challenges for exporting countries, as compliance will require strengthening the institutional capacity of central and subnational governments and implementing sustainable production practices (Calvo et al., 2023).

The EU is one of the world's largest importers of deforestation-intensive commodities on the international market. When global purchases are broken down by the origin of the products covered by the EUDR, the data reveals the bloc to be among the world's top three importers for almost all these products. Coffee and cocoa-based products are particularly significant: the EU is the largest global importer, absorbing 35% and 23% of the total supply, respectively (figure 6). The entry into force of the due diligence process will have an impact on i) international trade in raw materials by excluding international suppliers from the European market if their products represent a high risk of deforestation and/or if they do not adapt their production processes to the EUDR's requirements, and ii) the EU's capacity to obtain supply from countries that meet these requirements.

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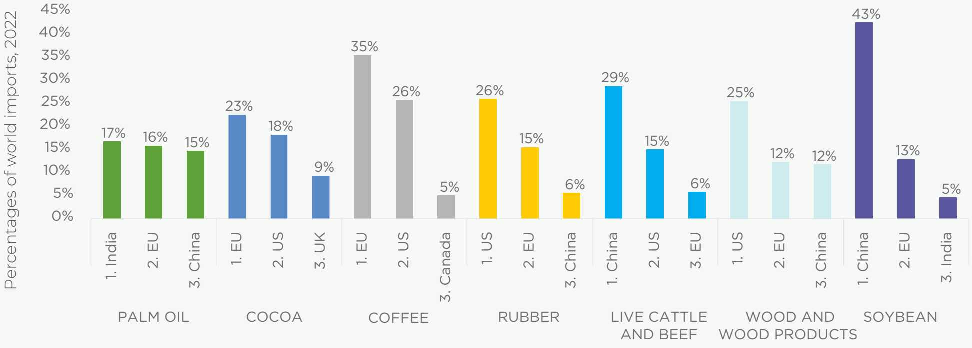
[14] · The Commission intends to propose a gradual extension of the list of products to be regulated and to review and update the list regularly on the basis of new data.

[15] · For more information on the due diligence process, see annex A.

Figure 6.

Main Importers of Products Covered by the EUDR



Source: INTAL (IDB) based on WITS.
Note: EU data does not include intrabloc trade.

The implementation of this measure will have an impact on LAC's foreign sales. This will vary according to each country's importance as a supplier, the relative significance of the different export sectors, the role of these in regional deforestation, and each country's capacity to comply with the due diligence process required by the law, among other factors. Based on the universe of tariff codes identified in annex I of the EUDR and the data on average export values for the same group of goods in 2019 and 2021,[16] the implementation of the standard will mean that more than US$23 billion in LAC exports to the EU will have to comply with the requirements of the due diligence process established by law to enter the EU market (table 1).[17] This represents 15% of LAC's foreign sales of these products. A breakdown of the value of exports by product type reveals the goods with the most significant shares to be soybeans (43%), wood (22%), and coffee (34%). LAC is one of the primary sources of EU imports of soybeans and coffee, accounting for 53% and 27% of the bloc's international purchases, respectively. The LAC export sectors most exposed to the EUDR are palm oil (54%), coffee (34%), and cocoa (18%), although the EU absorbs a high share of the region's global exports of almost all products.

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16 · Data for 2020 was excluded from the calculations because it was considered an atypical year, given the effects of the pandemic on international trade flows.

17 · Although not all LAC exports are deforestation-intensive, the EUDR does not indicate how the risk indicators to be published by the EU will be calculated and whether they will be differentiated by the different geographic areas of the countries. These aspects will be clarified when the regulations for the EUDR are published in the Official Journal of the EU. Regardless, to enter the EU market, all exports must be deforestation-free and comply with the due diligence process established by law. The stringency of this process will depend on the benchmarking system devised by the European Commission.

Table 1.

LAC Exports Covered by the EUDR

Millions of US dollars and percentages, averages for 2019 and 2021

PRODUCT	EXPORTS FROM LAC TO THE EU		SHARE OF LAC EXPORTS TO THE EU IN EXPORTS TO THE WORLD	SHARE OF EU IMPORTS FROM LAC IN IMPORTS FROM THE WORLD
	Millions of US$	%	%	%
Soybeans	9,948	43%	15%	53%
Coffee	4,247	18%	34%	27%
Wood	5,034	22%	13%	2%
Live cattle and beef	1,972	8%	9%	9%
Palm oil	1,311	6%	54%	9%
Cocoa	487	2%	18%	2%
Rubber	207	1%	5%	0%
Subtotal	**23,207**	**100%**	**15%**	**7%**

Source: INTAL (IDB) based on COMTRADE.

When the share of exports to the EU affected by the regulation is calculated in relation to the total export basket of each LAC country, the results show some variation between countries.[18] **On average, the MERCOSUR countries are more exposed than the rest of the subregions, as they are major global suppliers of beef, soybeans, and cellulose and have close trade relations with the EU (figure 7).** In Central America, Honduras (9%), Guatemala (5%), and Nicaragua (2%) are significant due to their comparative advantage in the production of coffee and palm oil and the importance of the EU market as a destination for these products. Finally, in the aggregate for LAC, exports covered by the standard account for 2% of total exports. Again, exports of soybeans, wood products, and coffee account for much of the exposure.

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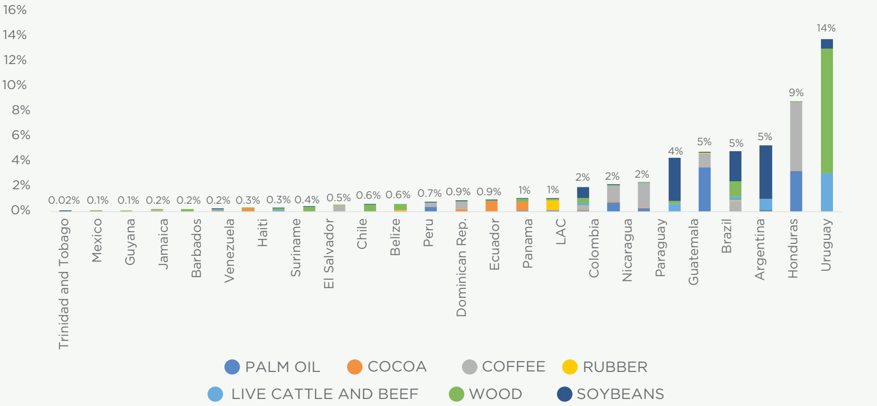
[18] · $X_{i,g,d}$ / *Xitot* where $X_{i,g,d}$ are the exports of products g (covered by the EUDR) from country i to destination d (EU); Xitot are the total exports of country i to the world.

Figure 7.

Exposure of LAC Countries to EU Deforestation-Free Requirements, Averages for 2019 and 2021

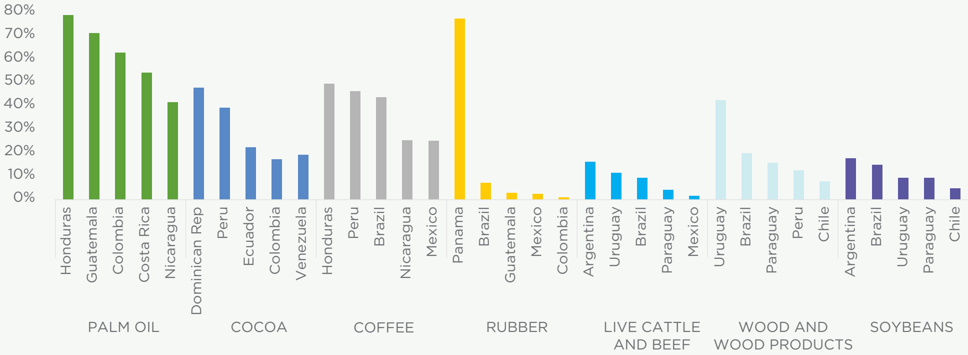


Source: INTAL (IDB) based on COMTRADE.

However, **analyzing the impact at the sector level reveals different results.** When we calculate the indicator as the share of exports of each of the products covered by the EUDR to the EU over exports of the same good to the world, countries whose export baskets are not significantly exposed[19] in general may reveal themselves to be more exposed. For example, while just 1% of Peru's export basket is exposed, the EUDR affects 46% and 39% of its total foreign sales of coffee and cocoa, respectively (figure 8). Similarly, 48% of the Dominican Republic's cocoa exports to the world would be affected.

Figure 8.

Sector-Specific Exposure of LAC Countries to EU Deforestation-Free Requirements, Averages for 2019 and 2021



Source: INTAL (IDB) based on COMTRADE.

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[19] · $X_{i,g,d}/X_{i,g,w}$, where Xi,g,d are exports of product g (covered the EUDR) from country i to destination d (EU); $X_{i,g,w}$

In this context, some countries in the region[20] have expressed their concern over the EU measure at the WTO Committee on Agriculture, arguing that "the country assessment criteria and benchmarking system are inherently discriminatory and punitive in nature. Its most likely effect will be to generate trade distortion and diplomatic tensions, without benefits to the environment. Furthermore, it imposes additional controls, entails reputational risks for companies, and is likely to penalize producers in developing countries, especially smallholder farmers and SMEs." They also expressed concern about "...uncertain and discriminatory nature of the scope of products; definitions that are not multilaterally agreed; retroactive cut-off date; burdensome due diligence mechanism and subjective risk assessment criteria; costly and impractical traceability and geo-localization requirements; and insufficient unilaterally defined transition period, which could increase costs and have negative social and economic consequences for developing countries."[21]

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20 · Argentina, Brazil, Colombia, Guatemala, Paraguay, Peru, Honduras, Ecuador, and Bolivia.
21 · https://docs.wto.org/dol2fe/Pages/SS/directdoc.aspx?filename=q:/G/AG/GEN213.pdf&Open=True.

5. CARBON-RELATED BORDER ADJUSTMENT MECHANISMS

In the absence of coordinated climate action, countries with more ambitious targets are resorting to border adjustment mechanisms. A border adjustment is a levy on the carbon embodied in products imported from an economy with a lower carbon price than the importing country or on imported products whose embodied carbon has not been priced. In other words, the price of certain carbon-intensive goods would be adjusted to reflect carbon emissions from their production processes (Dietrich Brauch et al., 2021). Adjustments could also be made by refunding the domestic carbon price that companies pay when they export their products to compensate for the higher carbon price they face domestically compared to companies in the country they are exporting to.

The EU is not alone in proposing a border adjustment mechanism: Canada, the United Kingdom, and the United States have also included instruments of this sort in their climate change agendas. Border adjustment is part of the EU's Fit for 55 policy package, which aims to reduce net GHG emissions by at least 55% by 2030 compared to 1990 levels. **To achieve this, the EU intends to gradually replace free emissions allowances with a carbon border adjustment mechanism (CBAM)[22].** The goal of the CBAM is to ensure that carbon is priced equally for imports and domestic products. In other words, companies that want to sell their products on the EU market will have to purchase carbon allowances equal to the carbon price that would have been paid to produce the goods in the EU. Although the CBAM came into force in 2023, the actual rates will not apply until 2026. According to UNCTAD (2022), the scheme will address carbon leakage and reduce emissions but will also lead to trade diversion: trade within the EU is expected to increase, while flows from trade partners will be diverted to other regions. Total exports from developing countries will fall more than those from developed countries, as the former use more carbon-intensive production methods in the targeted sectors. However, the CBAM also aims to encourage third-country producers trading with the EU to use more efficient technologies that produce lower GHG emissions (Cárdenas and Cazzolo, 2023).

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[22] · The EU Emissions Trading Scheme (EU ETS) provides free emissions allowances to domestic industrial manufacturers up to a certain limit, but they have to buy additional permits at auctions or on the secondary market. Sectors with a high risk of leakage are allocated most, if not all, of the allowances they need "for free," to avoid pressuring these industries to relocate outside the EU (Columbia, 2021). However, free allowances have been criticized for hampering the EU ETS by reducing the ambition of emissions-intensive trade-exposed (EITE) sectors to reduce emissions and for being inconsistent with the EU's goal of net-zero emissions by 2050 (Beaufils et al., 2023).

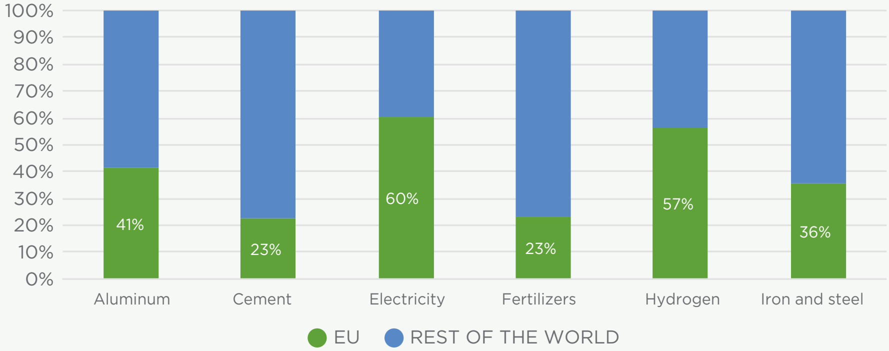
The products initially affected by the CBAM are aluminum (HS chapter 76), cement (heading 2523), electricity (heading 2716), fertilizers (chapter 31), hydrogen (subheading 280410), and iron and steel (chapters 72 and 73). By 2030, sectors such as petroleum refining, fuels, all metals, pulp and paper, glass and ceramics, organic acids and chemicals, air and sea transportation, and lime will be added (European Union, 2023). Agriculture is one of the other sectors that is becoming part of the EU climate agenda. In chapter 4 of this publication, Cabrini et al. analyze the potential impact of implementing a carbon adjustment mechanism at the EU border on the international beef trade, with a particular focus on Argentinian beef exports.

The EU is a major importer of the products included in the first phase of the mechanism (figure 9), accounting for 60% of global imports of electricity, 57% of hydrogen, 41% of aluminum, and 36% of iron and steel. However, these products account for only a small share of world trade: together, they represent 5% of global imports and 6% of the EU's.

Figure 9.

EU Share in World Imports of Products Included in the CBAM

Average percentages for 2019 and 2021

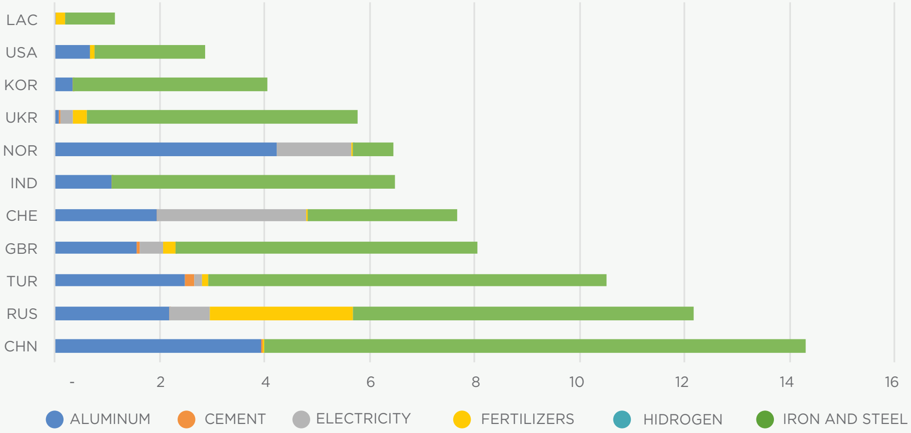


Source: Authors based on COMTRADE.

The CBAM therefore has important implications for the EU's trade relations. The first concern relates to the substantial costs that non-EU partners are likely to face due to increased tariffs on CBAM-affected products imported by the EU. Countries such as China, Russia, Turkey, and China are likely to be among the most affected in terms of the value of exports to the EU (figure 10).

Figure 10.
Main Exporters of Products Affected by CBAM and LAC

In thousands of dollars, averages 2019 and 2021



Source: Authors based on COMTRADE.

The initial impact for LAC is comparatively small because exports of this group of products to the EU do not account for a large share of the region's total foreign sales (table 2). In the average for 2019 and 2021,[23] LAC exports to the EU from the six sectors that will be affected by the CBAM amounted to US$1.76 billion, representing just 0.1% of the region's total external sales and 1.6% of exports to the EU. However, the CBAM will have a greater effect on some countries and sectors: in the case of fertilizers and hydrogen, the EU's share as a destination for LAC exports amounts to 14.0% and 23.1% of the total, respectively. Moreover, the region is not a major supplier of any of these products to the EU, which increases the likelihood of EU buyers seeking alternative markets if LAC products do not comply with the new requirements.

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[23] · We used data from these two years to ensure the information was up-to-date and had a high degree of coverage while avoiding the records for 2020, which include the impact of the Covid-19 pandemic on trade.

Table 2.
Exports from LAC Covered by the CBAM

Millions of dollars and percentages, average for 2019 and 2021

PRODUCT	EXPORTS FROM LAC TO THE EU		SHARE OF LAC EXPORTS TO EU IN TOTAL LAC EXPORTS	SHARE OF EU IMPORTS FROM LAC IN TOTAL EU IMPORTS
	Millions of US$	%	%	%
Aluminum	139	8%	3.6%	0.2%
Cement	21	1%	3.9%	0.8%
Electricity	0	0%	0.0%	0.0%
Fertilizers	494	28%	14.0%	3.4%
Hydrogen	0	0%	23.1%	0.0%
Iron and steel	1.100	63%	5.2%	0.5%
Total	1,755	100%	5.5%	0.5%

Source: Authors based on COMTRADE.

In some LAC countries, the sectors covered by the CBAM account for a high proportion of total exports to the EU. Although the amounts in question are mostly small, the exports of specific products from some countries are more severely impacted, such as aluminum for the Bahamas, Belize, Brazil, Mexico, and Venezuela; cement for Colombia; fertilizers for Chile and Trinidad and Tobago; and iron and steel for Brazil, Mexico, and Venezuela (table 3). According to Conte Grand et al. (2022), the potential expansion of CBAM products in the EU could increase the impact to 14% of LAC countries' exports.

Table 3.
EU's Share in Total LAC Exports in the Sectors Covered by the CBAM by Country

Average percentages of the total for 2019 and 2021

COUNTRIES	ALUMINUM	CEMENT	ELECTRICITY	FERTILIZERS	HYDROGEN	IRON AND STEEL
ARG	1.5%	0.1%	0.0%	0.0%	0.0%	2.8%
BHS	100.0%	0.0%	0.0%	100.0%	0.0%	67.4%
BRB	3.4%	0.0%	0.0%	0.0%	0.0%	25.7%
BLZ	79.4%	0.0%	0.0%	0.0%	0.0%	15.0%
BOL	5.2%	0.0%	0.0%	0.0%	0.0%	2.9%

BRA	4.6%	0.0%	0.0%	0.1%	1.4%	9.6%
CHL	0.6%	0.0%	0.0%	26.9%	0.0%	0.9%
COL	0.4%	41.2%	0.0%	3.0%	0.0%	2.5%
CRI	0.9%	0.0%	0.0%	0.4%	0.0%	3.1%
DOM	0.4%	0.0%	0.0%	1.4%	0.0%	0.5%
ECU	0.1%	0.0%	0.0%	0.2%	0.0%	1.0%
SLV	0.2%	0.0%	0.0%	0.2%	0.0%	0.1%
GTM	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
GUY	53.0%	0.0%	0.0%	100.0%	0.0%	32.9%
HTI	0.1%	100.0%	0.0%	0.0%	0.0%	1.7%
HND	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
JAM	0.0%	0.0%	0.0%	75.9%	0.0%	1.0%
MEX	1.1%	0.4%	0.0%	2.5%	0.0%	1.6%
NIC	4.0%	0.1%	0.0%	46.2%	0.0%	3.9%
PAN	0.1%	2.6%	0.0%	1.6%	0.0%	4.8%
PRY	0.1%	0.0%	0.0%	0.0%	0.0%	2.8%
PER	0.8%	0.0%	0.0%	0.1%	0.0%	1.4%
SUR	18.0%	14.5%	0.0%	0.0%	0.0%	53.4%
TTO	0.7%	0.0%	0.0%	14.8%	0.0%	2.3%
URY	0.4%	0.0%	0.0%	0.0%	0.0%	2.6%
VEN	18.0%	12.8%	0.0%	0.0%	0.0%	32.2%
LAC	3.6%	3.9%	0.0%	14.0%	23.1%	5.2%

Source: Authors based on COMTRADE.

Several aspects of the implementation of CBAM need to be taken into account **to minimize the impact of the measure on trade and the economy and to support the objective of reducing the impacts of climate change.**

First, it should be noted that the proposed methodology for calculating the CBAM is based on the carbon content of the product (including both direct and indirect emissions) and the difference between the EU carbon price and that of the trading partner.[24] Direct emissions are those from sources owned or controlled by the company. Indirect emissions include emissions from the generation of electricity purchased and consumed by the company. In this context, the first challenge facing LAC countries is **the availability of tools for measuring emissions.**[25] There are over 100 carbon accounting methodologies, each of which yields different results and presents these in different ways (Dietrich Brauch et al., 2021). One option is to use nationally determined benchmark emissions levels for the products in question. Another is to use benchmark levels specific to the exporting country needing to comply with the CBAM. Since emissions intensities for a given product may vary considerably from one country to another, this design feature may make the system less effective at achieving its objectives (WTO, 2022).

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[24] 24 · Two electricity-intensive sectors, aluminum and steel, will initially continue to receive carbon offsets in the EU to cover their indirect emissions, resulting in only their direct emissions being taken into account. However, this will only be the case until these offsets are phased out (Maliszewska et al., 2023).

[25] 25 · The CBAM does not include scope 3 emissions. This is an optional reporting category that allows the inclusion of the remaining indirect emissions that are a consequence of the company's activities but occur at sources that are not owned or controlled by the company, emissions that result from the value chain. https://ghgprotocol.org/sites/default/files/standards/ghg-protocol-revised.pdf..

For carbon-competitive developing countries—which is the case for most LAC countries—implementing **a comprehensive accounting system that accurately measures emissions** is essential to enable them to prove their competitiveness (Dietrich Brauch et al., 2021). However, it should be noted that this would entail considerable administrative and compliance costs for governments and companies. The CBAM requires producers to provide proof of the GHG emissions embodied in their products. This could impose a significant technical and administrative burden on less developed countries, which already face some of the highest trade barriers in the world, as they may not have the capacity to calculate these emissions and administer this mechanism (Berahab, 2022).

The CBAM also takes into account the carbon price in the country of origin in order to avoid double taxation. Between October 2023 and December 2025, data will only be collected to improve the methodology for calculating indirect emissions, but from then on, importers will present certificates and pay the carbon price difference. There are currently around 70 carbon-pricing initiatives worldwide, covering less than 24% of global GHG emissions.[26] Most have been adopted in developed economies.

Only five countries in LAC have implemented a national carbon-pricing system: Argentina, Chile, Colombia, Mexico, and Uruguay. The schemes take the form of taxes in all these countries except Mexico, which also has an Emissions Trading Scheme. In LAC, carbon prices are no higher than US$5 per metric ton (World Bank, 2023), while in the EU, allowances trade at between US$85 and US$105. In other words, all countries in LAC will have to pay the CBAM to enter the European market, to a greater or lesser extent. This has serious implications: the CBAM would imply a net financial transfer from LAC to the EU. To avoid this, LAC countries could adopt policy reforms focusing on the products covered by the CBAM to avoid or minimize these transfers. These could entail some form of carbon pricing to generate domestic revenues and use them to support countries' transition to a low-carbon economy or to compensate sectors for possible cost increases (World Bank, 2023; Cárdenas and Cazzolo, 2023).

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[26] · Data from https://carbonpricingdashboard.worldbank.org/map_data, accessed November 1, 2023.

In this sense, **the CBAM will have distributional consequences at the international level that are potentially important for LAC countries.** Given that many of the EU's LAC partners do not have a carbon-pricing system in place and the prices of those that do are generally much lower, the debate revolves around whether the revenues generated by the CBAM should be transferred to the budget of the implementing country or used specifically to support climate change mitigation actions, for example in developing countries.

In the absence of adjustment measures, carbon-pricing policies may have a greater impact on low-income regions and exporters of fossil fuels and emission-intensive products (WTO, 2022). In addition, smaller firms could be more affected by carbon-pricing schemes as they would be unable to invest in less carbon-intensive technologies and production processes (UNCTAD, 2022). Logic suggests that resources to support climate change mitigation should be allocated where the effectiveness of decarbonization spending is greatest—that is, in developing countries, which tend to use less efficient technologies and have more potential to replace high-carbon energy with low-carbon energy (Bellora and Fontagné, 2022; WTO, 2022).

An additional problem with the CBAM calculation is that **there is a wide variety of policies to reduce emissions,** ranging from carbon pricing to regulation or subsidies. An exporting country may claim that it can achieve the same GHG emissions reduction target with an instrument other than a carbon tax. Difficulties arise in comparing the implicit price of different instruments and considering whether this implicit price should be taken into account when calculating border adjustments (Bellora and Fontagné, 2022).

This difficulty also raises questions about **compatibility with multilateral international trade rules.** Border adjustments will inevitably be assessed differently depending on the extent of each country's environmental regulations, its technology levels, the availability of carbon pricing, and other issues. Consequently, even if the border carbon tax is applied through a common standard, some experts argue that there will still be discrimination in practice, which would violate one of the WTO's principles, most-favored-nation (MFN) treatment. This obliges member countries to refrain from discriminatory trade practices and to grant the same benefits that apply to a particular product to all similar products from member countries. In other words, a border adjustment could violate MFN treatment regarding other countries' exports of similar products (Byeongho et al., 2021).

In its review of the EU's trade policy, the WTO deemed that the CBAM was consistent with the established rules (WTO 2023a and 2023b). The mechanism was analyzed in the light of the MFN principle and GATT article XX, which contains provisions on exceptions, stating that countries may adopt policies that are inconsistent with GATT disciplines but are necessary to protect human, animal, or plant life or health (paragraph b), or relating to the conservation of exhaustible natural resources (paragraph g) (Conte Grand et al., 2022). Interventions must not constitute a means of arbitrary or unjustifiable discrimination or disguised protectionism (Liebreich, 2021; Dietrich Brauch et al., 2021). Moreover, Article II, paragraph 2 of the GATT allows parties to impose additional charges as long as they are equivalent to an internal tax on domestic products. Even if border adjustment were to be implemented and additional charges imposed on imported products to account for their carbon emissions, certain rules, such as national treatment and MFN treatment, would still have to take precedence. For this reason, it is critical that the carbon intensity of each imported product is accurately calculated to ensure that there are no miscalculations that could lead to less favorable treatment of foreign products (Dietrich Brauch et al., 2021). If the CBAM qualifies as an indirect tax, has the objective of protecting the health and life of humans and animals, preserving the life of plants, or conserving nonrenewable natural resources and does not discriminate against imports, it should comply with WTO law.

Border adjustments could lead to trade conflicts between regions imposing such levies and those that must pay them (Byeongho et al., 2021; WTO, 2022). More generally, some WTO members have called for a careful analysis of whether the CBAM risks becoming a barrier to trade (Cárdenas and Cazzolo, 2023). For example, some simulations have shown that it would be optimal for some economies to implement countermeasures against the CBAM to limit the adverse economic effects of this (Böhringer, Carbone, and Rutherford, 2016).

It would be more advantageous for large exporters of energy products to impose retaliatory tariffs than to accept the EU's unilateral tariffs. These reciprocal tariffs will offset the effect of the CBAM, thereby strengthening the price competitiveness of their energy products, which would increase non-EU exports of energy products (Byeongho et al., 2021).

Even if other countries choose to adopt such systems, the structure of each national system will vary, as will the many assumptions needed to calculate the appropriate tax in each case. Border carbon taxes will thus differ not only by product and destination market but also by country of origin and company, which would be a significant departure from the WTO's MFN principle (Berahab, 2022). While the proliferation of carbon-pricing schemes points to the urgency of addressing climate change, they may lead to a complex system of national and regional regimes, creating tensions that could derail climate diplomacy and trade liberalization at a time when these are needed more than ever (WTO, 2022).

In addition to WTO agreements, concerns have also been raised that border adjustments could violate the commitment to **common but differentiated responsibilities (CBDRs)** enshrined in the Paris Agreement. This is the principle of equity on which the global climate regime is based (WTO, 2022). According to this principle, all governments are responsible for addressing climate change but not equally so—economies that industrialized earlier have contributed more to environmental degradation than emerging or newly industrializing economies.

The CBDR principle also reflects differences in countries' economic capacity to contribute to mitigation and adaptation efforts. If a border adjustment mechanism is introduced by high-income economies with more ambitious mitigation targets, the adverse terms-of-trade effects would be concentrated in low-income regions, potentially bringing such schemes into tension with the CBDR principle (Böhringer et al., 2022). However, claims that border adjustment mechanisms are incompatible with the Paris Agreement could be successfully rebutted if it is argued that they would foster ambition at the national level and help address leakage, based on articles 2.1 and 4.1 of the Paris Agreement, which relate to strengthening the global response to climate change and achieving a balance between sources of carbon production and sequestration

(Dietrich Brauch et al., 2021).

However, border adjustment mechanisms could have other positive effects in the fight against climate change: most importantly, they could encourage foreign countries directly affected by border adjustment to adopt more ambitious carbon pricing to avoid it (Böhringer et al., 2022). They could also encourage countries to diversify their economies away from producing emissions-intensive goods. Compliance with border adjustments would require firms to report the amount of carbon emissions embodied in the products they trade in order to calculate the tariff associated with them. Compliance with this requirement could help increase the transparency of carbon emissions and motivate companies and individuals to make more climate-friendly investment and purchasing decisions (WTO, 2022).

6. CONCLUSION

Despite the global nature of climate change, mitigation responses are still only implemented at the national level. As a result, commitment levels vary significantly between countries. This disparity in policies can lead to carbon leakage, which means that reductions in domestic emissions may not lead to a fall in global emissions, and economies with more ambitious climate policies may lose competitiveness.

Given this context and the difficulty of achieving coordinated action at the global level, **possible strategies for offsetting carbon leakage include trade-related mechanisms** such as including provisions in trade agreements and applying unilateral trade policies. The proliferation of these measures will have a significant impact on international trade flows and will particularly affect some countries and sectors in LAC.

The inclusion of EPs in trade agreements has increased exponentially in recent decades, and most of the agreements in force now contain clauses related to climate change. This trend is also evident in LAC. Agreements between countries in the region and the United States include the highest number of such provisions; at the intraregional level, Chile's agreements are noteworthy in this regard.

The empirical evidence shows that countries with EPs in their agreements have lower emissions and that signing agreements may lead to changes to domestic environmental legislation. However, the unilateral measures implemented by some of the most environmentally ambitious economies reveal that agreements only have a limited impact on achieving climate goals, especially those set out in the Paris Agreement.

There are two noteworthy measures that the EU has recently implemented as part of its Green Deal and that are also on the agendas of other major economies: a ban on imports of goods from deforested areas and border taxes based on carbon content.

Given that LAC is the region with the second highest rate of deforestation in the world after Africa and that the comparative advantage of several countries in the region is concentrated in sectors such as agriculture and livestock, **the EU's ban on imports of goods from deforested areas requires special attention.** A significant proportion of goods affected by the EUDR are exported to EU economies, and these goods will need to be adapted to the new requirement in the immediate future to continue to access this market.

Another measure to limit carbon leakage at the heart of the trade agenda is the CBAM, which seeks to ensure that foreign competitors face the same carbon costs as domestic producers. The CBAM is expected to help reduce carbon leakage and provide incentives for carbon pricing in the rest of the world, thereby helping fight climate change. However, it could also have adverse effects on relatively less developed regions, which is true of most of LAC. Although the direct impact on exports is relatively small, some sectors in certain countries will be disproportionately affected, and the challenges will be greater for economies with fewer resources to meet the requirements of border adjustment systems. LAC countries will also need to consider the redistributive effects of the CBAM and the possibility of it triggering trade conflicts or issues with other multilateral agreements, such as the Paris Agreement.

To minimize the negative economic impact of these environmentally motivated trade measures and to maximize the impact on emissions reductions, **LAC countries should begin by monitoring the technical aspects of the measures applied by their partner countries** so that they are better prepared when measures start to be applied. Key capabilities to achieve this include capacity building, knowledge of methodologies, and data collection.

With regard to the provisions of trade agreements, **more evidence needs to be generated on whether the inclusion of environmental standards is a barrier to trade.** Berger et al. (2018) find that the more environmental measures are included in agreements, the greater the negative impact on trade; however, they find no evidence of "green protectionism" associated with these measures and against the competitiveness of developing countries. Second, while some agreements contain **provisions on implementation, this tends to be poorly documented**, and empirical evidence is limited to a few specific cases. Third, more research is needed on the long-term environmental effects of environmental standards in agreements and the transmission channels between these standards and the environmental impact of trade. Fourth, there is a need to **examine why countries adopt environmental regulations,** especially the more enforceable ones. Some researchers suggest that they were already in the process of implementing them; others argue that they do so in response to pressure from trading partners. Even if the same type of measures are at stake and the same country is a signatory to two environmental agreements with EPs, the scope, forms of implementation, and commitments enshrined in these may differ. Our analysis based on these databases is a useful starting point, but **more detailed studies are needed to compare the scope of each agreement and the commitments they entail.**

In the specific case of the due diligence process for the EUDR, one important factor is capacity-building in **certifying the traceability** of the value chains for commodities at risk of deforestation.

With regard to the CBAM, in addition to an **accounting, verification, and management system,** LAC countries need to implement a carbon-pricing system that limits the transfer of revenues to the EU (and other economies applying similar mechanisms) through border adjustments, thus raising revenues that can be reinvested at the national level for environmental purposes. Those that already have systems in place need to align their prices with those of the EU market. **Harmonizing carbon-pricing systems within LAC** would have the added benefit of avoiding the costs of multiple systems being in place, which not only creates tensions but also increases compliance and administrative costs. Regional coordination is essential in this regard.

At the same time, LAC countries need to **adapt commodity production to the challenges posed by climate change** by introducing technologies to increase productivity and sustainability. These include innovations such as bio-inputs, 4.0 technologies, and plant and animal genetics. International trade plays a fundamental role in this process and can have a positive impact on climate change through the technical channel.

In this regard, given the shortage of fiscal resources in the region, LAC's economies **need to access external financing,** which is increasingly available to support investments aligned with climate objectives, including reducing GHG emissions (Dietrich Brauch, 2021). To do so, the international community must find a multilateral solution to effectively mobilize the international financing for developing countries contemplated in the Paris Agreement. This needs to be used to support mitigation strategies, the productive transformation required by climate change, and the adaptation of value chains to increasingly frequent "green" trade restrictions.

At the 2023 Conference of the Parties of the UNFCCC (COP 28), the parties reached an agreement on implementing the loss and damage fund. Commitments to the fund totaled more than US$700 million. The second replenishment of the Green Climate Fund was boosted by six countries pledging new funds, bringing total pledges to US$12.8 billion from 31 countries, with more contributions expected. Multilateral development banks are also strengthening their capacity to invest in assets that are socially, environmentally, and economically sustainable. In particular, the IDB Group plans to triple direct and mobilize climate finance in LAC to reach US$150 billion over the next decade.

Another important factor at the multilateral level is maintaining the WTO's transparency mechanisms and its role as a forum for dialogue, as this could help mitigate possible trade frictions resulting from any of these measures. LAC would do well to present its position at these and other international forums, given that many of the countries in the region are among the lowest global emitters of GHGs but would be among the most affected by measures such as those being adopted by advanced economies. Similarly, LAC countries have only limited resources for transitioning to a low-carbon economy.

Addressing climate change through trade policy poses profound challenges for the region, which must adapt quickly to minimize negative impacts on the economy and maximize climate change mitigation. LAC could play a key role in promoting sustainable practices while fostering economic growth and resilience to climate challenges. This could be achieved by viewing trade not as a threat to the environment but as a vital tool in achieving the triple aim of increasing prosperity, reducing inequality, and mitigating climate change.

ANNEX

MAIN FEATURES OF THE DUE DILIGENCE PROCESS

There are three stages to the due diligence process for the DFP[27]

1) The first step entails collecting basic information on the product to be traded, including a product description, quantities, the country where it was produced, contact information for the trading partner, and so on. The new requirements include the geolocation of the land where the product has been grown or reared; conclusive, verifiable information guaranteeing the traceability of the finished products and/or raw materials used; and verifiable references that the goods have been produced in compliance with the relevant legislation in the producing country.

2) During the second stage, companies must analyze and assess the risk of potential noncompliance in the supply chain based on the information gathered during the first stage and a classification system to be developed by the Commission[28], which will classify countries according to the risk they present for producing products that are not deforestation-free. There will be three categories: low risk, standard risk, and high risk[29] Some additional assessment criteria will also be included, such as i) the forested area of the country of origin of the good and the region from which it is sourced; ii) the prevalence of deforestation or forest degradation in the country, region, and specific origin of the good; iii) corruption levels, the prevalence of falsification of documents and data in the country; iv) the risk of products being mixed with others of unknown origin or produced in areas affected by deforestation or forest degradation; and vi) additional information such as certifications. If the goods in question are found to be deforestation-free, they will be permitted to enter and leave the EU.

3) If the risk assessment is unsatisfactory, the third stage in the process will be set in motion, whereby companies will have to implement risk mitigation measures to reduce their risk levels to zero. These might

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[27] · For a more detailed analysis, see chapter 2 of this publication, “The Impact of the EU Green Deal on Exports from Argentina.” These obligations will apply from the end of December 2024 and the end of June 2025 for MSMEs.

[28] · This is referred to in the law as benchmarking system.

[29] · When the law enters into force, all countries will be classified as standard risk. The list of low- and high-risk countries will be published by December 30, 2024, and will be updated as new evidence becomes available.

include requests for additional information, data or documentation, surveys, or independent audits. If countries are classified as low risk according to the benchmarking system, they will be able to access a simplified due diligence procedure that will allow them to dispense with stages 2 and 3.

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